Launch Canada
2025 Challenge

Team 14, Carleton University InSpace

Carleton University, Ottawa, Ontario, K1S 5B6, Canada

August 29, 2026

Acknowledgements

Competing Team 14 of the 2025 Launch Canada Challenge would like to thank the following individuals/organizations for their contributions to the success of the team:

- Altium
- Apache NuttX
- Ansys
- Carleton Student Engineering Society
- Carleton University Department of Mechanical and Aerospace Engineering
- Carleton University Faculty of Engineering and Design
- Dassault Systèmes SOLIDWORKS Corp.
- Kostiuk Engineering Fund Collective
- Lockheed Martin Canada
- MDA Space
- Metal Pros Ottawa
- MHI RJ
- Protocase
- Seco Seals Inc.
- Swagelok Ottawa
- Reaction Dynamics

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Abstract

CU InSpace, Carleton University's Rocket Engineering Design Team, is participating in the Advanced Launch Competition of the 2025 Launch Canada Challenge. The target altitude is set to 30,000 ft. The main engineering goals of the rocket for this year are to further improve the reliability in the vehicle's wireless reefed recovery system and demonstrate our student-designed hybrid rocket engine in flight for the first time. The vehicle also aims to feature a decreased number of commercial components, as the team further develops their in-house manufacturing capability. The design and manufacturing of the rocket is detailed herein, documenting the notable achievements and challenges faced in the development of the rocket.

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Nomenclature

ADC Analog to Digital Converter.

ASME American Society of Mechanical Engineers.

BGA Ball Grid Array.

BTH Below-the-Hook.

COTS Commercial Off The Shelf.

CSV Comma Separated Value.

DMA Direct Memory Access.

EEPROM Electrically Erasable Programmable Read-Only Memory.

eMMC Embedded MultiMediaCard.

FEA Finite Element Analysis.

FOD Foreign Object Debris.

FSK Frequency Shift Keying.

FSM Finite State Machine.

GPS Global Positioning System.

HFD Hazardous Fragmentation Distance.

HRAP Hybrid Rocket Analysis Program.

I/O Input/Output.

ID inner diameter.

IMU Inertial Measurement Unit.

IPA Isopropyl Alcohol.

ISM Industrial, Scientific, Medical.

JSON JavaScript Object Notation.

LAN Local Area Network.

Li-Ion Lithium Ion.

LoRa Long Range radio.

MCC Mission Control Center.

MCU Microcontroller Unit.

MEOP Maximum Expected Operating Pressure.

MMPDS Metallic Materials Properties Development and Standardization.

OD outer diameter.

ORB O-Ring Boss.

PCB Printed Circuit Board.

PCC Pad Control Center.

POSIX Portable Operating System Interface.

PPE Personal Protective Equipment.

PTC Push To Connect.

PTFE Polytetrafluoroethylene.

PWM Pulse Width Modulation.

RF Radio Frequency.

RTOS Real Time Operating System.

RTV Room Temperature Vulcanizing.

SMA Subminiature A.

SNR Signal to Noise Ratio.

SPI Serial Peripheral Interface.

SRAD Student Researched and Designed.

TCP Transport Control Protocol.

TNT Trinitrotoluene.

UART Universal Asynchronous Receiver-Transmitter.

UC Urbanski-Colburn.

UDP User Datagram Protocol.

UI User Interface.

uORB Micro-Object Request Broker.

1 Introduction

Carleton University has a long-standing relationship with aerospace engineering, opening its first aerospace program in 1988, making it the first university in Canada to offer such a program. It now offers four different aerospace specializations; Aerodynamics and Vehicle Performance, Lightweight Structures, Avionics, and Space Systems Design. Carleton has always had a proud history of hands-on learning experiences as well, being the first Canadian university to develop a capstone project in the student's final year of study to showcase the skills learned during their time at Carleton. Additionally, Carleton has always endeavoured to excel in its extracurricular clubs, from FSAE cars to UAV Blackbird. Carleton has always provided support and financial assistance to students looking to further their education with hands-on experience. CU InSpace is one of those teams, bringing dozens of undergraduate students from various engineering disciplines together to design, develop, and test high-performance sounding rocket systems.

CU InSpace is a dynamic engineering competition team that specializes in the production of high-powered rockets. The team is comprised of undergraduate students that have a passion for rockets and design. Founded in 2014, CU InSpace works to compete in the annual Spaceport America Cup and Launch Canada challenges, launching to 30,000ft and 15,000ft above the surface of the Earth.

This year, CU InSpace is competing in the Advanced Launch (Hybrid) competition of the Launch Canada Competition with our rocket that has yet to be named, later referred to as CR25H "Quarter Pounder". CR25H is expected to be a 14.48 ft (4.412 m) tall high-power sounding rocket designed to reach an expected apogee of 25,779 ft [7857 m] . The vehicle weighs 110.57 lb [50.16 kg] when fully loaded and is propelled by a paraffin wax and nitrous oxide hybrid rocket motor (71% O-Class). Once the vehicle reaches its expected apogee, the student-developed reefed-main parachute is ejected from the rocket using two Raptor CO_2 ejection systems. The timing of the deployment and the relaying of telemetry is facilitated by the vehicle's avionics system, consisting of a CubeSat-standard avionics bay and custom software with several redundant COTS systems. The student-developed main parachute will disreef at an altitude of 1500 ft (457.2 m) and fully inflate.

2 Team Overview

2.1 Team History

CU InSpace has been competing in international rocketry competitions since 2017, starting with our first-ever high-power rocket, CU L8R. In past years, the team competed in the 10,000 ft COTS solid category; launching four rockets at competition and participated in one virtual competition. The team's inaugural year at SAC was back in 2017, the rocket was successfully launched, but not retrieved - perhaps alluded to by the name CU L8TER, it was never seen again. Over 2018 and 2019, the team proceeded with further improvements to the airframe and recovery systems - successfully recovering both the rocket as well as payloads. In 2022, the team once again competed in the 10,000 ft COTS competition with McNominal, this was a successful return to flight for the team after COVID. It was also the first time many of our members had the opportunity to see a rocket launch in person and provided a valuable learning experience on how to improve future iterations of the CU InSpace rockets.

Figure 1: 2017 Competition Team with CU L8R

Figure 2: 2018 Competition Rocket: Orbital Redenbacher

Figure 3: 2019 Competition Team with Chef Boy Apogee

Figure 4: 2022 Competition Team with McNominal

In 2023, InSpace progressed from the 10,000 ft COTS to the 30,000 ft COTS category at SAC with our CR23 rocket "NoName". The vehicle was the first in the team's history to break the sound barrier achieving Mach 1.4 along with a fully nominal flight. Placing 34th overall and 5th

in-category. In parallel to our SAC launch efforts, InSpace made its inaugural launch at the Launch Canada challenge in the summer of 2023 with “McDouble”, demonstrating the refurbishment and re-flight capabilities of a previous launch vehicle. The vehicle served as a testbed for the parachute reefing system, and a vessel for continuing improvement to lightweight aluminum structure design within the team. The team received the Team Spirit Award for both competitions in the summer of 2023.

Figure 5: 2023 SAC Competition Rocket: No Name

Figure 6: 2023 LC Competition Team with McDouble

Last year, InSpace flew our second 30,000 ft COTS vehicle at SAC with our CR24 rocket ”Red Bullistic”. Our fastest vehicle to date, achieving Mach 1.86 along with a fully nominal flight on the Cesaroni N5800 ”rocket killer”. However, we felt that much was left on the table due primarily to unfavourable weather and missing the recovery window on the final launch day, placing 91st overall and 17th in-category. At the Launch Canada challenge last summer, the team flew the third iteration of the “McNominal” series with “McTriple”, demonstrating re-flight of a previous launch vehicle for the second time. McTriple’s flight also marked the first successful reefed parachute recovery for the team, guiding our recovery architecture for this competition year.

Figure 7: 2024 SAC Competition Rocket: Red Bullistic

Figure 8: 2024 LC Competition Team with McTriple

Additionally, our team has ramped up our STEM outreach and event presence within the community and within Ottawa at large. As one of Carleton’s largest design teams, the team provides tangible hands-on engineering experience for over 150 design team members, and have brought over

40 students within the past year to both Spaceport America Cup as well as the Launch Canada challenge. CU InSpace regularly attends outreach events both internal and external to Carleton University, with the goal of getting more students involved in both the rocketry and general STEM community.

2.2 Team Composition

CU InSpace is composed of a wide variety of students from Carleton engineering and other programs at Carleton University. A breakdown of the team members can be seen below in Figure 9. The members of the team are also broken down into design subteams which helps with logistical organization and allows team members to develop and apply their skills in fields that most interest them. The subteams and their respective populations can be seen in Figure 10.

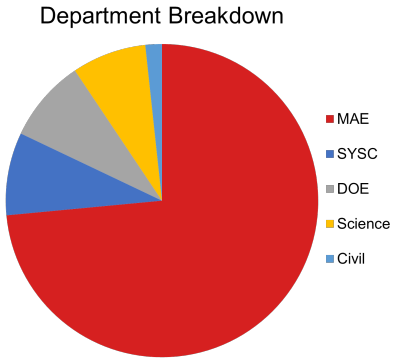

Figure 9: Departmental Breakdown

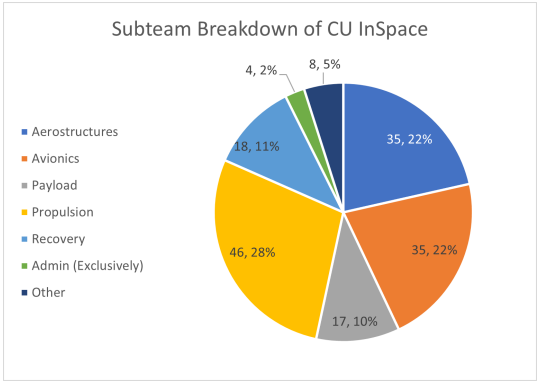

Figure 10: Subteam Breakdown

The team successfully navigated the COVID-19 pandemic and major leadership transitions, keeping membership numbers relatively high. The team has drastically increased its influence in the Carleton Engineering community, which has allowed the team to strive for the most ambitious goals in the team's history. Such goals as performing multiple launches in a single year, upgrading categories in the Spaceport America Cup, and finishing the development and flying the team's hybrid rocket motor at this upcoming Launch Canada challenge.

2.3 Team Structure

Figure 11 shows the current layout of CU InSpace, with the executives in each role named. Zack Renschler as team president performs the roles of Team Captain and Chief Engineer. Antoine Brule as the team's senior-most propulsion lead performs the role of Chief Safety Officer. The team's Faculty Advisor is Professor Jason Etele.

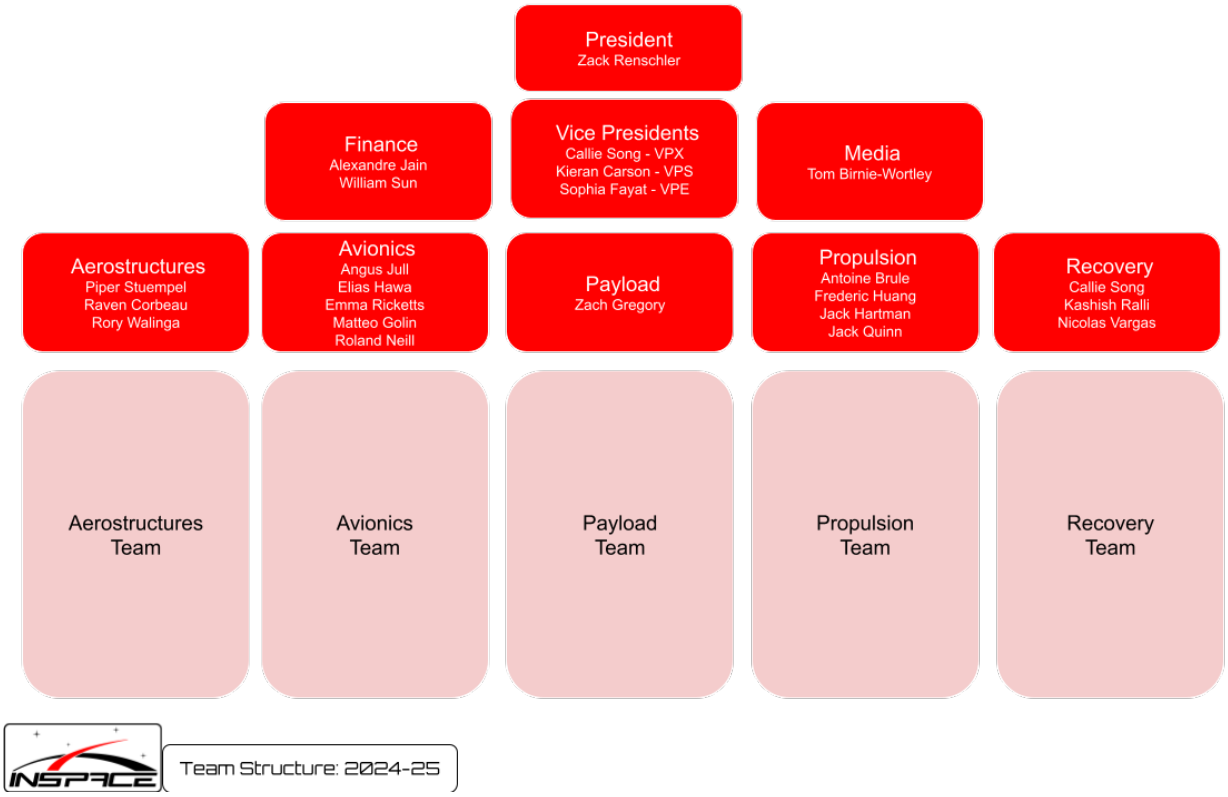

Figure 11: Layout of CU InSpace design team between 2024-2025

3 Outreach

Over the course of the 2024-2025 year, CU InSpace has continued to expand its presence in STEM youth engagement and academic outreach. The team regularly participates in school events such as open houses and STEM fairs, displaying rockets to prospective students and answering questions about Carleton University's aerospace stream and other engineering programs. CU InSpace has also maintained a strong relationship with Virtual Ventures, a Faculty of Engineering and Design organization that facilitates technology and engineering programs for youth. Team representatives have engaged with Ottawa youth through multiple events, reaching a broad audience from elementary to high school students. Additionally, members have presented at local high schools, inspiring students to explore aerospace and rocketry. These outreach efforts aim to not only educate young minds but also build a pipeline of future engineers passionate about space exploration and rocketry.

Figure 12: CU InSpace Presenting to SHAD

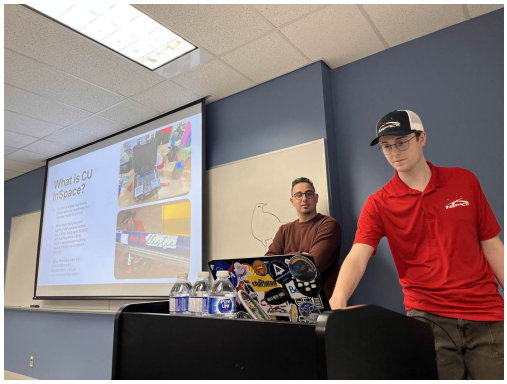

Figure 13: CU InSpace CS Students Talk

In the academic community, CU InSpace members have attended multiple conferences, including the Space Canada Horizons Conference and the SEDS Canadian Space Conference. In addition to this, members of CU InSpace's executive team presented research conducted by CU InSpace at the 2022 International Astronautical Congress in Paris, France. These opportunities allow team members to showcase their research, gain valuable insights from industry professionals, and build connections within the aerospace and rocketry community. Avionics team members have also given presentations to computer science students, highlighting how their involvement in CU InSpace has helped them develop key skills and secure employment opportunities. This event also served to introduce computer science students to the open-source development community by emphasizing CU InSpace's relationship with the Apache NuttX project. By engaging with students from diverse academic backgrounds, the team fosters interdisciplinary collaboration and knowledge-sharing.

CU InSpace also continues to host its annual hands-on events designed to engage new members and spark excitement for rocketry from the start of the year. The annual Small Rocket event was once again held this fall, bringing together over 75 participants who were divided into 18 teams. Each team designed and built a model rocket with a strong emphasis on safety, stability, and successful recovery. These rockets, powered by A and B class motors, provided participants with valuable hands-on experience in rocketry fundamentals. This event served as an introduction for new members, allowing them to familiarize themselves with key rocket components and the principles behind a successful launch, setting the stage for deeper involvement in the team's projects throughout the year.

Figure 14: Participants at Small Rocket 2024

This year, CU InSpace is collaborating with uOttawa Rocketry to support members in obtaining their Level 1 (L1) and Level 2 (L2) high-powered rocketry certifications. The team is facilitating material orders, hosting building workshops to teach the fundamentals of amateur rocketry, and planning a group trip to launch events later in the year. These hands-on experiences provide team members with invaluable practical skills in design, fabrication, and launch operations, further preparing them for careers in the aerospace industry, or just allowing them to go further into the amateur rocketry hobby. This initiative builds on the success of several team members who earned their first L1 certifications at Launch Canada 2024. As part of this effort, experienced members are mentoring new participants, guiding them through the certification process and ensuring they understand best practices for safe and successful rocketry operations.

In previous years, CU InSpace ran the McNth program, which focused on iterative rocket design and continuous learning outside the annual design cycle for IREC. The McNth program allowed members to refine previous airframe designs, analyze past successes and failures, and implement improvements in subsequent iterations. Additionally, the team held "Thank God It Flew Fridays" (TGIFs) as dedicated sessions for studying past rocket designs and refining engineering processes. While these programs are not continuing this year, they played a crucial role in advancing the team's technical expertise and providing members with real-world engineering challenges to solve.

This year, the Avionics subteam has begun using the NuttX Real Time Operating System (RTOS)

Figure 15: L1 Certification Successful Recovery

Figure 16: L1 Certification Rocket on the Rail

to support our flight computer and hybrid engine control system (described further in Sections 9.1.1 and 9.2). The NuttX community has been extremely receptive to the teams inquiries, so much so that the team has developed somewhat of a rapport with the NuttX core contributors and the wider NuttX community. Given the open-source nature of the NuttX RTOS and the ease at which contributors can implement support for sensor peripherals, several Avionics members of CUInSpace have been able to write code that is now officially included in the NuttX repository. This is extremely beneficial as this has not only given team members the rare incentive and opportunity to contribute to open-source software but has also given CUInSpace a favorable reputation in terms of contribution to open-source software which is rare among amateur rocketry teams.

4 Architecture

4.1 Architecture Overview and Layout

The primary target of the CR25H "Quarter Pounder" airframe development is to implement the team's first SRAD hybrid motor into a flight vehicle. As this is the team's first time flying a hybrid motor, many new design considerations must be made, especially with regards to the oxidizer tank and respective plumbing system. The airframe is organized into four main sections.

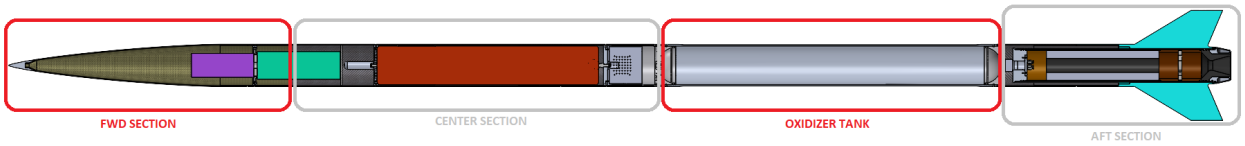

Figure 17: Main Airframe Components and Sections Breakdown, CR25H, Launch Canada 2025

The airframe is composed of SRAD roll-wrapped carbon fiber body tubes mated using COTS carbon fiber couplers, an SRAD fiberglass and aluminum nose cone, carbon fiber layup over a 3D-printed fincan, and a hybrid motor. Key dimensions of the airframe are provided in Table 1. Furthermore, preliminary breakdown of the airframe based on the diagram in Figure 17 are presented in Table 2.

Table 1: Key CR25H Airframe + Hybrid Motor Dimensions

Components	Dimension [Imperial]	Units
Tip-to-Tail length	4412 [173.7"]	mm
Fin Semi-Span	129 [5.08"]	mm
Airframe OD	151.5 [5.96"]	mm
Oxidizer Tank Diam.	152.4 [6"]	mm
Oxidizer Tank Len.	1212 [47.7"]	mm
Combustion Chamber Diam.	124 [4.88"]	mm
Combustion Chamber Len.	786 [30.94"]	mm

Table 2: Airframe Breakdown by Section

Section	Structural Components	Manufacturing
Forward (Black)	Nosecone Tip	MAE Machine Shop - 6061-T6 Aluminum Alloy
	Bulkhead and Support Ring	MAE Machine Shop - 6061-T6 Aluminum Alloy
	Nosecone	SRAD Fiberglass Layup Nosecone
	Nosecone Coupler	COTS Fiberglass Coupler
Center (Green)	Bulkhead and Support Ring	MAE Machine Shop - 6061-T6 Aluminum Alloy
	Hybrid Retention Mount	MAE Machine Shop - 6061-T6 Aluminum Alloy
	Body Tube	SRAD Carbon Fiber Roll-Wrapped Body Tube
	Break Coupler	COTS Carbon Fiber Coupler
	Ox Tank Coupler	MAE Machine Shop - 6061-T6 Aluminum Alloy
Ox-System (Red)	Oxidizer System	MAE Machine Shop
Aft (Blue)	Stringers	MAE Machine Shop - 6061-T6 Aluminum Alloy
	Motor Mount	MAE Machine Shop - 6061-T6 Aluminum Alloy
	Combustion Chamber	MAE Machine Shop - 6061-T6 Aluminum Alloy
	Body Tube	SRAD Carbon Fiber Roll-Wrapped Body Tube
	Fincan Coupler	COTS Carbon Fiber Coupler
	Fincan	SRAD Carbon Fibre Fincan - Wet Layup over 3D-Print

5 Mission Concept of Operations (CONOPS)

The mission concept of operations (CONOPS) for this rocket is made up of nine stages, ranging from pre-flight activities to post-flight recovery. A flowchart of the progression of these stages may be seen in Figure 18. An illustration of each portion of flight is also provided in Figure 19.

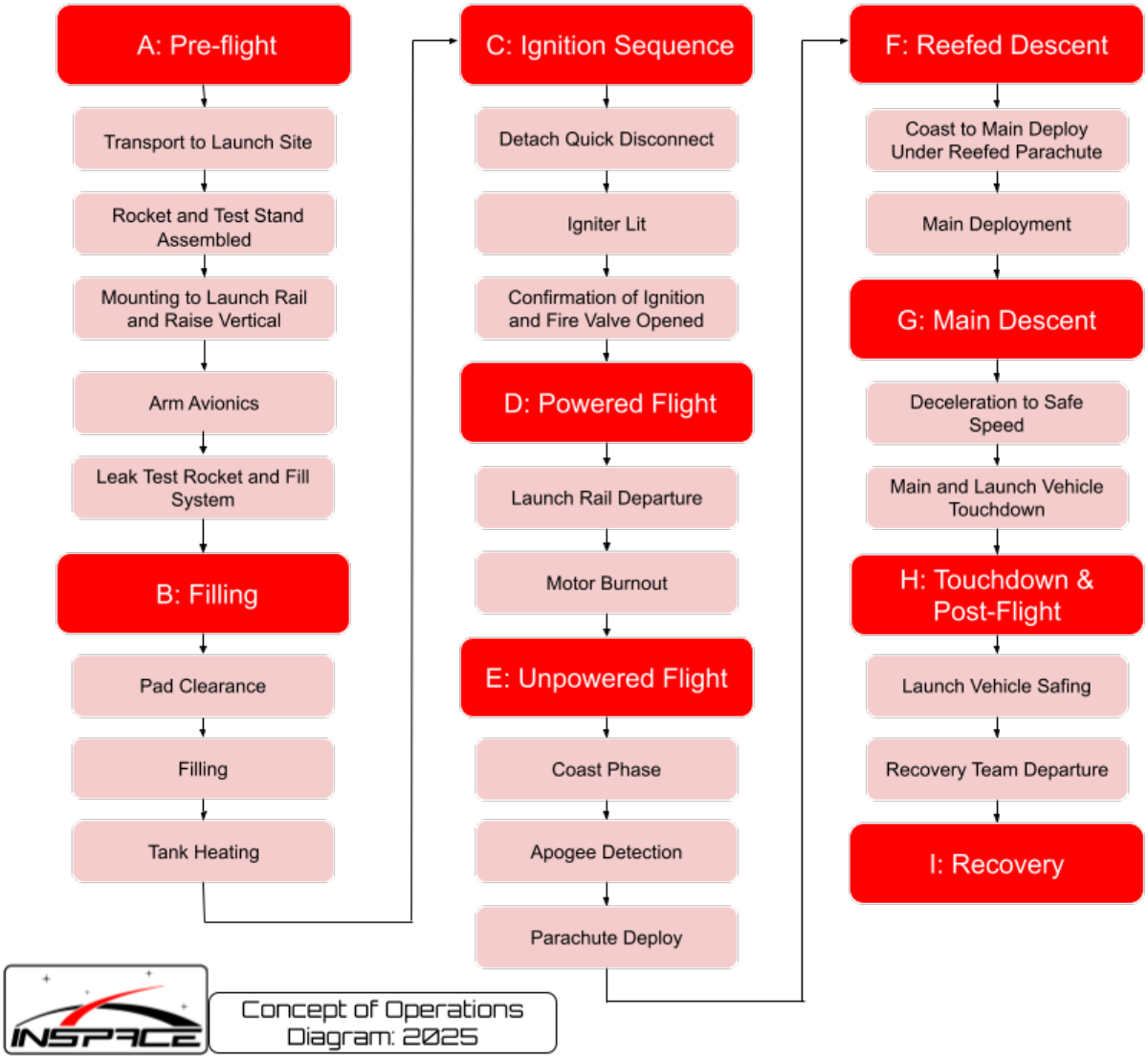

Figure 18: Mission Concept of Operations Flowchart

A. **Pre Flight**

Start: Assembly has been completed and the vehicle and has been approved for transportation to the launch site.

End: The vehicle is on the launch rod, raised to a vertical position, avionics armed, and fill

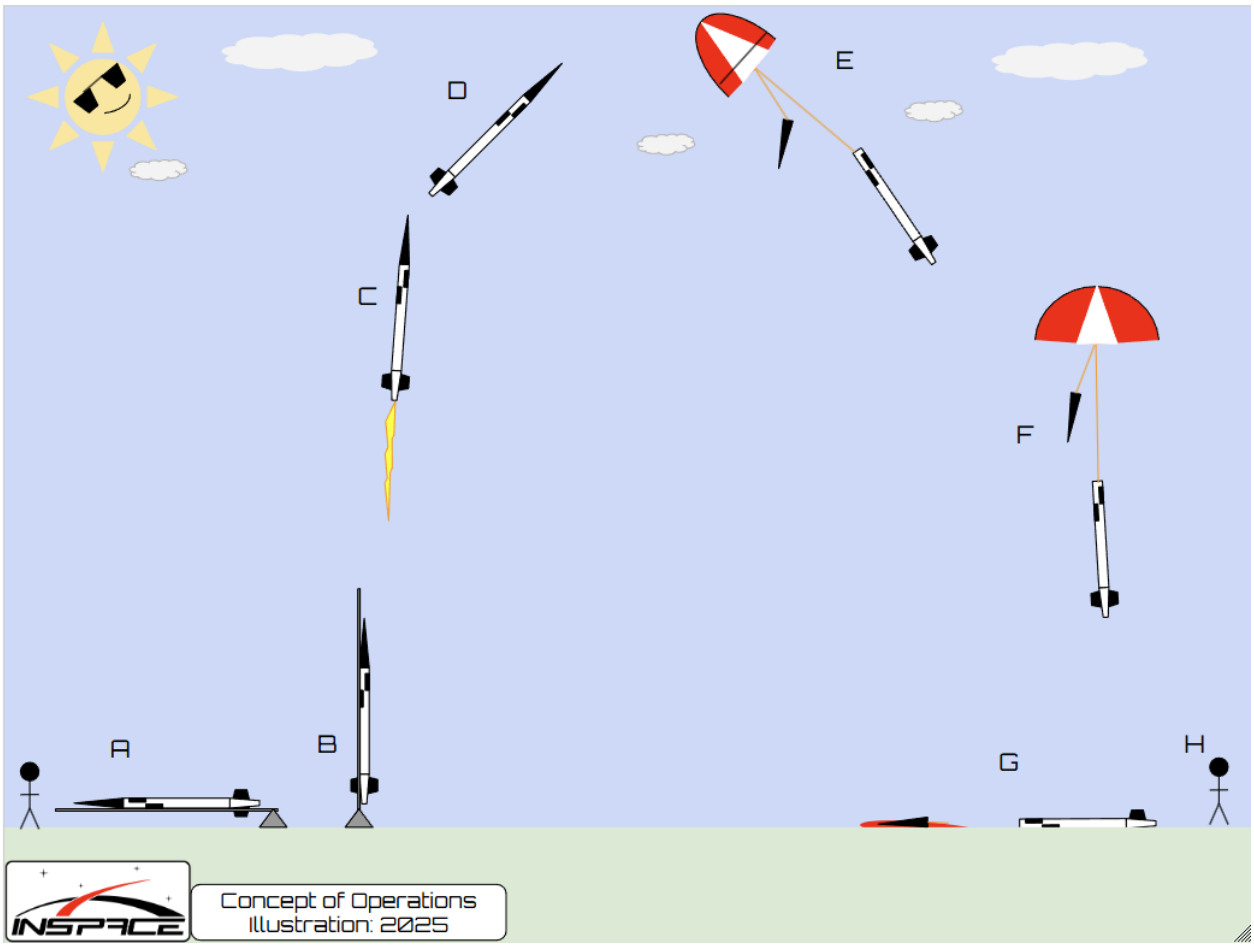

Figure 19: Mission Concept of Operations Diagram

system leak tested.

Once the test stand, launch vehicle and payload assembly has been completed at the assembly site, the integrated vehicle and test stand are transported to the launch site. At the pad, the launch rail is taken horizontal, and the rocket is carefully loaded onto the horizontal launch rail through its rail attachment buttons located on its side. Once secure, the vehicle may then be raised to the vertical position and connected to the fill system. Rocket and fill system are leak tested by pad personnel prior to filling. Key is placed in the rocket side control box to lockout ignition, and avionics are armed.

B. **Filling**

Start: Pad personnel clear the pad and return to the mission control centre.

End: The vehicle's oxidizer tank is full and at temperature.

Pad personnel depart the pad and return to the mission control centre, inserting the key to arm it. Filling of the rocket begins and lasts 20 minutes. A 10-minute tank heating phase begins. Rocket is ready to be launched at any moment.

C. **Ignition Sequence**

Start: The quick disconnect to the vehicle is disconnected.

End: The ignition signal has been sent and the igniter has been lit and the fire valve opened.

When permission is acquired from LC staff, the quick disconnect is detached and the ignitor is lit. Upon visual confirmation of the ignitor being lit, the main fire valve is opened. As the ignitor has successfully fired and the fuel grain successfully ignited, the combustion chamber begins to come up to pressure to accelerate the vehicle up the launch rail. The avionics detect the change in acceleration and altitude to indicate that powered flight has begun.

D. **Powered Flight**

Start: Acceleration is detected by the avionics.

End: The avionics detect motor burnout at nominal burn duration.

As the rocket accelerates up the launch rail, the avionics bay begins to log sensor and telemetry data to the local SD card. The structures of the vehicle transmit the thrust from the engine casing to the surrounding airframe to lift the entire vehicle. Once the launch rail has been cleared, the fins maintain stability of the vehicle through the entire powered and unpowered upwards flight. The vehicle remains in powered flight until the avionics detect the acceleration and altitude is consistent with the coast period. Once this threshold is crossed, the motor has burned out and unpowered flight begins.

E. **Unpowered Flight (Coast)**

Start: The motor has burned out and the rocket is coasting upwards.

End: Apogee is detected and the recovery system activates.

As the motor burns out, the launch vehicle decelerates as it coasts to an expected apogee of 25,779 ft [7857 m] . The avionics and payload continue to collect data, the avionics continually check the altitude to determine when apogee has been detected. The recovery system is activated for deployment once apogee is detected.

F. **Reefed Descent**

Start: The vehicle has detected apogee and the recovery system deploys the reefed parachute.

End: The rocket has descended to an altitude of 457 m (1500 ft) above the surface as the main parachute deployment begins.

Once apogee is detected, the avionics bay sends the signal for the recovery system to activate. The Raptor recovery system then bursts a CO2 cartridge. Approximately two seconds later, a second CO2 cartridge is burst for redundancy. This pressurizes the internals of the rocket and facilitates nose cone deployment and parachute ejection. The parachute then inflates to the extent it is able when reefed, pulling the payload out with it as well. The descent rate is slowed to a constant speed of approximately 100 ft./s. At an altitude of 457m (1500 ft) AGL, two Easymini altimeters sewn into the skirt of the parachute each send a signal to the two Mako Line Cutters also sewn into the skirt for main parachute deployment. The payload will also send out signals from their own altimeters at this altitude in order to detach themselves from the main parachute riser.

G. **Main Descent**

Start: The rocket has descended to an altitude of 457m (1500 ft) AGL, the parachute has been successfully disreefed.

End: The launch vehicle touches down on the surface.

Once the main parachute deploys, the launch vehicle descends at approximately 30 ft/s awaiting touchdown. The avionics register that the main parachute has deployed and continues to log data.

H. **Touchdown & Vehicle Post Flight**

Start: The launch vehicle contacts the ground.

End: Recovery team departs to the last transmitted coordinates of the launch vehicle.

Once the launch vehicle detects touchdown, it begins the process of safing vehicle's systems. The avionics location is transmitted to the team's ground station and the recovery team begins planning the trip to recover the vehicle. Once launch vehicle location is known, routes are planned, and clearance is attained, the recovery team departs to the expected location of the launch vehicle.

I. **Ground Recovery**

Start: The recovery team has been cleared to depart for the recovery of the launch vehicle.

End: The recovery team returns the launch vehicle to the mission control centre for inspection.

Once obtaining permission to depart for the landing zones, the recovery team proceeds towards the last known location of the vehicle. Once at the landing zone, the vehicle is safely recovered. This includes disconnecting unignited charges for which protective equipment must be worn. The vehicle is then transported back to the control centre for inspection of damage to the vehicle and any hazards to the team. The rocket is then returned for post-flight evaluation.

5.1 Mission Analysis

CU InSpace's paramount goal this 2024-25 competition season is the inaugural flight of our SRAD hybrid motor, which has been under development for several years. Therefore, the design of the CR25H air-frame was largely driven by the mission requirement of flying our in-house developed hybrid motor. In summary the system level requirements (SLRs) for this year's rocket are as follows:

- Successfully launch using our SRAD hybrid propulsion system
- Reach an altitude of 30k ft +/- 3k ft
- Successfully recover the airframe in a re-flyable state
- Work within a total annual budget of $ 85k CAD
- Complete functionally required CR25H airframe and associated SRAD motor by the first week of August, 2025.

Several high level requirements (HLRs) and low level requirements (LLRs) were developed to achieve these SLRs. In summary, a combination of calculations, simulations (ANSYS and SolidWorks FEA, OpenRocket and RAS Aero), and physical testing was used to verify that our design meets the defined SLRs. More details on specific derived HLRs and LLRs can be found in Appendix A. In terms of budgetary requirements, spending was tracked through custom spreadsheets throughout the year and adjusted in real-time. These spreadsheets allowed us to track the team's current liquid capital as well as make predictions about upcoming purchases to ensure we fall within our budget. Two VPs of Finance were appointed this year to superintend the team's financial aspects. Furthermore, a project management tool, ProofHub, was implemented this year with the goal of improving timelines and communication between subteams while working on collaborative projects. We plan to continue using a similar project management tool, Jira, moving forward. Furthermore, the team has been working on implementing a 'wiki' to culminate data, lessons learned from previous seasons, and general information. Some subteams are already effectively using knowledge bases, but the goal for next season is to build a database that covers the scope of the whole team.

6 Aerostructures

6.1 Introduction

CR25H is Carleton's newest primarily composite airframe. Carleton's No Name was the first to focus on the integration and development of primarily structural composites, moving beyond EFRIS design. Then, Redbullistic saw major improvements with weight optimized components and composite manufacturing techniques that pushed the teams boundaries. CR25H looks to improve upon system integration, assembly time, and flight performance while taking inspiration from Carleton's previous composite airframes. Use of Carleton's first SRAD body tubes are a showcased project in the aerostructures section of this report.

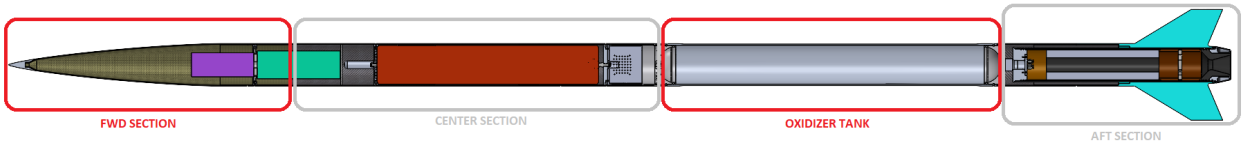

Figure 20: Structure Sections

6.2 Nose Section

The purpose of the nose cone system is threefold; to reduce drag during flight in Mach numbers of interest, provide a structure to mount the upper bulkhead and support ring that attaches the payload and avionics stack to the airframe, and a method to permanently attach the composite nose cone to the forward body tube. The nose cone profile will follow a Von Kármán curve with a 5:1 ratio of length to diameter. This profile was chosen as it is known to have better performance in the transonic flight regime compared to most nosecone shapes. The shape provided the greatest improvement to our altitude when running simulations in RASAeroII. The nose section will be composed of a fiberglass body, a fiberglass coupler, and the forward internal bulkhead. This material was chosen instead of carbon fiber to allow for radio transparency for our telemetry. Attached to our forward internal bulkhead is our avionics stack and 3U cubesat payload. Figure 21 is a CAD model of the nose section assembly. The purple box pictured is a temporary representation of the avionics enclosure. The turquoise green box is a temporary representation of the 3U cubesat payload. Sandwiched between the avionics enclosure and payload are the forward bulkhead and support ring. For manufacturing reasons, it is difficult to obtain a sharp nose cone point. To overcome this, a machined 6061-T6 aluminum nose cone tip is epoxied to the tip of the nose cone body. This nose cone tip follows the optimized Von Kármán geometry outline above.

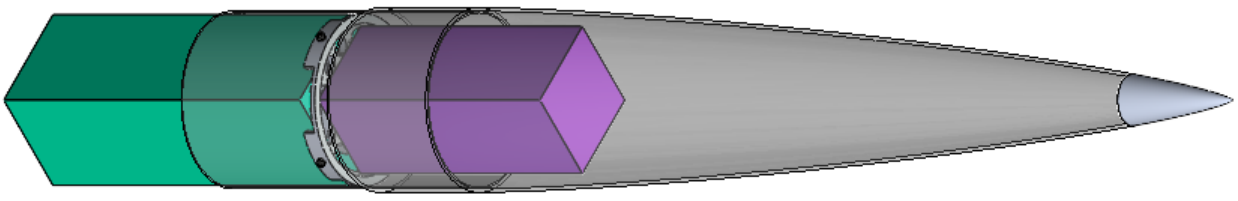

Figure 21: Nose Section Assembly

6.2.1 Nosecone Integration with Center Body Tube

A 100mm linear section is part of the main fiberglass nose cone layup. Inside of that is a fiberglass coupler (permanently fixed to the main nose cone) with holes for 4 radial bolts. As seen in Figure 22, these four bolts pierce the center body tube, through the nose cone coupler, through the support ring, and finally thread into the bulkhead. It is important to note that the support ring and nose cone coupler are epoxied onto their respective mating surface in addition to radial bolts to be permanently attached during launch and recovery events.

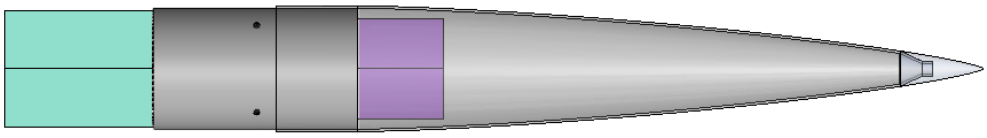

Figure 22: Nose-Center Coupler

6.3 SRAD Body Tubes

This is the first year the team will be using SRAD body tubes. The tubes consist of 8 layers of 3k plain weave that were roll wrapped with plies in a (0, 90) orientation. This orientation was chosen to maximize the bending resistance of each tube. The number of layers was chosen based on a crush test which is outlined in Subsection 14.1 as per the DTEG. Resin was applied to the carbon weave in a 1:1 ratio by mass during the layup to ensure sufficient impregnation of the weave.

Figure 23: Roll-Wrap of a Body Tube

After the layup concluded, wax paper was wound over the tube to provide compression and ensure a good surface finish. Minimal post-processing was required to achieve a successful body tube.

6.4 Twist and Lock Bulkheads

Our bulkheads aim to provide ease of integration for both the avionics and recovery systems, marking improvements to previous recovery bulkhead designs. They accomplished this objective by using a “Twist-and-Lock” system. The bulkhead is placed on the fixed support ring within the rocket, aligning its flanges to a slot within a guide. The bulkhead is then rotated within the guide until its holes align with the support ring holes. External fasteners are inserted that retain the bulkhead in its ”locked” position. The goal of this system is to allow the other subsystems within the rocket (i.e. the avionics and recovery) to be assembled outside and then lowered into the airframe with all connections made. Both bulkheads in addition to their embedded support rings are manufactured using 6061-T6 aluminum, and the retention bolts are 4 M4 fasteners.

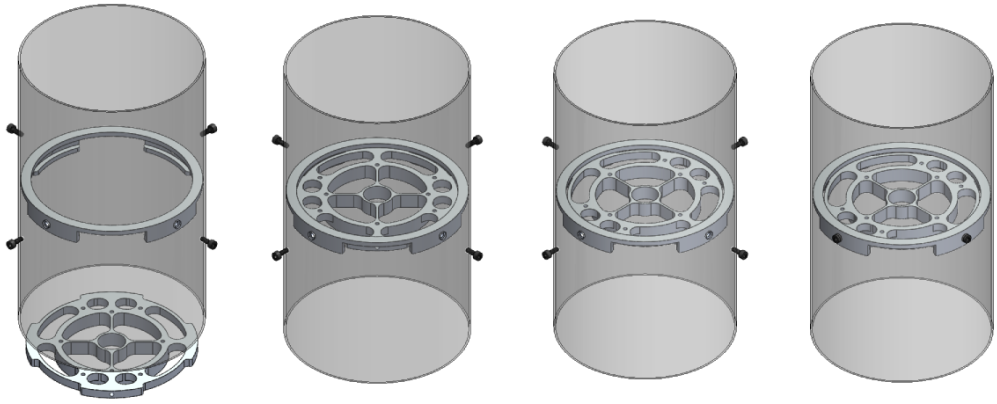

Figure 24: Steps to Assemble Twist and Lock Bulkhead

6.5 Avionics and Payload Mount

The upper mount carries the avionics stack and payload and joins with the upper support ring, which is internally fixed to the nose cone and nose cone coupler. The mount and support ring were designed with ease of assembly in mind. The mount has four radial tabs which interface with four adjacent slots in the support ring. Once the mount tabs are inserted into the support ring, it is rotated into the locked position, where four radial screws fix it to the support ring and nose cone. Shear calculations for the four radial screws can be found in Figure 171. One of the mount tabs and one of the support ring slots are shorter than the rest–this is to accommodate an angular positioning screw which stops the mount from rotating once it has reached the locked position. Statically, the upper mount and support ring pass by inspection due to the fact the lower mount and support ring were statically substantiated to far greater stresses.

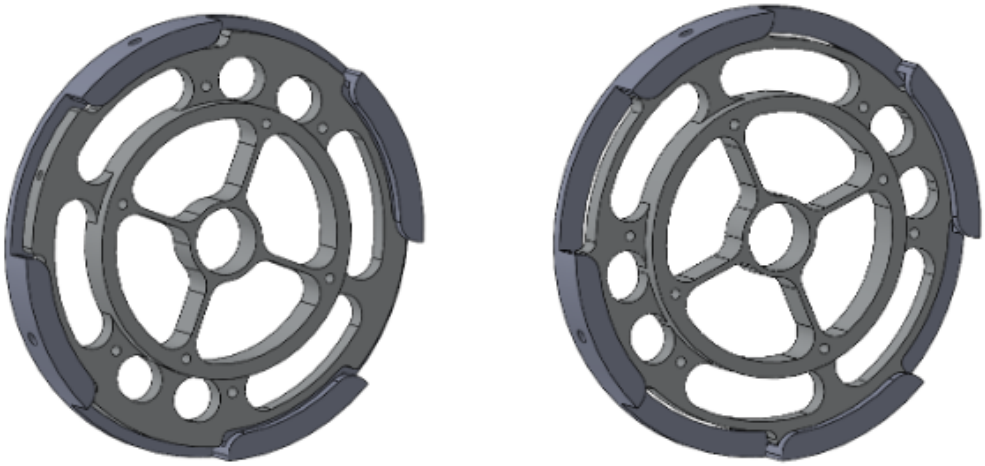

Figure 25: Avionics and Payload Bulkhead

6.6 Recovery Lower Mount and Support Ring

The lower mount carries the recovery systems–parachute, avionics, and raptor ejection system, and is where one of the parachute shock cord eye bolts are fastened. The lower mount is joined to the lower support ring, which is internally fixed to the main body tube. The lower mount and support ring must withstand the stresses generated by the parachute opening shock cord load and, as with the upper mount and support ring, are designed with ease of assembly in mind. The lower mount and support ring use identical tab-slot features for assembly. The lower mount has several machined pockets on its upper surface which remove weight, while having some extra thickness around the eyebolt hole for added resistance to bending stress.

Maximum stress on the mount is assumed to be shear stress of the tabs while maximum stress on the support ring is assumed to be shear stress of the corresponding areas where the mount tabs interface. Simulation results matched this assumption, showing that bending stresses between the eyebolt and the tabs were sufficiently below design limits. However, the simulations showed signs of mesh divergence in some of the elements where the mount tabs were supposed to be under shear, so the shear calculations were performed analytically. As shown, average shear stress calculations show that the mount and support rings can withstand the shear stresses generated by the shock cord opening load with safety factors well above 1.5.

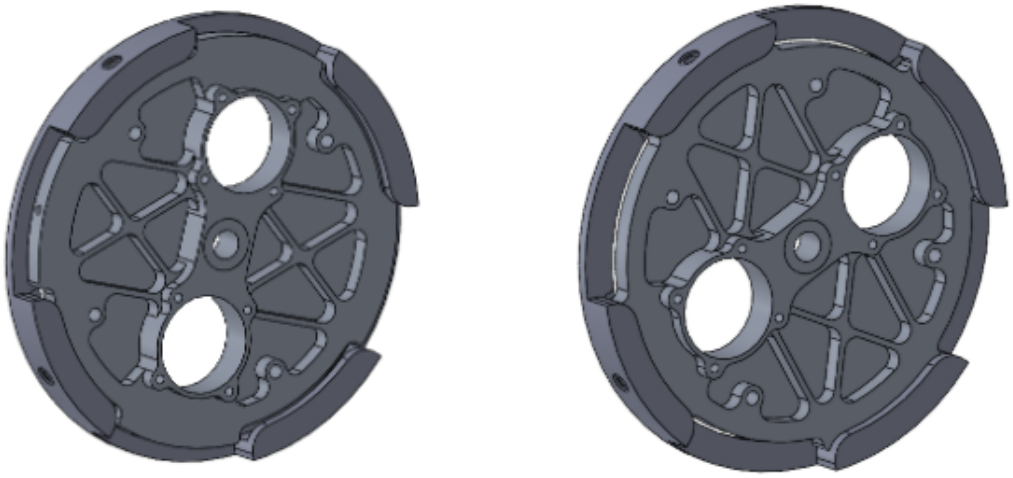

Figure 26: Recovery Mount and Support Ring

Found in Figure 172 are the shear calculations done for the 4 radial fasteners securing the recovery support ring and bulkhead with the center section bodytube. Assumptions for these calculations can be found in Appendix I.3. The resultant factor of safety was found to be 8.8.

6.7 Oxidizer Tank Coupler

The upper body tube and the oxidizer tank are joined together with an AL 6061-T6 coupler. The coupler is joined to the oxidizer tank using six radial countersunk M6 bolts. The counter sunk bolts help minimize drag and create a precisely aligned joint. The coupler extends into the upper body tube by one body caliber and is permanently joined using epoxy.

The coupler will be machined out of a 5.5" ID and 6.5" OD peice of tube stock by Engineering department's machine shop. Before applying epoxy to the coupler the mating surface will be degreased and sanded to allow the epoxy to bond properly.

Figure 27: Center-Oxtank Coupler

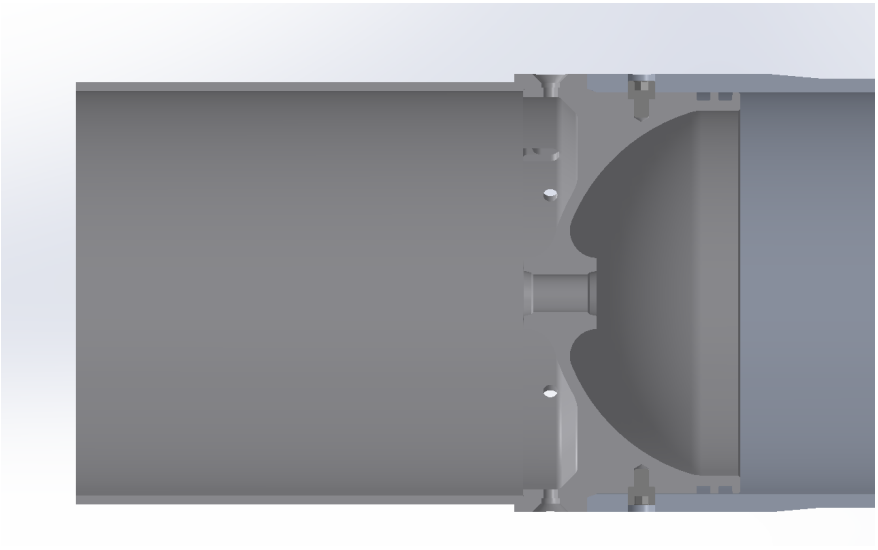

Figure 28: Center-Oxtank Coupler Cross-section

6.8 Nose Section Bulkhead

The nose section bulkhead is an attachment point for our payload and telemetry avionics. It is housed within the nose-center section coupler. The steps to assemble the bulkhead are shown in

Figure 24. Simulations were run for both the bulkhead and support ring to observe the inertial effects of the predicted weights of the avionics and payload on this bulkhead. The simulation results yielded safety factors of 9.32 and 52.1 for the bulkhead and support rings, respectively.

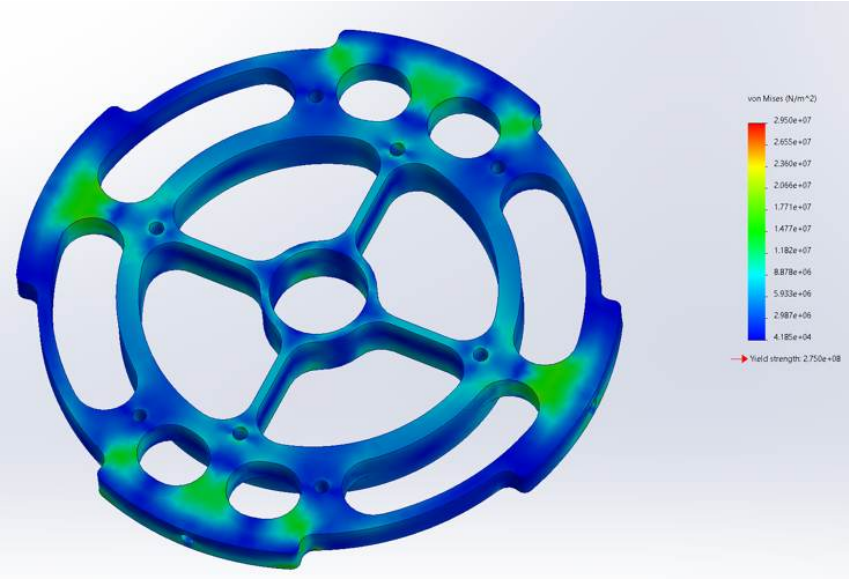

Figure 29: Nose Section Bulkhead Stress

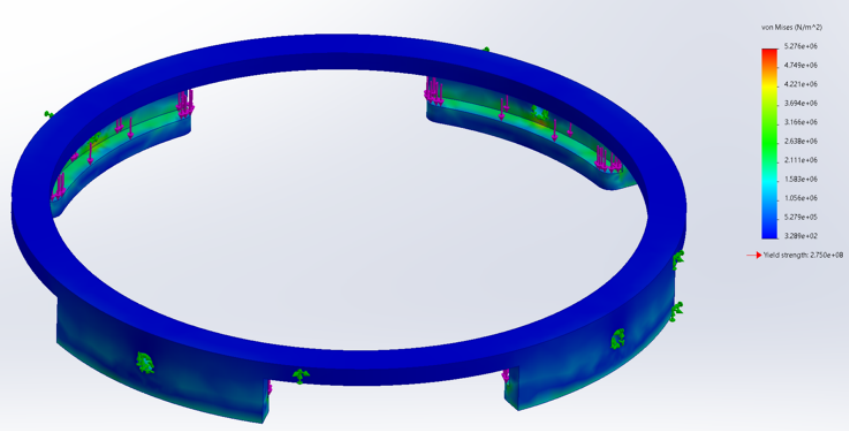

Figure 30: Nose Section Support Ring Stress

6.9 Recovery Bulkhead

The recovery bulkhead houses our recovery system and deployment energetics. It sits just below the acrylic pressure sealing plate within the center section body tube. This plate will be epoxied into the center section body tube, acting as the pressure sealing surface for our energetics. This ensures that only the recovery side of the center section is pressurized. Simulations have been conducted on both the bulkhead and the embedded support ring. Results from these simulations using the predicted shock forces from the recovery event yielded factors of safety of 1.34 and 20.6

for the bulkhead and support ring, respectively.

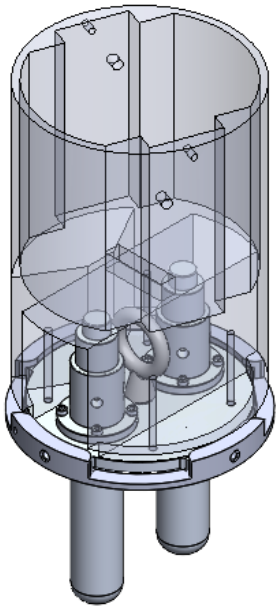

Figure 31: Recovery Bulkhead Assembly and Parachute Bay (Transparent)

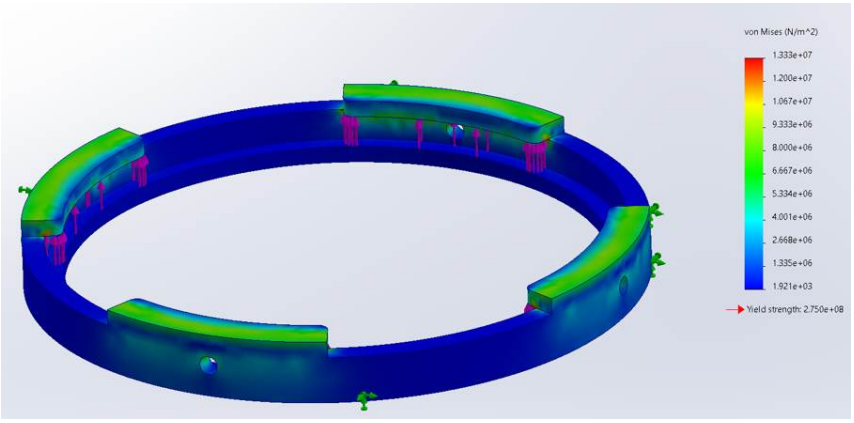

Figure 32: Recovery Support Ring Stress

6.10 Adaptor Bulkhead

The adaptor bulkhead's role is to mount an eyebolt for the reefing parachute and provide a mount for avionics components forward of the ox tank. The bulkhead is epoxied to its respective body tube and is secured with 6 radial M4 bolts for a redundant load path. The bulkhead was static structurally simulated in SolidWorks to a conservative recovery load of 5kN. The simulations suggested the bulkhead has a safety factor of 2.75 after mesh refinement around stress rising areas and result convergence.

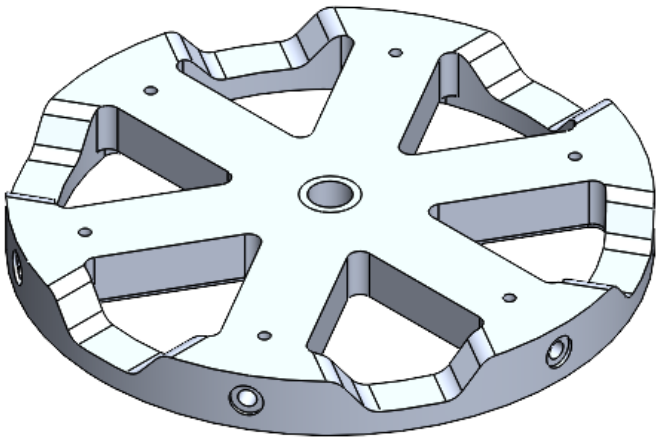

Figure 33: Adaptor Bulkhead

6.11 Aft Section

The aft section begins at the aft end of the oxidizer tank and ends at the metal boat tail, the aft-most part of the rocket. It features a smaller SRAD body tube and fincan, both of which house the combustion chamber. A general overview of the aft section can be found below.

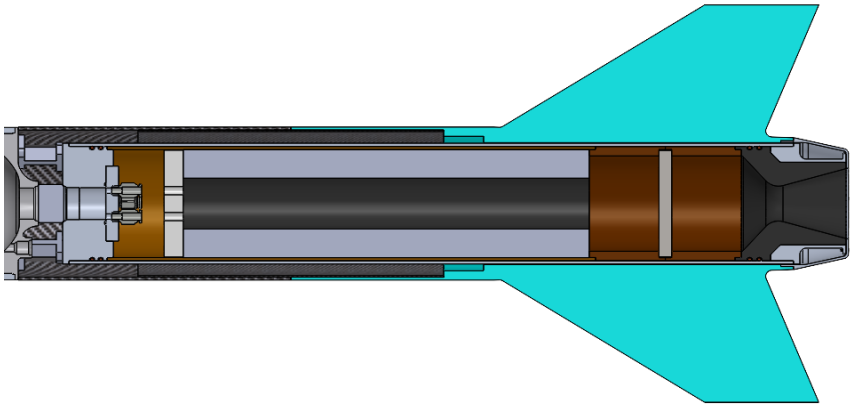

Figure 34: Aft Section Assembly

The aft section is fixed to the oxidizer tank with stringers (Figure 35) which retains the combustion chamber, as well as the aft body section and fin can.

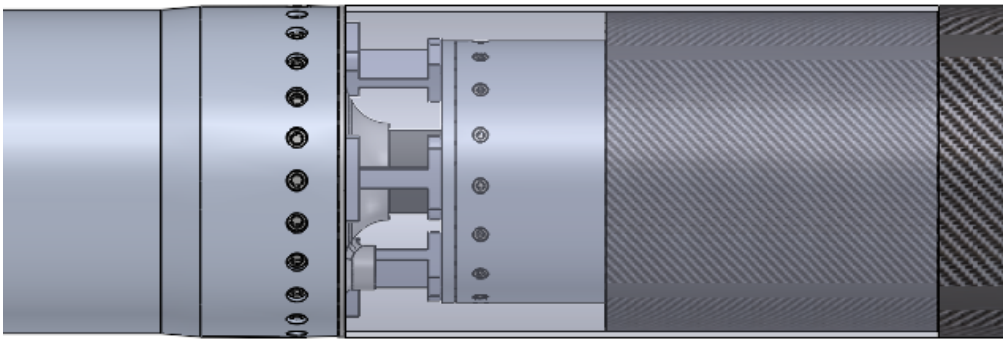

Figure 35: Stringers Attaching the Aft Section to the Oxtank

Below the stringers is our SRAD fin can and aft body tube section which will be manufactured to integrate both our fin can design and aft body tube section in one piece. The goal of this is to improve stability and integration of the aft structure. This section (Figure 34), including the fins, will be made from 3K plain weave carbon fiber. We manufacture our fin can with a tip-to-tip wet layup over a 3D-printed core (Figure 36). As the 3D-printed core remains in the finished part,

it is printed from Aero ASA, minimizing weight. The layup consists of four tip-to-tip layers with plies in the (0, 90), (45, -45), (45, -45) and (0, 90) directions respectively. These tip-to-tip layers are alternated with four "I" shaped pieces with similarly oriented weave angles to improve bending resistance in the fin fillets. These pieces also serve to increase the thickness of the boat tail, ensuring the thrust of the motor is transferred to the aft body tube. The third fincan in 36 shows 3D printed aft and forward sacrificial couplers that are remove after the layup cures to provide a space for the most aft coupler and interface between the fincan and aluminum boat tail.

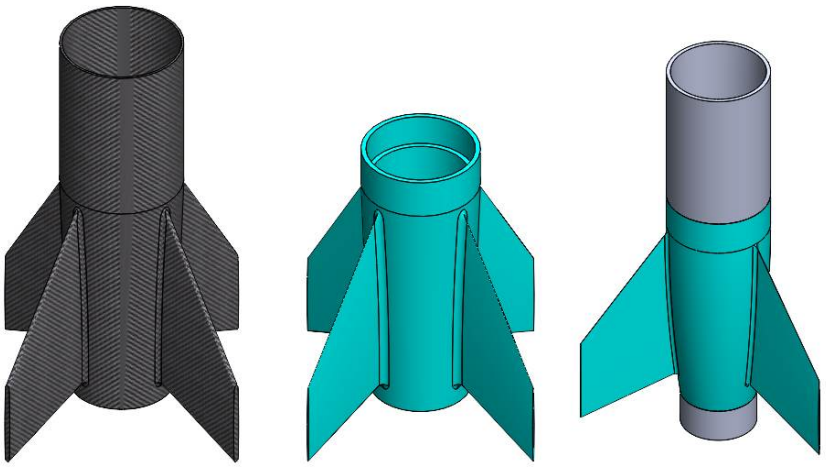

Figure 36: Fincan

7 Propulsion

7.1 Overview of Propulsion

The propulsion sub-team is responsible for the selection and handling of the rocket's motor for competition and the development of propulsion systems for future flights. Thus far, InSpace has flown COTS solid motors at both the 10,000ft, 15,000ft as well as the 30,000ft categories at competitions. While solid motors have been in use, the team has also been working to develop a hybrid rocket engine.

7.2 System Design and Performance Summary

CR25H will be Carleton's first vehicle to fly on a student-developed hybrid motor. This has been the result of many years of development and the team is excited to see it come to fruition. Being the team's first Student Researched and Designed (SRAD) motor, the goal for the project has been ease of testing and operations. While changes that increase performance have been made where they were perceived to be trivial, performance is not currently the team's primary goal. In future years the team will become more performance-focused as the hybrid engine project matures.

The propulsion system of the rocket has been designed with the goal of achieving a simple yet effective motor capable of propelling a vehicle to an apogee of 30,000 ft. The rocket uses a Urbanski-Colburn (UC) valve, removing the need for separate fill control and fire valves, while also keeping full abort capability up until the moment of the motor igniting. The rocket has been designed for a nominal tank pressure of 700 psi, corresponding to an average chamber pressure of approximately 320 psi. The performance parameters for the propulsion system are tabulated in Table 3.

Table 3: Hybrid Motor Performance Parameters

Parameter	**Value**	**Unit**
Average Thrust	3600	N
Peak Thrust	5351	N
Impulse	29065	N·s
Burn Time	8.07	s
ISP (estimated)	185	s
Average Chamber Pressure	324	psi
Peak Chamber Pressure	600	psi
Oxidizer Consumed	13.69	kg
Fuel Consumed	2.367	kg

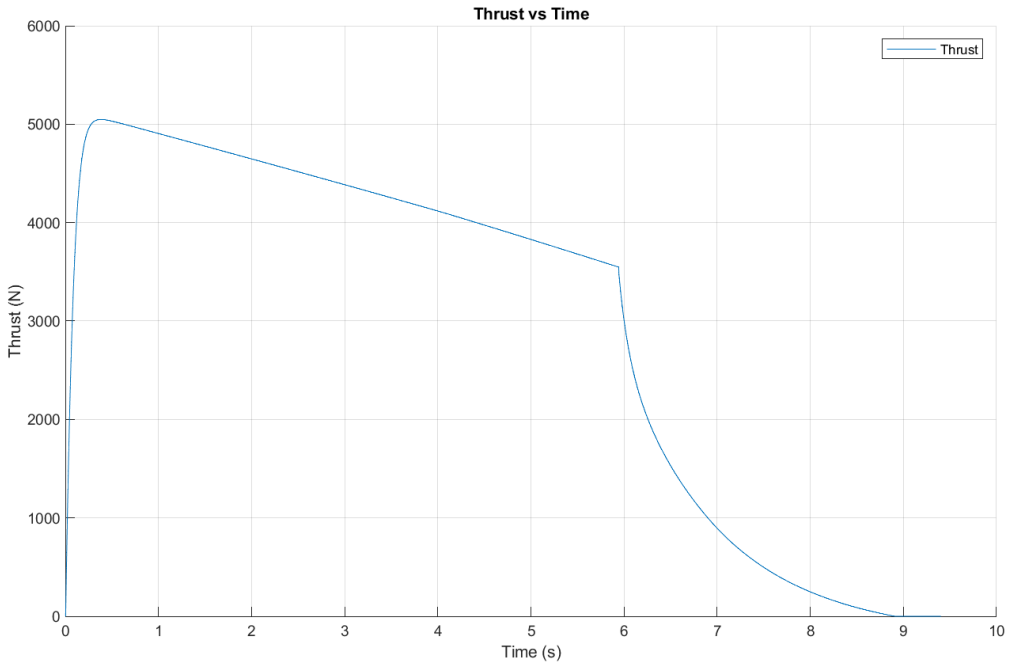

Figure 37: Estimated Motor Thrust Curve

As of the writing of this report, a small scale static fire has been carried out, achieving a peak thrust of 5.7 kN and total impulse of approximately 11800 N*s. Other measured performance values have been tabulated in Table 7.2 below.

Figure 38: Hybrid Motor Small-Scale Static Fire

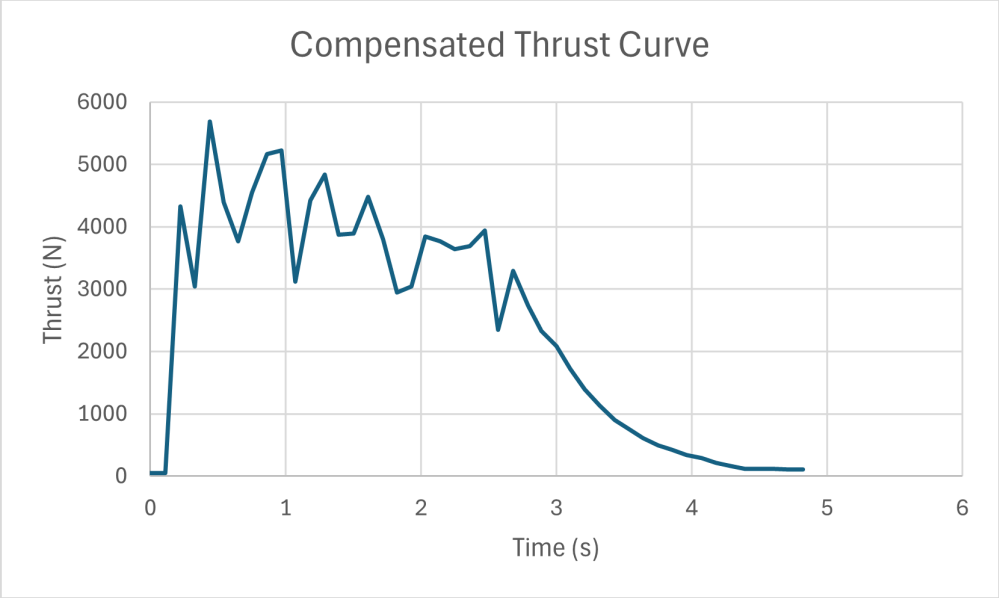

Figure 39: Measured Motor Thrust Curve

Parameter	Value	Unit
Average Thrust	2643	N
Peak Thrust	5691	N
Impulse	11165	N·s
Burn Time	4.5	s
ISP (estimated)	TBD	s
Average Chamber Pressure	292	psi
Peak Chamber Pressure	495	psi
Oxidizer Consumed	TBD	kg
Fuel Consumed	TBD	kg

Between this static fire and the attempted full-scale static fire, The fire valve and injector were changed from an SRAD poppet valve and showerhead injector to a U/C Valve, due to the previous design requiring that the valve remain pressurized to stay closed. The oxidizer tank was stretched from 15.5in to almost 48in, and the chamber liner and fuel composition was modified to accomodate a longer burn. With all of these design changes, a second static fire was attempted. However, the e-matches burst the nylon tubes but failed to light the ignition puck, resulting in a cold flow and loss of all nitrous. Due to the time of day and loss of all nitrous, the team was not able to re-attempt a fire.

7.3 Combustion Chamber and Nozzle

7.3.1 Overview and Requirements

The hybrid engine's thrust chamber is constructed using an extruded aluminum tube, and capped at each ends with bulkheads fastened by radial bolts. Designed to withstand a Maximum Expected Operating Pressure (MEOP) of 625 psi, the structural design of the chamber is largely inspired by

Half Cat Rocketry's Pressure Vessel Design Guide [1]. Another requirement of the thrust chamber is to house the engine's paraffin wax fuel, igniter, graphite nozzle, and SRAD thermal liner. The ultimate goal of the combustion chamber is to survive a relatively long burn time of about 10s. This section gives a general overview of the design of the thrust chamber and nozzle; safety related calculations can be found in Appendix I. Assembly procedures can be found in Appendix D. An overview of the thrust chamber is shown below in Figure 40.

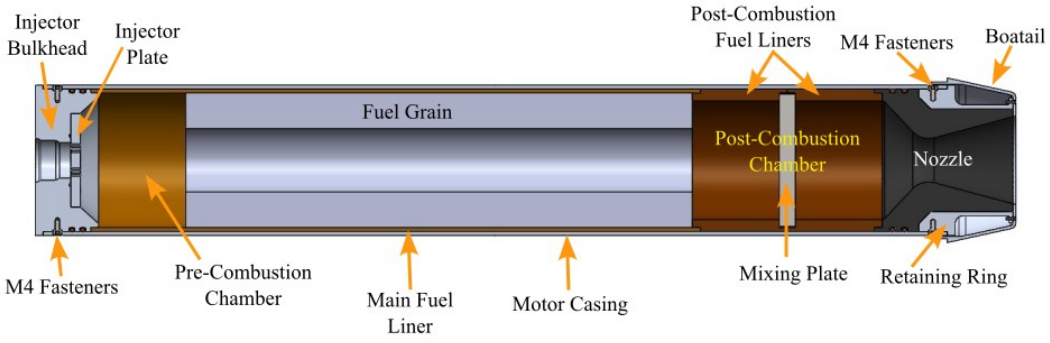

Figure 40: Thrust Chamber Overview

Key dimensional parameters of the thrust chamber are given below in Table **??**.

Table 4: Thrust Chamber Key Dimensions

General Dimensions	**Value**	**Unit**
Pre-Combustion Chamber Length	2.75	in
Post-Combustion Chamber Length	6	in
Injector Parameters	**Value**	**Unit**
Injector Hole Diameter	0.25	in
Injector Hole Length	1.2	in
Number of Injector Holes	4	N/A
Fuel Grain Parameters	**Value**	**Unit**
Grain ID	2.1	in
Grain OD	4.25	in
Grain Length	16	in
Nozzle Parameters	**Value**	**Unit**
Throat Diameter	1.37	in
Exit Diameter	2.75	in

7.3.2 Structural Design

The thrust chamber's structure is comprised of various aluminum 6061-T6 components. The casing is an extruded tube with a 4.5" inner diameter (ID) and a 1/8" wall thickness. With this thickness, the yielding hoop stress safety factor is 3.37 and the ultimate hoop stress safety factor is 3.79.

Figure 41: Thrust Chamber Casing

Both ends of the chamber are retained using radial bolts. The injector bulkhead bolt circle uses 16 M4 fasteners and the retaining ring bolt circle uses 12 M4 fasteners. The forward side of the chamber is capped by an injector bulkhead. The aft portion of the chamber houses the nozzle which is held into the chamber using an aluminum retainer.

Figure 42: Injector Bolt Circle

Figure 43: Nozzle Retainer Bolt Circle

These bolts are recessed into the bulkhead and retainer such that the heads of the bolts are in shear. The bolted joint design is identical between the two sides of the chamber, with the exception of there being less bolts on the nozzle side.

The injector bulkhead is attached to the casing using 16 grade 12.9 M4 alloy steel fasteners. Half Cat Rocketry's Pressure Vessel Design Guide was used to calculate the strength of the bolt circle [1]. The safety factors relative to the MEOP of various failure modes are given below. See Appendix I for more detailed calculations and explanations.

Table 5: Injector Bolt Circle Factors of Safety

Failure Mode	**Yield**	**Ultimate**
Bolt Shear	N/A	7.81
Casing Bearing Failure	2.59	3.07
Bulkhead Bearing Failure	3.24	3.85
Bolt Tear-Out	N/A	3.24
Casing Tensile Failure	7.74	8.70

The nozzle rests on a 6061-T6 aluminum retainer, which is radially bolted into the chamber casing using 12 grade 12.9 M4 fasteners.

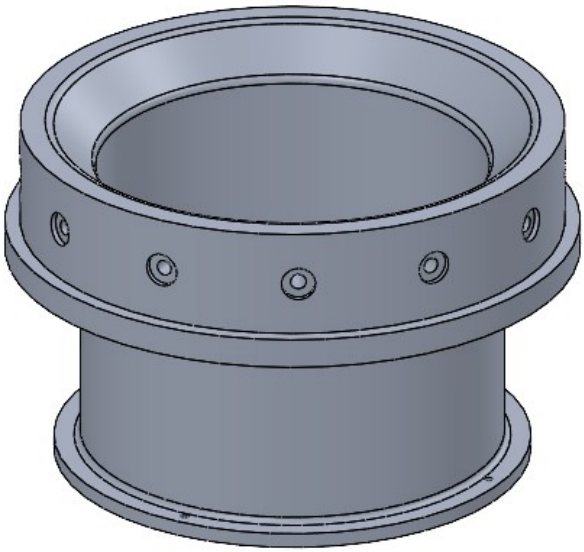

Figure 44: Nozzle Retainer

The nozzle retainer uses 12 grade 12.9 M4 fasteners to fasten to the chamber casing. The bolts are the exact same as the injector with the exception of the number of bolts. Because the nozzle retainer uses less bolts than the injector, the retainer bolt circle is weaker than the injector one. This is by design, so that in the event of over-pressurization of the chamber, the nozzle bolts tear out and eject the nozzle. This failure mode is safer than a forward closure failure or casing failure. The safety factors relative to the chamber's MEOP for various nozzle bolt circle failure modes are shown in the table below. See Appendix I for more detailed calculations and explanations.

Table 6: Nozzle Bolt Circle Factors of Safety

Failure Mode	**Yield**	**Ultimate**
Bolt Shear	N/A	5.86
Casing Bearing Failure	1.95	2.31
Retainer Bearing Failure	2.44	2.89
Bolt Tear-Out	N/A	2.43
Casing Tensile Failure	5.62	6.32

7.3.3 Thermal Protection

The primary form of thermal protection for the thrust chamber is a custom linen phenolic liner from Franklin Fibre. The liner is 19.5” in length and 1/8” thick spanning the pre-combustion chamber and fuel grain. The post-combustion chamber consists of two 3” long phenolic liners with a 3/8” thickness.

The main liner houses the fuel grain and the igniter. The post-combustion chamber liners retain the mixing plate in between them. The shoulders in the liners allow for an overlap to eliminate a direct path for hot gases to pass through. A thin layer of Room Temperature Vulcanizing (RTV) will also be used to seal the gaps in the faces of the shoulders of the joined fuel liner sections.

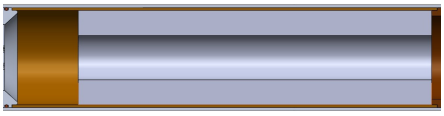

Figure 45: Main Liner

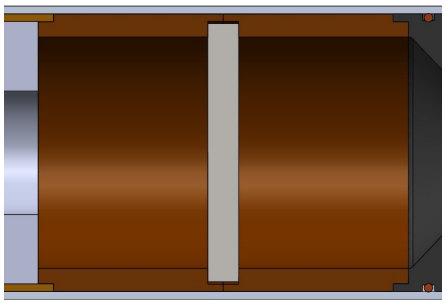

Figure 46: Post-Combustion Liner

7.3.4 Fuel

The thrust chamber will have one 16” long fuel grain cast directly into the fuel liner. The fuel grain uses CWC 140 paraffin wax from Canada Wax Corporation, 3%w/w A-C6A from Honeywell, 7%w/w Vybar, and 5%w/w carbon black. The A-C6A and Vybar 103 increases the tensile strength and slows the regression of the paraffin wax. The carbon black is added to reduce the IR transparency of the fuel to reduce excessive regression. The fuel grain is spin casted with a port diameter of 2.1”. The 2.1” port diameter keeps the port area larger than the mixing plate while also providing enough fuel for what is expected to be consumed in the hot fire based on Hybrid Rocket Analysis Program (HRAP) simulations and data from previous hot fire testing.

Figure 47: Fuel Grain

The fuel grain is manufactured by spin casting the wax into the fuel liners. The paraffin wax is weighed and melted in a pot at 60°C. The temperature is then increased to 90°C. The additives A-C6A and Vybar 103 are then weighed to the necessary quantities to achieve the specified concentration and added into the melted paraffin wax. The temperature is increased to 120°C so the additives can fully melt into the wax. The necessary quantity of carbon black powder is then weighed and added into the liquid wax. The mixture is stirred to ensure there are no clumps of carbon black powder. The end caps for the fuel liners are preheated to the same temperature of the liquid wax to prevent rapid cooling when the wax is poured into the liner. The enclosed liner filled with liquid wax is then installed onto the spin casting machine and is spin casted for roughly 6 hours to cool. An insulated box with a heater is used over the grain during the spin casting process is used to slow down the cooling rate to prevent cracking. Due to a tapered port diameter where the ends have a smaller diameter than the center, the grain is casted with excess fuel to allow the port to be bored to the nominal diameter afterwards.

7.3.5 Mixing Plate

The thrust chamber implements a static mixing device designed to increase combustion efficiency. The mixing plate is 1/2” thick, machined from 304 stainless steel. It is centered in the post-combustion chamber between the two post-combustion chamber liners. A thick layer of RTV will be used around the perimeter of the mixing plate to insulate the steel mixing plate from the liner. Additionally, the forward face of the mixer (facing the fuel) is coated in RTV to reduce the radiative heat transfer from the steel mixer back up into the fuel grain.

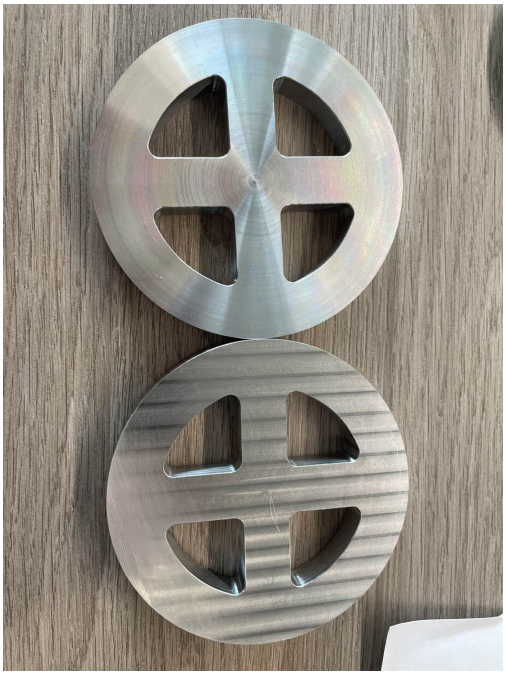

Figure 48: Mixing Plate

7.3.6 Nozzle Design

The nozzle of the thrust chamber is retained inside the combustion chamber using an aluminum 6061-T6 retainer radially bolted into the casing. The chamber's nozzle is machined on a CNC lathe from a superfine isomolded graphite rod. A full graphite nozzle was chosen due to its superior thermal performance, which allows for the nozzle to be re-used multiple times. The throat diameter of the nozzle is 1.37" and the exit diameter is 2.75", yielding an expansion ratio of 4.03. The nozzle was sized using isentropic flow relations, and verified in HRAP [2].

The chamber side of the nozzle features a shoulder to slot into the post-combustion chamber liner. This eliminates the direct path for hot gases to pass between the liner and nozzle interface. The faces in contact with the liner will be coated in red RTV silicone to fill any gaps between the liner and the nozzle. This acts as the primary seal in the aft end of the chamber. The nozzle also has two -244 O-rings to act as a secondary seal to hot gas leaking out of the sides of the chamber. The O-ring seal geometry was designed using table 4-2 in the Parker O-Ring Handbook. The O-rings are made of high temperature silicone.

Figure 49: Graphite Nozzle

7.3.7 Nozzle Geometry Sizing

The nozzle geometry was sized using the following methodology:

The throat area A^* is calculated using the inputted mass flow. The following expression exists to determine the theoretical maximum mass flow ($\dot{m}_{max}$) based on a given area where the flow is choked:

$$\dot{m}_{max} = A^* P_o \sqrt{\frac{k}{RT_o}} \left(\frac{2}{k+1}\right)^{\frac{k+1}{2(k+1)}}$$

Where k is the fluid's specific heat ratio, R is the fluid's ideal gas constant, and T_o is the chamber temperature. This expression is rearranged to solve for A^* and converted into the throat diameter D^* as shown:

$$D^* = 2\sqrt{\frac{\dot{m}\left(\frac{k+1}{2}\right)^{\frac{k+1}{2(k-1)}}}{\pi P_o \sqrt{\frac{k}{RT_o}}}}$$

With the throat area determined, the exit area A_e can be determined using the outlet Mach number Ma_e and the following expression:

$$A_e = \frac{A^*}{Ma_e}\left[\left(\frac{2}{k+1}\right)\left(1 + \frac{k-1}{2} Ma_e^2\right)\right]^{\frac{k+1}{2(k-1)}}$$

The exit area is then converted into the exit diameter D_e. To use the above expression, the exit Mach number is required, given by:

$$Ma_e = \sqrt{\frac{2\left(\frac{T_o}{T_e} - 1\right)}{k-1}}$$

This expression introduces another unknown parameter, the exit temperature T_e. This parameter is calculated using the following relation:

$$T_e = T_o \left(\frac{P_e}{P_o}\right)^{\frac{k-1}{k}}$$

Where P_e is the exit pressure and P_o is the chamber pressure. At this point, all the required parameters are known to size the nozzle exit.

7.3.8 Testing and Validation

The structural design of the combustion chamber has been tested and validated through a hydrostatic pressure test. The purpose of this test was to validate that the thrust chamber would withstand 1.5x MEOP, as stipulated by the DTEG. The chamber was pressurized to approximately 950-1000psi and was held at pressure for 1 minute before depressurizing.

Figure 50: Combustion Chamber Hydrostatic Test Setup

7.4 Injector

The thrust chamber's injector design features a bulkhead and a removable plate. The plate is removable to allow for different injector geometries to be tested.

7.4.1 Bulkhead

The bulkhead features two M5 mounting holes on the top for use in cold flow testing, where it is mounted to the test stand. The bulkhead also has six M6 mounting holes on the top face to mount three intertank struts that connect the chamber and oxidizer tank together. The oxidizer runs through a large channel through the middle of the bulkhead, to the injector plate. The top of the bulkhead has a thread designed for a -16 O-Ring Boss (ORB) fitting, to attach a 1 inch feed line to it. There is a secondary port on top for a pressure transducer line, which measures chamber pressure.

Figure 51: Top of Injector Bulkhead

The opening for the pressure transducer is 1/32” in diameter on the fire face of the injector, and widens further into the bulkhead for ease of manufacturing. For hot fire testing, the pressure transducer will be attached to a long pig tailed 1/4” line. Both the line and the port in the bulkhead will be filled with white lithium grease to protect the pressure transducer from the heat of combustion. A section view of the pressure transducer port geometry is shown in Figure 52.

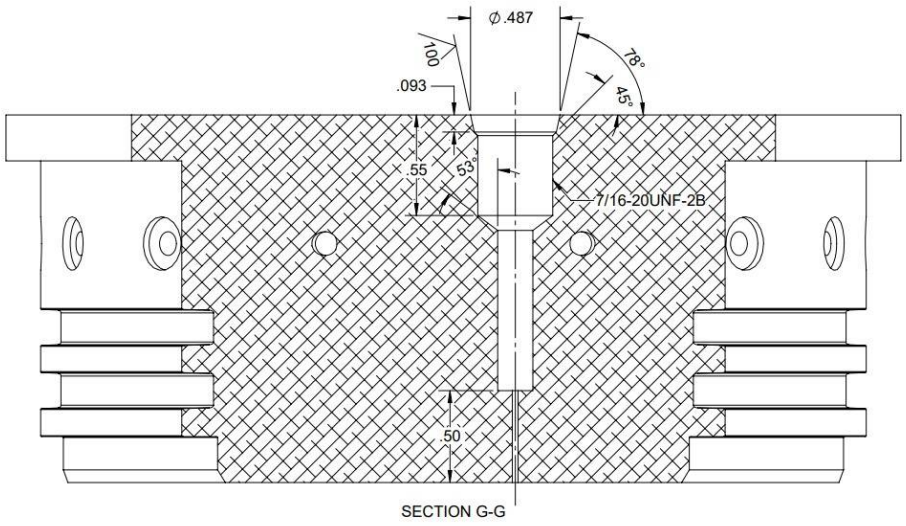

Figure 52: Injector Bulkhead Pressure Transducer Port

The injector bulkhead has two O-ring seals to keep combustion gases within the chamber. The O-rings are -244 high temperature silicone. The seal geometry was designed using table 4-2 of the Parker O-ring handbook. These O-rings are backup to a seal created by the liner and red RTV silicone. The injector bulkhead has a step where it slots into the liner, which can be seen in Figure

53. The O-ring grooves are also visible.

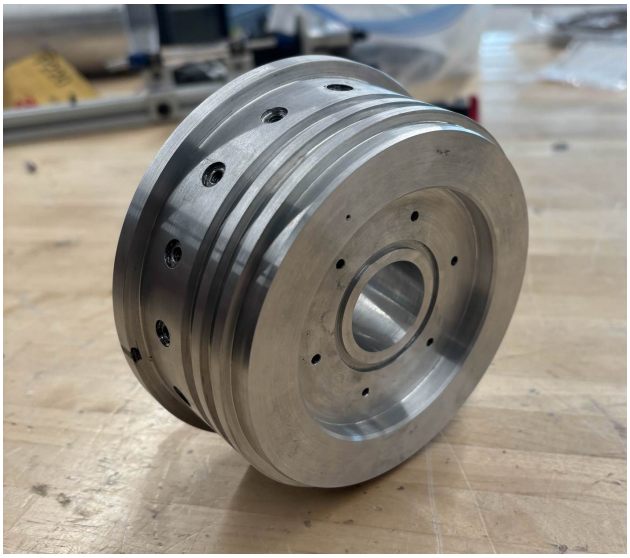

Figure 53: Injector Bulkhead

7.4.2 Urbanski-Colburn (UC) Valve

The injector features a UC valve to fill the run tank and to inject the nitrous oxide into the combustion chamber. The valve uses four 5/16" Push To Connect (PTC) fittings which nylon lines will connect to, through the combustion chamber and out the nozzle. Two of these lines will be used to fill the tank, while the other two PTC fittings will be connected together to plug them during the fill. When the engine is ready to fire, an igniter puck will be lit in the pre-combustion chamber, which will melt the nylon lines and trigger the release of nitrous oxide into the chamber.

7.4.3 Injector Plate

The injector is designed to have a removable aluminum 6061-T6 plate that fastens into the bulkhead with 6 black oxide M4 fasteners. The plate has 4 holes in it, 0.34" in diameter. The 4 holes are used to mount the 5/16" PTC fittings. The plate is 0.3" thick. The plate and bulkhead has been hydrotested to 1,500 psi. To prevent the nitrous from escaping around the sides of the plate, a -029 Viton O-ring face seal is implemented in the injector bulkhead.

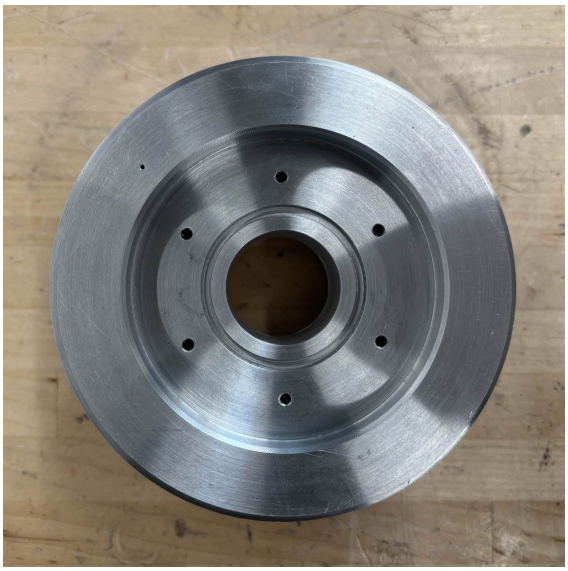

Figure 54: Plate Installation Area

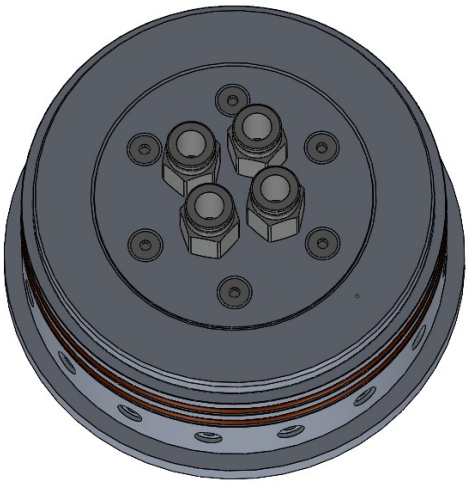

Figure 55: Installed Injector Plate CAD

To protect the fire face of the injector, the face of the plate, fasteners, and bulkhead will be covered in red RTV silicone.

7.5 Propellant Tank

7.5.1 Overview

The propellant tank was sized so that it would be able to provide enough oxidizer for a relatively inefficient motor to propel a rocket to roughly 30000 ft. The outer diameter (OD) of the propellant tank was sized based off of the OD of previous rockets flow at Launch Canada. This was done to minimize the impact of the tank on the design of the flight vehicle and for it to be a structural component. The propellant tank is constructed entirely out of 6061-T6.

The tank has a volume of roughly 18L allowing it to carry 13.4kg of N2O at 20°C and 5% ullage volume. The propellant tank was designed to have a MEOP of 1000 psi allowing it to handle condensed N_2O at temperatures of up to 34°C. The tank is proofed to 1500 psi using a hydro static pressure test. The yield and burst pressures are roughly 2100 psi and 2400 psi respectively.

7.5.2 Detailed Design

The propellant tank uses radial bolts for bulkhead retention. The tank has a volume of approximately 18 L. Except for the bolts, it is built entirely of aluminum 6061-T6. The casing is 48” long, has an ID of 5.5”, and an OD of 5.875”. The ends of the run tank are thicker with an OD of 6”. Both the fill and vent bulkheads are retained to the casing with 24 ASTM A574 $\frac{1}{4}$-28 bolts. Each bulkhead is sealed using two 1/8” 252 Viton o-rings. A cross-section view of the assembled tank is provided below in Figure 56.

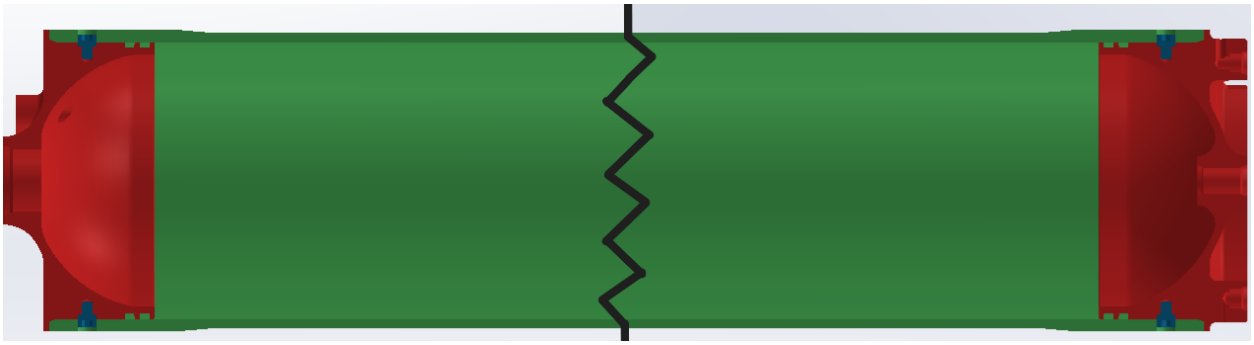

Figure 56: Cross-Section of the Propellant tank Assembly

The bottom bulkhead of the tank can be seen on the left side of the figure above. Along the center-line of the propellant tank on the bottom bulkhead is a -16 ORB fitting port to connect to the injector bulkhead. The bottom bulkhead also includes a -6 ORB fitting port which was originally intended to be the fill and dump port but will remain capped for flight. The bottom bulkhead is mounted to the combustion chamber through a -16 ORB female-female fitting and three stringers designed to handle the thrust and bending loads. Each stringer is attached to the bottom bulkhead using a pair of M6 bolts. The outside face of the bottom bulkhead is presented below in Figure 57.

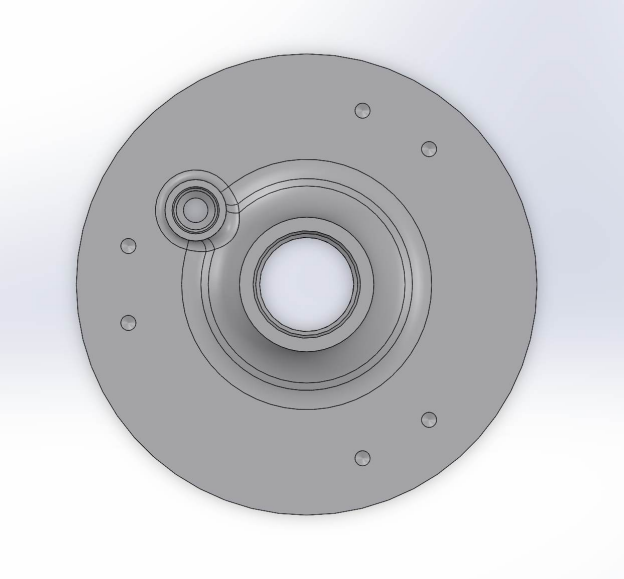

Figure 57: External Face of Bottom Bulkhead

As shown above the pairs of stringer mounting holes are equally space and located 2.375” from the center line of the tank.

The top bulkhead seen on the right of Figure 56 has two -6 ORB fitting ports along the center-line of the tank. One is located on the wetted side of the bulkhead for connecting a XX” long 3/8” dip tube. The length of the dip-tube was sized so that 5% of the tank volume is filled with nitrous oxide vapour and to prevent overfilling. When the liquid nitrous oxide reaches the bottom of the

dip tube it gets sucked up into the vent line. The vent line is routed to the outside of the vehicle thought one of two -4 ORB fitting ports on the top face of the bulkhead. The vent line is connected to the port that leads to atmosphere through a 1mm orifice to prevent the tank from being filled to quickly. The top face of the top bulkhead is shown below is Figure 58.

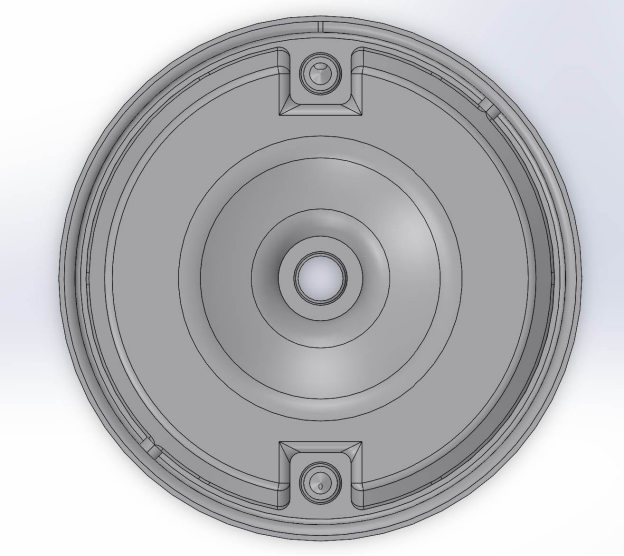

Figure 58: Top Face of Top Bulkhead

The top side of the top bulkhead includes a ring with six tapped M6 holes which acts as the coupler to the upper body tube of the flight vehicle. The ring also includes two slots so that cables for the a pressure transducer and vent valve can be routed to ground support equipment.

As previously mentioned the bulkheads are retained to the casing using 24 1/4" bolts. The bolt circle for each bulkhead are the same. The heads of the bolts are recessed into the casing and bulkheads to increases the bearing area on the casing and bulkhead. The recessed bolts also prevent the threads of the bolt from cutting into the casing and improve the aerodynamics of the propellant tank. The ends of the casing are 1/16" thicker than the center of the tank to reduce the weight of tank while keeping the ends of the tank thick enough for the bolts. To limit the stress concentrations from the change in diameters the transition follows a 5deg slope.

Dimension drawings of the propellant tank are available in Appendix F.2.

7.5.3 Manufacturing Process

The bulkheads and casing were machined by the machinists of the Engineering departments machine shop. Both bulkheads we machined from pieces of 6" 6061-T6 round bar stock. The casing was machined from a 6" OD and 5.5" ID extruded AL 6061-T6 round tube. During the machining of the casing the machinist noted that the stock was slightly oval. The center of the tank was turned down less than 1/16" to ensure that thickness of the casing stays above 3/16". Due to the ovalness of the stock the transition between the diameters of the casing are not of uniform length.

7.5.4 Analysis

Analysis for the pressure stresses of the tank were done with hand calculations for the bolt circles and casing. These were completed using the Half Cat Pressure Vessel guide for the bolt related calculations. Thick walled pressure vessel theory was used for the casing, rounded parts of the bulkheads, and for the bolt heads to ensure that they would not collapse under the bearing stress. Stress concentrations in the casing from the presence of the bolt holes were also considered using empirical stress concentration factors for an infinite row of holes with an axial force acting perpendicular to the row of holes. The hand calculations performed for the propellant tank are summarized in Appendix I.1.

7.5.5 Testing and Validation

To prove the design and construction of the propellant tank it was hydro-statically tested to a pressure of 1500psi. The tank was first pressurized up to 1500psi for a short duration and then depressurized to allow the bolts to settled in the casing's bolt holes. The gap between the casing and the bulkheads was measured before and after being re-pressuized to 1500psi. After waiting one hour at 1500psi the gap was measured before and after being depressurized. The casing boltholes and bulkheads showed no signs of having yielded during the test. The test report for the hydro-static test is provided in Appendix **??**.

7.6 Oxidiser System

7.6.1 Requirements and Design

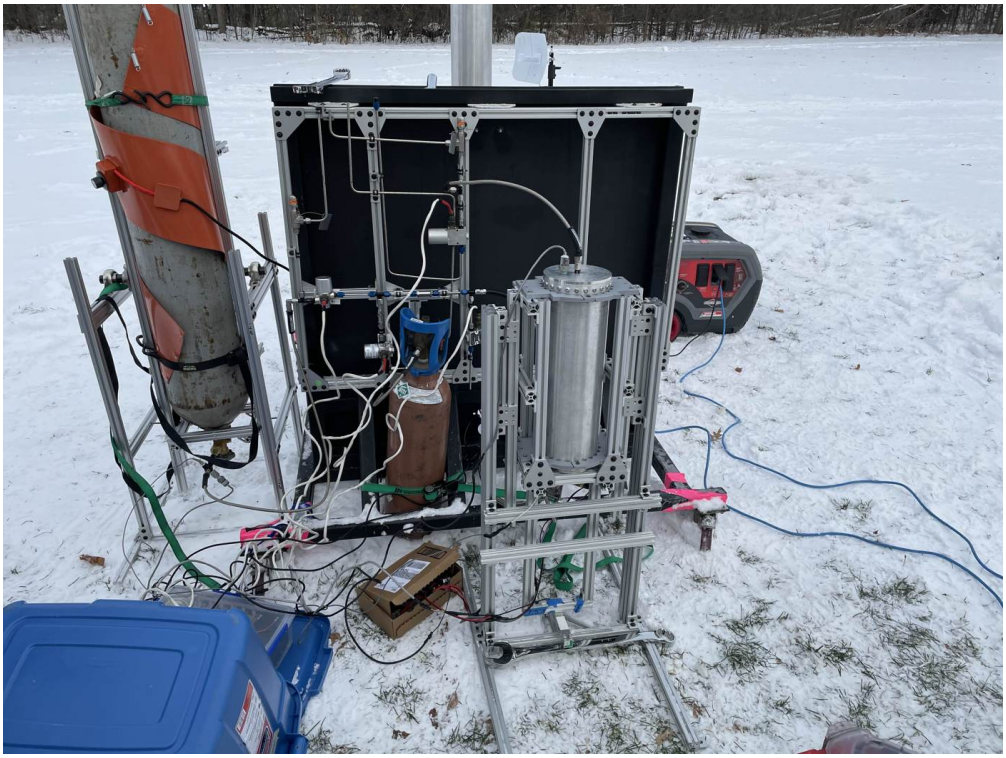

Figure 59: Fill System Mounted On Test Stand Prior to Cold Flow

The oxidizer system is designed with the primary requirements of allowing remote filling and vent-

ing of the system, and ensuring that the system will return to a known safe state in the event of a loss of power. A system MEOP of 1000 psi was chosen to accommodate a larger range of temperatures that the team may experience during testing, particularly that of CO_2 around 30°C. From this MEOP, a working pressure of 700 psi was decided upon, aiming to increase nitrous density in order to increase impulse. To ensure that the system could be filled remotely and fail save, solenoid valves are used as the fill valves. Marotta MV-100 solenoids are used as normally closed fill valves, cutting off pressure from the tank in the case of a power loss. An MV-74 three way valve is used as a normally open vent valve on top of the rocket in order to control pressure, as well as offer a worst case scenario abort option to slowly depressurize the tank. In most abort scenarios, a larger Swagelok valve connected to the tank through 3/8" or larger equivalent diameter lines is used to abort the tank in approximately a minute.

The majority of fittings on the oxidizer system are AN fittings. These fittings were chosen due to their ability to be easily disconnected and re-assembled, their low costs compared to alternatives such as compression fittings and the wide selection of fittings available. While many parts do not come in AN fittings, adapters are used to convert these other standards into a common AN standard. All AN fittings present on in the fill panel and vent system are either made from anodized aluminum alloys or 316 stainless steel for corrosion resistance. 1/4" tubes are used for the initial sections of the fill panel, due to their low cost and ease of assembly compared to larger sizes. The comparatively small orifice area of the Solenoid valves prevents much gain from filling from larger line sizes. 3/8" tubes are used for the fill line to the rocket, and for connecting to the main dump valves XV-3 and HV-3. This large line size allows faster dump times than a 1/4" line would, and better matches the orifice area of the large main dump valves. The run line is approximately 1" OD, having been sized up from the 3/4" lines used in the fall by the team after cold flow testing showed they would be inadequate to achieve the required mass flow rate for a 5kN engine.

The sizing of the overpressure devices was performed for the case of a failed-open regulator. Using the flow coefficient of the regulator and relief valves, it was determined that two valves would be required in order to prevent an overpressure in this scenario. To address this, two relief valves have been placed on the nitrogen full system, while one relief valve is present on every other volume in the system. Because of this, there will always be at least two relief valves in the path of the flow of a failed open regulator to prevent an overpressure of the system. Calculations used to determine relief valve sizes can be found in Appendix K.

7.6.2 P&ID

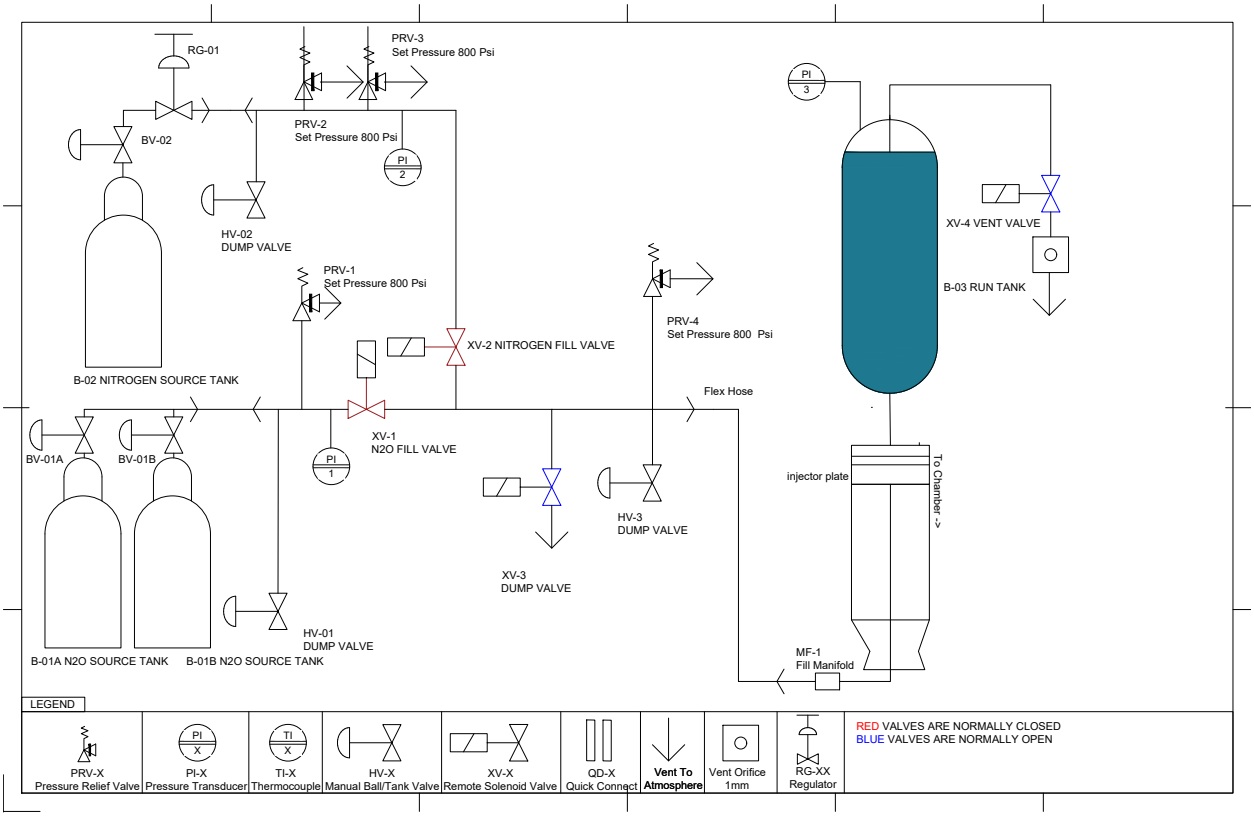

Figure 60: P&ID Diagram for the Test Stand Oxidizer System

Table 7: Oxidizer System Quick Reference Table

Component Name	**Purpose**	**Other Info**	**Wetted Materials**
XV-1	Nitrous Fill Control Valve	Normally Closed Solenoid	Aluminum Alloy, PTFE, Viton
XV-2	Nitrogen Fill Control Valve	Normally Closed Solenoid	Aluminum Alloy, PTFE, Viton
XV-3	Dump Valve	Servo Actuated Ball Valve	316 Stainless Steel, PTFE, Alloy X
XV-4	Vent Valve	Normally Open Solenoid	Aluminum Alloy, PTFE, Viton
PI-1	Pressure Transducer	0-1000 psi pressure transducer	316 Stainless Steel
PI-2	Pressure Transducer	0-1000 psi pressure transducer	316 Stainless Steel
PI-3	Pressure Transducer	0-1000 psi pressure transducer	316 Stainless Steel
PRV-1	N_2O Fill System Relief Valve	800 psi setpoint	316 Stainless Steel, PTFE
PRV-2	N_2 Fill System Relief Valve	800 psi setpoint	316 Stainless Steel, PTFE
PRV-3	N_2 Fill System Relief Valve	800 psi setpoint	316 Stainless Steel, FKM Seals
PRV-4	Run Tank Relief Valve	800 psi setpoint	316 Stainless Steel, FKM Seals
HV-01	Manual N_2O Fill System Dump Valve	Ball Valve	304 Stainless Steel, PTFE
HV-02	Manual N_2 Fill System Dump Valve	Ball Valve	304 Stainless Steel, PTFE
HV-03	Manual Run System Dump Valve	Ball Valve	316 Stainless Steel, PTFE
RG-01	N_2 Fill System Regulator	Manual Pressure Regulator	Brass, PCTFE
B-01A & B-01B	N_2O Source Tank	K Bottle	N/A
B-02	N_2 Source Tank	K Bottle	N/A
B-03	Run Tank	SRAD Pressure Vessel	6061 Aluminum Alloy, Viton
MF-01	Fill Manifold	Aluminum block for PTC fittings	6061 Aluminum, Brass, Nylon

The rocket's oxidizer system is split between a fill panel and a small vent section mounted on top of the rocket's oxidizer tank. The oxidizer system is designed to allow remote loading of nitrous oxide as well as purging. This panel has connections for flex hoses to ground based source tanks on one end, with fluid exiting from a single outlet which can then be connected to a fill manifold via flex hose. The majority of tubing is 1/4" 304 stainless steel tubing from mcmaster-carr, with the exception of 3/8" tubing running from the fill manifold to XV-3 and HV-3. The larger flow area between the run tank and these valves allows for a quicker dumping of the tank in an abort. The

motor uses a Urbanski-Colburn (U/C) Valve, and 2 5/16" nylon fill lines run from the fill manifold to the injector plate to fill the tank prior to launch. A full list of all components, burst pressures and orifice sizes can be found in Appendix K.

7.6.3 Component Selection

Relief Valves Swagelok R3A relief valves (PRV-1 through 4) are used by the team due to the team having purchased these valves in previous years, as well as their excellent documentation. The relief valves are configured to have a set pressure of 800 psi, corresponding to a full flow at the team's selected MEOP for the system of 1000 psi. Fluorocarbon seals are used to ensure chemical compatibility with nitrous oxide. Due to the high price of these valves the team will likely begin seeking suitable replacements in future years however.

Dump Valves Swagelok SS-63 series ball valves with PTFE seats are used as main remote/manual dump valves (XV-3 and HV-3). The large .5" orifice coupled with 3/8" tubing allows the team to dump a full oxidizer tank in approximately a minute as discussed in 7.6.6. A String is fastened onto HV-3, allowing for remote dumping of the rocket's oxidizer tank from outside the safety radius (pending a suitable path to pass a string from the fill panel to outside the radius) should a failure occur that prevents remote depressurization of the system.

Single piece process valves valves (HV-01 and HV-02) made from 304 stainless steel are used to depressurize the nitrogen tank and the N2O tank connections during leak testing and once the pad is being safed.

Fill Vent Valves Marotta MV-100 and MV-74 solenoids are used as fill control and vent valves on the rocket. These valves were purchased as military surplus, and have previously been used in many historical space programs and aircraft. All the valve has been cleaned from their previous lives before entering nitrous service, and all o-rings have been replaced with viton to ensure nitrous compatability. The simplicity of actuating solenoids, as well as known off positions for the valves.

Pressure Transducers TE connectivity MSP300 pressure transducers were used, with a range of 0-1000 psi [3]. From testing experience, these transducers can work at ranges at least 100 psi over 1000 psi without losing accuracy, however the team does not plan to operate above MEOP and should such conditions occur an abort will be called.

7.6.4 Redundancies and Failure Modes

The oxidizer system has been designed to mitigate many common failure modes for student designed rockets. Every section of plumbing is equipped with relief valves capable of venting the flow from a failed open regulator, as well as venting pressure that may occur from nitrous overheating in the run tank from the sun. The run tank is equipped with two .5" orifice dump valves, one able to be remotely actuated though a servo motor and one able to be remotely actuated though pulling a string which has been tested for pulling outside of a 180m safety radius in static fires. Due to the UC valve design, these valves retain the ability to dump the tank all the way until ignition occurs and the nylon tubes break, venting the tank contents into the chamber during ignition. In the case both ball valves fail, the tank can be slowly vented through XV-4, however this option is only a last-minute abort mode due to the long time required to vent a tank.

In the case of a loss of power to the control system, the normally closed solenoids XV-1 & XV-2 will close, cutting the run tank off from the source cylinders. The normally open solenoid XV-4 will then open, slowly depressurizing the tank. HV-03 can then be used to rapidly depressurize the tank by pulling the string.

In the case of loss of communications, XV-1 and XV-2 will be closed and XV-3 and XV-4 will be opened automatically by the control system in order to vent pressure from the run tank. More information on how this auto-abort works can be found in section 9.2.6. Should this auto-abort fail, HV-3 can be opened to depressurize the system using a string.

Because of these fail safe designs, many other common off-nominal scenarios such as loss of power to specific valves, fill valves freezing shut and serious leakage can be handled by an abort of the launch. The system is then safed and ground crews can approach to troubleshoot before another attempt can be made.

7.6.5 Material Compatibility

The components of the fill system were selected with material compatibility in mind. This section gives a brief overview of the compatibility of materials present in the system. Aluminum and 304/316stainless steel are metals and are commonly used and available for sale in nitrous systems, however polymers for sealing have been selected more carefully.

Viton/FKM Viton is used in all o-rings contacting in the oxidizer system, and similar FKM O-rings are given a "satisfactory" rating by the Parker O-ring guide for nitrous oxide compatibility [4]. While o-ring puffing is a concern, frequently replacement of o-rings is performed to prevent this, coupled with relatively short exposure times to nitrous oxide.

PTFE For non o-ring seals, PTFE is used due to it's known oxygen compatibility and availability as seals for components such as ball valves.

Krytox All components requiring lubrication in contact with N2O have been lubricated with Krytox, due to it's known oxygen compatibility.

7.6.6 Testing

The oxidizer system has been under testing since the spring of 2023, beginning with an initial hydro static test. Over two years, the design has been simplified and improved as a result of hydrostatic testing, leak testing and numerous cold flows.

Hydrostatic Testing Multiple articles have been hydrostatic tested to prove the capabilities of the oxidizing system. This involved pressurizing the respective test article to 1.5x MEOP and was determined to be successful based on the criteria of: no observable leaks over the course of the pressure hold; no permanent large deformations.

The first test article was the oxidizing system. The piping was tested twice in the summer of 2023 to validate early designs of the oxidizing system. In the fall of 2023, the valve assemblies were tested to validate that they had been properly re-assembled after being taken apart for inspection and

cleaning. One valve assembly and one piping test were deemed failures on the account of improper tube flaring and a defunct hydrostatic pump. The other two tests met the success criteria. After the successful tests, any further validation of the oxidizing system was conducted via leak tests and cold flows. A picture of the old hydrostatic testing setup can be seen below in figure 61.

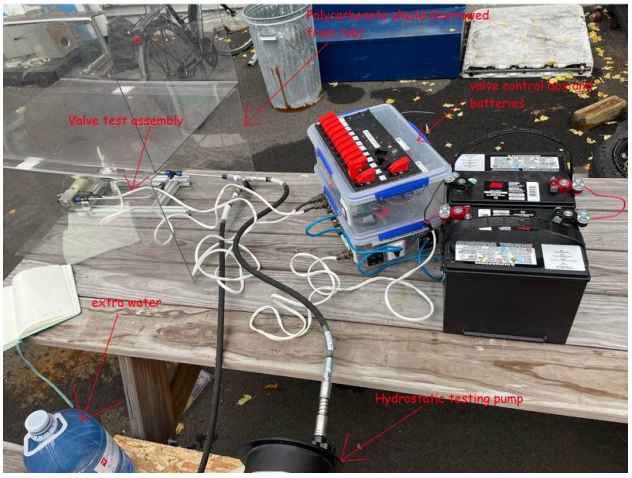

Figure 61: Valve Hydrostatic Test Setup

The next test article was Proto Tank. This was a small scale oxidizer tank. Proto Tank was successfully hydrostatic tested twice. One of the tests can be pictured in Figure 62 below.

Figure 62: Proto Tank Hydrostatic Test Setup

The SRAD Run Tank has been successfully hydrostatic tested with water to 1500 psi for a duration

of over an hour [5]. This pressure was chosen according to a proof pressure of 1.5x MEOP. The gap for both bulkheads before pressurization was 0.015". During peak pressurization the gap was 0.020", and then returned to 0.015" after depressurization. There were no observed leaks and the deformation was only elastic, therefore the success criteria was met and the test was determined successful. An example hydrostatic test procedure can be found here: Hydrostatic Testing Procedures. The test setup of the most recent hydrostatic test of the Run Tank is shown below in Figure 63.

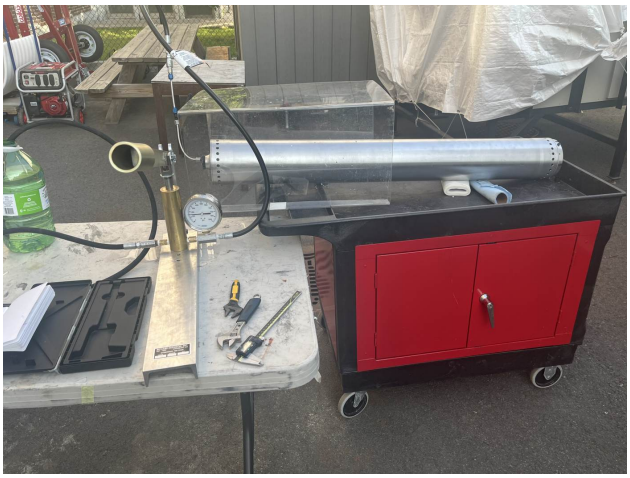

Figure 63: Hydrostatic Test Setup

Leak Testing Leak testing of the Fill, Run, and Vent systems have been completed both as independent tests, as well as during cold flow tests to verify assembly prior to fill of the oxidizing tank. These tests serve as practice for operating the system in later more complex tests as well as identifying and reducing leaks in the system to acceptable levels determined in test procedures.

The test is connected to the electrical control systems and conducted at 0.25x MEOP with nitrogen. All personnel conducting the test wear 30 dB hearing protection, safety glasses, face shields, gloves, and wear appropriate clothing such as long pants and closed toed shoes.

The test involves systematically checking each valve and fitting with liquid leak detector for any signs of leaking in the form of bubbling. Leak rates are also verified with the control systems User Interface (UI). Once the leak has been identified the proper corrective action can be taken, such as but not limited to: adding crush seals, redoing poorly made tubing, and properly tightening or torquing loose fittings. The leak test procedure can be found here: Leak Test Procedures.

Figure 64: Leak Test Setup and Configuration

Cold Flow Testing CU InSpace has conducted many cold flow tests to validate the oxidizing system and injector plate, with the first taking place in early May 2024. These tests aimed to model static fire conditions with a stand-in oxidizer of liquid CO2 and similar pressures and temperatures. These also served to test the functionality of the electronic control systems. The cold flow procedure can be found here: Cold Flow Procedures. A summary of all cold flow tests completed to date is listed below:

Table 8: Cold Flow Test Summary

Cold Flow Test #	**Date**	**Result**
1	2024-05-08	Successful, COTS run tank
2	2024-10-09	Unsuccessful, leaky ball valve prevented testing
3	2024-10-23	Successful, 3 discharges and first cold flow on SRAD tank
4	2024-12-06	Unsuccessful, low battery charge lead to abort
5	2025-02-01	Partial Success, Low tank pressure resulted in gas flow
6	2025-03-04	Successful, Static Fire Configuration, 3 discharges
N/A	2025-03-29	SRAD Valve Actuation Test, 3 successful discharges
7	2024-05-31	Successful, First cold flow with SRAD valve, 5 discharges.
8	2025-07-05	First Flight tank cold flow. 1 Successful discharge & Abort Test

The most recent cold flow test occurred using the U/C valve. a separate fire valve was used to limit the flow to the injector until the tank was filled, and then CO2 was dumped from the injector plate to simulate a fire.

Figure 65: July 5th cold flow test

Abort Testing One full-tank abort test has been conducted, with the flight tank filled with CO2 through the injector with nylon lines and then dumped through XV-3. The abort took 1 minute, approximately 30 seconds of that abort having been liquid dumping, shown in Figure 66.

Figure 66: Flight tank during abort test

Static Fire Testing Two Static Fire tests have been attempted. Both times the oxidizer system has successfully filled the tank with N2O, with some leaks. More information on static fire testing can be found in Section 14.4.1.

Ops Rehearsal Two ops rehearsal tests have been attempted with a flight configuration oxidizer system. More information on the ops rehearsal can be found in Section 14.4.2.

7.6.7 Oxygen Cleaning

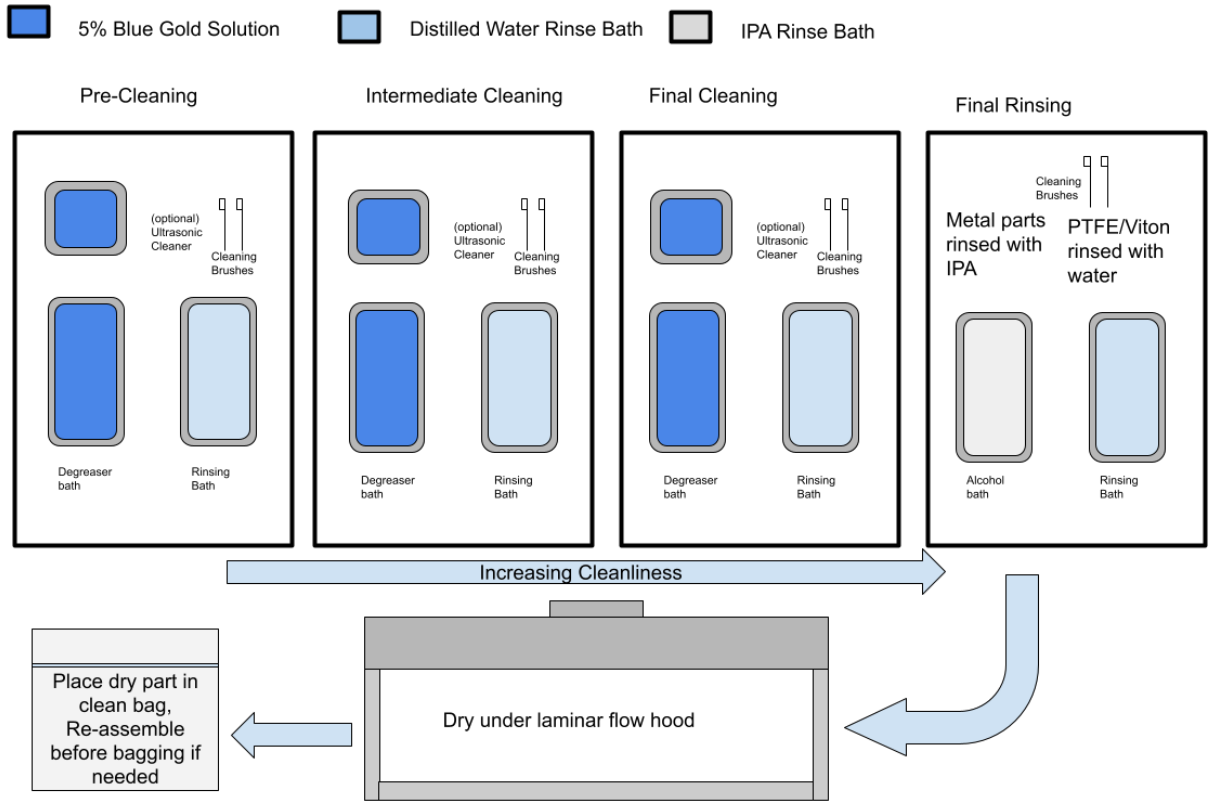

Figure 67: Oxygen Cleaning Summary

All oxidizer system components are cleaned using a 4 step process, with 3 steps of cleaning and rinsing with Blue Gold solution and distilled water. Sealing parts such as PTFE seals are then washed a final time with distilled water and all other parts are rinsed with Isopropyl Alcohol (IPA) before being set to dry for 24-48 hours under a laminar flow hood. After drying, parts are double bagged for storage. This cleaning is undertaken to remove contaminants such as non-oxygen compatible greases and oils from handling from parts before exposure to an oxidizing environment. Once cleaned, parts are only handled for brief times before installation with nitrile gloves and the system is kept clean with Foreign Object Debris (FOD) caps and sealed. The cleaning process is summarized in Figure 67 above. A full procedure for cleaning can be found linked here.

7.7 Ignition System

The thrust chamber houses an igniter to trigger decomposition of the nitrous oxide in the combustion chamber. The igniter is comprised of potassium nitrate and epoxy resin. The mix is 70% potassium nitrate and 30% epoxy resin by weight. The igniter is positioned on top of the fuel grain to be close to the nitrous entering the chamber and so it can transfer heat to the fuel grain. The igniter is a puck that has four holes drilled in the middle that the four nylon lines pass through.

Figure 68: Igniter System Prior to Assembly

The choice of igniter comes from flight heritage; this igniter composition successfully triggered combustion in a previous hot fire test, and this exact igniter composition was used by Waterloo Rocketry in the 2023 competition year [6]. The igniter is triggered by two Aerotech FirstFire e-matches that are placed into holes in the igniter and tacked in place with JB Weld. These are wired in parallel to ensure ignition of both e-matches. The igniter has been burn tested using the electronic control system. Most recently, the ignition system was tested in a hot fire attempt. Unfortunately, the ignition system failed to trigger combustion in this test. More information on this failure is described in Section 14.

8 Engine Test Stand Design and Test Plan

8.1 Test Stand

The load bearing structures of the test stand were built entirely out of welded 44W steel square tubing sections. The welding was performed by a welder from the Carleton Mechanical and Aerospace Engineering Machine Shop. The test stand was designed to safely withstand thrust load of 5 kN applied horizontally. An isometric view and annotated side view of the test stand with all its components are provided below in Figures 69 and 70.

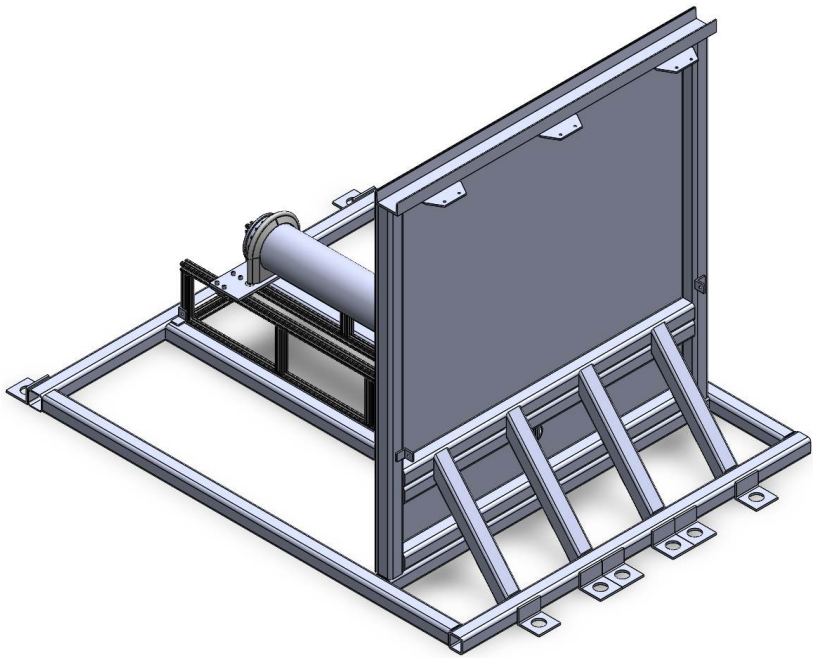

Figure 69: Test Stand Isometric View

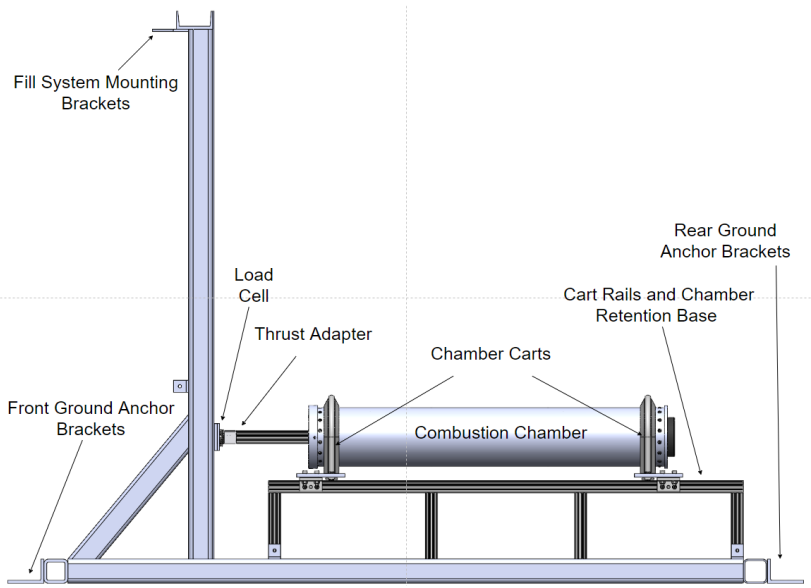

Figure 70: Test Stand Side View

8.1.1 Structural Analysis

Hand calculations for the safety factors provided in this section can be found in Appendix J.

The structure of the test stand was analysed using Finite Element Analysis (FEA) in Ansys. To simplify the analysis of the test stand, only the 5 kN thrust of the engine was considered. The thrust load was assumed to be acting straight ahead at the load cell location. To further simplify the analysis, the chamber, chamber retention, thrust adapter, load cell, and anything above the diagonal beams are excluded. Since the test stand is symmetric, only one half of it was analysed using symmetry constraints. This means that only half of the thrust load, 2.5kN was applied to the FEA model. The welds on the test stand are not included in the model presented above so a copy of the model was made with the welds on the real test stand. The welds of the test stand were inspected and measured so that they could be approximated using fillets [7]. The model and mesh used for the analysis can be seen in Figure 71 below.

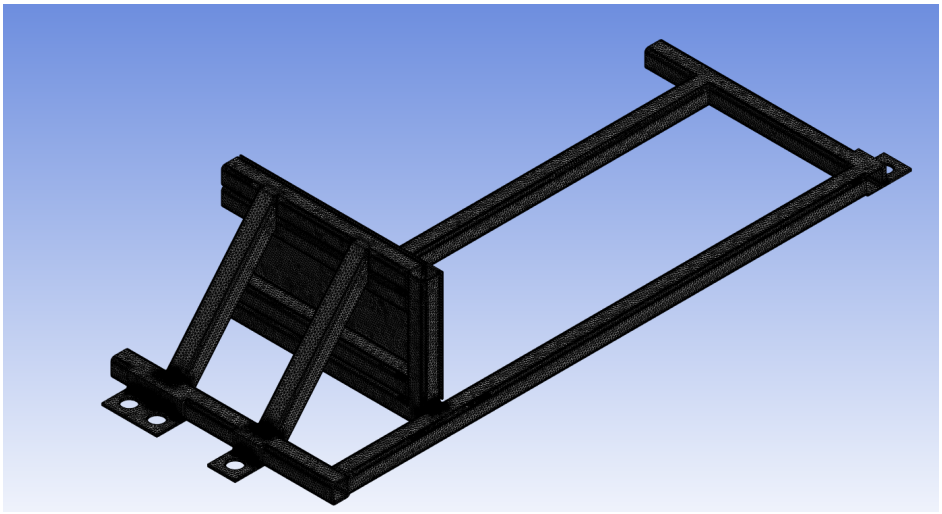

Figure 71: Test Stand Mesh Isometric View

The test stand is constrained using elastic constraints with a high foundation stiffness. Elastic constraints were used instead of fixed constraints to prevent stress singularities from occurring at the boundary of the constraints. The elastic constraints are applied on the top surfaces and holes of the anchor brackets. The contour plot of the von Mises stress can be seen below in Figure 72.

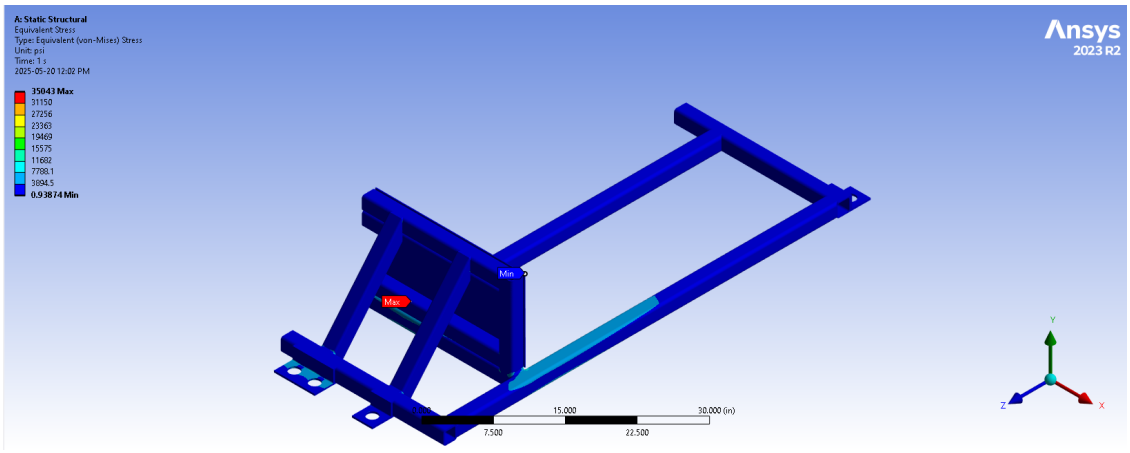

Figure 72: Test Stand von Mises Stress Contour

A close up of the peak stress location is shown below in Figure 73.

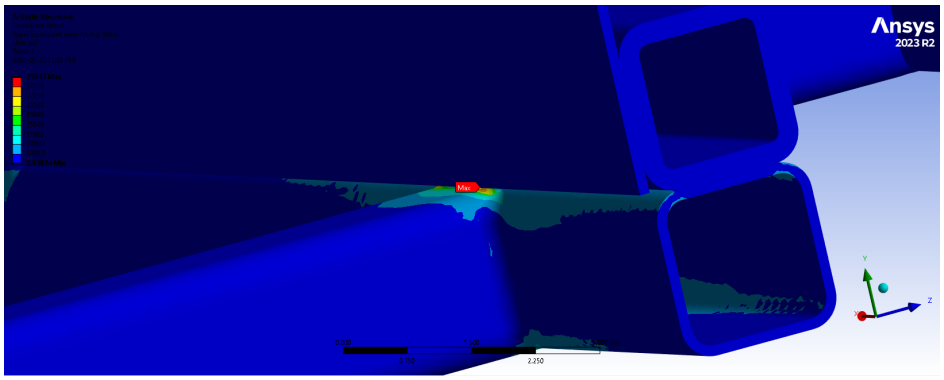

Figure 73: Test Stand Peak Stress Contour

The peak stress is roughly 262 MPa and occurs at a weld joint between three steel members. This corresponds to a safety factor of 1.14 for a yield strength of 300 MPa for 44W steel [8]. On the real test stand this weld is not readily accessible so it cannot be observed or measured to improve the model. Due to the location of the ground anchors on the rear of the test stand, this member should not be carrying the majority of the overturning load. To show that the test stand would not fail if this joint were to fail, it was simulated with this member cut off. The mesh of the test stand with out this member is shown in Figure 74 below.

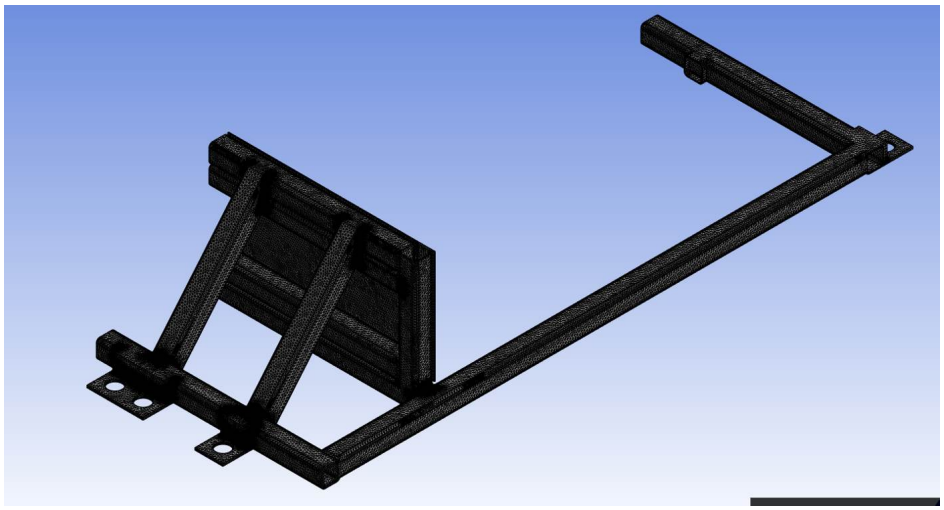

Figure 74: Test Stand Mesh with Cut-off Member

Apart from the missing member the only difference with this mesh is that the area around the welds is meshed finer than previously. The stress contour of this model is shown below in Figure 75.

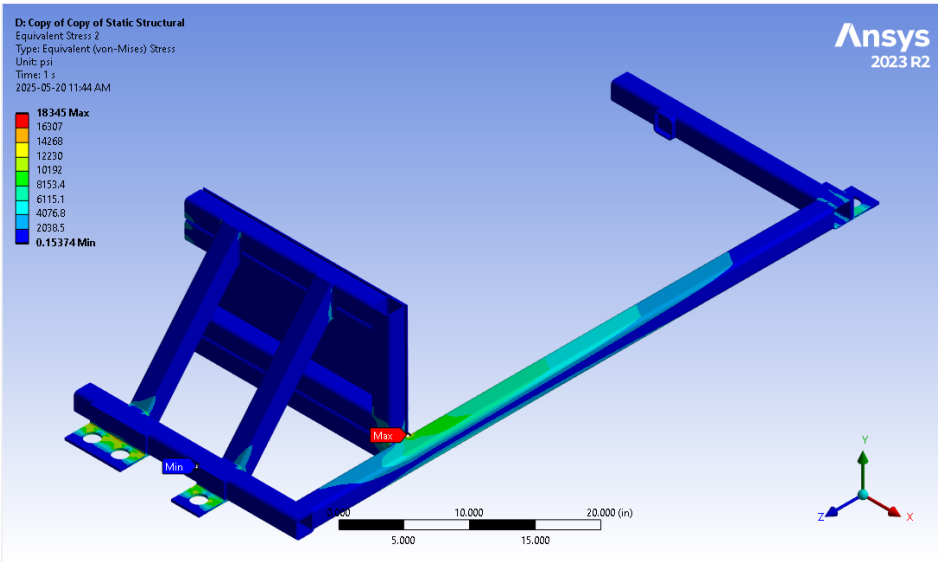

Figure 75: Test Stand with Cut-off Member Stress Contour

The peak stress location occurs at a similar location but at a weld along the outboard longitudinal member. The peak stress at this location is 126.5 MPa. The factor of safety contour is shown below in Figure 76.

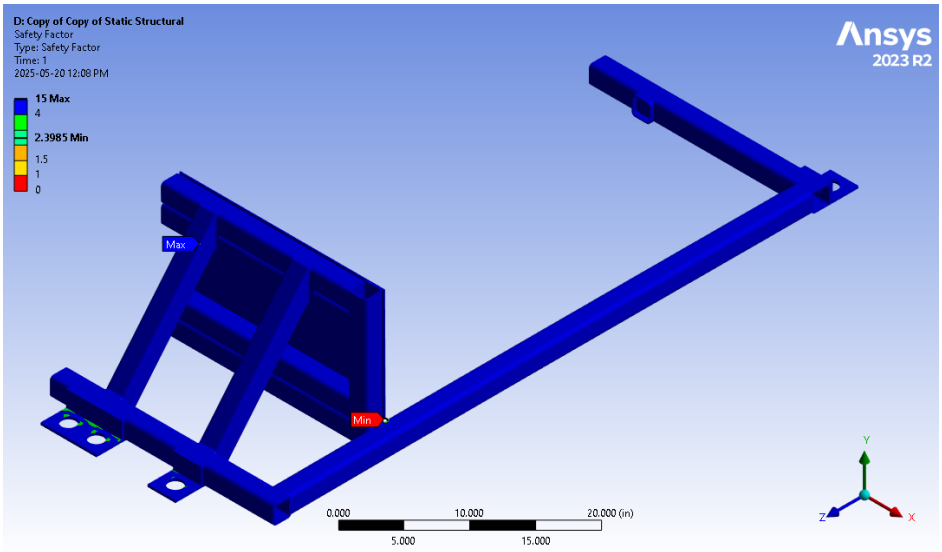

Figure 76: Test Stand with Cut-off Member Safety Factor Contour

As seen above the only other location that has a safety factor below 4 is at the anchor brackets. In these areas the safety factor is still above 3. The minimum safety factor is 2.39 at the previously mentioned weld.

To ensure the mesh on the test stand model was fine enough, nine different mesh sizes were simulated. The coarsest mesh had element sizes of 3.175 mm around the welds and 12.7 mm everywhere

else. The finest mesh had elements sizes of 0.254 mm around the weld and 3.175 mm everywhere else. The maximum stress and node count for each iteration is plotted in Figure 77 below.

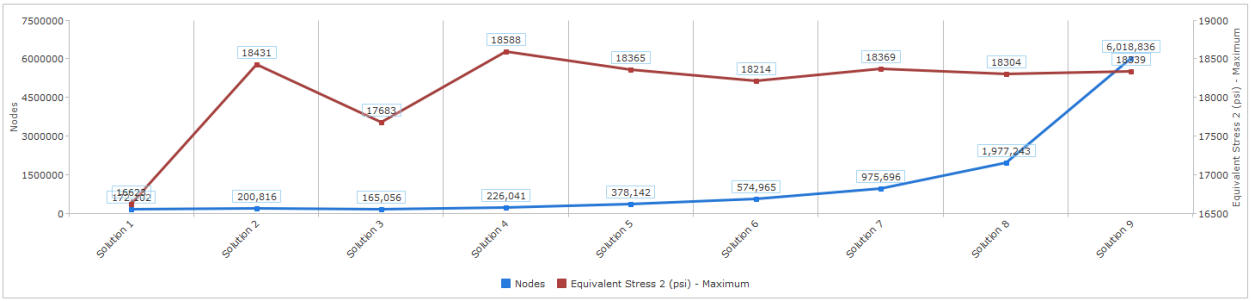

Figure 77: Test Stand Mesh Convergence

As can be seen above, the simulation converges after about 7 iterations.

The results presented in Figure 76 are conservative since they do not include the contribution of the inboard longitudinal members. When these members are removed, the stress on the outboard longitudinal members increases by roughly 36% at the welds and 29% along the center of the top face. If the yield strength of the inboard welds was to be exceeded, they would plastically deform until enough of the load has been redistributed to the outboard longitudinal members.

The previously mentioned safety factor of 2.39 assumes that the welds have the same strength as the 44W steel. This is a conservative approximation since the strength of the weld is controlled by the weld metal [9]. Linde Prostar S6 welding wire was used as filler metal for the welds of the test stand. This welding wire falls under the American Welding Society classification of ER70-S6, which according to the specification has a minimum “as welded” yield strength of 400 MPa and tensile strength of 480 MPa [10][11].

For the reasons listed above, the safety factor of 2.39 determined from the FEA of the test stand structure is conservative. Even if the motor were to over-perform by 20%, which is believed to be very unlikely, the safety factor would be about 2.0 assuming that the stress scales linearly. Furthermore, yielding of the welds due to the highlighted stress concentration will not cause catastrophic structural failure of the test stand.

8.1.2 Ground Anchoring

The test stand is anchored to the ground using eight PE26 ground screws from American Earth Anchors. The anchors are threaded through holes in the anchor mounting brackets using a large impact wrench. To prevent the test stand from overturning, two anchors are located at the rear of the test stand. The remaining anchors are placed at the front of the test stand, directly in front of the diagonal load bearing members. The inner two diagonal members have two anchor mounts in front of each since they sustain larger loads than the outboard diagonal members.

The pullout force of the PE26 ground anchors are dependant on the soil they are used in, shown below in Figure 78, as provided by the manufacturer.

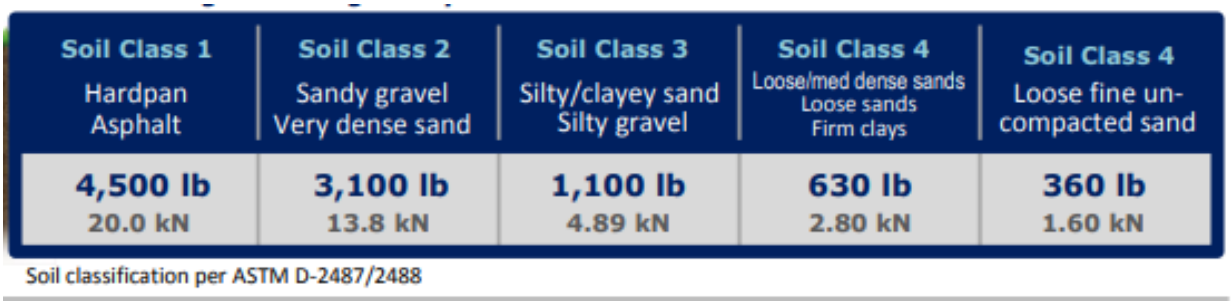

Figure 78: Ground Anchor Pullout Strengths

The lateral load capacity of the PE26 provided by the manufacturer is 950 lb (4.22 kN) for class 2/3 soils. Excluding the use of the test stand for static fires on loose class 4 soils, this gives a maximum pullout strength of 4.89 kN. Due to the unpredictable nature of soil and our limited ability to classify soils, the ground retention system was designed with the constraint of having a minimum factor of safety of four. Table 9 below gives the required reaction forces to keep the test stand in place and its corresponding safety factor.

Table 9: Test Stand Reaction Forces and Safety Factors

	Lateral	Overturning
Load [kN]	5	1
# of Anchors	6	2
Anchor Capacity [kN]	4.89	4.22
Safety Factor	5.868	8.44

Since the ground anchor brackets at the front of the test stand are loaded purely in compression, only the bearing failure mode is considered. The brackets are also made of 44W steel while the ground anchors are made of A365-T6 aluminium which has about half the yield strength so the bearing of the anchors must be considered instead. The bearing capacity of the ground anchors was determined using the American Society of Mechanical Engineers (ASME) Below-the-Hook (BTH) lug analysis method [12]. This results in a factor of safety of over 32 for an applied force of 833 N per anchor.

The anchor brackets on the rear of the test stand are assumed to not carry any of the thrust load. Instead, they carry a vertical load which stops the test stand from overturning. This load can be obtained by taking the sum of the moments at the location of the front anchors. As previously mentioned the force required for this is 1 kN or 0.5 kN per anchor. For the rear brackets, there are two possible failure modes: shear and bending. These both have safety factors well over 10.

8.1.3 Chamber Integration

To constrain the combustion chamber in the vertical and lateral axis, it is secured using vibration damping U-bolts on a pair of sleds. Each sled slides along the top of an aluminium extrusion cage which is bolted onto the test stand. A thrust adapter is mounted on the injector end of the chamber which transfers the load to the load cell without interfering with the feed line into the chamber. The assembly is pictured independent of the test stand in Figure 79 below.

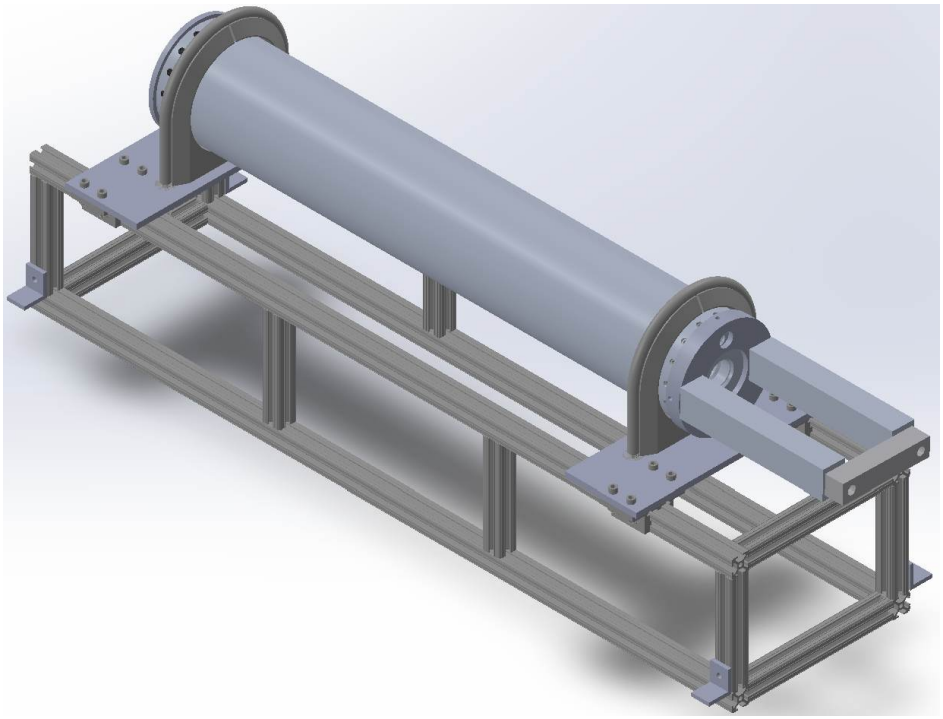

Figure 79: Chamber Sleds and Cage Assembly Isometric View

The small brackets at the corners of the cage seen above are welded onto the test stand. These are placed at the corners on vertical aluminum extrusion members, allowing the height and pitch of the chamber to be adjusted. This ensures that the thrust adapter can be properly aligned with the load cell. The assembly is installed onto the test stand in Figure 80 below.

Figure 80: Test Stand Chamber Integration

Ratchet straps are added to help align the thrust adapter with the load cell as precisely as possible

and provide additional lateral support in case the chamber is not perfectly aligned. Assuming a worst case misalignment of 5 degrees this would create a lateral thrust load of 435 N. Each of the four sleeve bearings have a static load capacity of 889 N, the U-bolts are rated for 10 kN, and the ratchet straps are rated for 1.3 kN. Therefore, there is no danger for the structure to fail due to lateral loading from a misaligned chamber.

8.1.4 Thrust Adapter

The thrust adapter was designed to handle thrust loads of 5 kN, while allowing adequate space for the feed lines. The thrust adapter is held together using four M6 bolts which screw into the center of the columns. These bolts are lubricated and installed with 8 Nm of torque using a torque wrench. The thrust adapter is held onto the chamber with 4 set screws so that it does not fall off during installation or operation. A cross section view of the thrust adapter is pictured below in Figure 81.

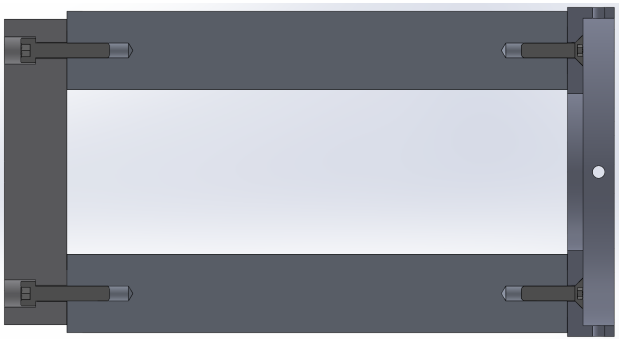

Figure 81: Thrust Adapter Cross-section View

The following failure modes were considered for the the thrust adapter: bending of the beam, bending of the columns, failure of the column-base bolts, load cell contact stresses, and the thrust adapter slipping off the load cell. The safety factor for each of these failure modes are summarized in Table 10 below.

Table 10: Thrust Adapter Failure Modes Safety Factors

	Safety Factor
Beam Bending	6.1
Column Bending	8.54
Bolt Failure	1.48
Joint Separation	1.5
Bolt Pullout	3.0
Load-cell Contact Stress	2.85
Slipping	4.59

A comprehensive overview of the calculations performed to obtain the results above are included in Appendix J.2.

8.2 Test Plan

8.2.1 High-Level Concept of Operations

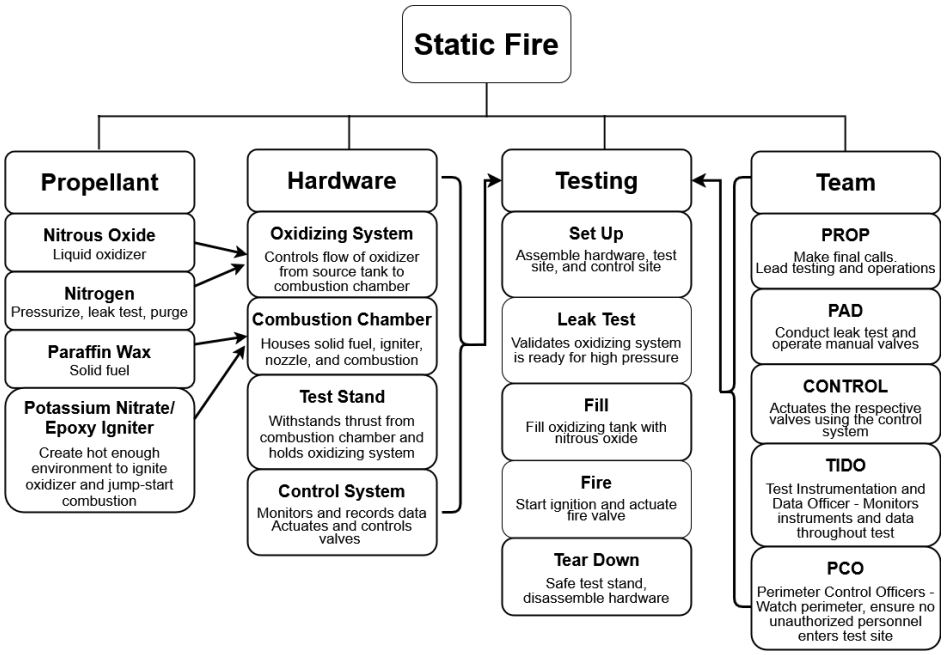

Figure 82: Key Operational Relationships

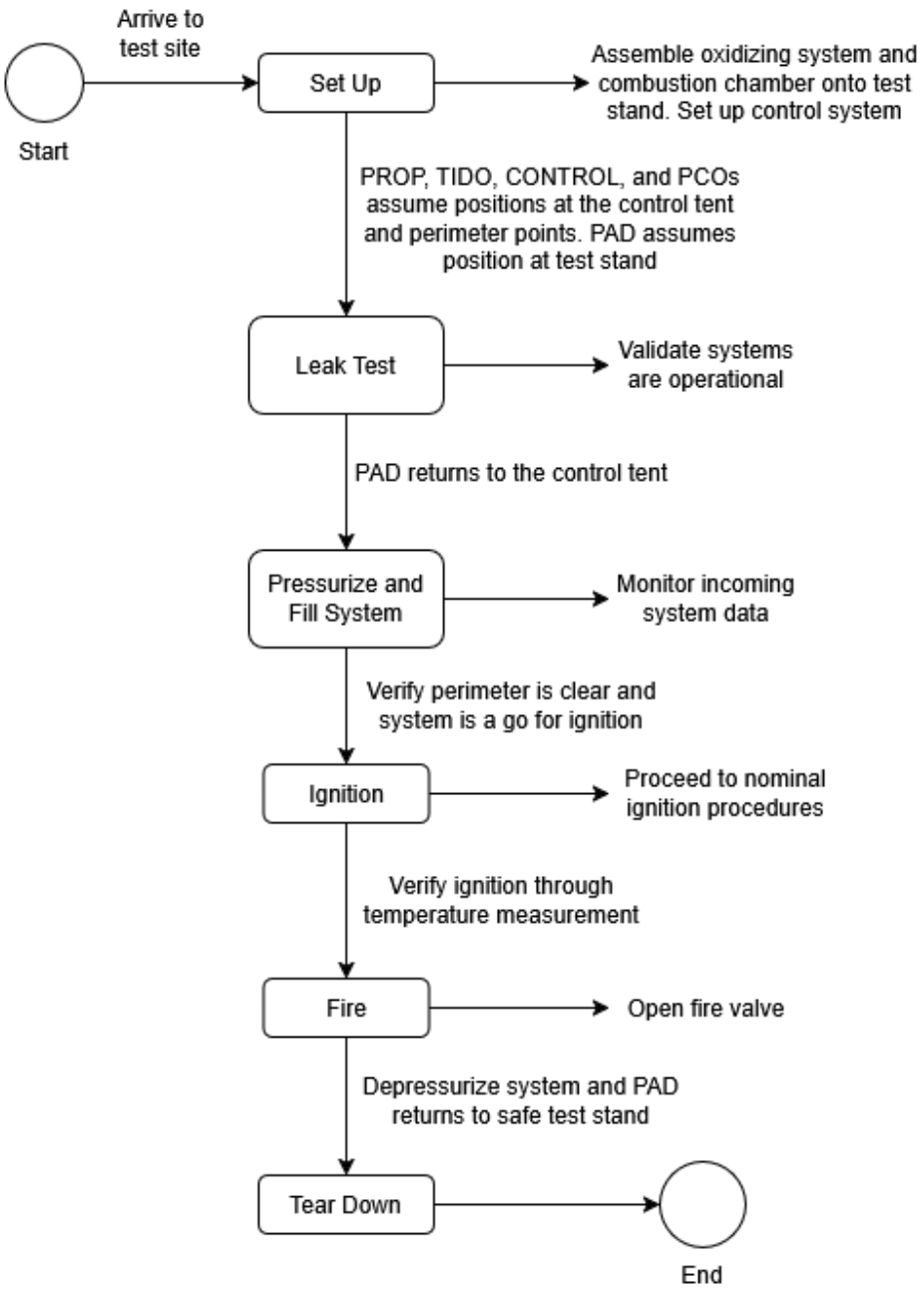

Figure 83: High-Level Concept of Testing Operations

The above figures demonstrate the high-level relationships, key project milestones, and nominal testing operations to be expected of the static fire. Not captured is the expected test campaign duration and potential setback points.

8.2.2 Mission Duration

The static fire is expected to take anywhere from 7 to 10 hours to complete from set up to tear down. The typical cold flow duration can be seen in Figure 84. The potential for time setbacks increases during the set up of electrical systems, leak testing, filling, and ignition. Each step presents the

opportunity for a failure mode that requires trouble-shooting. Further, the test is expected to take longer than the average cold flow due to set up of the fuel grains and igniter puck. Should anything during the ignition sequence fail, resetting will add additional hours onto the test duration estimate.

Operation	Expected Duration
Unpacking and set-up of hardware and electronics	1 ½ - 2 ½ hours
Leak testing	½ - 1 ½ hours
Systems check before pressurization	½ - 1 ½ hours
Pressurization before fill	⅓ hour
Fill	⅓ hour
Ignite, fire, safe test stand	⅓ hour
Tear down	1 - 1 ½ hours

Figure 84: Typical Cold Flow Duration

8.3 Static Fire Safety

8.3.1 Personnel Safety

During any and all operations, the necessary personal protective equipment is identified and implemented to keep personnel safe. This takes shape by being informed on the type of operations we need to carry out, potential hazards (hazards matrix), and the involved materials or chemicals. CU InSpace is equipped with safety glasses, 30 dB hearing protection, 3M respirators, face shields, gloves, and lab coats. Having access to proper Personal Protective Equipment (PPE) has enabled the team to safely carry out a variety of tests.

8.3.2 Testing Safety

It is the responsibility of the team and team leads to understand the procedures, associated risks, and mitigation strategies before conducting any tests. Before tests can begin, pre-test safety talks are conducted by team leads to ensure every participating member is aware of what is about to happen. These talks discuss the purpose of testing, related risks, the need for PPE, and prescribe a general awareness. Signatures are then collected after members have listened to the talk. This also includes the need, if determined, to set up perimeter control officers to enforce public safety.

More specifically, safety radii are utilized during SRAD valve testing, cold flows, and static fires to ensure all personnel and public members are not within immediate harms way. Please refer to appendix L - Blast Fragmentation to review blast fragmentation and safety distance calculations.

8.3.3 Guidelines and Regulations

CU InSpace adheres to guidelines and regulations set by both Launch Canada and Carleton University, to ensure the highest standards of safety are being met. This is possible through constant review and comparison of our systems with the Rules and Requirements, and Design, Test, and Evaluation Guide as stipulated by Launch Canada. At the same time, the team has been is close communication with Carleton University's Environmental Health and Safety, Fire Safety, Engineering Design Centre faculty, Campus Safety, and Parking Services. This has allowed us to receive experienced peer review and advice, safety checks from extra sets of eyes, and to make sure any testing site on campus has been booked in advance to minimize risk to public safety.

9 Avionics

The avionics subteam is responsible for three primary subsystems within the team's rocket: an SRAD telemetry system, an SRAD control system for the hybrid rocket and COTS electronics. This year, the recovery subteam is handling the integration of COTS altimeters into their reefed system, but avionics maintains responsibility of COTS GPS tracking and a COTS altimeter that is not in control of any energetics.

9.1 SRAD Telemetry System

The avionics subteam is responsible each year for the development of a SRAD telemetry system. This telemetry system gathers data about the rocket during flight and both logs it to onboard storage as well as transmits it over long range radio. The telemetry system does not have any connections to energetics, it is purely for data logging and transmission.

9.1.1 Flight Computer

This year the avionics subteam has created a single-board flight computer for obtaining and recording in-flight telemetry data (Figure 85). This is an improvement over the 2024 system which consisted of multiple PCBs connected via a back-plane (ribbon cable bus connecting each board to one another). The single-board flight computer, colloquially referred to as "Josh" uses the STM32H743 MCU and runs the Apache NuttX RTOS. NuttX is a mature POSIX-compliant RTOS with an emphasis on standards compliance and a small memory footprint. [13] This makes it an excellent choice for low-power, MCU-based systems such as Josh. The student designed software that runs on Josh is described in Section 9.1.2, while the software that receives the in-flight telemetry data is described in Section 9.1.4.

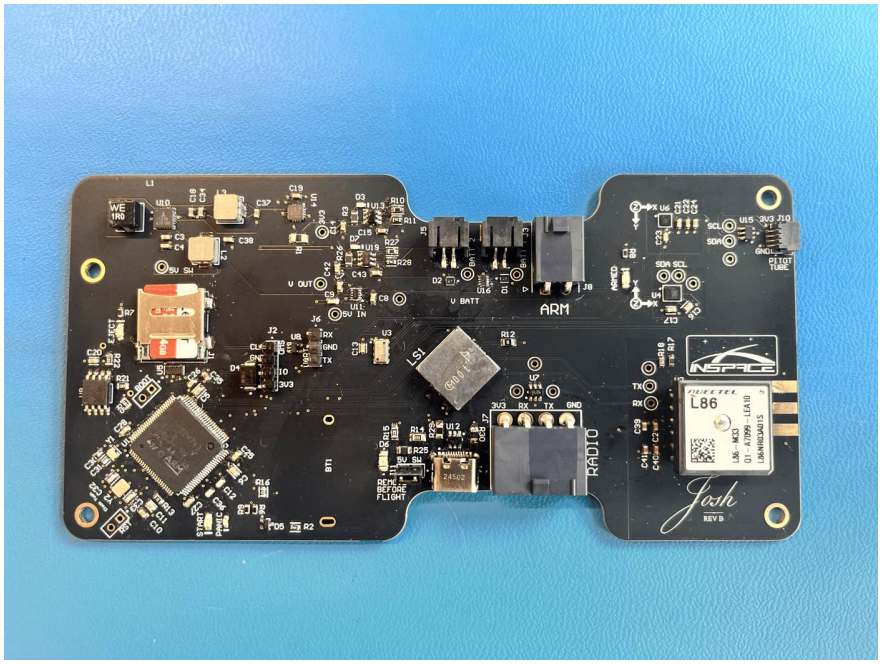

Figure 85: Top-down view of the "Josh" flight computer

In addition to the necessary electrical components required for the flight computer to function, the flight computer consists of the following sensors for gathering telemetry data:

- MS5607 barometric pressure sensor: Measures pressure and temperature (25Hz)
- LIS2MDL magnetometer: Measures 3-axis magnetic field (100Hz)
- LSM6DSO32 6-axis IMU: Measures angular velocity and linear acceleration (100Hz)
- L86-M33 GNSS module: Measures position, altitude and velocity (10Hz)
- PAC1951 current sensor: Measures current and power consumption (5Hz)

Data from these sensors are collected at various rates during flight and fused together using sensor fusion algorithms to determine the rocket's position, orientation, speed, acceleration and altitude during flight and at landing.

The flight computer has two types of on-board storage. The first and primary one is a micro SD card for recording all of the data measured in flight from the sensor array. The team initially considered Serial Peripheral Interface (SPI) flash chips to avoid vibrations disconnecting removable media temporarily. NOR-flash packages had too little storage space for logging applications, and unfortunately, Apache NuttX does not have a NAND-flash driver at the time of writing either, only a simulated implementation without much hardware support yet. Embedded MultiMediaCard (eMMC) was considered as well, but it was only available in Ball Grid Array (BGA) packages which are difficult for students to solder by hand. As a result, Josh uses a removable micro SD card for easy driver compatibility with NuttX and the convenience of being able to remove the media by hand for immediate analysis post-flight. To overcome vibration issues, the micro SD card slot selected for Josh is the MEM2067-02-180-00-A, which is shock resistant up to 50 G and vibration resistant at 10-55Hz with 1.52mm peak-to-peak amplitude, with only 1 microsecond of discontinuity. [14, pp. 3–4] In addition to the vibration resistance provided by the slot, special software support is used to guarantee that flight logs will not be corrupted (see Section 9.1.2).

The other storage on Josh is a 4KB Electrically Erasable Programmable Read-Only Memory (EEPROM). This is used to store board-specific but unchanging configuration parameters, such as calibration values to be set on the Inertial Measurement Unit (IMU) and magnetometer when the board is powered on. The EEPROM also stores the user HAM radio call-sign to use for radio transmissions (this allows the call sign to be changed without re-flashing firmware). Finally, the EEPROM is used as a permanent way to store a small amount of information that needs to be retained in case of power loss. Currently, this only includes a single byte indicating the flight state (described further in Section 9.1.2).

Another peripheral on-board Josh is the piezoelectric buzzer. The buzzer is used to indicate the arming state when Josh is powered on when mounted in the rocket on the rail. This provides an easy method to identify proper arming and boot without having a visual of the board itself.

Josh provides two battery connections via JST connectors. With two 8800mAh Lithium Ion (Li-Ion) batteries, it is predicted that Josh will maintain nominal operation for roughly 48 hours before requiring a recharge. The battery connections also provide charging support from the 5V supply coming from the USB-C connector. InSpace plans to include a small USB-C dongle inside the rocket attached to Josh, which can be connected to externally from the rocket with a magnet connector (pictured in Figure 86. This allows Josh's batteries to be charged before the rocket's flight without requiring a disassembly. The charging circuitry does not require Josh to be armed to work. Battery voltage is sensed through a voltage divider to the MCU's Analog to Digital Converter (ADC), and

is a measurement which is transmitted as part of the telemetry system in order to verify that enough charge is left in the batteries for flight. In addition to simple ADC voltage sensing, the PAC1951 sensor records current consumption and energy consumption information, which provides information for future years about the power requirements of the system.

Figure 86: Magnetic USB-C charging cable

Not found on the flight computer is the radio transceiver used for data transmission. The radio transceiver is found on a separate board and connects to the flight computer via a 4-pin Molex connector. The transceiver was designed as a separate board since two are used in the system: one in rocket as a transmitter, and another on the ground station as a receiver. The radio transceiver is described in Section 9.1.3.

9.1.2 Flight Computer Software

When armed, the flight computer starts a SRAD program that gathers sensor data, performs sensor fusion, and packages/queues data for transmission. The program is written in the C programming language and consists of a number of subsystems that run concurrently. The telemetry software has been redesigned this year with the goals of making it faster, more reliable, and adding the ability to detect and use flight state information.

Using the C programming language and NuttX RTOS for our flight computer software were choices primarily aimed at maximizing efficiency, allowing our software to run faster than if it was written for a programming language like Python and with a lower power consumption and more reliable task scheduling than with a desktop operating system like Linux. Last year the team used the Blackberry QNX RTOS, which required larger, power-hungry processors and did not provide the access to hardware that the team has realized is necessary to maximize performance.

Using the NuttX RTOS also offered a number of advantages over a bare-metal (software running without any operating system) system for our flight computer software. By making use of Portable Operating System Interface (POSIX) threads in NuttX, errors in the different subsystems could be isolated from each other. Additionally, the use of threads allows the different subsystems to be scheduled independently by the operating system, meaning the team did not need to design our own scheduling solution (usually a complex problem for real-time systems). The use of an RTOS allows the team to make performance guarantees by assigning a higher priority to more critical tasks, as well as providing robust mutex handling. This will ensure that high priority tasks (such as transmitting Global Positioning System (GPS) position) are performed first and are given all of

the resources they need to operate before other tasks. Due to NuttX being open-source under the Apache 2.0 license, the team was able to read the source code and extend existing drivers to fit our use case for rocketry. InSpace also contributed over 10 device drivers and several other feature implementations throughout the year for the use of other hobbyists.

The design of the flight computer software is similar to last year, with its overall structure remaining almost the same. The components that make up the flight computer software include a data collection subsystem, a sensor fusion subsystem, a data transmission subsystem, and a data logging subsystem. Each runs independently and communicates with the other subsystems as depicted in Figure 87 below.

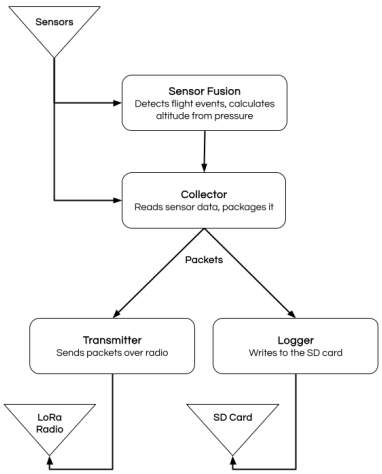

Figure 87: Telemetry Application Data Flow

The collection subsystem, depicted in figure 87, is the first step towards logging/transmitting data. It collects measurements from sensors, as well as fused data published by the sensor fusion thread, and packages them into a radio packet format that the team developed (the specification sheet is linked in H). The collection subsystem has been designed to include at least one type of measurement in each packet, to prevent sensors with a higher data rate from filling up radio packets. Every packet includes the most recent GPS coordinates of the rocket and the current battery voltage. The collector thread then passes the packets to the transmission and logging subsystems through a shared queue.

The logging subsystem has been designed to ensure reliable retrieval of data from the rocket. The first way it does this is by writing to a power-safe file system (`littlefs`) while the rocket is recording sensor data. If the flight computer loses power mid-flight, data corruption of this file system is impossible [15]. When power is restored, logging can continue as normal. Upon landing, the logging subsystem will copy data from the power-safe file system (which cannot be interpreted by regular computers without special tooling) onto a FAT32 file system which allows any computer to browse the flight logs. The logging subsystem also prevents overfilling the SD card on the pad by only recording up to 30 seconds of data until liftoff is detected. The 30 second window ensures that data recording of the lift-off event will not be lost due to a delay in detecting launch. However, overwriting the last 30 seconds of data prevents long delays of being armed on the pad from filling up valuable space on the SD card. Logging packets to the SD card is not throttled by waiting for the radio to transmit the packet, which means that even if all data cannot be transmitted via radio (due to low bandwidth), it will all be logged to the SD card.

The sensor fusion subsystem, also depicted in figure 87, is a new subsystem added this year with the special task of detecting changes in flight state. This task first passes sensor measurements through a digital low pass filter to obtain a less noisy estimation of the rocket's state. The natural choice for this task would normally be a Kalman filter [16]. Due to limited development time, InSpace has elected to filter the altimeter and accelerometer data using simpler methods this year, such as a median filter and a moving average filter. Median filters are great at reducing sudden spikes while still preserving sharp edges in data [17]. Moving average filters on the other hand is good for reducing random noise [18]. Each of these filters introduce a lag between input and output, but by keeping them smaller and with the high sample rate of the accelerometer and altimeter, this should not have much of an impact. InSpace is also recording the raw data measurements alongside filtered ones to the onboard storage. This will allow the team to perform sensor fusion post-flight to tweak filters using real flight data.

The sensor fusion subsystem then uses the filtered accelerometer and altimeter data to detect liftoff, apogee, and landing. These states and the requirements for the transitions between them are listed below in Figure 88.

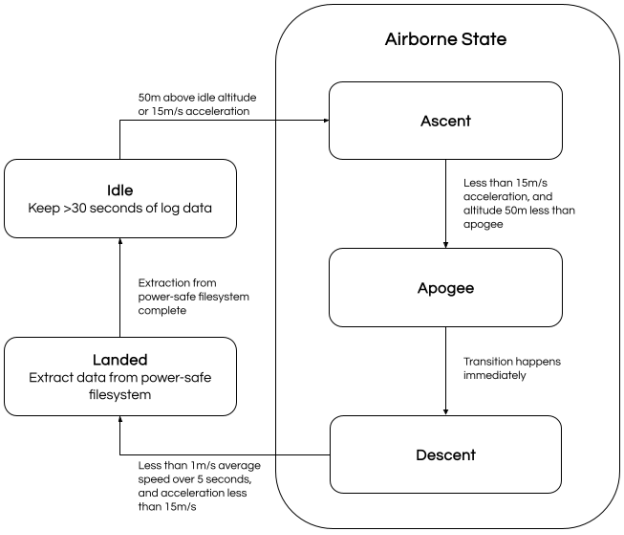

Figure 88: Telemetry Flight States

These flight states and transitions were chosen based on previous flight data and on the logic that the Altus Metrum altimeters use, and can be easily tuned after further testing to verify their effectiveness.

To further ensure resilience against power loss during flight, when the system's state is updated, it is saved to an EEPROM module on the flight computer. This saved information includes an 8 bit cyclic-redundancy check, preventing invalid data on the EEPROM from confusing the application. Importantly, if loading the previous flight state fails, the rocket assumes it is airborne and begins collecting data immediately. This is because power loss and corruption are more likely to occur in the flight state, and this assumption ensures that the flight computer loses as little data as possible. If the flight computer is truly in the landed state, it will detect that quite quickly since the altitude and acceleration conditions there are static.

One final, but very important note about the telemetry system is the methods available for it's

testing. First, NuttX provides a simulator that allows the telemetry system in its entirety to be run on a regular computer, complete with mocked sensor data to simulate a flight. The team can therefore run the system using data fed from a Comma Separated Value (CSV) of previous flight data (either the team's own previous flights, or flight data records published online) and observe how the latter behaves under load. The simulator also allows the subteam to write unit tests for each individual part of the telemetry system, making it much easier to catch bugs before flight. These tests are run on every software change that is merged into the software repository, preventing errors from being included in the final deployment. This functionality has been a critical aid in testing due to InSpace's very limited ability to flight test the avionics systems, it also allows every individual member to test their changes without the actual flight computer on their desk, especially since InSpace's budget limits the number of flight computers that can be manufactured each year.

9.1.3 Radio Communication

The radio transceiver was designed around the RN2483 radio transceiver chip, a low-power chip that provides Frequency Shift Keying (FSK) or Long Range radio (LoRa) modulation in the 433MHz band. The team's SRAD radio transceiver PCB, colloquially referred to within the team as "Talking Tuah", can be seen in Figure 89.

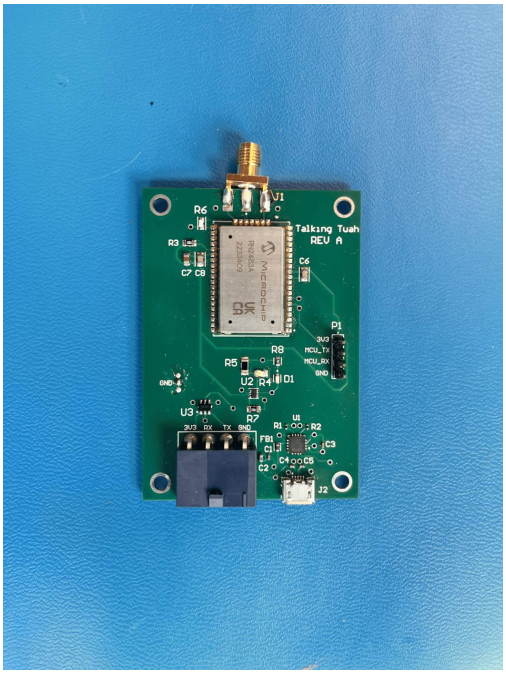

Figure 89: "Talking Tuah" radio board, Rev A

The RN2483 was selected because of its low power consumption of 38.9mA when transmitting [19, Table 2-3] and long-range capabilities, transmitting up to a distance of 15km in suburban environments [19, p. 1]. It also has a Commercial Off The Shelf (COTS) twin, the RN2483 Pictail board. This has been especially useful for comparison against our SRAD design during range testing.

The transmitter's sole responsibility is to relay packets from the flight computer aboard the rocket

to the ground station receiver. The PCB was designed to be used either on the rocket or from the ground station. This was done by providing a direct Universal Asynchronous Receiver-Transmitter (UART) connection from a Molex connector to the flight computer, while also implementing a USB-C interface so that the transceiver can be controlled from a general purpose computer. In this case, the Raspberry Pi 4B is being used as the ground station.

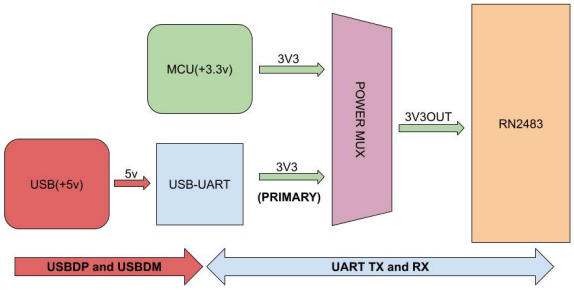

Figure 90: Power and data line architecture for transceiver PCB

The RN2483 transceiver operates on the 433MHz Industrial, Scientific, Medical (ISM) band. This frequency band was selected because it has a longer range at the same transmit power as the 868MHz band. Additionally, the longer wavelength is superior at penetrating obstacles (such as nearby trees) than 868MHz. The 433MHz band is a HAM radio band. InSpace has several certified amateur radio members who can operate our radio systems and provide their call-sign for use in transmissions.

The RN2483 is capable of both FSK and LoRa modulation, but InSpace will be using LoRa modulation since it provides better long-range performance due to better noise resistance. Additionally, LoRa modulation can robustly handle small shifts in frequency caused by the Doppler effect [20]. Further testing is underway using FSK modulation to determine if it would be a feasible option for rocket flight, as this type of modulation would increase the bandwidth of the radio link by 10x.

The radio's spread factor determines how much of the signal is spread across the frequency band used by the radio to distinguish the signal from noise [21]. It is possible to set spread factors from 7 to 12 [19]. A lower spread factor increases data rate whereas a higher spread factor is more tolerant to both noise and frequency shifts. For our purposes, a lower spread factor of 7 is preferred to maximize the data rate of our packets. Range testing has confirmed at least a range of 9 kilometers using this spread factor.

The team has selected a bandwidth of 125KHz to improve transmit speed and lower susceptibility to noise. InSpace also enables the cyclic redundancy check configuration of the RN2483 to provide error detection on transmitted packets. This allows packets corrupted in flight to be ignored by the receiver.

The coding rate determines how many bits of information are transmitted as compared to the total

number of bits [22]. The extra redundant bits are for error correction and detection. Since InSpace enables the cyclic redundancy check configuration of the RN2483 for tracking errors, our coding rate can be set to a high ratio of 4/5 to allow for the maximum amount of information to be sent as well as the fastest transmission rate. A coding rate of 4/5 means that for every five bits, there are four information bits with the fifth bit being redundant. Our transmit power is currently set to 15 dBm, which is the maximum possible transmit power for the longest possible range.

9.1.4 Ground Station Software

The ground station is a set of applications that receive, parse and display telemetry data transmitted by the flight computer in real time. It is comprised of three separate services that can be deployed using Docker (a virtual containerization tool, described below). These services are:

- A Python backend application that receives and parses transmitted data
- A React frontend web application that displays parsed data from the backend
- A map tile HTTP server for providing offline maps

Docker is an open-source application that allows one to create a custom environment for other pieces of software to run, entirely separated from the host operating system. By using Docker, setting up the ground station suite of applications for usage and development is much simpler and makes it easier to ensure redundancy in our system. Furthermore, using plugins for Docker such as Docker Compose, we can ensure that services automatically restart if they run into an error they can't recover from.

The ground station Python backend application is responsible for processing incoming data packets from the rocket into a format that the React frontend application can process. Keeping these two services separated means that there could be multiple frontend views of the same flight information, such as on team members' phones.

The ground station backend application is split further into three processes. The first of these processes is responsible for connecting to, configuring, and receiving data from the radio receiver module that is connected to the host machine via USB. The second process is responsible for transforming the packets received from the flight computer into a simpler format that makes it much easier to work with the received information. The third process is responsible for publishing data to a web socket server in JavaScript Object Notation (JSON) format, allowing communication with frontend views. In addition to spawning, managing, and publishing data to a web socket server, this third process also spawns a simple HTTP web server that allows for a simple debugging interface to the backend. This debugging interface makes it easier to diagnose issues or interact with hardware.

The React frontend web application allows the ground station operator and other users to view live telemetry data from a regular browser window. Users can send commands through the frontend to be able to perform operations such as connecting to the radio module, starting a recording of received data, or beginning a replay of a recorded flight. These commands first require that the user authenticate with the backend, to prevent simultaneous access from multiple users.

When launched, the React frontend establishes a connection with the web socket server provided by the Python backend process. It displays received data in a number of charts and visuals, such as through quick readouts at the top of the page, time series charts and gauges in the center of the "Dashboard" tab (as seen below in figure 91), and a live map displaying the path and last-known

location of the rocket in the "Map" tab (as seen below in figure 92). These readouts should give the team a complete picture about where the rocket is and what state it is in.

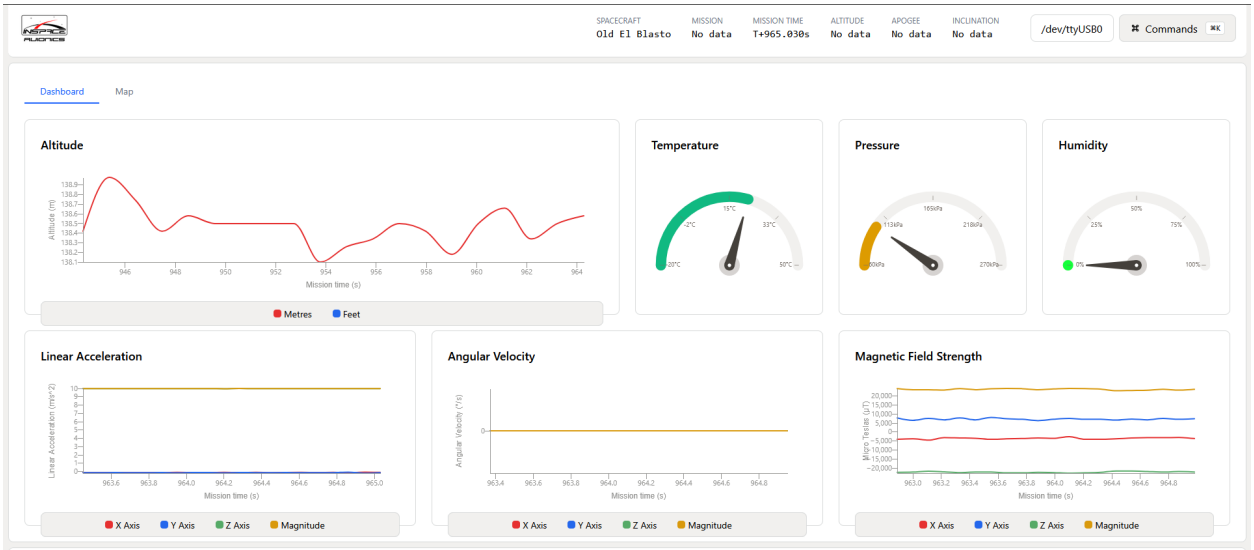

Figure 91: Ground station web interface

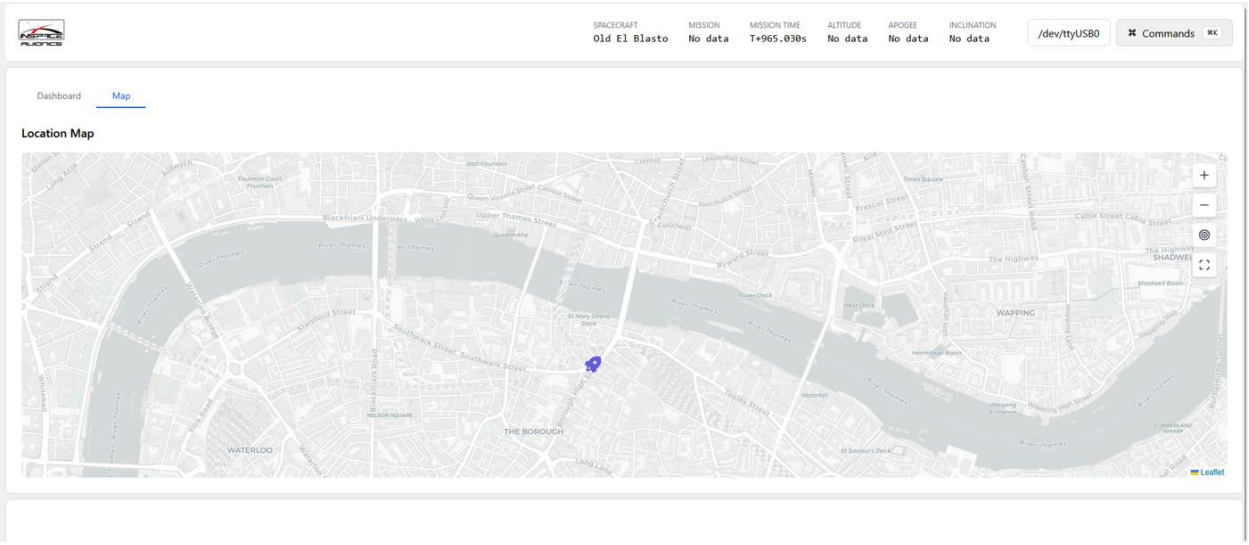

Figure 92: Ground station web interface map tab

The map in the "Map" tab is provided by querying a web server for images or "tiles" according to the area being displayed. Seeing as there will not be access to an internet connection at the launch site, the map on the "Map" tab will use tiles provided by a locally-hosted web server that provides downloaded images of the competition area. The images used by the team were obtained using Geographic Information Systems software like QGIS.

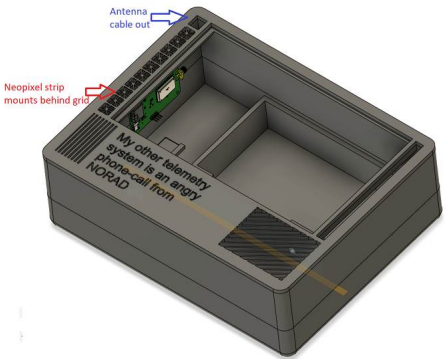

Figure 93: Launch console interior with radio board pictured

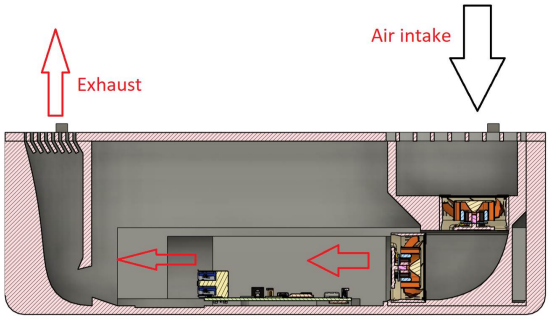

Figure 94: Launch console airflow cross section

Figure 95: Launch console interior

Figure 96: Launch console exterior

The UI can be displayed on any team member's laptop, but the team has also designed a "launch console" in a small Pelican case format to hold a Raspberry Pi 4B and OLED display. The launch console contains active cooling using two CPU fans with dust filters, as well as an additional HDMI output to provide to live-streaming services at competition. The Raspberry Pi runs the ground station software and displays it through the primary HDMI output to the OLED display, protected by an acrylic panel which is attached to the main frame using magnets. The radio board can also be included within the launch console case, and a coaxial cable is run from within the case to an external antenna (which is necessary for the long monopole antenna being used by the team). The entire launch console is powered off a large battery, pictured in Figure 97. The launch console frame is designed to be easily removable from the outer case for the purpose of swapping out the battery for charging as this was a major limitation in last year's design. The team has 2 backup batteries for the launch console in the event that the primary battery is depleted while waiting for the rocket to launch.

Figure 97: The launch console battery

9.2 Hybrid Engine Control System

The avionics and propulsion sub-teams worked in close coordination this year to accomplish an SRAD hybrid engine control system for the team's first hybrid rocket, CR25H.

The hybrid engine ground control system has three major components:

- The MCC: a switch control panel for remotely operating the control system
- The PCC: the control electronics system that remains at the pad with the rocket
- The Hybrid Engine UI: a native Python program for visualizing the telemetry data from the PCC in real time on multiple devices

In addition to these ground systems, there is also a single PCB that flies in the rocket. This PCB is responsible for switching power to the vent valve (XV-4) from the ground systems to an on-board battery after the rocket leaves the launch rail. This ensures that all of the oxidizer passes through the combustion chamber instead of venting.

9.2.1 Ground Systems Network

The ground systems architecture is based around a Local Area Network (LAN) which connects all of the devices in the control system together.

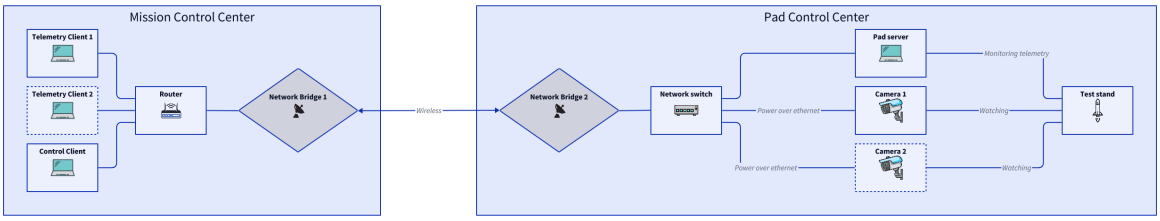

Figure 98: Ground systems network topology

Figure 98 shows the network topology of the ground systems, which is how all of the devices in the system are able to communicate with one another. On the pad side, the PCC and network cameras are connected to a Ubiquity LiteBeam through a switch. The Ubiquity LiteBeam provides a wireless network connection to a corresponding twin LiteBeam at the mission control side, which allows the MCC to connect with the PCC. The mission control side hosts a wireless capable router which provides DHCP reservations to the devices on the LAN and allows the control operators (and spectators) to wirelessly connect to the system to visualize telemetry.

The system is designed to leverage the robust communication options that networking hardware provides. It takes on a traditional client/server architecture, where the PCC acts as the server, and the MCC is a client. The telemetry spectators (the hybrid engine UI) are clients as well, but they are only allowed to view telemetry data and not send commands to the PCC. The control client reserves the ability to control the operations of the ground system.

The first design choice that leverages networking capabilities is the consumable telemetry stream. Telemetry is generated by the PCC, which reads sensors and publishes data over the telemetry

stream. The stream is implemented using User Datagram Protocol (UDP) multicast. This has several major advantages. The first advantage is that UDP is a connection-less protocol, which means it does not incur any additional overhead for acknowledging receipt of packets or re-transmitting dropped packets. This is suitable for a LAN based telemetry system, where packets are unlikely to be dropped. If a packet is dropped, it will be milliseconds before another is sent containing new telemetry information. There is no point to requesting re-transmission of outdated telemetry information, since this will incur more delay than simply waiting for the latest packet. The second major advantage is in multicast addressing: the UDP packets need to only be sent once by the PCC to the multicast address, but an unlimited number of subscribers (at least, up to the network bandwidth and LAN capabilities of the router) are able to receive the telemetry data at no additional overhead to the PCC. This allows the hybrid engine UI to run on multiple devices connected to the wireless router at the mission control side, and have all of them monitor the system in real-time.

The second design choice leveraging networking capabilities is the use of Transport Control Protocol (TCP) for the connection between the MCC and the PCC. This connection is used for the MCC to send control commands to the PCC for actuating valves and arming sub-systems. Unlike UDP, TCP is a connection-based protocol which is used for networked systems that require reliability. It requires all packets to be acknowledged in a three-way handshake (both the sender and receiver acknowledge messages), and also provides re-transmission of dropped packets. This is very important for the control connection, where command packets are much less frequently sent (compared to telemetry packets) and where it is not acceptable to drop packets (it would not be safe to "drop" a command to open a valve). TCP, being connection-based, also provides the ability to detect connection loss between the sender and receiver. If the PCC detects a loss of communication with the MCC, a 5 minute timer is started that automatically saves the system by opening the dump valve if the connection is not re-established.

Typically, TCP will only detect the connection break if a message is not acknowledged. This means that a message must be sent as a precursor to detecting a broken connection, which is not acceptable for the control connection where commands are infrequently sent. For instance, the operator may be monitoring the fill process and not actively sending commands. To resolve this problem and guarantee that both sides of the connection are able to detect a disconnect within 15 seconds, TCP "keepalive" was used to monitor the connection. This protocol extension sends "ACK" probes at a set interval on the connection automatically. If the probes are not acknowledged within the time limit, the TCP connection is considered disconnected. The current implementation sends 1 probe at 5 second intervals for a total of 3 probes which allows detection of a disconnect within 15 seconds.

9.2.2 Ground Systems Architecture

The ground systems are designed using a client-server architecture. There are several "nodes" in the system, of which there are three distinct "node" types.

1. **Server:** Generates telemetry and processes command requests
2. **Control client:** Generates command requests
3. **Telemetry client:** Consumes telemetry data (i.e. logs to a file, visualizes it, etc.)

The system is restricted to only one server node and one control client node. An unlimited amount of telemetry client nodes are permitted. All three types of nodes communicate using a message specification described in the Hybrid Control System Packet Format.

The simplest node is the telemetry client. The telemetry client must only implement functionality to receive UDP multicast messages and decode them into telemetry data. From that point, the telemetry client may do anything with the data it receives, such as logging it to a file or displaying it visually. This is what the hybrid engine UI does. Any number of telemetry clients are able to do this simultaneously on the same network, since they can all hook into the same UDP telemetry stream. This ensures redundancy across a new telemetry client with the appropriate software can easily substitute a telemetry client that has failed.

The next node is the control client node. This node is simply responsible for sending commands to the server node. The control client node is not responsible for maintaining any internal state about the actuators it is controlling (i.e., it should not "remember" if the fire valve is open or closed). It is also not required to know anything about the correct arming sequence or which arming levels give permission to actuate certain actuators. This information is all held by the server node, which is the single source of truth. If the control client node sends a command which it does not have permissions to perform, the server node will respond with a permission error. Only one control client node is permitted to be connected to the server at once, which is enforced by the server node. This prevents conflicts between control points and ensures that only one operator is in control of the system, and their connection cannot be hijacked during that time. In the current hybrid system, the control client node is a simple switch panel, with each switch associated with a certain network command. If the switch is flipped, the command is sent. It is up to the server node if the command is executed.

Finally, there is a server node. This node is responsible for generating a stream of telemetry which includes both sensor measurements as well as information about the current state of all actuators in the system. This allows the server node to be the single source of truth for the system state, preventing a scenario where nodes store their own state and get out of sync with one another. The server node is also responsible for processing all commands from the control client node. Commands are checked against the current arming level to determine if they are allowed to be executed, and if permitted, are turned into actuations of hybrid control hardware.

The control and telemetry client nodes are designed in such a way that it would be possible to create a node which combines the functionality of both. Such a node could perform decision-making tasks and send commands based on the current system state being reported (i.e. open the dump valve if pressures are spiking). This is not implemented on the current system, but was a design choice to leave room for improvements in further development.

At a high level, this described the architectural design for the control system. These design choices eliminate several sources of error (out of sync state, conflicting control operations, third party control) and are guiding principles for the implementation of the control system hardware.

9.2.3 Mission Control Center

The MCC is implemented as a network connected switch box.

Figure 99: MCC closed

Figure 100: MCC opened

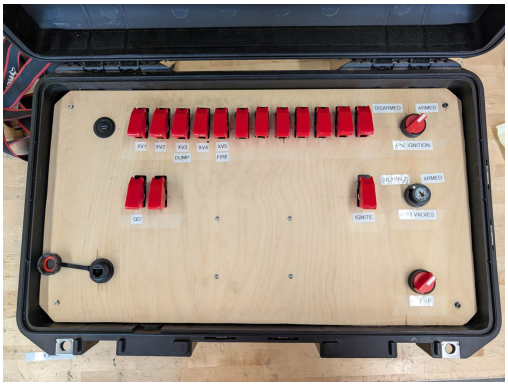

Figure 101: MCC switches

There are several key design choices to the build of the MCC. First, the Pelican case and wooden panel for switch mounting allow it to be transported without damaging the electronics inside. It is also weatherproofed this way.

The switches themselves are used for actuating the solenoid valves, the dump valve and the ignitor circuitry. They are each covered with a red cap which makes it visibly obvious when they are opened or closed.

Arming takes place on this box with several switches. The first arming level, used for actuating the valves, is obtained by placing the arming key in the key switch and turning it to the right. This arming key is the same key which arms the ignition circuitry on the PCC. This scheme ensures that the operator is not able to control the system at the pad until the team members at the pad return with the arming key, keeping them safe. A rotary switch is used to arm the PCC for ignition, and then certain actuators must be actuated in sequence to advance beyond that level, up to the point of firing. The arming progression is detailed in Section 9.2.5.

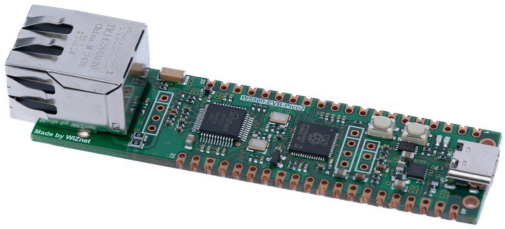

Figure 102: W5500-EVB Pico

The MCU used for the MCC is the W5500-EVB Pico, pictured in Figure 102. It provides an Ethernet interface and plenty of Input/Output (I/O) for switch connections while remaining low-power. All of the switches wired to the MCU trigger an interrupt when opened or closed, which keeps switching response latency in the microseconds. The MCU is powered via the micro USB port on the switch panel, which also provides the ability to re-flash the firmware easily and obtain a serial debug console to the MCU. The firmware for the MCC is written using the Apache NuttX RTOS. The firmware is open-sourced and can be found in Appendix H.

9.2.4 Pad Control Center

Figure 103: Front of PCC

Figure 104: Back of PCC, showing connectors

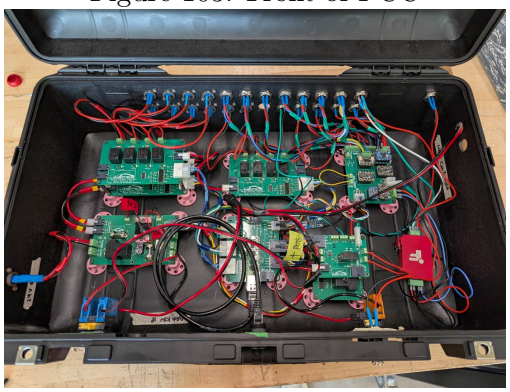

Figure 105: PCC interior layout

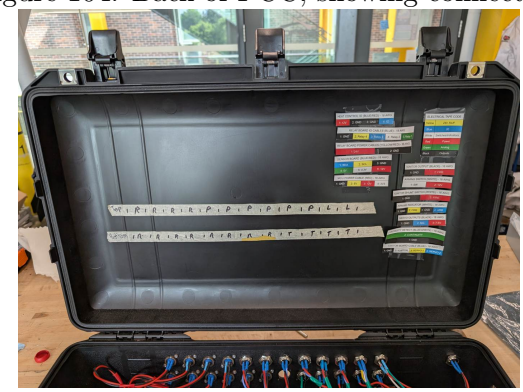

Figure 106: PCC wiring information

The PCC is implemented as a network connected array of sensors and relays. The entire pad control center is composed of 5 distinct types of boards, which are fixed inside of a large Pelican case for weatherproofing (shown in Figure 105):

1. Power board (1x)
2. Relay board (3x)
3. Ignitor board (1x)
4. MCU board (1x)
5. Sensor board (1x)

The PCC has inputs for the following sensors, connected using screw-on connectors:

- Pressure transducers (6x)
- Thermocouples (2x)
- Thermistors (2x)
- Tank mass load cell (1x)
- Thrust load cell (1x)

Outputs on the PCC are provided using military spec connectors (except the dump valve, which uses a screw-on connector). They exist for:

- Solenoid valves (11x)
- Dump valve servo motor (1x)
- Ignitor (1x)

All the sensor inputs and valve outputs can be seen on the back of the PCC in Figure 104. Some cables connecting to the box require an additional connection between the cable and the included cable on certain sensors (such as the pressure transducers). These connections use screw-on PLT circular connectors (pictured in Figure 107) to prevent them from accidentally dislodging. [23]

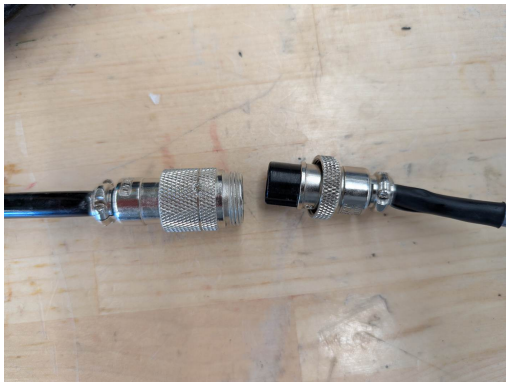

Figure 107: Connector unscrewed

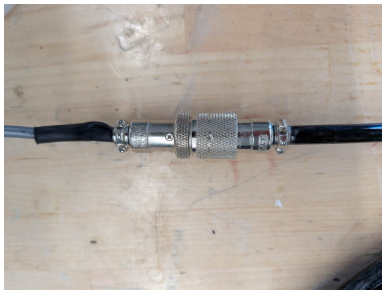

Figure 108: Connector screwed together

The PCC takes two power inputs: 24V power, which is supplied by two car batteries wired in series, and 12V power, which is supplied by a small lead-acid battery external to the case. A power switch

is used to supply the 12V input to the 12V rail powering the MCU and sensor network. This switch can be seen on the left-hand side of Figure 103.

The solenoid valves are powered using the 24V rail, which is fused at 15 amps. This provides enough current for all 12 solenoid valves to draw their required 1 amp without overdrawing beyond the rated amperage of the power board (20 amps). The 24V volt rail is not used to power any other devices in the PCC.

The 12V input is stepped down to 5V using a switching regulator, which is then stepped down to 3V3 by a buck converter. The 5V and 3V3 rails are used to power the network of ADCs, thermocouple amplifiers and tank mass load cell amplifier. The 5V rail is also used to power the MCU. All sensors are supplied power from the PCC. Pressure transducers and the thrust load cell amplifier are powered using 12V, thermocouples are powered from the thermocouple amplifier at 3V3, thermistors are powered at 5V and the tank mass load cell is powered using its amplifier at 3V3. The 12V rail used to power the aforementioned sensors, as well as the 5V and 3V3 rails, is fused at 5 amps. This is sufficient to power all the devices on this power rail while protecting circuitry during a short circuit. The dump valve is given a 3V3 Pulse Width Modulation (PWM) signal from within the PCC, but it is powered externally from a 7V5 LiPo battery.

Finally, the 12V rail is also used to power the ignition circuitry. This circuitry is given a separate 12V rail, fused at 10 amps, which is the limit of the ignitor relay. The ignitor used for CR25H draws only 7.5 amps. This circuitry is connected to the 12V supply through an arming key switch (pictured on the right-hand side of Figure 103). The ignition circuitry is only powered when the key is removed from the switch, which will happen when the pad team leaves the rocket and returns to mission control.

Both the 24V and 12V rails have primary and secondary inputs. The primary input will fall back to the secondary input if the voltage drops too low. In the current setup for CR25H, only the primary inputs are used.

9.2.5 Arming Progression

The arming progression for the entire hybrid control system is enforced using a combination of hardware arming circuitry and software enforced arming progression by the PCC. The full sequence of software-enforced arming progression can be seen in Figure 109. When moving from a lower arming level to a higher arming level, privileges are only additive (i.e., none of the privileges from the previous arming level are lost). It is possible to ”safe” to a lower arming level from any higher arming level, the sequence does not need to be performed in the reverse order. This allows the operator to rapidly safe back to the lowest arming state at any time to prevent accidental actuation of switches. The arming sequence must be obeyed when moving from a lower arming level to a higher one though, or the PCC will deny the arming request.

The arming sequence is started by placing the arming key in the PCC and turning it such that it cannot be removed. This prevents the ignition circuitry from receiving power on the 12V rail, therefore keeping the ignition circuitry in a safe state.

The next step is powering on the PCC. At this point, the PCC is considered to be in the `ARMED_PAD` state. This state only allows operation of the dump valve and vent valve, nothing else. The PCC still transmits sensor telemetry at this time.

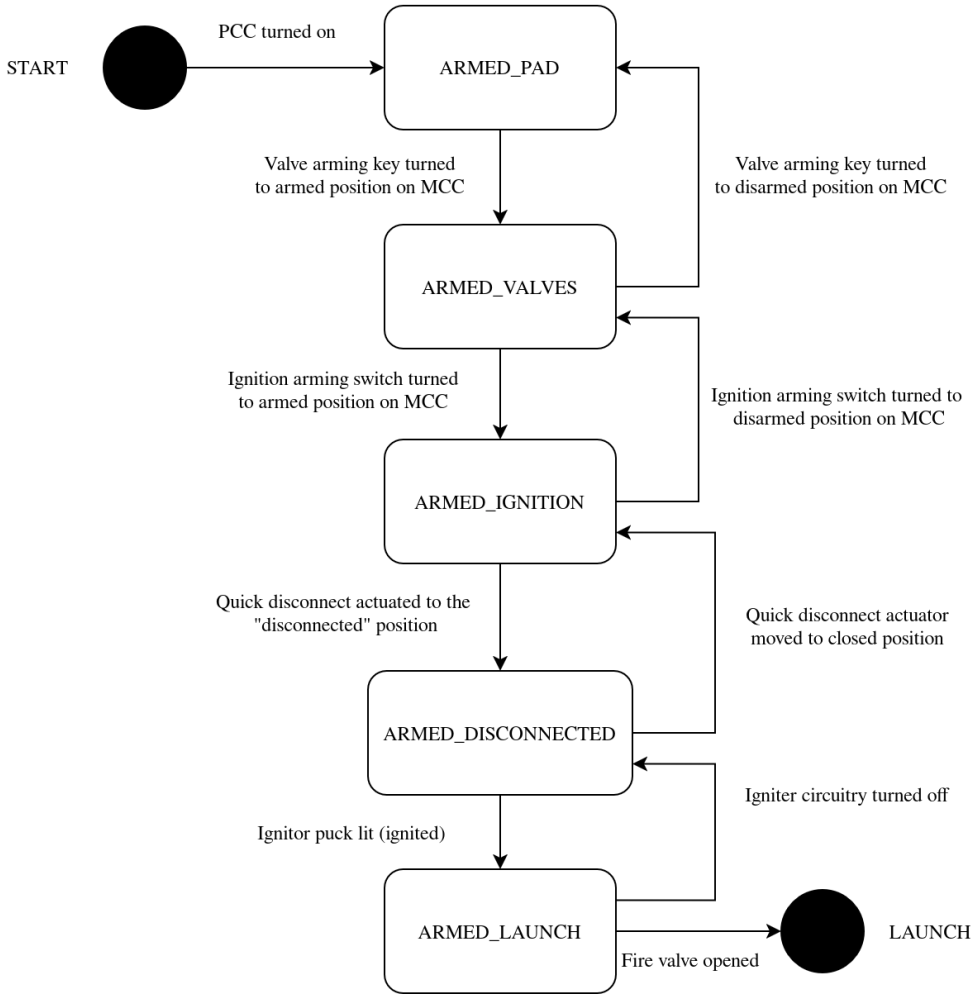

Figure 109: The arming progression of the hybrid control system

The next step in the sequence is arming the valves. This is done by placing the same arming key used for the PCC into the MCC (after it is powered on) and turning it such that it cannot be removed. Using the same arming key ensures that the pad team has safely powered the PCC ignitor circuitry and returned to mission control (a safe distance away from the pad) before the operator can actuate any of the valves or start the ignition process. Once the arming key is used to arm the valves on the MCC, the request to advance arming states is sent to the PCC over the network and approved, moving the PCC to the `ARMED_VALVES` state. From this point, all solenoid valves and the dump valve can be actuated.

From the `ARMED_VALVES` state, the operator can turn the ignition arming switch (pictured in Figure 101). This will send a request to the PCC to advance to the `ARMED_IGNITION` level, which will be approved. In this state, the operator gains the ability to switch the quick disconnect actuator. This actuator has since been removed from CR25H.

Once the quick disconnect has been actuated (again, this is a vestigial feature that simply provides another step in the ignition sequence for safety), the PCC updates the arming level to `ARMED_DISCONNECTED`. This arming state allows the operator to actuate the ignitor.

Once the ignitor has been actuated on by the operator, the PCC advances the arming state to `ARMED_LAUNCH`. This allows the operator to open the fire valve, dumping nitrous onto the lit ignitor and launching the rocket. The fire valve has since been removed from CR25H; the ignitor will instead burn through nylon tubing containing the nitrous, performing the ignition and launch simultaneously.

9.2.6 PCC Software

The PCC software is written on top of the Apache NuttX embedded RTOS. Using a RTOS allows InSpace to make guarantees about the response time and robustness of the system. NuttX provides several features that make the design of the PCC robust.

A primary feature is the ability to assign priorities to running threads and processes on the PCC. The PCC software is composed of two primary threads: one for handling telemetry, and the other for handling actuators. The priority of the thread handling actuators is higher than the thread handling telemetry, since it is critical that an actuation command is immediately handled when received. Telemetry can be paused for the short duration that it takes to actuate a solenoid valve, which is achieved through preemption. Priority settings also allow low priority threads to have full CPU access if they need to release a mutex that is required by the high priority thread, preventing blocking. Priority selection on NuttX is what allows a guarantee of very low latency on actuation requests.

Another feature that is leveraged by the PCC box is panic notifier call-chains. On NuttX, if a program behaves unexpectedly or there is a critical error in the system, the RTOS kernel will "panic". This is a crash. When this happens, a "panic handler" is called. Panic handlers are short routines to perform some task at the time of the crash. NuttX allows these routines to be registered through a panic handler interface. [24] On the PCC, there are two routines that get registered when the MCU boots up: one which turns off power to all of the solenoid valves, and another which opens the dump valve. This feature means that if the system crashes unexpectedly, it is guaranteed to enter a safe, de-pressurized state. This is the automatic-abort routine.

The panic notifier feature is further leveraged by the deliberate use of assertion statements throughout the PCC software. Assertion statements are logical statements which must evaluate to true, otherwise they trigger a panic. The PCC software is filled with assertion statements in locations where the program should not reach, or on statements that cannot fail. For instance, during operation the control handling thread should never exit. An assertion statement is used to guarantee that, or cause a panic which will safe the system. These assertion statements allow the PCC software to guarantee a safe system state on any logical failure or unexpected behaviour in the program.

As mentioned in Section 9.2.1, NuttX implements a TCP and UDP networking stack which allows the use of UDP multicast for telemetry and TCP keepalive for monitoring connection status. TCP keepalive allows the PCC software to guarantee that if the control connection disconnects for longer than 5 minutes, the system will be de-pressurized by powering off all valves and opening the dump valve.

All of the sensors in the PCC sensor network, including continuity detection for the ignitor leads, are sampled by the telemetry thread through a combination of the ADC and Micro-Object Request Broker (uORB) drivers at a total rate of 20Hz. Several measurements have been put in place to maximize the latter. The first one of which unifies the data from all the sensors in the collection

cycle to a single packet to only then transmit it, this is because when the packet size exceeds 128 bytes, it is copied to the Ethernet chip via Direct Memory Access (DMA) directly, saving significant CPU usage. The second optimization is about minimizing waiting time during sampling, to do so we trigger conversion for all ADCs as early in the collection cycle as possible, after the fact we check whether the conversion has been completed and if so retrieve the data. Every measurement is sent with a timestamp that has millisecond accuracy, in case of uORB drivers the timestamp reflects the driver retrieval time, further increasing accuracy.

9.2.7 Hybrid Engine UI

The Hybrid Engine UI is a native Python application that serves as the primary telemetry sink for the hybrid engine control system. It is built on top of the PyQt UI framework and displays the most recently recorded value for each of the sensors in the sensor array, the connection status of the system, arming and continuity state, the state of all actuators in the system, and time series graphs of all of the telemetry data whose range of displayed values can be configured dynamically.

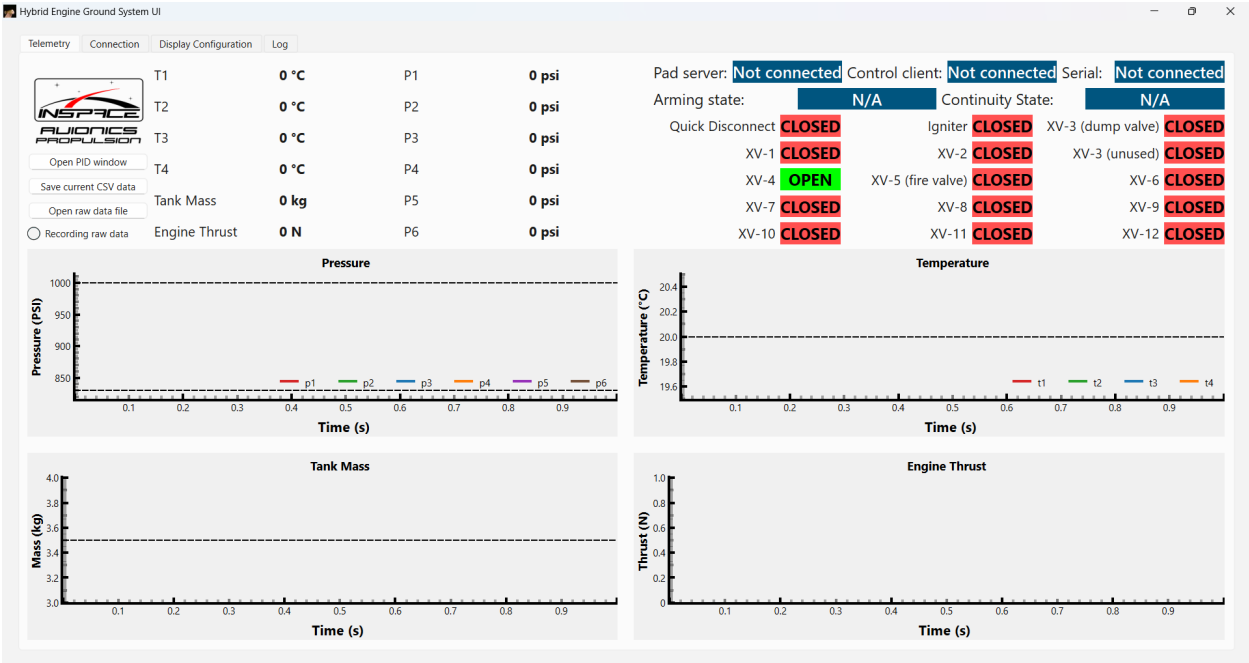

Figure 110: The Hybrid Engine UI Python application sitting idle

The user interface is also capable of opening a second window which displays the P&ID diagram for the hybrid launch system (Figure 111). This window displays readings from sensors within their position as represented by the P&ID diagram.

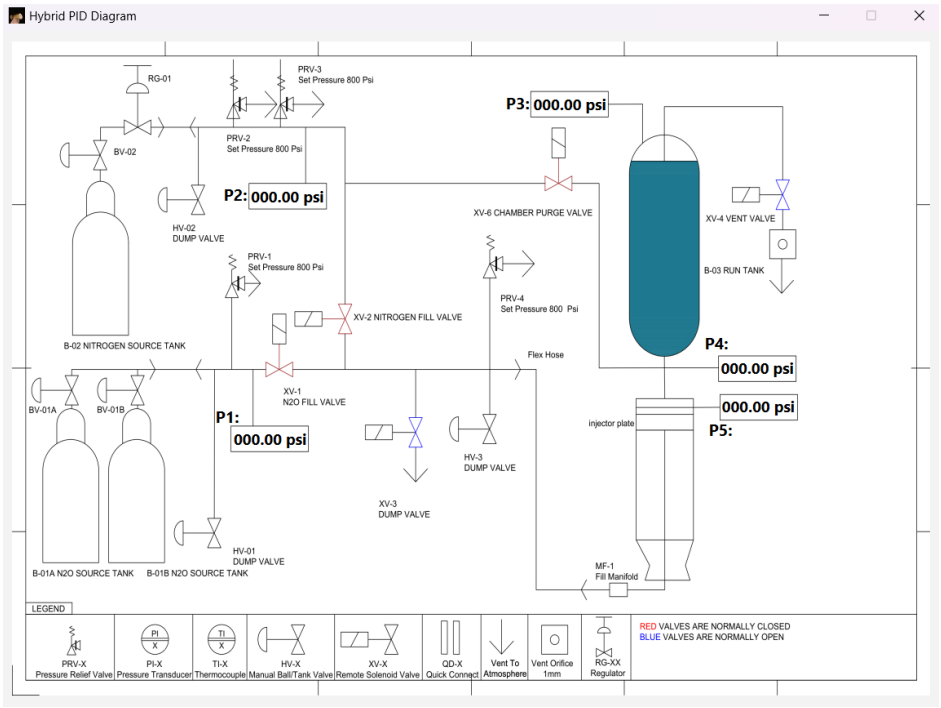

Figure 111: The P&ID window of the Hybrid Engine UI Python application

The PyQt framework was chosen for the Hybrid Engine UI for a number of reasons. Firstly, the framework is extremely feature-rich and provides robust libraries for critical features such as UDP multicast packet handling. The framework is also straightforward to develop with, offering a graphical UI builder which helped with the rapid development of the Hybrid Engine UI. The choice to use the framework also took into consideration the varying skill levels of team members as many of the students who had worked on the project will have either taken or will take courses that use this framework to teach subject material.

The Hybrid Engine UI application is run on the laptops of the operators at the MCC and optionally by spectators. For ease of use, a cross-platform executable is compiled and distributed to all team members helping with the launch of the rocket beforehand. The executable packages all dependencies required by the application, meaning that no additional software needs to be downloaded or configured. This ensures that this aspect of the hybrid control system is redundant, as should one laptop fail to be usable (battery is depleted, etc.), another laptop with the application downloaded can easily take its role. This has shown to be extremely useful during testing

In order to connect the Hybrid Engine UI to the PCC, the laptop running the Hybrid Engine UI must first connect to the router located at the MCC either via a wired connection (Ethernet) or wirelessly (Wi-Fi). Once connected, an operator can launch the application and join the UDP multicast group that the PCC is transmitting data to by navigating to the "Connection" tab (Figure 112) and clicking the button labeled "Join multicast group". The associated options are pre-set for ease of use. If connected correctly, the operator should begin to receive data and have it displayed on the UI. The operator can also modify how data is displayed on the main tab by changing and saving options found in the "Display configuration" tab (Figure 113) and view system logs in the "Log" tab (Figure 114).

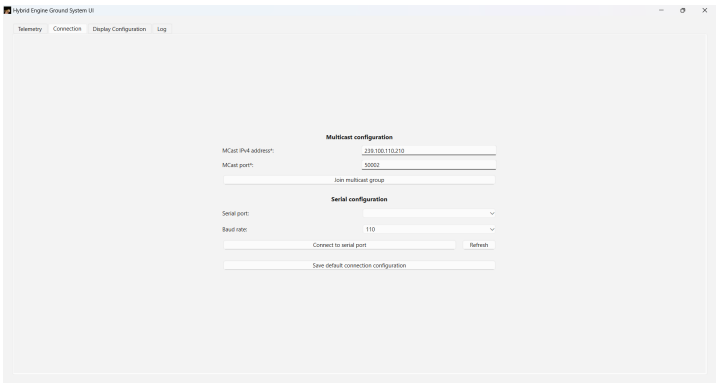

Figure 112: The Hybrid Engine UI "Connection" tab

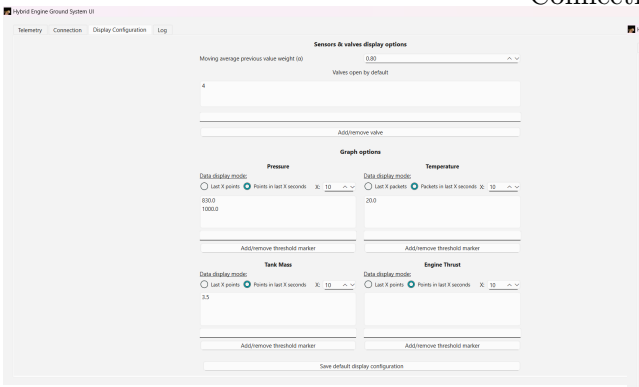

Figure 113: The Hybrid Engine UI "Display configuration" tab

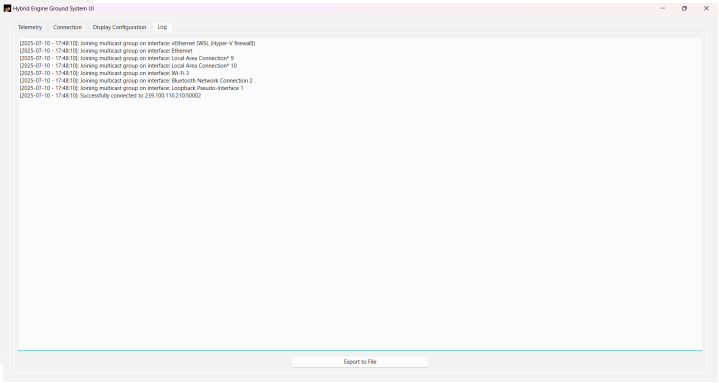

Figure 114: The Hybrid Engine UI "Log" tab

If at anytime the MCC loses connection to the PCC the Hybrid Engine UI will begin flashing to make it easier for an operator to notice. Additionally, to prevent an erroneous mouse click from terminating the application, safeguards have been put in place such that a user must confirm that they want to close the application.

The Hybrid Engine UI is capable of logging all data received in both its raw unparsed format and human-readable data as timestamped CSV files. Raw data can be re-displayed on the Hybrid Engine UI for post-test analysis, and human-readable timestamped data can be analyzed in applications such as Excel.

Saving human-readable data in particular is implemented in such a way that no data is lost. Upon receiving and parsing data, an in-memory buffer is written to with the parsed data. To ensure that the Hybrid Engine UI runs smoothly, a limit is set as to how big this buffer can be. The application will write the data in this buffer to a file on the host's file system once this buffer has reached this limit. This occurs on a separate execution thread spawned by the application. At the same time, the application will switch to a secondary in-memory buffer to ensure that no data is lost while writing to the file system. This is especially important to take care of considering the rate at which the Hybrid Engine UI receives and parses data. Operators of the Hybrid Engine UI can create sections of the human-readable timestamped data by clicking the "Save current CSV data" button. This allows operators to more easily organize data recorded using Hybrid Engine UI.

The Hybrid Engine UI application is open-source and available through the link in Appendix H.

9.2.8 Flight Hardware

The electronics flight hardware for CR25H consists of a single PCB, the *Valve-O-Matic*. This board acts as a "smart" switch, capable of supplying battery power to up to 3 24V solenoid valves during flight to keep them from venting nitrous.

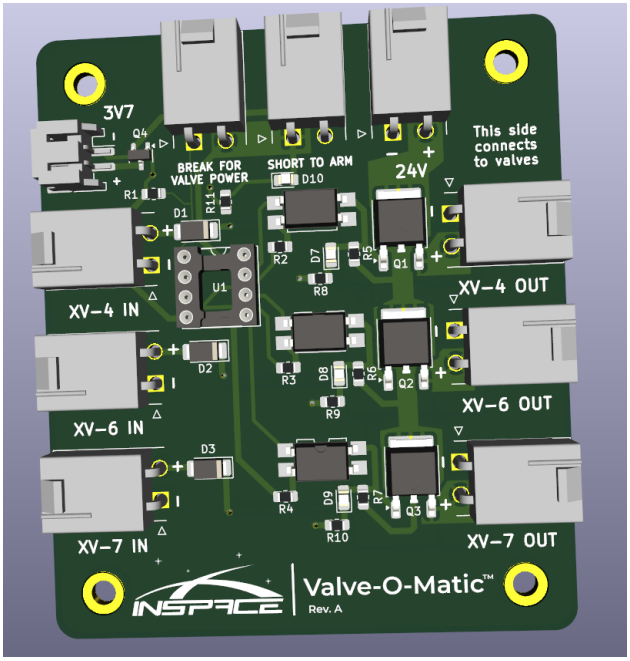

Figure 115: A render of the *Valve-O-Matic*

The board has 3 input connections where the valve control connection from the ground systems (PCC) can connect. There are then 3 outputs which connect the ground systems inputs to the solenoid valves that are flying in the rocket. During the time that the rocket is on the rail (for the entire fill sequence), the *Valve-O-Matic* acts as a wire from the ground systems inputs to the solenoid valves, providing normal control of the valves. A loop of wire is connected to the board via the continuity output connector. This wire runs external to the rocket with a quick-disconnect connector in the center of the loop. This connector will be pinned to a stationary object, such as the launch rail or ground. Upon launch, the force of the rocket leaving the rail will sever the quick-disconnect connector. The *Valve-O-Matic* will detect the loss of continuity in the cable, indicating lift-off, and flood the solenoid valve outputs with power from the battery supply, keeping them open during flight. A flight timer is started on the board's MCU to turn off battery power after 2 minutes, well beyond the expect burn duration of CR25H. This allows the flight tank to safely depressurize before the recovery team approaches CR25H.

Figure 116: ATTiny85-20PU MCU

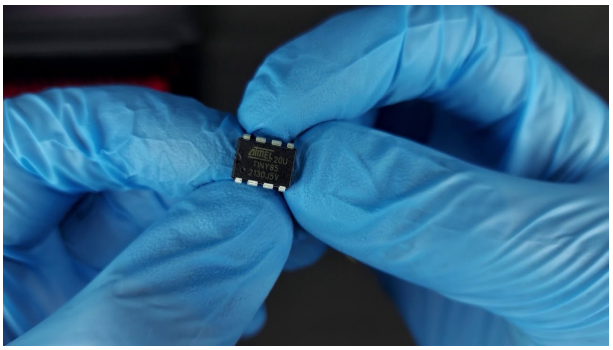

Figure 117: ATTiny85-20PU scale reference

The MCU used for valve control is the ATTiny85 (pictured in Figure 116), an automotive grade 8-bit micro-controller. The continuity line is connected to a pin on the MCU which triggers an interrupt on discontinuity. This allows the MCU to switch power to the solenoid valves with microsecond latency, ensuring that the valves do not vent any nitrous during the power switch-over.

The MCU is electrically isolated from the 24V power rail by using optocouplers for switching power. The MCU will be powered using a LiPo battery, while the 24V rail will be powered using a 1.7Ah DeWalt drill battery (with an adapter to router power to a Molex connector). This provides sufficient power to the valves for the total flight duration. The MCU draws less than 1mA of current during operation, so a 500mAh battery will provide over 500 hours of battery life to the device. The battery power is supplied directly to the MCU itself without need for an external voltage regulator, since the MCU is rated to be powered this way between 1.8V and 5V. An arming connector is used to prevent the flow of battery power to the MCU prior to arming.

The board provides several additional protections:

- Reverse polarity protection on the MCU battery input
- Flyback diodes on the solenoid connections to prevent transient current from damaging circuitry
- Electrical isolation of the two voltage rails using optocouplers to prevent a short from destroying the MCU
- Diodes on the ground systems input lines to prevent back-flow of current from the battery supply to the ground system in case of pre-mature opening of the battery MOSFETs

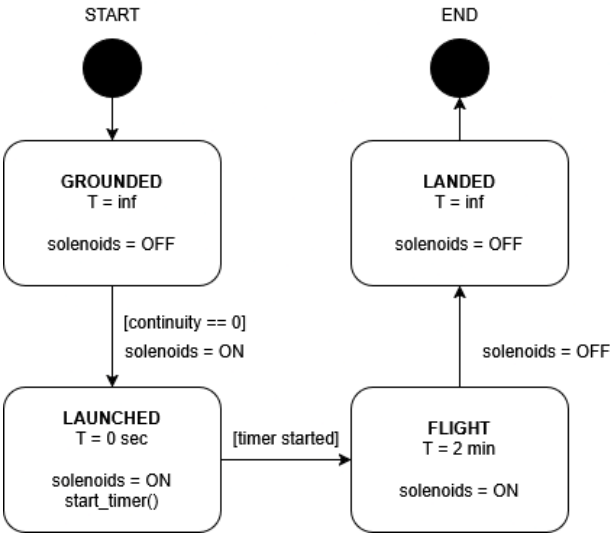

Figure 118: *Valve-O-Matic* control logic in FSM notation

The firmware used to control the *Valve-O-Matic* follows FSM logic, shown in Figure 118. The firmware is open-source and available through the link in Appendix H. The program is bare-metal C and is very small, allowing it to be easily analyzed for making guarantees about the upper limits of latency. The entire schematic and board layout can be found linked in Appendix G.

9.3 COTS Avionics

The COTS avionics components are mounted in a radio transparent fibreglass nosecone. The COTS components consist of:

- 2x Featherweight GPS trackers
- 1x Featherweight Blue Raven altimeter

Figure 119: 18650 lithium ion battery used for powering COTS electronics

9.3.1 Blue Raven

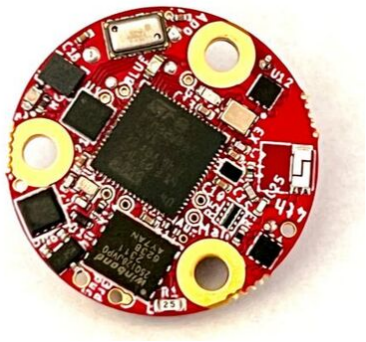

Figure 120: The Blue Raven altimeter

The Blue Raven altimeter does not control any energetics in CR25H. It is being flown purely to log altitude data as an initial test, since this is the first time the team will be using it. The altimeter is powered with its own 2000mAh 18650 Li-Ion battery (Figure 119), and it will be armed using its built-in magnet switch by waving magnets near it from the rocket exterior.

9.3.2 Featherweight GPS Trackers

Figure 121: Featherweight GPS tracker and its ground station

The two Featherweight GPS trackers provide dual redundancy on the rocket's GPS location in addition to the SRAD telemetry system. The COTS trackers are each powered with their own 2000mAh 18650 Li-Ion battery. Each tracker also has its own corresponding Featherweight ground station used to receive its telemetry broadcast.

The trackers operate on the 915MHz unlicensed band. In order to prevent conflict between the two trackers, they will each be operated on a different center frequency and channel, as assigned by Launch Canada. Prior to launch, both trackers will be configured with a unique call-sign corresponding to the device serial number. The Featherweight trackers will be armed separately using one screw switch each (pictured in Figure 9.3.2), connected to the arming terminal block on the Featherweight tracker. The screw switches will be accessible externally via a small hole in the rocket body tube.

Figure 122: Arming screw switch

The SRAD telemetry system will also be armed using a screw switch. All of the screw switches in the rocket will include a surrounding 3D-printed casing to prevent the screw from falling out of the switch in the case that it is backed out too far as seen in Figure 9.3.2.

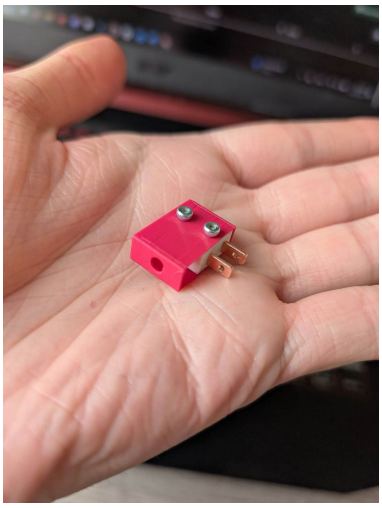

Figure 123: 3D printed casing for screw switch

All of the COTS electronics and SRAD avionics will be mounted on the enclosure (pictured in Figure 124), with batteries mounted on the interior. This enclosure sits inside the radio transparent nose cone on top of the avionics bulkhead.

Figure 124: The aluminum avionics enclosure (duplicates)

10 Recovery

The recovery sub-team is responsible for the design, assembly, and testing of the reefing parachute system and its deployment mechanisms.

10.1 Recovery Overview

This year's rocket will use a reefed recovery system. The system will use wireless communication between Raspberry Pi Pico W's to perform dis-reefing at 1500 feet. For Quarter Pounder, the entire recovery system will be packed inside a 3D printed SRAD parachute bay, aimed at improving mounting and organization of the parachute along with its reefing components. The reefing components have been re-designed from McTriple's resounding reefing success for more efficient use of space. COTS CO2 Raptors remain as primary and redundant deployment mechanisms.

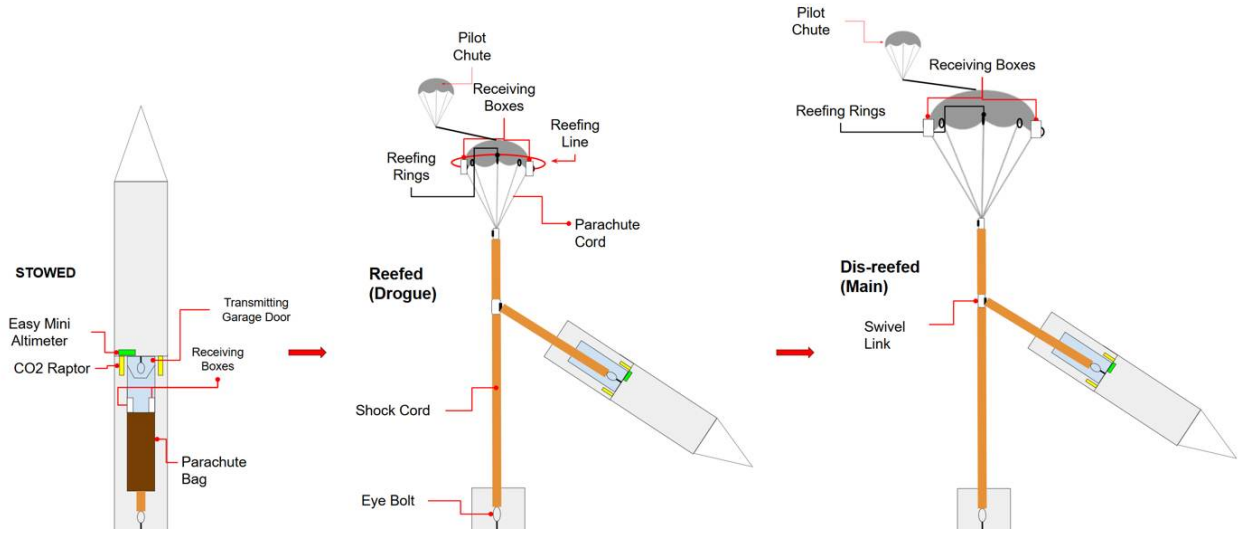

Figure 125: Recovery Wireless Reefing Configuration

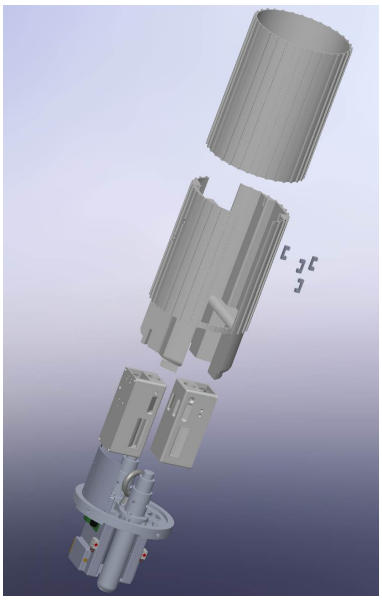

Figure 126: Recovery Full System CAD Assembly

Figure 127: Recovery Full System Assembly

10.2 Reefing System Configuration

There are multiple methods that accomplish reefing. The system developed and successfully flown by CU InSpace - as observed in Figure 125 - uses a wireless Raspberry Pi Pico connection and altimeters to initiate reefing. These electronics are stored within Receiving Boxes (RB) along the skirt of the parachute on opposing ends. The first box contains a Raspberry Pi Pico, screw switch, lipo battery and a COTS Mako line cutter. The second box contains an easymini altimeter, screw

switch, and the other lipo battery. The line cutter can be initiated by either the Raspberry Pico charge or the altimeter's main charge, making it a redundant system. At 1500 ft, the Pico in one of the RB receives a signal from a second Pico located in the Transmitting Garage Door (TGD) in the parachute bay on the recovery bulkhead, which is connected with a secondary easymini on the avionics stack to detect altitude. The second RB hosts our redundant easymini to send a charge to the mako line cutter at 1400 ft to allow for a delayed backup, which would be effective in understanding which of our systems worked nominally.

10.2.1 Receiving Boxes (RB)

The RBs act as the housing for the Mako line cutter, used to dis-reef the parachute. As previously mentioned, it has been modified from its successful flight at Launch Canada 2024 to be more compact. Embedded hex nuts have also been implemented for the threaded screw attachments. One RB will house our custom receiving Pico-PCB boards and the second one will house our redundant EasyMini altimeter.

Figure 128: Receiving Box

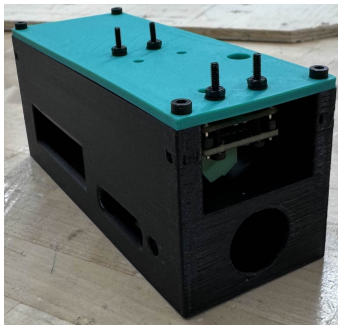

Figure 129: Receiving Box

The RBs have additionally undergone line cutter tests to ensure structural integrity of the prints and have withstood three charges, as seen in Figure 130 below:

Figure 130: Recovery Line Cutter Test

10.2.2 Transmitting Garage Door (TGD)

In order to transfer signal from the avionics stack mounted on the opposite side of the recovery bulkhead, our custom transmitting Pico-PCB is housed in a slotted mount that clips into the side of the parachute bay, along with a lipo battery and screw switch to arm the board.

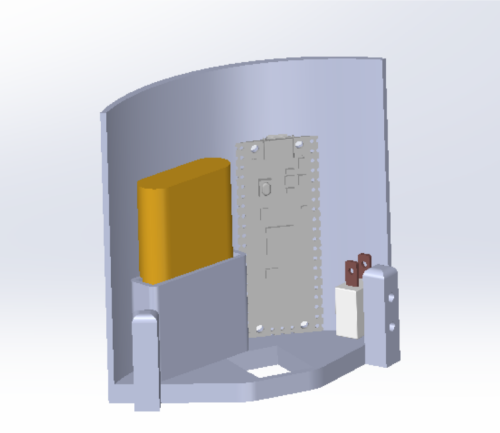

Figure 131: Transmitting Garage Door (TGD)

10.2.3 Custom PCB

The wireless reefing system, which uses a Raspberry Pi Pico connection, is facilitated through Recovery custom-made PCBs. These PCBs are designed to be attached to COTS Raspberry Pi Pico W's and provide a current boost to the dis-reefing charge. For Launch Canada 2025, the PCB design has been updated: 1) the addition of a buzzer to provide an auditory aid on the launch pad for successful arming, and 2) a flight timestamp logger has been implemented to keep track of certain recovery events (e.g. deployment charge), and PCB functions (i.e. successful wireless connection). This information is useful for the post-flight report. After multiple design cycles, the final board design can be seen in Figure 132.

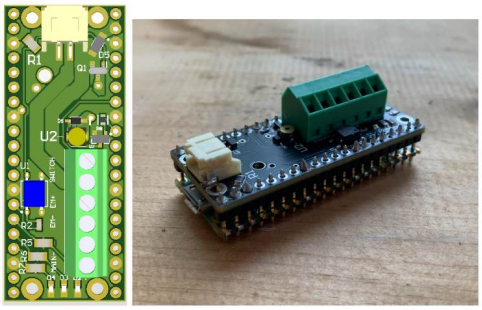

Figure 132: Recovery PCB

10.2.4 Parachute Bay

The majority of the recovery system will now be organized inside the parachute bay housing. It was designed to contain the RBs, TDG, and lower section of the folded parachute. The design is to be mounted onto the bulkhead with embedded nuts inserted in the base for easy threading during

assembly. Due to 3D printer size constraints, the parachute bay is two pieces. The upper section clips into the lower section. Note that the diameter of the parachute bay is just shy of the inner diameter of the rocket, and that it has ridges along the outside to keep it properly aligned against the body tube. This was done to ensure sufficient ventilation once the deployment charges are set off, increasing the odds of a successful deployment.

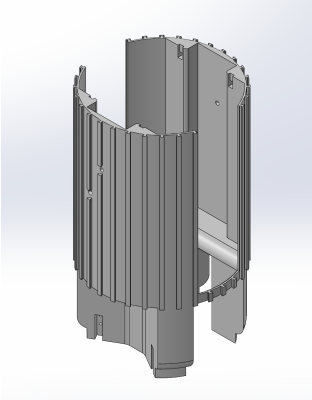

Figure 133: Parachute Bay CAD

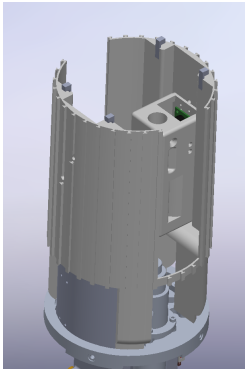

Figure 134: Parachute Bay CAD

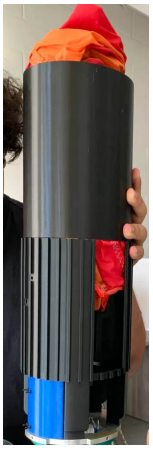

Figure 135: Parachute Bay Printed

10.2.5 Recovery Avionics Stack

The recovery and deployment avionics consist of two redundant Altus Metrum EasyMini altimeters. Each Raptor is wired to its own altimeter from its apogee terminal. One of the two altimeters has an additional connection between its main terminal and the transmitting Pico-PCB. Both altimeters are powered by manufacturer supplied 900 mAh lithium-ion polymer (Lipo) batteries and are mounted on a 3D printed housing. Along with the altimeters and batteries, connectors for the two screw switches and ignitors run through the housing mounted on the opposing side of the recovery bulkhead.

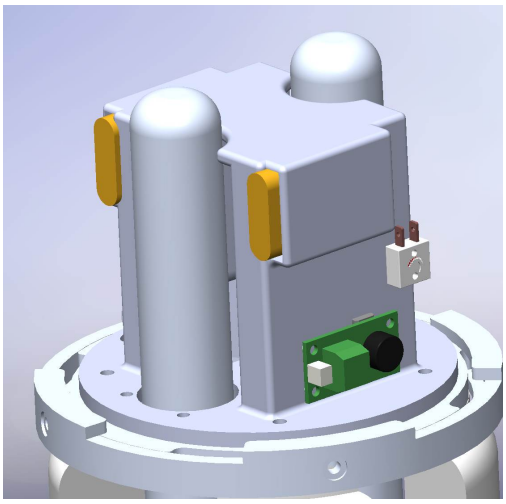

Figure 136: Recovery Avionics Stack

10.2.6 Recovery Wiring Schematics

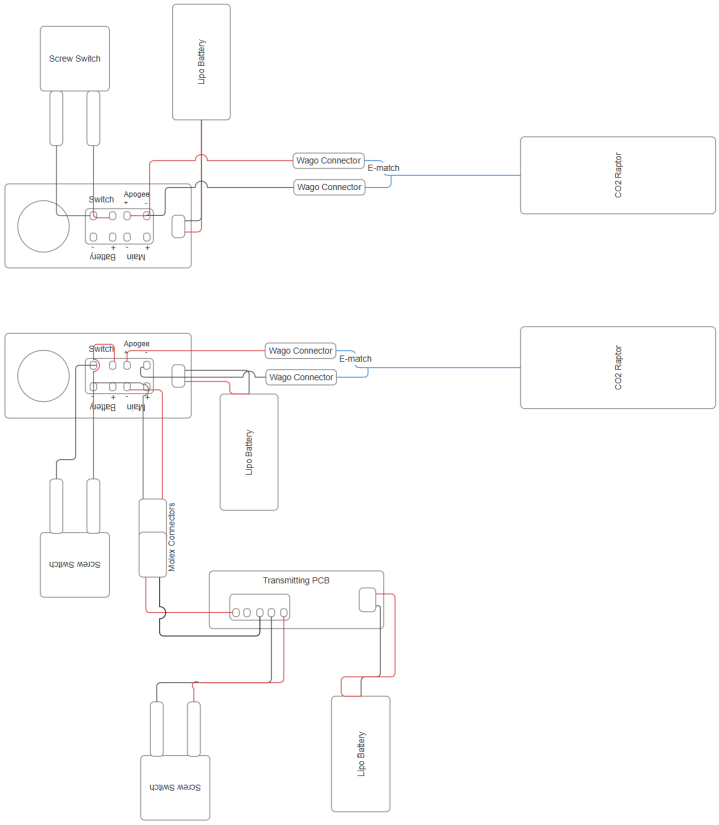

Figure 137: Recovery Wiring Schematic - Avionic Stack/Bulkhead/TGD

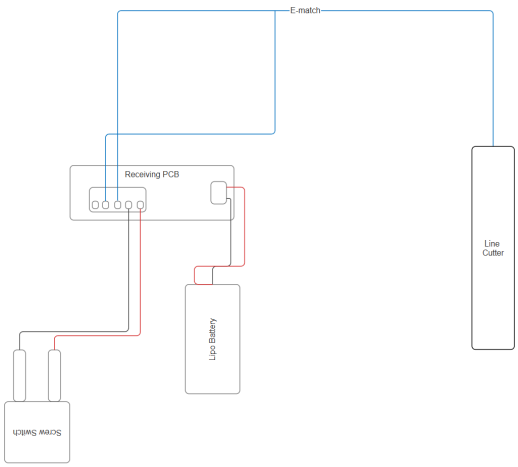

Figure 138: Recovery Wiring Schematic - Receiving Box 1

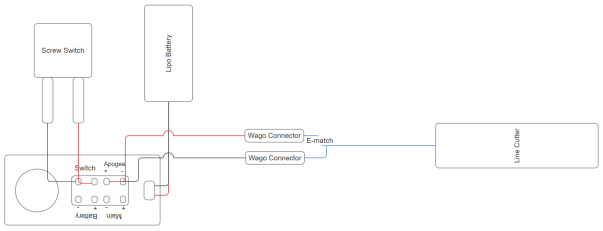

Figure 139: Recovery Wiring Schematic - Receiving Box 2

This system is a more refined version of the previous Reefing Recovery system, used in Launch Canada 2024 in a number of aspects as listed below:

A. Reduced receiving box sizes

B. Elimination of purple/transmitting box to allow for transmitting components to be attached to parachute bay directly thus reducing space consumption.

C. Elimination of dual altimeter and PCB redundancy to only one PCB and one altimeter

D. Elimination of threaded screws in 3D models by implementing embedded nuts, to improve structural rigidity.

E. Parachute bay designed to house all recovery components

10.3 Reefed Parachute Configuration & Design

10.3.1 Introduction

The main constraint for this year's parachute design was the opening shock force it would experience during reefing and full-chute deployment, since the dry weight (weight at motor burnout) of the hybrid rocket is approximately double as past designs. To reach the target descent velocities for both deployment stages, a full gore design would introduce shock forces too severe for the airframe's design. Ultimately, a disk band-gap parachute of 22 gores was chosen, as it provided a lowered opening force, while minimizing manufacturing complexity and parachute size.

10.3.2 Descent Velocity Calculations

The aforementioned shock forces and expected descent velocity were calculated in MatLab via the Pflanz Method. The entirety of the code is listed in Appendix A. It takes as input parameters:

- Drag Coefficient
- Opening Force Coefficient
- Canopy Fill Constants (listed as *Pflanz Opening Method Constants* in the code)
- Desired Descent Velocities (Drogue and Main)
- Rocket Mass
- Deployment Altitude (Drogue and Main)
- Ground Altitude (ASL)
- Reefing Line Force Percentage (listed as *Reefing Line Constant* in the code)

The code references diagrams in the Parachute Recovery Systems Design Manual (1991) by Theo W. Knacke. These calculations also relied on experimental data from the Manual for opening force coefficients (Table 5-2), canopy fill constants (Table 5-6), and the drag coefficient (Table 5-2).

The code used to simulate the results above can be found in these folders:

- Gore Stencil Calculator
- Parachute Forces Calculator

Parameter	Value	Units
Drag Coefficient	0.58	
Opening Force Coefficient	1.30	
Canopy Fill Constant	8	
Desired Descent Velocity (Reefed)	26	m/s
Desired Descent Velocity (Disreefed*)	8	m/s
Rocket Mass	34	kg
Deployment Altitude (Reefed)	9144	m
Deployment Altitude (Disreefed*)	457.2	m
Ground Altitude (ASL)	295	m
Reefing Line Force Percentage	7.5	%

Table 11: Parachute Parameters

*'Disreefed' refers to the main parachute.

Some assumptions were made regarding the environment, such as the gas constant of air (287 J/kg*K), acceleration due to gravity (9.81 m/s), and sea level pressure (101325 Pa) having typical values. With the above parameters, the code produced the following results:

```
------------- Reefed Chute -------------
Reefed area (m^2):   0.403873
Reefed diameter (m):   0.717097
Reefing line lenght (m):   2.25283
------------- Disreefed Chute -----------
Disreefed area (m^2):   11.1876
Disreefed diameter (m):   3.77419
landing speed (m/s):   9.29121
------------- Shock Forces -------------
Reefed opening shock force:   3052.56
Disreefing shock force (N):   452.979
Reefing line shock force (N):   33.9734
```

Figure 140: Recovery Parachute Sizing MATLAB Results

10.3.3 Dimensional Design

With the parameters above, the dimensional proportions for the canopy, gap, and band can be determined. The method is taken from the Recovery Systems Design Guide. [25]

The canopy has two components: the primary flat- circular disk that comprises the majority of the canopy, and the cylindrical band, each separated by a gap. Below is an illustration of the gore's dimensional layout, profile dimensions, and equations for determining each term (taken from the Guide, page 100).

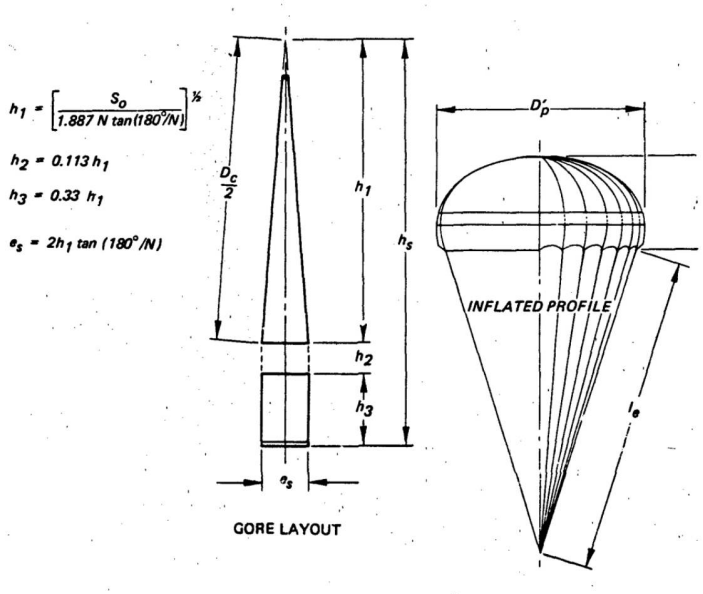

Figure 141: Primary Flat-Circular Disk

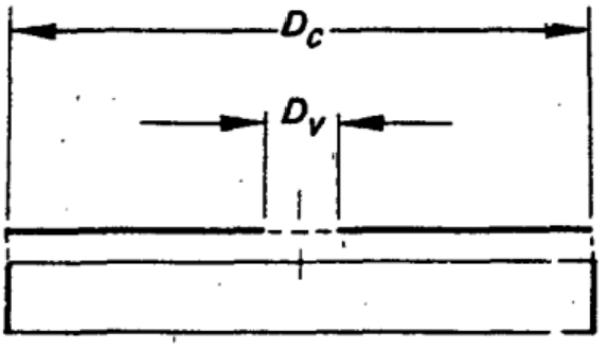

Figure 142: Cylindrical Band

The disk, gap, and band areas are 53%, 12%, and 35% of the total canopy surface area (denoted So), respectively. A summary of each term, its description, and calculated value is given below.

Although not calculated in the Guide, the radius of the vent hole is given. As a good rule of thumb, the vent hole diameter is sized to be considerably smaller than the reefed diameter, to avoid a collapse of the canopy.

Term	**Description**	**Value (m)**
h_1	Primary canopy gore height	2.23
h_2	Gap height between the primary gore and the band	0.25
h_3	Band height	0.74
e_s	Band width	0.64
D_c	Inflated Diameter	4
D_v	Vent hole diameter	0.32
N	Number of gores	22

Table 12: Parachute Geometric Dimensions

The shock cord length, l_e, is discussed in the rigging section.

10.3.4 Manufacturing

Foundations

The manufacturing of the parachute was laborious, but relied on a strict, consistent method. The primary stitch type used was the standard flat- felled seam, both because it's flat and incredibly strong. Flat- felled seams can be found on the side and inner seam of jeans (typically with two lines of yellow thread), employed there for identical reasons.

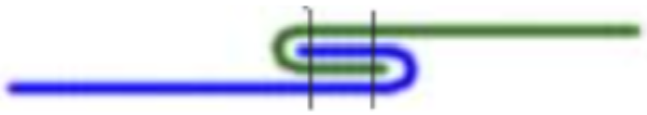

Figure 143: Flat- Felled Seam interlocking with stitch indications

Each edge of the fabric is 'wrapped' one over the other to interlock them. This creates a strong seam, as it distributes the load orthogonally (relative to the black stitch lines), across two stitches, thus greatly reducing the risk of material tearing during canopy inflation. Sewing through all 4 layers is essential for consistent force distribution. For additional reinforcement, a layer of ribbon was also inserted along the length of the seam.

Generally, straight stitches with a stitch length 3.5mm were employed, but typical and cross bar tacks were also employed to attach components or reinforce areas of high stress.

The fabric chosen was a typical nylon ripstop fabric with a waterproof coating with a reinforcement grid. For calculations, we assumed zero fabric porosity given the waterproof coating. It has minimal stretch along its warp and weft, but does deform slightly on the bias, which is typical for woven materials.

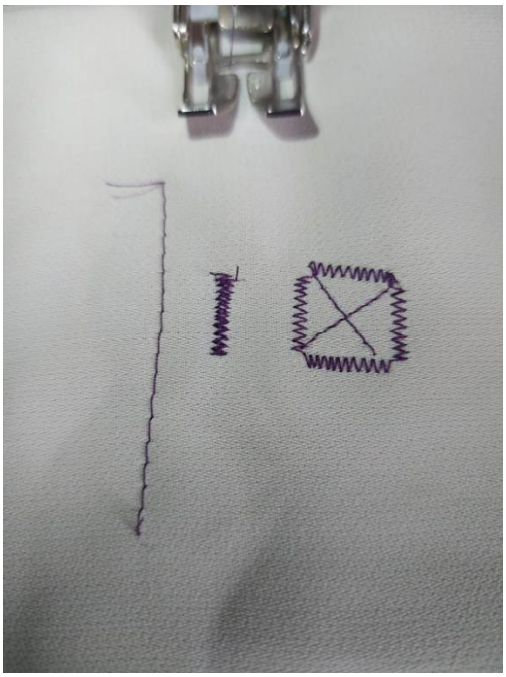

Figure 144: From right to left: 3.5mm straight stitch, bar tacking, and cross- bar tacking

Gore Stencil

Using the dimensions calculated in the previous section, a triangular stencil for the main gore section was made with a height of 2.07 m ($h_1 - \frac{D_v}{2}$) and base width of 0.64 m (e_s). Seam allowances of 0.08 m and 0.06 m were added to the height and base respectively, which were used for the seams and hems of the parachute, rather than adding to the surface area of the parachute. The subtraction of $\frac{D_v}{2}$ from the height is to account for the vent hole. An additional rectangular stencil was made for the band gore sections, with the same seams allowances of 0.08m and 0.06m for the height and width, respectively.

Figure 145: The finished gore stencil

Process

- Trace the 22 gores on the fabric, ensuring they're all oriented along the grain line.
- Cut the gores out
- Hem the gores at the skirt and vent hole to the seam allowance
- Press the side seams of the gores in a single fold, within the seam allowance

Note: Pressing, unlike ironing, applies concentrated heat over a longer period (than ironing) with the goal of creating a crease.

- Measure and cut 22 pieces of reinforcement ribbon (that will run the length of the seam)
- Interlock the side edges of 2 gores, with a ribbon running in the centre, pinning.

From top to bottom, the order should be top fabric (green), bottom fabric (blue), ribbon (pink), top fabric (green), then bottom fabric (blue).

Figure 146: Diagram of interlocking edges

- Sew each line of the flat felled seam (as shown in the initial flat felled seam diagram)
- Repeat steps 6 and 7 until all gores are attached into pairs

Figure 147: Side edges interlocked and pinned

- Attach the next set of gores (i.e. attach two 2x2, to make a 4x4) following steps 6 and 7.
- Repeat step 9 until all gores are attached to form the canopy
- Repeat steps 1- 10 for the band. A separate rectangular stencil was made for the band.
- Insert the suspension lines down the length of the canopy seams and fix the hem. That is, between the two lines of stitching. Reinforce with bar- tacking

Additionally, the receiving boxes had to be attached to the skirt. This was done by threading shock cord through it, then securing it with cross bar tacking. The surrounding fabric was reinforced with parachute tape.

10.4 Deployment

CU InSpace makes use of black powder paired with CO2 canisters as the deployment mechanism for the rocket. CR25H will be armed with 35g CO2 canisters, which have been proven to function well via ejection tests conducted with the rocket body tubes. The rocket's deployment system is armed with EasyMini altimeters which detect apogee and send a charge via a SRAD wiring system to an e-match attached to the CO2 Raptors.

In order to allow for barometric detection by the altimeter, the rocket body tubes are drilled with vent holes allowing sustained access to the atmospheric pressure for the altimeters and preventing any loss of gas volume when deployment charges are fired.

The entire recovery system has under gone ejection testing multiple times throughout the design cycle, with four successful results in the month of July. The four tests were run with 6 shear pins respectively and varying canister sizes to ensure 35g will function as required.

10.4.1 Rigging

The rigging of this year's recovery system has been summarized in the following table. Note that shock cord lengths are proportional to the rocket length.

Table 13: Recovery System Parameters

	Deployment Type	Reefing
	Deployment Method	COTS Raptor CO_2 Ejection System (Forward Section Deploy)
Parachute	Total Number	1
	Reefed Diameter	approximately 2.4 ft
	Dis-reefed Diameter	approximately 25 ft
	Material	Ripstop Nylon
Suspension Lines	Total Number	22
	Length	approximately 34 ft
	Diameter	1/8 in
	Material	550 paracord
	Strength	1200 lb
Shock Cords	Total Number	2
	Length	Main riser: 50 ft, Nosecone riser: 5 ft
	Diameter	1 in
	Material	Tubular Nylon
	Strength	**4000** lb

10.5 Next steps

Design & manufacturing phases have now been completed and the team is in the testing phase. Next steps will be to ensure all tests are successfully completed and any changes required are completed before August to ensure highest reliability of the rocket's landing.

10.6 Challenges

While no longer new, the system has been re-engineered and optimized for more efficient use of space and simplified arming procedures. However, there have been challenges relating to the connection between the Raspberry Pi Picos due to the addition of an auditory device for arming detection and a flight timestamp logging feature. A full system test with the completed hybrid parachute is still in the works and is posing a challenge in terms of packing, which will be resolved within the month of July.

11 Payload

11.1 Overview of Payload

The Payload sub-team constructs a scientific payload to compete in the Payload Competition at Launch Canada. This payload follows a 3U CubeSat form factor (10 cm × 10 cm × 30 cm). This year, the team is developing a payload to house cultured neurons, which will be observed under a microscope. The payload will also contain a Micro Electrode Array (MEA) to study the neurons' electrophysiology before, during, and after rocket flight.

11.2 Rat Cortex Neurons

The experiment will use primary rat cortex neurons sourced from Fisher Scientific, which will be cultured directly on the MEA. These cells are highly sensitive to environmental conditions, necessitating a life support system. The neurons will be cultured at Carleton University and given time to develop a healthy neural network and adhere to the MEA before departure for Launch Canada. The culturing procedure is based on a guide provided by Fisher Scientific, with modifications for use on an MEA [26].

Safety data sheets indicate that the dye used in the experiment is the only potentially hazardous substance, as it may cause skin and eye irritation. Therefore, whenever the dye is not in a sealed container, it will be handled using nitrile gloves and lab safety goggles. Although not strictly required, the same precautions will be applied when handling the neurons and other reagents. To prevent contamination, the neurons will only be exposed to air under a laminar flow hood, which directs filtered air onto the neurons to prevent particulates from causing harm.

11.3 Life Support System

The life support system is responsible for maintaining the strict environmental conditions required for neuron survival. Cultured neurons require a temperature of 36–37.5°C, approximately 5% CO2 content, and at least 65% humidity. Additionally, minimizing vibrations is essential for improving microscope imaging and preventing the neurons from detaching from the MEA.

Temperature control is achieved using a thermoelectric heat pump that can heat or cool based on sensor feedback. Humidity will be adjusted manually by spraying water mist before sealing the life support compartment. CO2 will be introduced into the sealed compartment using a CO2 injector. The compartment is thermally insulated and equipped with shock absorbers, while the microscope is isolated from the compartment walls using rubber vibration dampers to further reduce vibrations.

11.4 Micro Electrode Array

The MEA is a device used to measure neuronal action potentials and stimulate neurons with electrical pulses. The design is based on the OpenMEA open-source MEA [27]. To reduce cost and complexity, the OpenMEA design has been modified to use 16 channels/electrodes instead of 64. By using an MEA, it is possible to assess both the viability and electrical activity of the neurons. The user interface of the OpenMEA software can be seen below in Figure 148.

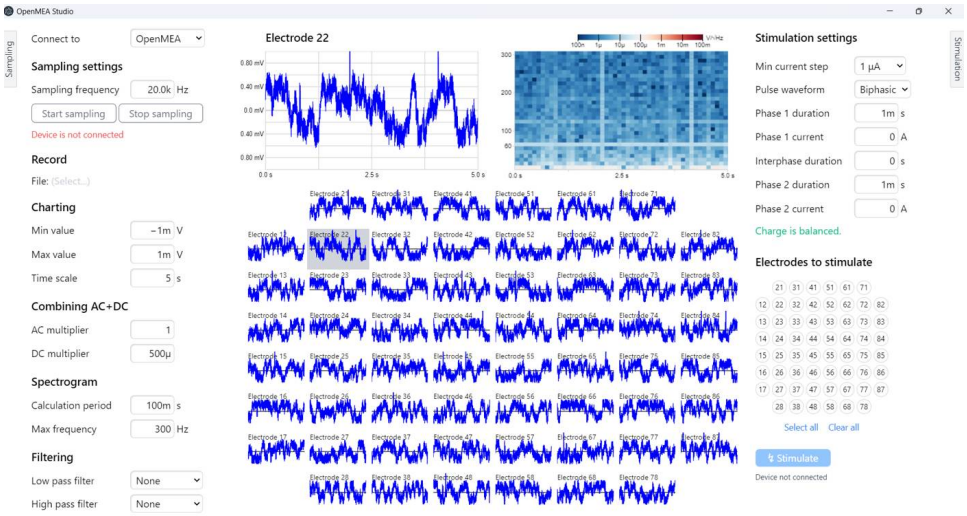

Figure 148: User interface of OpenMEA Studio [27] displaying simulated electrode signals

11.5 Fluorescent Microscopy

Inside the life support compartment, the microscope observes neuron viability using fluorescence from an applied dye. The experiment will use Calcein Blue, which fluoresces blue when cells are alive. This dye has an excitation wavelength in the ultraviolet range, requiring an LED that emits in that spectrum.

The microscope uses a Raspberry Pi to capture images and store them on an SD card. A dichroic mirror is used to separate excitation and emission light; it reflects ultraviolet light from the LED onto the dyed neurons while allowing the emitted blue light to pass through to the camera. A diagram of the fluorescent microscope can be seen below in Figure 149.

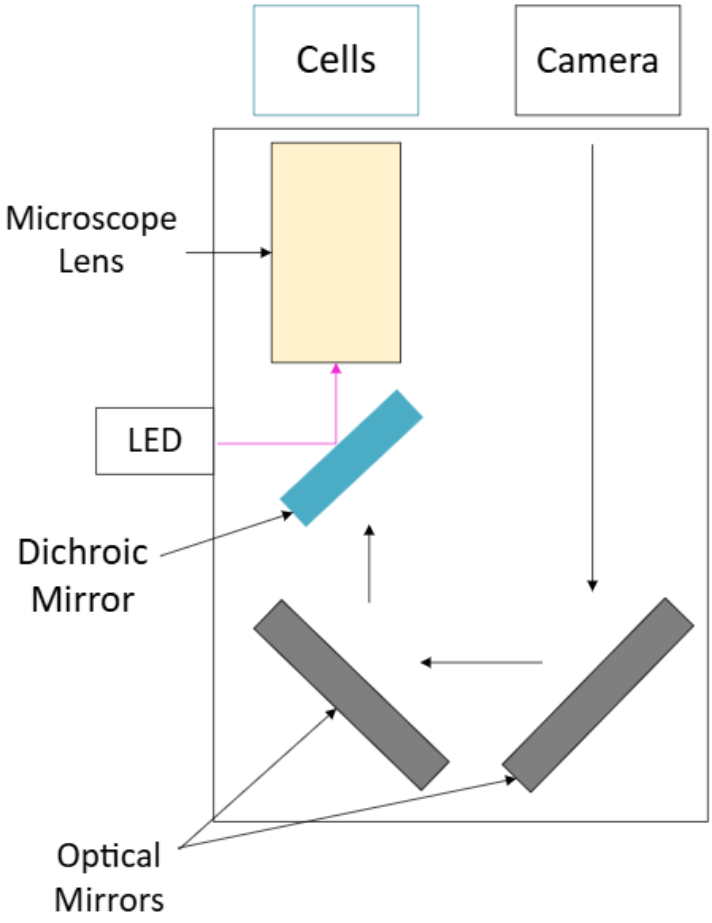

Figure 149: SRAD Fluorescent Microscope Block Diagram

12 Ground Support Equipment and Architecture

12.1 Launch Tower

12.1.1 Tower

The main supporting structure of the launch tower are 10' antenna tower sections. This modular system will allow for varying launch rail heights. The launch rail is comprised of 1.5" aluminum extrusion supported by a 1.5" C-channel. The C-channel is secured to the antenna tower section using carriage bolts. Quarter-inch steel mounting plates are used where two 10' sections overlap to create the most overlap in the rail sections and maximize the structural integrity of the rail during launch.

The four pieces of aluminum extrusion seen in Figure 150 will allow our vehicle to have a simple and standard sliding track system. The second piece is an aluminum channel that holds the extrusion in place and adds strength to the 10' sections of extrusion by having its own joints off set from the extrusions. Thirdly is the main tower sections themselves which are made from steel tubing and are bolted together to create the full 40' tower at make height. Lastly is the bolt plate which is simply a .25" steel plate with bolts, these are positioned to overlap the connections between pieces of aluminum extrusion and aluminum channel.

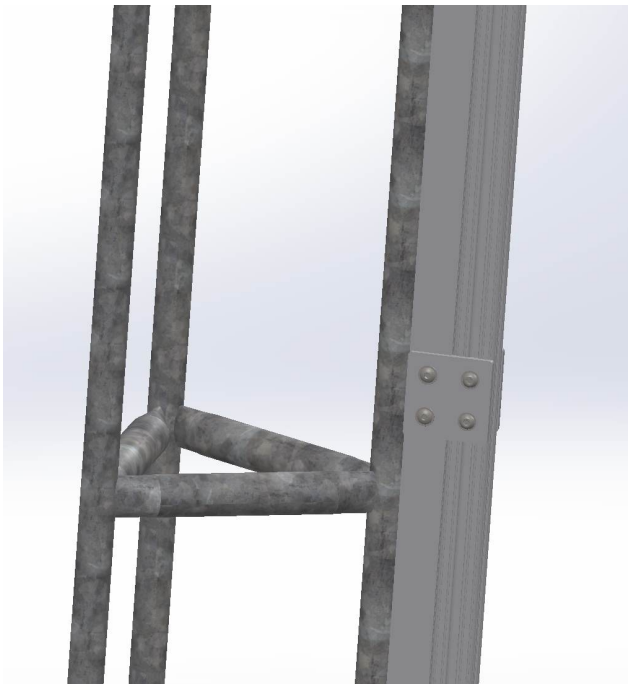

Figure 150: Side View of Tower Section

12.1.2 Base

The base of our launch tower system is compromised of three "legs" each around 3 feet in length, made from square steel tubing. At the end of each leg we have a down turn in the tubing to lift the whole base off the ground. This extra height allows us to level the tower more efficiently on more uneven terrain. At the end of each leg we have added a 1.5" hole to secure a helical ground anchor through. Each leg assembly is slotted and bolted into a larger piece of square tubing in the

centre of our tower base. The centre assembly is made of the three pieces of square tubing and two truncated triangular pieces of .25" steel. Connected to the central base of our tower is a riser bar made from more square steel tubing. Bolted to the riser bar is our four bearings that support the full range of motion our tower is built for. The tower uses a rack and pin system to allow for launch angles between 81 and 90 degrees.

The tower is raised via a gin pole and an electric winch, which will be anchored using more helical ground anchors. The tower will also be aided in raising with human interaction as the force needed to lift the tower the first few degrees is exponentially higher than later in the raising process. The main tower sections are bolted securely to the moving leaf assembly and gin pole allowing for ease of mounting of the rocket, then easy lifting of the tower and rocket prior to launch.

12.1.3 Loads

The tower has been designed to withstand winds of up to 35 kts (65 kph) with a safety factor of at least 2. This is accomplished through the use of guy wires, which are all anchored with more helical ground anchors. There are three guy lines placed three quarters of the way up the tower allowing for not only stabilization during use but allowing extra force to aid in the raising of the tower.

12.2 Fill System

Details for the propellant fill system can be found in Section 7.6.

13 Flight Simulations

The vehicle performance was simulated using RASAero II. While the team has simulated the performance of many past rockets, this year was the first to simulate a rocket employing an SRAD motor, in addition to it being the team's first hybrid motor. This was achieved by compressing the thrust curve of the engine down to 32 data points, and appending it to the RASAero II engine configuration file along with the tank and combustion chamber length, total wet mass and total propellant mass (sum of fuel grains and oxidizer). An off-rail stability of two body calipers was found using the Rogers Modified Barrowman method, which is nominal. The results of the flight performance simulation are tabulated below:

Table 14: Predicted Vehicle Performance

Apogee *[m(ft)]*	7857.4 (25 779)
Max Accel *[m/s^2 (ft/s^2)]*	85.6 (280.8)
Max Velocity *[m/s(ft/s)]*	461.2 (1513.1)
Off-rail Velocity *[m/s(ft/s)]*	40.6 (133.2)

The rocket trajectory data was also modelled using the RASAero II. The following graphs depict the preliminary flight data; the coordinates used are those of Timmins, Ontario; where the Launch Canada competition was held last year. A max nominal launch angle of 6° was assumed as per R10.1.1 in the Spaceport America Cup Design Test and Evaluation guide [28]. A similar value is assumed for Launch Canada. The simulation also takes into account a reasonable wind speed of 24 km/h to help predict the maximum distance downrange the rocket would drift under a nominal parachute deploy [29]. The final assumption in the ground track simulation was that the launch rail is pitched into the wind, which in theory would minimize the distance the rocket would travel relative to the launch site. The predicted ground track of the flight can be seen below in Figure 155.

A series of plots were generated from RASAeroII, which may be seen to visualize the placement of events and the altitude, velocity, acceleration, and stability over the time of flight. These may be seen below in Figures 151-153.

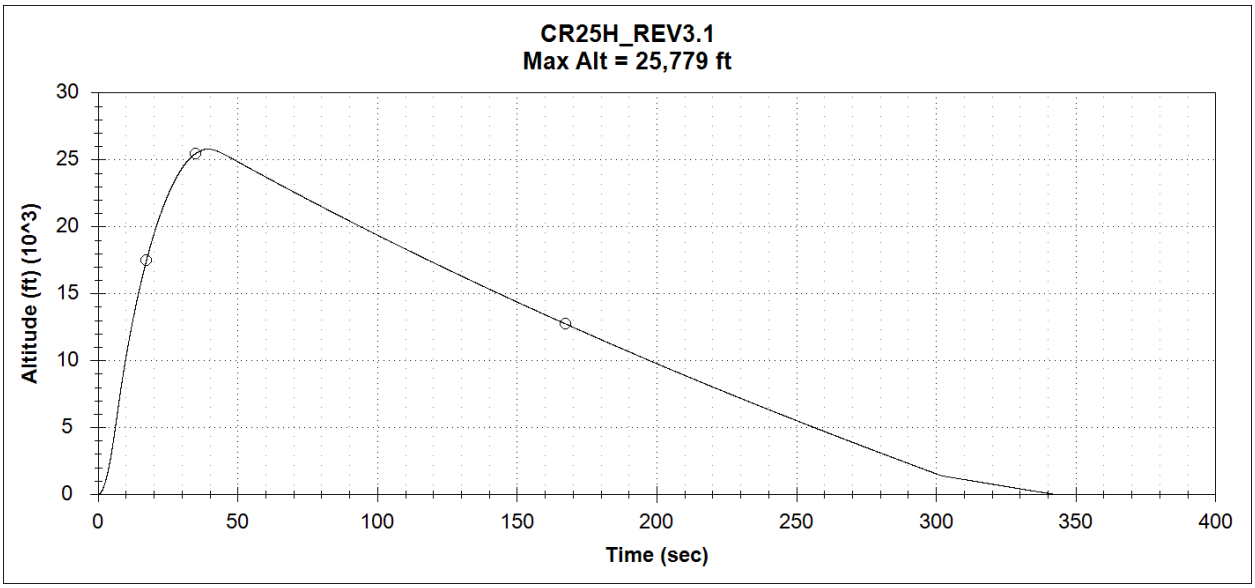

Figure 151: Predicted Altitude Data of Flight

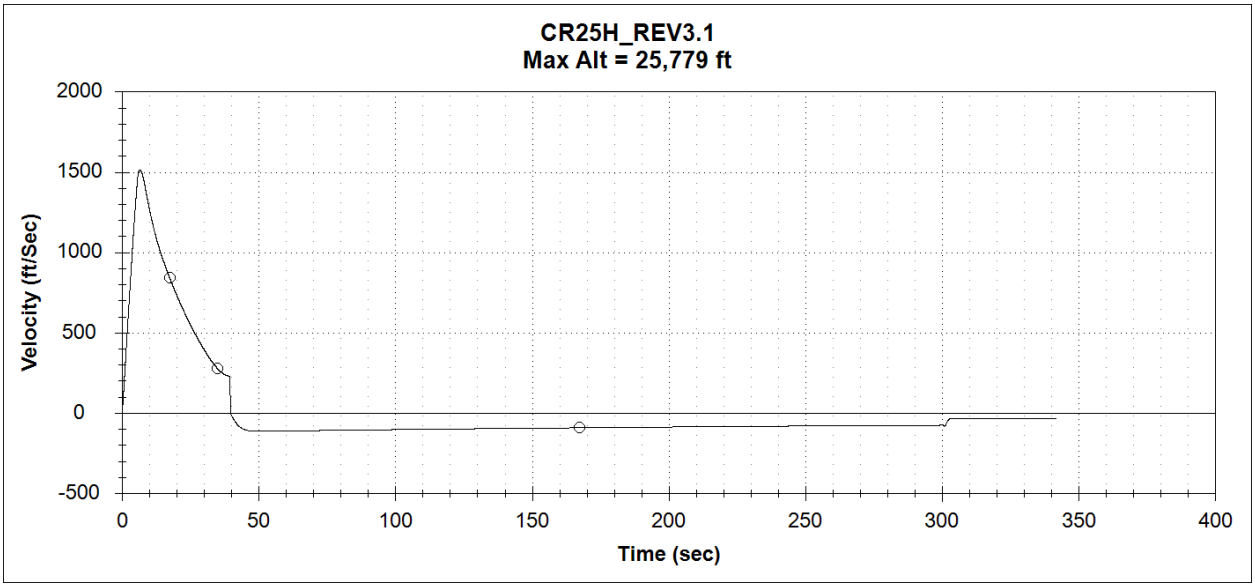

Figure 152: Predicted Velocity Data of Flight

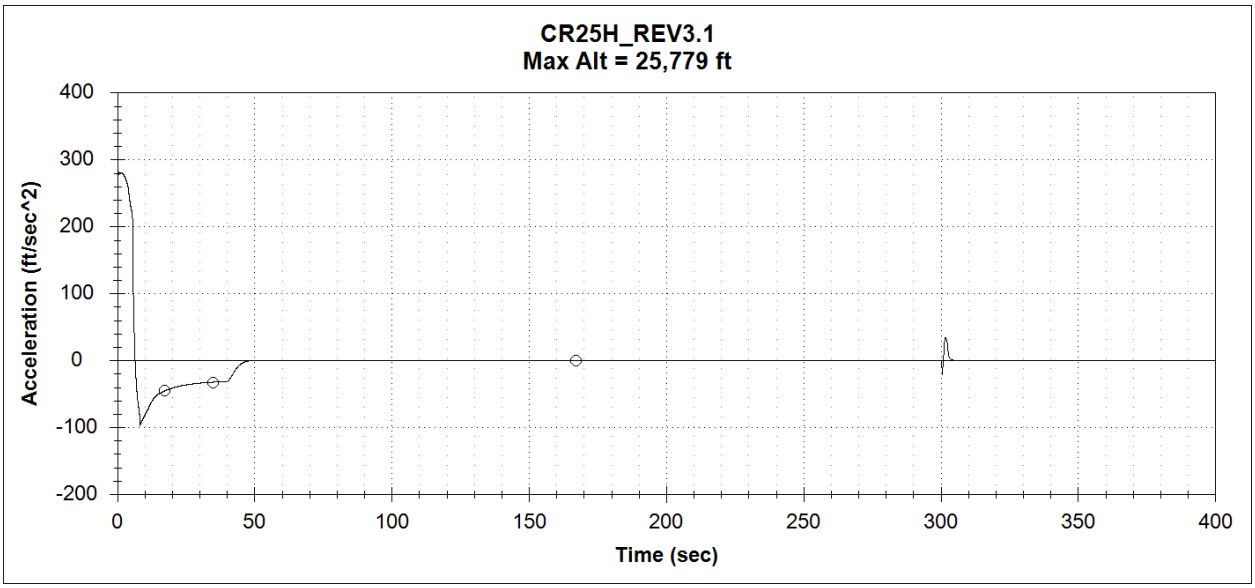

Figure 153: Predicted Acceleration Data of Flight

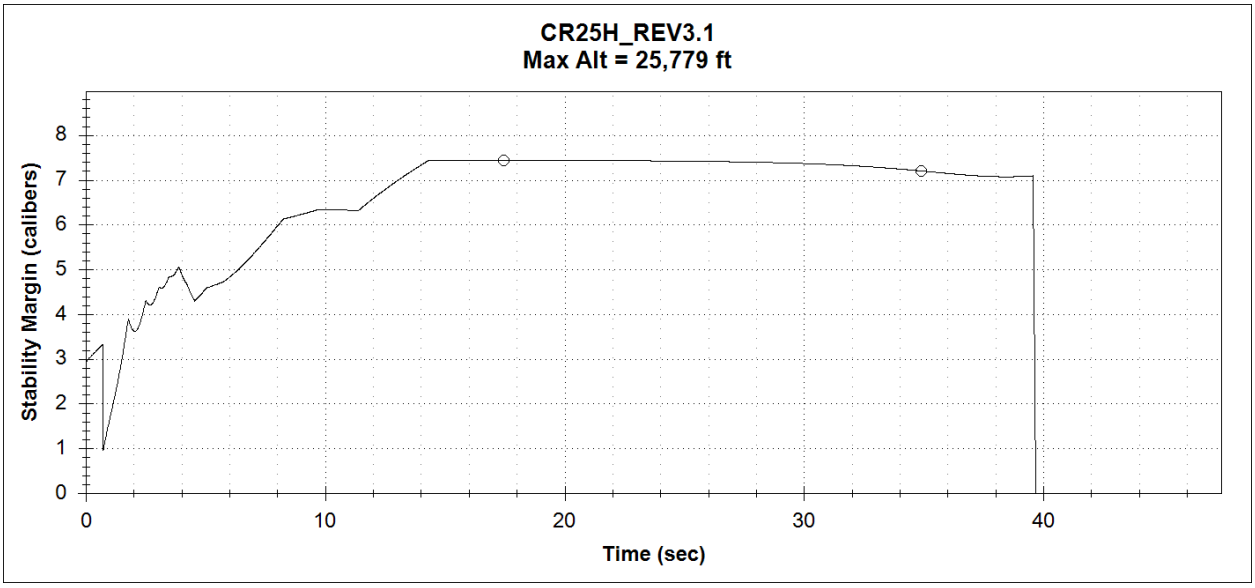

Figure 154: Predicted Stability Data of Flight

A ground track was also generated using RASAero II, which may be seen below in Figure 155

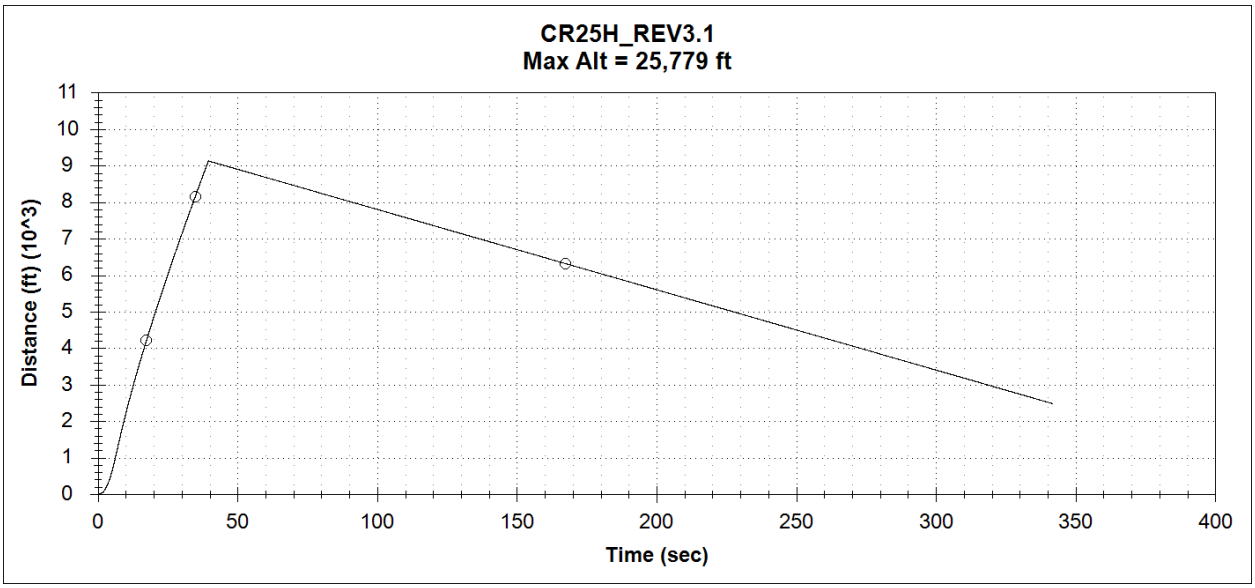

Figure 155: Predicted Distance from Launch Site vs. Time

14 Project Testing and Updates

The CR25 Hybrid rocket has plans to undergo various tests to verify its functionality during launch and landing. These plans are outlined in the sections below.

14.1 Aerostructures

As mentioned in Section 6, it is the team's first year manufacturing SRAD body tubes. While the team has sufficient manufacturing experience through previous nosecones and fin cans, testing was required to ensure the body tubes would support the bending and shear loads experienced during flight. Six tubes of increasing wrap layers were cut into 3 test sections of equal length. From there, the sections were sanded to have perpendicular edges for use in the UTM. Note that sample 6 is the thinnest sample and sample 1 is the thickest (most number of layers). The following tables summarize information regarding testing and the sample's geometry.

Tube Set Number	Thickness 1 (mm)	Thickness 2 (mm)	Thickness 3 (mm)	Thickness 4 (mm)	Thickness 5 (mm)	Average Thickness (mm)
6	1.56	1.37	1.16	1.12	1.32	1.306
5	1.35	1.72	1.63	1.8	1.87	1.674
4	1.8	2.11	2.11	2	1.83	1.97
3	2.24	2.01	2.25	1.96	1.74	2.04
2	2.29	2.63	2.51	2.21	2.46	2.42
1	2.63	2.06	2.32	2.46	2.7	2.434

Figure 156: Average Tube Set Thicknesses

Tube Set Number	ID 1 (mm)	ID 2 (mm)	ID 3 (mm)	ID 4 (mm)	ID 5 (mm)	Average ID (mm)
6	146.79	146.43	146.91	147.44	147.38	146.99
5	147.39	147.04	147.24	147.7	147.51	147.376
4	146.99	148.03	147.46	146.34	146.37	147.038
3	147.04	148.34	147.08	147.87	146.92	147.45
2	146.84	147.12	147.57	147.86	146.65	147.208
1	146.35	147.46	147.41	147.33	146.5	147.01

Figure 157: Average Tube Inner Diameter

The UTM machine was set to a rate of 2 mm per minute and to max out at 30kN. The test samples were placed between two parallel plates that had a diameter larger than that of the tube. At this strain rate, the testing was largely successful. All tubes passed the 7.6 kN factor of safety design load. Tube set 5 and 6 failed at roughly 15 kN and samples with more layers survived past 30 kN. Overall, this testing method and strain rate would suggest that 3 layers would be enough to satisfy pure compressive loads during launch. The results of the UTM testing are shown below.

However, a shock load of unknown strain rate amounting to 8 kN after a small UTM operation error where the grip pressure was incorrectly released was applied to a test section. This test piece cracked under only 8 kN while a similar test piece survived to 15 kN with a slower strain rate. It was at this moment that the team realised that the strain rate of the test samples had a major influence on the load that samples could withstand. Unfortunately the team concluded that this test process was not representative of flight conditions. Additional layers needed to be added in order to account for this lack of information of shock loading.

The team concluded that 8 layers was appropriate, keeping in mind testing results and the load estimations outlined in this section. Note that in some samples there seems to be a sudden drop

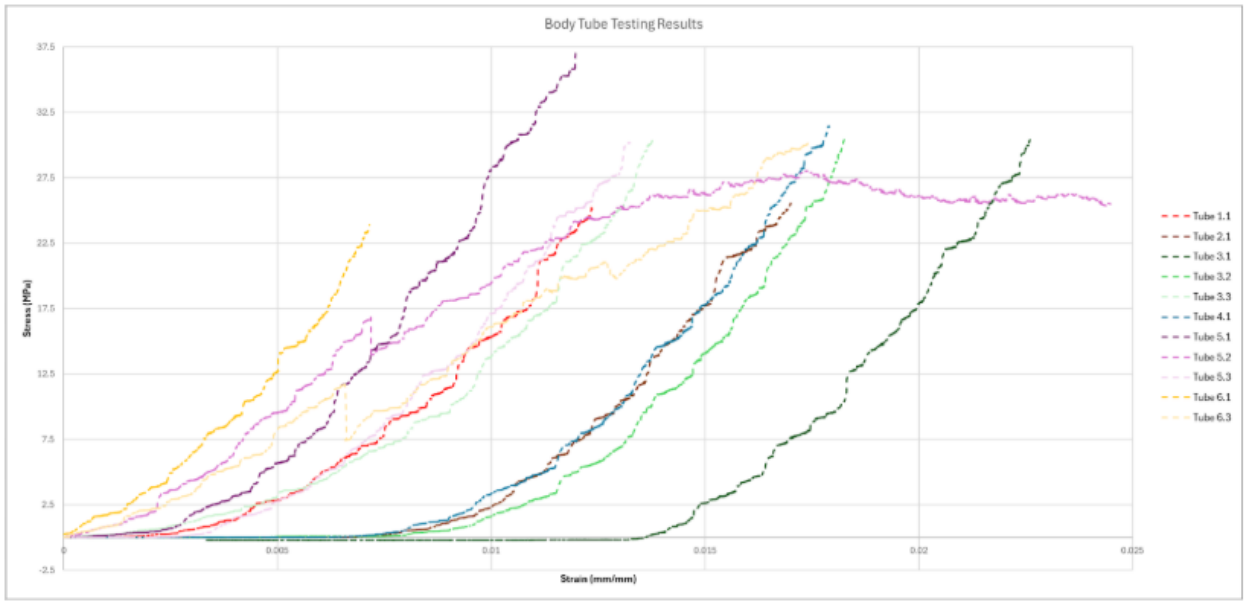

Figure 158: Body Tube Crush Test Data

in the stress. This is hypothesized as either a crack or machine error due to grip sliding. The true reason should be determined for future testing to see if this discontinuity is the result of an impurity or manufacturing error.

Future testing must involve determining the necessary strain rate to recreate the stress rate that the body tubes will see during a launch. Only when that rate is determined can accurate and representative loading be obtained. As well, the tests used two parallel plates. Plugs on either end that protrude from the parallel plates should have been used to recreate coupler interfacing during loading. Most lessons learned during this process were a result of inexperience with composite testing and operation of UTM's, however, the team was pleased with the progress made in the body tube program. Next year's results should allow the team to further optimize our SRAD body tubes.

14.2 Avionics

Testing the avionics subsystem for telemetry is difficult without the ability to flight test throughout the year. Many of the sensors on board this year's flight computer, as well as the general radio design, has flight heritage which has allowed this years designs to be improved. This year, a few test methods were employed to guarantee certain performance aspects. These included both tests of hardware and software. These tests are detailed in Section 14.2.1.

In addition to the telemetry system, avionics was responsible for the testing of the hybrid engine control system alongside the propulsion subteam. Tests were performed for both hardware and software, both in a bench-testing environment (in the lab), at several cold flows, operations rehearsals and static fires. These tests are detailed in Section 14.2.2.

14.2.1 Telemetry Testing

The first type of tests performed on the telemetry subsystem this year were radio range tests. These tests were performed to assess the maximum transmission range when both the transmitter and the receiver were on the ground, as well as when the receiver was at a high elevation. These tests are critical, as Radio Frequency (RF) systems are susceptible to large gain losses if the circuit design is incorrect or sub-optimal. They are also sensitive to their environment, such as objects in the transmission path or even metal objects beneath/around the transmitter (such as the avionics enclosure it is mounted to).

Prior to range testing, the avionics team performed significant research about 433MHz RF systems to determine the sub-optimal range achieved last year during ground testing. It was determined that the 433MHz band is more susceptible to ground interference due to an effect called Fresnel zones [30]. These zones are envisioned as large ellipsoids between the receiver and transmitter (Figure 159, and objects within this zone can greatly affect the received signal strength. In the case of the 433MHz band with the transceivers 1 kilometer apart, the maximum radius is 13.1 meters. At ground elevation, this puts approximately half of the Fresnel zone within the ground, significantly increasing path loss.

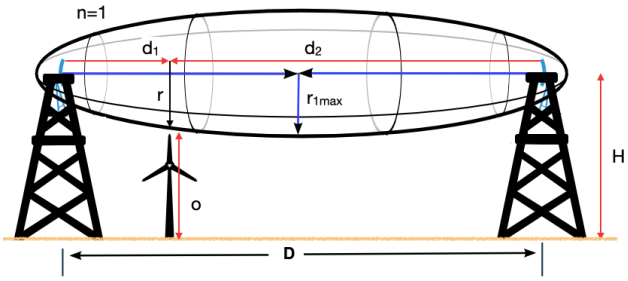

Figure 159: Fresnel zone visualization

The team performed several RF transmission simulations using a Matlab program, including parameters for different types of antennas the team owns (whip, monopole, fin, etc.). The simulations were performed for two different range testing locations: a large hill at Gatineau park, and the Carleton University campus Dunton Tower (77m elevation). Simulations predicted a high receiver signal strength even at up to 10 kilometers with line of sight between the transceivers, since the transmitter at an elevation prevented most obstructions within the Fresnel zone. The heat map of path loss from the simulation for both sites can be seen in Figures 160 and 161.

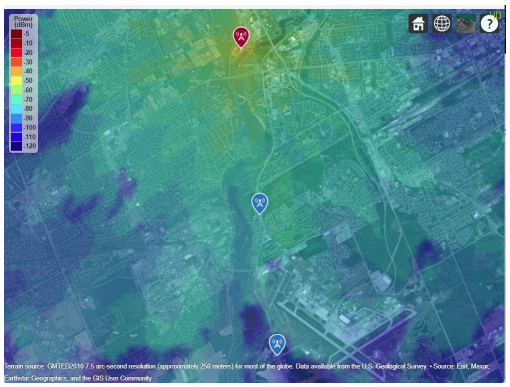

Figure 160: Dunton Tower range test simulation in Matlab

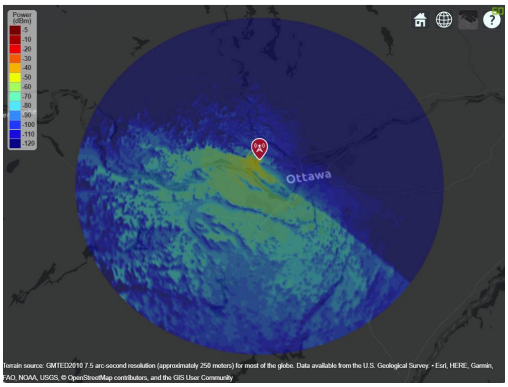

Figure 161: Gatineau park range test simulation in Matlab

The results from range testing from Dunton Tower coincided with the predicted results from the Matlab simulation. Ground elevation tests were also performed and confirmed data reception up to 1.5km, which is sufficient for recovery given that that GPS coordinates will have ample time to transmit during the rocket's descent before the range is reduced to 1.5km. With a transmission elevation of 77m, distance of 9km, receiver antenna sensitivity of 5dBi, and a transmit power of 13.88dBm, we were able to receive packets at a Signal to Noise Ratio (SNR) of 3. During competition, Fresnel zone interference will be significantly less of an issue due to the elevation of the rocket (during ascent and descent). Range testing has allowed InSpace to select the optimal antennas for both the rocket's transmitter and the ground station receiver to maximize signal strength. InSpace hopes to bring a secondary receiver to compare the signal strength from the monopole receiver antenna and a Yagi receiver, since pointing the Yagi antenna at a stationary target was quite easy during range testing but will likely be more of a challenge with a moving rocket, despite providing a higher gain if directivity is correct.

Figure 162: Field range testing with monopole receiver antenna

14.2.2 Hybrid Control System Testing

The was hybrid control system was extensively tested via both isolated bench tests and during cold-flows, operations rehearsals and hot fires.

The business logic of the hybrid control system was tested both on the real control system hardware (the PCC and MCC) as well as on the virtualized system on desktop computers. The POSIX design of the system allowed primary control flows to be verified on desktop computers before the hardware was fully built. InSpace was able to confirm that the control system had correct enforcement of the arming sequence (see Section 9.2.5), correct interpretation of the packet specification (see Appendix H, hybrid communication format) and correct running of the telemetry and control tasks. Actuator commands were abstracted as console messages (i.e. an actuation of XV-1 would result in a "actuated XV-1 to ON" message) to verify that the correct routines were called for each actuator, and only when the correct arming privilege was active. Sensor telemetry was mocked using randomly generated data, and later on using recorded CSV data from prior cold flows. This allowed the entire hybrid engine UI to be prototyped before the control hardware was complete.

Once hardware was available, bench test were performed by connecting the PCC to the fill panel and checking that all sensors could be powered and read and their measurements streamed over telemetry. Each solenoid channel was tested by actuating one of the system's real solenoids from each relay. Ignitor continuity testing was tested by short-circuiting the ignitor leads during bench testing. Firing was simulated by sending current through a 1.5 ohm resistor in series with a multi meter in ammeter mode and verifying that the supplied current exceeded the 6 amps required to blow two parallel e-matches. All initial hardware bench testing was performed between the real MCC and real PCC directly connected to just a router. Once all functionality was confirmed to work, connection tests were performed with the full network setup (Ubiquity LiteBeam network bridge, switch, router and video cameras connected in the production configuration). Latency

in actuation requests was quantified by monitoring network traffic using WireShark and taking the timestamp difference between the request and the success response. Actuation of all solenoid valves and the dump valve solenoid was tested, as well as the automatic abort routine triggered by a connection loss for 5 consecutive minutes.

Pressure transducer measurements were tested by using an air compressor to pressurize a volume connected to the transducer to a known pressure and comparing the digital measurement to the analog gauge. The mass load cell was calibrated using aluminum stock of a known weight (measured with a high-accuracy scale in the lab) and then verifying the weight of the ox tank in flight configuration. Finally, the button load cell used for engine thrust measurements was calibrated using an MTS machine and marking down the amplifier voltage output against the known force being applied by the MTS machine.

Once all bench testing was completed to a satisfactory level, further tests were performed during several real operations of the system. The team performed 3 cold flow tests using the PCC and MCC systems developed for competition (prior tests used an older system). During these tests, the network infrastructure behaved nominally, with no issues in the live video stream or control system connection. Over the three tests, the primary changes to the system were visualization changes on the hybrid engine UI to make it easier to interpret the system state (colour coding for continuity status, threshold markers for high pressures on time series graphs, connection status indication, etc). Some improvements were added to the system, such as optimizations that increased the sampling rate of each analog sensor input from 10Hz to 20Hz, as well as the addition of automatic abort routines (see Section 9.2.6) that as both bench tested and tested in a valve test prior to being used in a cold flow. Across all tests so far, the automatic abort has not been necessary. It was only tested deliberately.

Figure 163: The hybrid control system in use at the second static fire test

The ignition circuitry was tested successfully during the team's ignitor burn test, where it was able

to supply 8 amps to the two e-matches embedded in the ignitor puck and cause it to catch fire. The entire system, including the ignition circuitry, was later tested successfully during the team's first hot fire at the Reaction Dynamics test site in Quebec where the hybrid engine ignited and fired. The latest test of the control system occurred on July 20th at the team's second static fire, where the e-matches unfortunately caused the nylon tubing in the chamber to burst before the ignitor puck could catch on fire, resulting a dump of all the nitrous oxide through the combustion chamber. The control system had a nominal operation throughout this entire test, with the exception of the continuity indicator failing. The team has investigated and solved the issue since this test.

14.3 Recovery

Recovery has performed multiple reefing tests for our custom PCB's as well as multiple ejection tests throughout the year. A complete full system ejection test with the finished Hybrid parachute is the final remaining test as of the current date. Below test reports from all conducted tests can be found:

- Line Cutter Test
- Full System Ejection Test
- PCB Code Test
- PCB Connection Test
- PCB Disconnection Test
- Full System Ejection Test 2
- Full System Assembly Test

14.4 Propulsion

14.4.1 Static Fire Testing

Two static fire test attempts have been made with the rocket, one small-scale reduced duration and test and one full-scale full duration test. The static fire procedure can be found here: Static Fire Procedures. The small scale test was the team's first ever static fire, and culminated in a fully successful test shown in Figure 164 below.

Figure 164: Static Fire Test

From this static fire, a thrust curve was collected showing the engine had over performed it's initial estimations. Tank, injector and chamber pressure readings were also collected, and some feed system instabilities appear to be present. The determination of the ISP is still being worked on, as a good measurement of fuel and oxidizer consumed was not obtained. It is possible that the thrust load cell was giving inaccurate readings as well, however it was tested in a recently calibrated MTS machine the week after the hotfire, and was found to have been reading 10% below the values on the MTS machine. The values shown in this report have been corrected for this error.

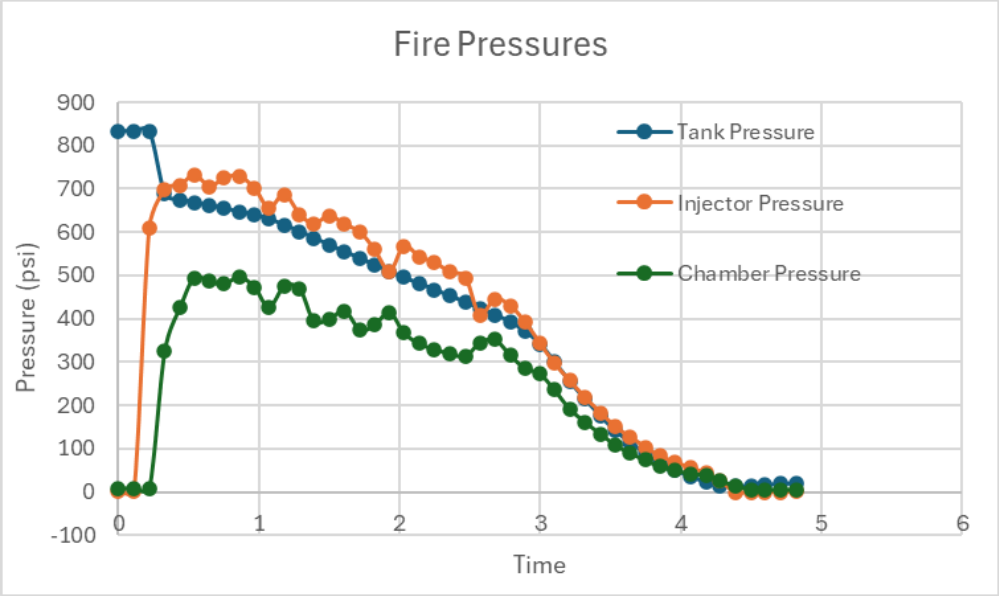

Figure 165: Static Fire Pressures

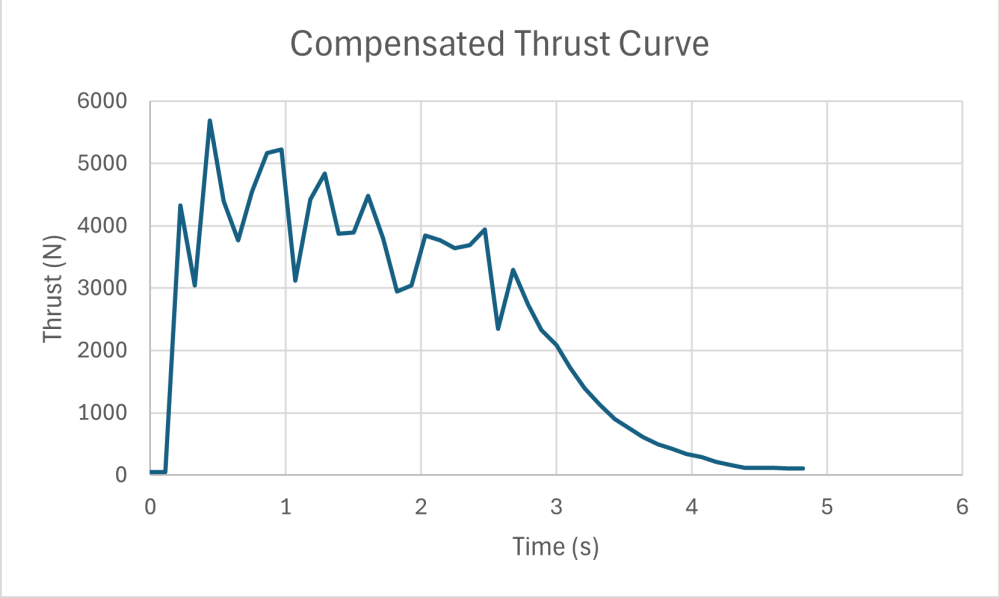

Figure 166: Static Fire Thrust Curve

Table 15: Theoretical vs Measured Performance Comparison

Parameter	**Theoretical**		**Measured**	
Average Thrust	3025	N	2643	N
Peak Thrust	5000	N	5691	N
Impulse	9463	N·s	11165	N·s
Burn Time	3.12	s	4.5	s
ISP (estimated)	176	s	TBD	s
Average Chamber Pressure	300	psi	292	psi
Peak Chamber Pressure	590	psi	495	psi
Oxidizer Consumed	4.594	kg	TBD	kg
Fuel Consumed	0.844	kg	TBD	kg

For the full scale hotfire, the motor was switched to using a U/C valve and the injector area was slightly increased. After troubleshooting leaks coming from PTC fittings in the injector plate, the team completed a fill of the engine and lit the motor. Unfortunately, the Aerotech First Fire ignitors had enough energy to puncture the nylon tubes before igniting the preheater puck, causing a cold flow to occur. Due to the limited amount of time in the day, another test could not be completed with a different ignitor configuration.

Figure 167: Epic Ignition Fail

14.4.2 Ops Rehearsals

Two attempts at an ops rehearsal test have been made prior to the writing of the report. The first test resulted in a successful setup, however an approaching thunderstorm prevented the team from completing the fill and dump portion of the test. While the inability to carry out the testing was disappointing, the team still gained valuable experience leak testing the flight configuration motor, as well as performing a quick setup and tear down.

Figure 168: Assembled flight configuration oxidizer system beside rocket

A second test was attempted three days later, and the nylon tubing which had been recommend to the team for use with the U/C valve burst at approximately 700 psi. Upon re-testing of the tubing and covering the tubing from the sun it again burst at the same pressure, confirming the team had been given tubing which could not take the pressures required.

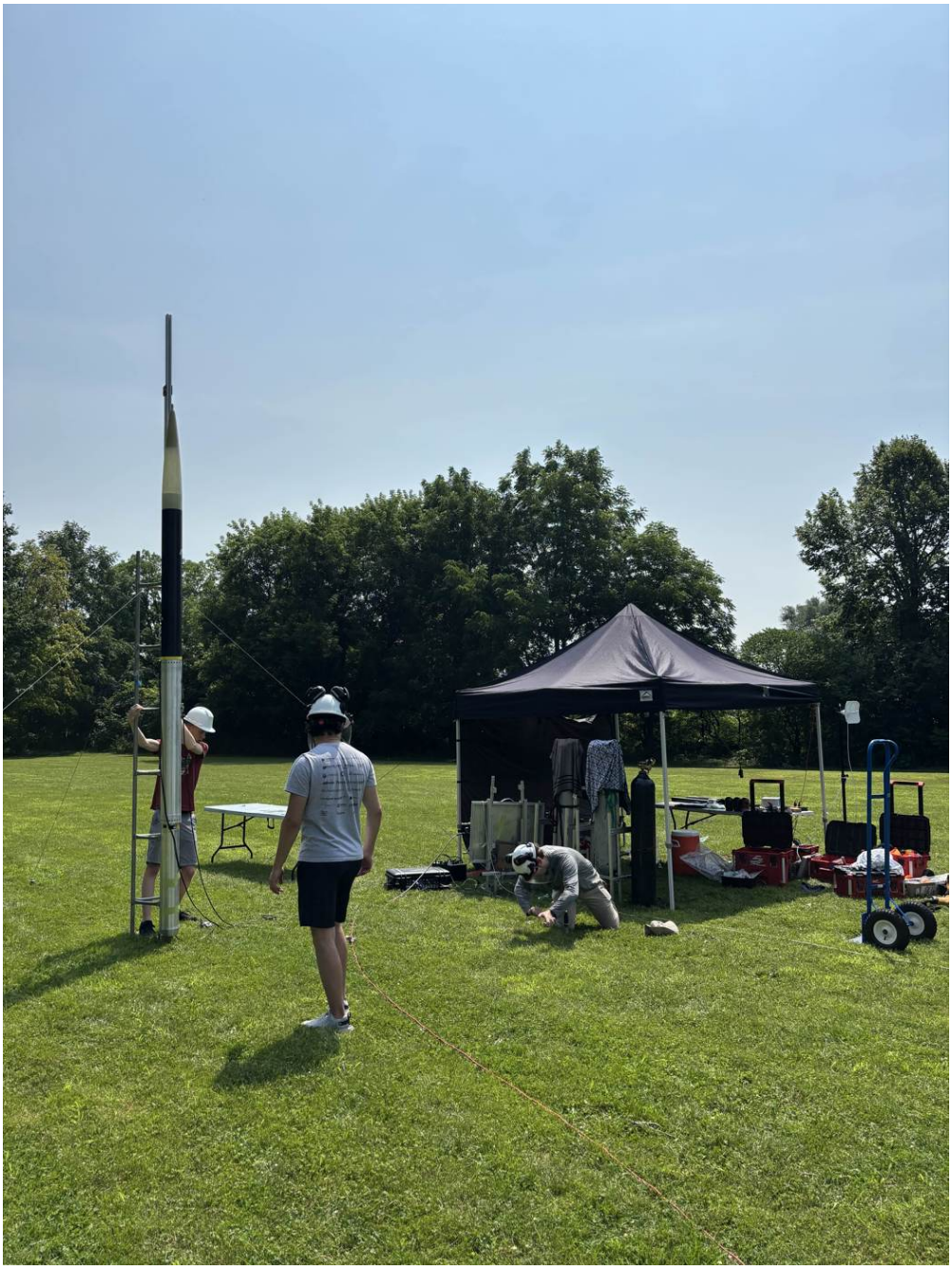

Figure 169: Flight configuration motor on rail

Figure 170: Burst tubing during second ops rehearsal

15 Risks and Mitigation Strategies

CU InSpace faces a number of risks to the successful operation of its CR25H launch vehicle. Some of these risks are analyzed in Sections 15 and 15. The risks in these sections have been ranked according to severity. Our severity index is as follows:

- 1 - Minor Inconvenience, delay under 1 week
- 2 - Moderate delay, 1-2 week delay
- 3 - Considerable delay, 3-4 week delay
- 4 - Severe Delay, 1-2 month delay
- 5 - Vehicle can not launch

Programmatic Risks

The programmatic risks presented in Table 16 are some of the risks that could disrupt CU Inspace's ability to launch at the Launch Canada challenge.

Table 16: CR25H Programmatic Risk Matrix

Pre-Mitigation					Post-Mitigation			
Program Risk	Description	Probability of Occurrence	Severity of Risk	Risk Level	Mitigating Action	Probability	Severity	Risk Level
Launch vehicle unmanufactured due to lack of suitable workspace	The lack of student workspaces on campus may cause delays or shortfalls in manufacturing	10.00%	5	0.5	1) Risk accepted, Carleton's composites manufacturing facility is not open yet and team can not control anything with regards to this leading to delays in repair of critical composite components	10.00%	5	0.5
Missing components at competition	Inadequate checklists and packing procedures may result in a forgotten component	10.00%	1	0.1	1) Pre-shipping checklists of all components to be completed and signed off by team executives 2) Check lists to be completed upon arrival to allow for shipping of missing components should they not be packed	5.00%	1	0.05
University does not permit travel of design teams to competitions	Carleton University does not insure team or permit team to leave for competition	25.00%	5	1.25	1) Risk accepted, team can not control the university's travel policy	25.00%	5	1.25
Team experiences budget cuts	Carleton University's funding stream for design teams does not fund critical components for launch vehicle	100.00%	4	1.75	1) Risk accepted, team can not control the university's club operation and funding policy	100.00%	4	1.75
Insufficient team member numbers at critical period	Team membership may drop off at a critical point in manufacturing, leading to a strain in the vehicle's development	15.00%	2	0.3	1) Team to continue outreach to assure member levels do not drop below the threshold for operation	10.00%	2	0.2

Technical Risks

The technical risks presented in Table 17 are some of the risks that could disrupt CU InSpace's ability to successfully bring a launch vehicle to the Launch Canada challenge.

Table 17: CR25H Technical Risk Matrix

Pre-Mitigation					Post-Mitigation			
Technical Risk	Description	Probability of Occurrence	Severity of Risk	Risk Level	Mitigating Action	Probability	Severity	Risk Level
Inability to satisfactorily manufacture composites	Composite components are difficult to manufacture consistently outside of carefully controlled environments. With the closing of the school's composites lab, the team must manufacture these off campus with more complex designs than previous years	50.00%	4	2	1) COTS Alternatives Available for Critical Components 2) Composites testing is performed as early as possible	25.00%	4	1
Inability to locate recovery test grounds	Recovery testing is critical before flight; the reefing system is complex and requires enough time to inflate and then activate the reefing system. This is complex to test.	80.00%	3	2.4	1) Independently verify each component 2) Continue to discuss the use of wind tunnel facilities with Carleton. Wind tunnel is a location to bring parachutes to steady state before charge deployment	35.00%	3	1.05
Unexpected component failure at critical period	Flight components may be damaged in the weeks leading up to departure for competition	25.00%	5	1.25	1) Perform testing as early as possible without using intended flight hardware 2) If not possible to use test articles in lieu of flight hardware, test at conditions component will experience and shield components from conditions outside expected flight envelopes (ie. assure unshielded wires are not brought close to electronics, as this is not a flight condition to test)	25.00%	3	0.75
Sensitive components damaged in transport	Transportation of the launch vehicle and payload components is a significant risk due to many uncontrollable factors (other road vehicles, road conditions, environmental conditions)	10.00%	5	0.5	1) Components to be placed in protective containers during transport to prevent uncontrolled movement and potential damage 2) Containers storing flight-critical components should be water-tight 3) Components to be inspected before departure and upon arrival for damage 4) Bringing duplicates or COTS alternatives to critical high risk components (ie. additional fins, COTS altimeters, etc.)	5.00%	5	0.25

Pre-Mitigation					Post-Mitigation			
Program Risk	Description	Probability of Occurrence	Severity of Risk	Risk Level	Mitigating Action	Probability	Severity	Risk Level
Manufacturing bottlenecks	The school has a single machine shop used by all Carleton Capstone teams and design teams, this leads to high lead times, especially when students are not permitted to enter the machine shop to perform their own machining	30.00%	3	0.9	1) Assure school machine shop has months of lead time in schedule to produce components 2) Source COTS replacement components for high-lead time components as contingency 3) Be in contact with external machine shops to produce components	25.00%	3	0.75
Electronics failure due to adverse launch site conditions	Flight vehicle avionics must be robust to withstand the hot and dry conditions in New Mexico and the potential wet conditions at Launch Canada	25.00%	5	1.25	1) Shield electronics against water ingress and assure components are rated for high heat environment in launch vehicle 2) Have backups for flight-critical components (ie. batteries, altimeters, etc.)	10.00%	5	0.5
Unexpected initial structural testing results	Initial structural testing may yield unexpected results when compared to simulations, these may require redesign of specific load bearing components for the rocket to be able to fly; this is combined with the potential bottlenecks from the school's machining resources	50.00%	2	1	1) Assure designed components have been analyzed in finite element analysis software and meet a predetermined safety factor 2) Communicate designs with team mentors prior to manufacturing 3) Produce test articles prior to flight hardware to avoid breaking flight hardware	50.00%	1	0.5
Flight component damage during testing	Many flight components are not currently planned to have separate test articles due to needing to construct a second airframe. Thus, the flight component may become damaged prior to competition, requiring remanufacturing and potential redesign	25.00%	5	1.25	1) Assure designed components have been analyzed in finite element analysis software and meet a predetermined safety factor 2) Communicate designs with team mentors prior to manufacturing 3) Produce test articles prior to flight hardware to avoid breaking flight hardware	25.00%	1	0.25
Unacceptable vehicle manufacturing tolerances during assembly	Should the vehicle's airframe not meet the bending tolerance requirements, it may be prevented from launching at competition as the flight characteristics may be unexpected	10.00%	4	0.4	1) Assure tolerances are checked with previous team leads 2) Compare tolerances with previous nominal toleranced parts 3) Communicate designs with team mentors prior to manufacturing	5.00%	4	0.2
Vehicle mass outside of predicted mass budget	The rocket may be heavier or lighter than predicted using the mass budget, causing issues with meeting altitude and stability requirements	25.00%	1	0.25	1) Regularly update mass budget with measured values 2) Hold sub-teams to mass limitations to assure they do not exceed them	12.50%	1	0.125

Pre-Mitigation					Post-Mitigation			
Program Risk	Description	Probability of Occurrence	Severity of Risk	Risk Level	Mitigating Action	Probability	Severity	Risk Level
Vehicle stability caliper outside of predicted window	Should the vehicle's stability caliper change prior to launch, the vehicle may not be permitted to launch	25.00%	5	1.25	1) Regularly update aerodynamic simulations as per the mass budget to assure fin sizing does not change 2) Perform computational fluid dynamics simulations to verify centre of pressure position 3) Perform physical testing to verify centre of gravity position prior to departure for launch	12.50%	5	0.625

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Appendix

A Requirements

The requirements used to develop the launch vehicle and re-certify it for launch may be seen below in Table 18.

Table 18: Vehicle Requirements

No	Description	Specification	Source	Rank	Chosen design explanation	Planned way to show compliance
1.0	Avionics System					
1.1	COTS GPS Tracker	A COTS GPS tracking system will be incorporated which is able to locate the rocket at a distance of at least two miles.	Launch Canada	1	We will incorporate 2 Featherweight GPS trackers.	Flight heritage.
1.2	Parachute Deployment	The rocket will have a COTS avionics system capable of detecting apogee in order to deploy the reefed parachute.	Launch Canada	1	We will incorporate an Altus Metrum EasyMini Altimeter.	The EasyMini's specifications is a competition approved altimeter and has significant flight history
1.3	Official Altitude Logging	All rockets must carry a COTS altimeter for the purposes of official altitude measurement.	Launch Canada	1	We will incorporate an Altus Metrum EasyMini altimeter.	The Altus Metrum EasyMini specifications meet our requirements and was recommended to us as the preferred altimeter at SAC 2022-2024. We are also flying a Blue Jay altimeter as a redundant logger.
1.4	SRAD Tracking	Our SRAD avionics will have GNSS based tracking and the ability to relay that location data to our ground station.	Launch Canada	1	Our SRAD avionics system includes a GNSS receiver module and a LoRa radio for communication with our ground station.	We will perform range testing and location tracking testing on the ground. There is also flight heritage for this system.
1.5	SRAD Parachute Deployment	Our avionics will be capable of detecting apogee and providing a redundant parachute deployment capability.	Launch Canada	1	Our SRAD avionics system includes an altimeter for apogee detection and the capability to trigger a CO2 parachute deployment system.	We will perform vacuum and flight testing (2023 SAC) of our SRAD deployment board to ensure function
2.0	Propulsion System					
2.1	SRAD Propulsion	All SRAD motors shall be static fired, well characterized and tested before arrival at the competition.	Launch Canada	1	SRAD Motors are well developed, validated and verified before competition.	The SRAD hybrid has undergone numerous tests to characterize its performance and validate its safety and operations.
2.2	Impulse Limitations	The total combined impulse of all COTS components shall remain under 40,960 Newton Seconds (O class)	Launch Canada	1	The designed motor for use at competition is under the maximum impulse limits.	The SRAD hybrid has 29,065 Newton seconds of impulse, well below the competition maximum

Table 18: Vehicle Requirements

No	Description	Specification	Source	Rank	Chosen design explanation	Planned way to show compliance
2.3	Non-toxic Propellants	Propellants that propose an immediate danger to personnel and have significant environmental dangers shall not be used	Launch Canada	1	The motor for use at competition does not contain toxic propellants.	The propellant in the SRAD motor is a paraffin wax and nitrous oxide motor which does not pose an immediate risk to people or the environment.
2.4	Sealed Systems	Any sealed systems or segments shall have a pressure relief device.	Launch Canada	1	If any section of plumbing can be closed or isolated by an actuated valve or check valve, there must be a method to relieve pressure.	Each closed section can be vented by a pressure relief valve. There is also a vent, manual and controlled dump valve.
2.5	Oxidizer System Cleanliness	All oxidizer system hardware shall be thoroughly cleaned to an acceptable standard for oxygen service.	Launch Canada	1	The oxidizing system must be cleaned to an oxygen-safe level.	A thorough cleaning process using distilled water, a blue-gold solution, IPA, and a fumehood are employed to ensure a cleaned system.
2.6	Proof Testing	The tank shall be designed such that it does not exceed 95% of the yield strength of the material with a proof pressure not less than 1.5x MEOP.	Launch Canada	1	SRAD pressure vessels are able to withstand 1.5x MEOP during proof testing such that the gross stress level doesn't exceed 95% of the material.	The SRAD pressure vessels are designed such that 1.5x MEOP will not compromise the structural integrity.
2.7	Overpressure Protection	All pressure relief devices shall be sized to provide relief at full flow capacity at 110% MEOP.	Launch Canada	1	The system must not exceed 110% MEOP under any circumstances.	Relief valves are sized to crack at 800 psi and fully open at 1000 psi.
2.8	Valves	Piloted valves shall abort in the depressurized state.	Launch Canada	1	Piloted valves shall abort in the depressurized state.	The UC valve design removes the need for a pressurized pilot line.
2.9	Valves	Control valves which provide the pressure to operate piloted valves shall depressurize the pilot line when the control valve is de-energized.	Launch Canada	1	The system shall be depressurized when it is de-energized.	The UC valve design allows for the system to be depressurized when the system is de-energized via the dump valves.
2.10	Ignition	The action that arms the igniter shall be independant of the action that fires it.	Launch Canada	1	Ignition should not be possible with the press of one button.	The hybrid control system has multiple hardware and software lockouts that prevent the accidental trigger of ignition.
2.11	Ignition and Arming	All arming and igniting sequences shall not pose a risk to personnel, fail safe and shall have redundant systems	Launch Canada	1	To accommodate all arming and ignition requirements (Lockouts, shielding, insulation etc.) the competition's provided launch tower and system will be used (Section 11.2)	The competition vehicle will not have onboard ignitions systems and the engine will be unable to ignite without the provided ground systems
3.0	Flight Dynamics					

Table 18: Vehicle Requirements

No	Description	Specification	Source	Rank	Chosen design explanation	Planned way to show compliance
3.1	Launch Rail Velocity	The vehicle shall leave the launch rail at an absolute minimum of 50 ft/s with a desired minimum of 100 ft/s	Launch Canada	1	Using the SRAD hybrid motor the vehicle is expected to leave the launch rail at approximately 100 ft/s . This exceeds the recommended minimum, with further simulation planned.	Empirical analysis and simulations are to be performed to confirm the off-rail stability.
3.2	Target Apogee Range	The SRAD motor shall lift the vehicle to an apogee of 30,000 ft, plus minus 3000 ft.	CU InSpace	2	Simulating the most recent airframe using RasAero II with the created motor file, the vehicle reaches an apogee of 25,779 ft [7857 m] .	This apogee was found to be the closest achievable value to the target given the current airframe design
4	Aerostructures					
4.1	SRAD composites	The vehicle should be primarily composed of SRAD composite airframe components.	CU InSpace	1	The current airframe was designed with tolerance margins for student-created composites.	Nosecone, body tubes, and fincan were manufactured entirely by student composite layup.
4.2	SRAD Body Tubes	SRAD Body Tubes shall, at a minimum, be tested under a static load equal to that of the maximum thrust of the hybird motor.	Launch Canada	2	Roll wrap body tube test sections are made to test, with multiple data sets, the static strength of the composite.	The strength of the body tubes were tested using a UTM and demonstrated to have a high margin of safety compared to ultimate loads to account for strain rate effects of composite strength.
5	Recovery System					
5.1	Dual-event	Each independently recovered launch vehicle body anticipated to reach an apogee above 1,500 ft (457 m) above ground level (AGL) shall follow a "dual-event" recovery operations concept (CONOPS).	Launch Canada	1	To facilitate main deployment, the parachute mounted redundant altimeters will fire a pyrotechnic line cutter, which will release tension from the skirt of the parachute, thereby opening the parachute to its full diameter at 1,500 ft	A series of ejection tests, line cutter tests, and integration tests will be facilitated to ensure the system functions as intended.
5.2	Initial deployment	Shall occur at or near apogee.	Launch Canada	1	A COTS altimeter located in the avionics bay will be used to eject the nose cone and the recovery systems from the rocket at apogee.	This COTS altimeter has now been flight proven at Spaceport America Cup and in ground testing where it ignited e-matches under an equivalent vacuum.
5.3	Drogue/Reefed chute	Shall stabilize the vehicle's attitude (i.e., prevent or eliminate tumbling).	Launch Canada	1	A redundant CO2 ejection charge is present to ensure drogue deployment.	Both CO2 ejection charges, the primary and the redundant canisters, will activate with a slight delay for redundancy.
		Shall reduce its descent rate enough to permit the main deployment event yet not so much as to exacerbate wind drift (e.g., between 75 and 150 ft/s [23-46 m/s]).	Launch Canada	1	The main parachute will be reefed to slow the rocket down to a comfortable speed below 100 ft/s, but not opened to such an extent as to take its velocity below 75 ft/s.	Wind tunnel testing and scale testing will be used to ensure that the parachute will operate within the competition parameters.

Table 18: Vehicle Requirements

No	Description	Specification	Source	Rank	Chosen design explanation	Planned way to show compliance
5.4	Main deployment	Shall occur at an altitude no higher than 1,500 ft (457 m) AGL.	Launch Canada	1	Both a COTS and a SRAD altimeter will be used to ensure redundancy on main release.	Deployment tests will be performed to ensure the primary system will release prior to the redundant system.
		Shall reduce the vehicle's descent rate sufficiently to prevent excessive damage upon impact with ground (i.e., less than 30 ft/s [9 m/s)]).	Launch Canada	1	A 25 ft diameter main parachute will be used to slow the rocket down to a comfortable speed below 30 ft/s.	Simulations and scale model testing were performed using conservative coefficients of drag that ensure the drogue will operate within the competition parameters.
5.5	Ejection gas protection	Shall implement adequate protection (eg., Fire resistant material, pistons, baffles etc.) to prevent hot ejection gases (if implemented) from causing burn damage to retaining chords, parachutes, and other vital components as the specific design demands	Launch Canada	1	The use of CO2 ejection charges in the main deployment event minimizes the risk of hot gases damaging the recovery hardware during the initial deployment event. During skirt reefing at 1,500 ft, the line cutter exhaust will be oriented towards a vent in the fabric and packed with wadding to limit the charring of the parachute fabric	CO2 ejection does not typically damage recovery hardware through the generation of heat. Skirt reefing employs methods of reducing the likelihood of parachute damage during main deployment
5.6	Swivel links	The recovery system rigging (e.g., parachute lines, risers, shock chords, etc.) shall implement swivel links at connections to freely spin.	Launch Canada	1	The parachute system uses swivel links at connection point to ensure soft connections are present.	Chosen swivel links have been simulated to comfortably comply with the maximum torsion during decent.

B Hybrid Development Hazard Matrix

Table 19: Potential Hybrid Development Hazards

Hazard	**Causes**	**Risk of Mishap and Rationale**	**Mitigation Strategy**	**Risk After Mitigation**	**Severity**	**Risk Score**
Damage to hearing of members	Loss of power in the electrical system causes normally open valves to discharge high-pressure gas	Medium: control box relays have burnt out in previous tests, and while the issue has been fixed before the test the possibility still exists	Medium: control box relays have burnt out in previous tests, and while the issue has been fixed before the test the possibility still exists.	Very Low	High	Medium Low
Personnel hit by flying debris from component bursting	Failure of nitrogen tank regulator causes overpressure of system	Low: the system has 2 Normally open solenoid vent valves that can be opened if pressure is seen to be rising above operating pressure, as well as a mechanical relief valve that will open before the burst pressure of any component is reached	All Personnel will be over 80m from the test stand when at operating pressure(1000 psi), which is the standoff distance for hazardous fragments calculated byDoD 6055-09-M for the TNT equivalent energy of the system at operating pressure. All components have a safety factor of at least 2x operating pressure as well as redundant venting methods.	Very Low	Very High	Medium
Uninvolved personnel entering the standoff zone	Bystanders may wander into the test standoff distance while the system is pressurized	Low: if bystanders are seen inside the standoff radius the test will be paused and the system will be depressurized	Two members will patrol the standoff radius to notify any passers-by to stay clear of the test stand while pressurized.	Very Low	Medium	Low
Harm to a member by cold temperatures in plumbing post-cold flow	After the cold flow, the carbon dioxide flowing out of the tank may have cooled the plumbing to a point where it could give frostbite from touching	Medium-low- temperature on the system will be monitored by a temperature sensor so any members approaching the test stand will be notified of any low temperatures present	In case the temperature sensor is inaccurate or not representative of the entire system, members approaching the stand after cold flow will make use of insulated gloves to protect their hands from any cold temperatures that may exist on the plumbing.	Very Low	Medium	Low

Table 19: Potential Hybrid Development Hazards

Hazard	**Causes**	**Risk of Mishap and Rationale**	**Mitigation Strategy**	**Risk After Mitigation**	**Severity**	**Risk Score**
Harm to personnel due to improper Personal Protective Equipment	Personnel are injured or harmed due to improper PPE. That is, a risk as outlined above, resulting in bodily damage to personnel.	Low: ALL personnel will be required to wear the appropriate PPE before they are allowed near the test site.	To prevent hearing damage: Earmuffs MUST be worn within the minimum safety distance of 80m. Around the perimeter of which, ear plugs MUST be worn. To prevent damage to skin, feet, or limbs: Long sleeves, long pants, and closed-toe shoes MUST be worn. That is, NO loose articles of clothing are allowed and any loose hair MUST be tied back. To prevent facial damage: Face shields and safety glasses MUST be worn within the minimum safety distance of 80m. Around the perimeter, safety glasses MUST be worn. To prevent damage to hands: Insulated gloves MUST be worn when within the minimum safety distance as well as while handling any pressurized vessel.	Very Low	High	Medium Low

C Hazard Matrix

Table 20: Potential Hazards

Hazard	Causes	Risk of Mishap and Rationale	Mitigation Strategy	Risk After Mitigation	Severity	Risk Score
Outbreak of illness within the team	Transmission through material transfer. Infection during manufacturing	Low: All team members doing manufacturing are fully vaccinated and have been screened for symptoms. Contact tracing applied for personnel doing in person work	Mandatory PPE during all periods of manufacturing contact. Suspension of all in person manufacturing unless absolutely necessary	Very Low	Medium	Low
Injury of team members during SRAD part development	Improper use of tools. Not using PPE	Very low: All team members are aware of the manufacturing procedure and should have proper training on tools/equipment	Define dangerous goods and equipment to all personnel before manufacturing begins. Use proper PPE. Do a dry run of any dangerous procedures to define areas/steps that may have increased risk	Very Low	High	Medium Low
Launch vehicle falls from launch rail or elevated position and comes in contact with personnel, causing injury	Improperly installed launch lugs. Launch rail lifting is done too quickly or with incorrect personnel position	Low: Launch rail attachments are installed onto aluminum load bearing bulkheads designed to withstand the weight of the rocket	All mechanical components such as rail attachment and launch rail installation are checked by experienced members prior to installation on launch rail. Rail attachments are directly connected to main structural members to assure rigidity	Low	High	Medium Low
High temperature electrical failure while personnel are working on the system, causing injury	Shorted Electrical Components	Low: Battery and power will be disconnected from the system while personnel are working. Potential injuries are minimal	All safety switches will be safed and power disconnected while electronics modifications are being done. Static bracelets will be worn when applicable. Personnel will wear personal protective equipment while working with the avionics	Very Low	High	Medium Low
Rocket weathercocks off launch rail during boost and comes into contact with personnel at high speed	High winds at launch. Improper launch rail or lug alignment	Low: Launch rail alignment will be checked prior to launch. Wind speed will be verified acceptable before launch.	Launch rail will be between be at 84 +/1 degrees prior to launch. Wind speeds will not exceed Tripoli guidelines prior to launch. Personnel will be a safe distance and behind cover	Low	Very High	Medium High
Premature activation of recovery system during assembly of the rocket while personnel are working on it, causing injury	Improperly installed CO2 Cartridges. Ejection system is prematurely armed	Low: Recovery system is only fully armed on the launch rail within the launch range	Electronic systems are designed and tested to avoid such events. Armed at the launch site only by experienced executives involved in the system.	Very Low	Very High	Medium

Table 20: Potential Hazards

Hazard	**Causes**	**Risk of Mishap and Rationale**	**Mitigation Strategy**	**Risk After Mitigation**	**Severity**	**Risk Score**
Ignition of unexploded charges during recovery of the rocket, causing injury	Spontaneous ignition of propellant during recovery	Low: Avionics has redundant backups, charges are proven commercial products.	All electrical connections are to be checked prior to launch. Recovery personnel have proper training to check for unexploded charges. Pyrotechnic devices are to be securely installed and checked by experienced members, assuring wires are not stressed	Very Low	Very High	Medium
Personnel dehydration or injury during rocket recovery	Improper PPE/Equipment taken during recovery procedure	Low: Personnel should be well informed of the weather and environment, ensuring adequate measures taken in PPE and supplies	Ensure all personnel in recovery of the rocket is well dressed and equipped. Equipment includes PPE, adequate equipment and water.	Low	Medium	Medium Low
Premature Ignition of ignitor during motor loading	Power is applied to the ignition system prematurely. Ignitor is improperly grounded/shorted	Low: Personnel responsible should be well versed in installation of ignitor. Ignition system are proven commercial products	Ignitor system checked at the pad prior to installation. System only armed once personnel is well clear of the pad at a predetermined safe distance set by the competition	Low:	Very High	Medium High
Recovery system fails to properly deploy, rocket and/or payload comes in contact with personnel.	Improper parachute packing and installation. Nose cone fails to deploy. Parachute caught on deployment. Drogue parachute fails to pull main	Medium: Recovery system ejection testing is performed in a static environment without the presence of conditions in flight.	Parachutes should be packed and installed by recovery executives. Clean and smooth inner surfaces of rocket in path of recovery system deployment. Electronic systems are checked, tested, and armed at the launch site by senior avionics or recovery executives	Low	Very High	Medium High
Rocket does not ignite when the command is given, ignites when personnel approaches.	Faulty ignitor. Wiring fault. Failure in electronics	Low: Engine and engine ignitor systems are proven commercial solutions	Igniters are ensured to be securely installed mechanically and electrically. Initial installation of ignitor performed by experienced members	Low	High	Medium
Rocket suffers a CATO after ignition, causes injury.	Personnel are too close. Improper motor assembly. Damage to motor during assembly	Very low: All team members and other personnel are at well understood safe distances set by competition. COTS solid motor suffering such event is extremely unlikely, motor assembly process well documented by manufacturer.	Personnel are to comply with safe viewing distance requirements set by the competition. Motor assembly is to be performed by HPR certified team members and/or experienced members	Very Low	Very High	Medium

D Pre-flight and Launch Checklists

CR25H: Quarter Pounder
Carleton University

Rocket Assembly Procedures

2025 Launch Canada

Revision A: 2025/07/17

CR25H Assembly Procedure

This document acts as a step-by-step guide on how to properly assemble CR25H. All precautions to not strip threads or force components together should be taken. Use caution and critical thinking when assembling components. Visually inspect all components before assembly for noticeable damage.

Special Notes

1. All radial fasteners making either: composite-composite, or composite-metallic connections are NOT to be over-tightened. Inspect surrounding area of composite for fraying, delamination or other damages upon installation.
2. Special attention when installing and connecting electronics corresponding to energetics. Ensure under no circumstances shall one work with hands inside the body tube when the system is loaded and live.

Forward Section Assembly (A for Upper, B for Lower)

Check: Bulkheads

- ☐ Ensure the god-nut A is attached to bulkhead A.
- ☐ Ensure god-nut B is attached to bulkhead B.
- ☐ Ensure eyebolts for attachment to god-nuts are ready with shock cord installed to each eyebolt.

Bulkhead B

- ☐ Ensure personnel installing are wearing appropriate PPE - eye protection, cut proof gloves, long sleeve, etc.
- ☐ Install forward shock cord (main not attached) and eyebolt assembly to the bulkhead. Apply Loctite to god-nut where necessary.
- ☐ Load and prep the raptor systems onto recovery bulkhead following the raptor system assembly manual. (printed and in binder).
- ☐ Assemble deployment avionics (2 easy minis, 2 LiPo, 2 screw switches. Do not attach to the bulkhead yet.
- ☐ Fit LiPo, screw switch, and Raspberry Pi Pico W PCB into the TGD and slot into the parachute bay, ensuring battery-negative and main-negative wires pass through the hole.
- ☐ Attach the parachute bay and deployment avionics housing to opposite sides of the bulkhead using 4 M4x30 bolts. Thread shock cord through parachute bay and attach quick link to parachute suspension lines.
- ☐ Ensure passthrough wires are connected to raptor charges and the deployment system is disarmed. Screw in CO2 cartridges.
- ☐ Insert receiving boxes into slots in the parachute bay, ensuring that arming holes on both components are correctly aligned.
- ☐ Twist bulkhead into support ring and secure with M4x10 bolts. Ensure the bolt head is flush with the body tube. Insert two bolts to keep receiving boxes in place, making sure to tape the bolt heads (these will be removed once boxes are armed).

At this point you should have:

- ☐ Bulkhead B is in the rocket, Bulkhead A is outside
- ☐ Bulkhead B fully assembled and loaded, attached to the deployment board but DISARMED.
- ☐ Deployment board IS NOT CONNECTED to bulkhead B passthroughs
- ☐ Passthroughs should not be connected to bulkhead B raptors until final step

Final Bulkhead B Arming

- ☐ Ensure personnel installing are wearing appropriate PPE - eye protection, cut proof gloves, long sleeve, etc.
- ☐ Ensure wires from both ends - passthrough wire, and the charge cup wire are of sufficient length to be connected OUTSIDE the vehicle's aft opening.
  - ☐ Attach charge-cup wires to bulkhead passthru, ensure connection is made outside of the vehicle's aft opening in the event of misfire.
  - ☐ Secure excess length of wire with cable ties and tie downs
- ☐ Arm avionic stack screw switches
- ☐ Arm TGD Screw switch
- ☐ Arm receiving boxes
- ☐ Detach bolts holding receiving boxes

The deployment system is now ready to be armed. Ensure the system stays disarmed. Secure pull tabs to side of body tube, apply tape for extra security.

You may now load the main chute 🙂

Bulkhead A

- ☐ Attach the mounting plate to the avionics enclosure using M4x5mm screws and hex nuts

- ☐ Attach four M3x5mm standoffs to the Josh (flight computer) side of the avionics enclosure with 4 M3 hex nuts.
- ☐ Apply lock-tight fluid to the hex nuts
- ☐ Attach Josh to the standoffs using M3x4mm socket cap screws with lock-tight fluid
- ☐ Attach two Featherweight GPS trackers to their mounting holes on the avionics enclosure using their provided mounting screws and standoffs, as well as lock-tight fluid
- ☐ Attach the Blue Raven altimeter to its mounting holes on the avionics enclosure using 3 nylon standoffs and screws
- ☐ Attach four M3x5mm standoffs to the Talk Tuah (radio board) side of the avionics enclosure with 4 M3 hex nuts
- ☐ Apply lock-tight fluid to the hex nuts
- ☐ Attach Talk Tuah to the standoffs using M3x4mm socket cap screws with lock-tight fluid
- ☐ Attach screw switches in a stack within their 3D-printed enclosure to the small arming switch shelf using M2x30mm socket cap screws and M2 hex nuts
- ☐ Apply lock tight fluid to the M2 hex nuts
- ☐ Attach the arming switch shelf to the enclosure mounting holes with M3x5mm socket cap screws and M3 hex nuts
- ☐ Apply lock-tight fluid to the M3 hex nuts
- ☐ Place 2x 8800mAh Li-Ion batteries and 3x 2200mAh 18650 Li-Ion batteries into the battery enclosure
- ☐ Insert the battery enclosure down the middle of the avionics enclosure and fasten it in place
- ☐ Plug the UART interface cable into Josh’s 4-pin Molex connector and Talk Tuah’s 4-pin Molex connector
- ☐ Remove the two-pin header shunt on Josh, next the USB connector and labelled “remove before flight”
- ☐ Plug the first arming switch’s molex connector into Josh’s two-pin Molex connector labelled “ARM”
- ☐ Connect the second arming switch’s wires to the arming terminal block on the first Featherweight GPS tracker

- ☐ Connect the last arming switch's wires to the arming terminal block on the second Featherweight GPS tracker
- ☐ Screw the 915MHz whip antenna onto both Featherweight GPS trackers
- ☐ Screw the 433MHz whip antenna onto Talk Tuah's SMA connector
- ☐ Screw the orange GPS antenna onto Josh's SMA connector
- ☐ Plug the two 8800mAh Li-Ion battery JST connectors into Josh's JST connector ports
- ☐ Plug one 18650 battery into each Featherweight GPS tracker
- ☐ Plug the last 18650 battery into the Blue Raven altimeter
- ☐ Attach the magnetic USB-C connector to Josh's USB-C port
- ☐ Arm all electronics and ensure that they power on and communicate with ground station electronics successfully
- ☐ Disarm all electronics
- ☐ Attach the enclosure to the avionics twist and lock bulkhead using 4 M4 screws through the mounting plate

The enclosure is now assembled and ready to be installed in the rocket.

Final Forward Assembly:

- ☐ Insert Bulkhead A into the nosecone and slot into the support ring
- ☐ Twist Bulkhead A until bolt holes are aligned with the support ring, and insert the alignment pin into the alignment hole
- ☐ Remove the alignment pin and insert the nosecone assembly into the forward body tube, ensuring bolt holes are aligned with the body tube
- ☐ Secure nosecone assembly with 4 M4x10 bolts

Avionics Telemetry Arming

- ☐ Arm Josh by inserting a 3/32 imperial hex key into the arming hole labelled "Josh" on the rocket exterior and screwing clockwise until the arming buzzer begins beeping (it will sound like the OLG jingle prior to "Winner, gagnant": https://www.youtube.com/watch?v=Y5H3Edu6Vz4)
- ☐ Arm the first Featherweight tracker with the same hex key into the second arming hole labelled "GPS 1" on the side of the rocket, screwing clockwise until the screw no longer turns

- ☐ Arm the second Featherweight tracker with the same hex key into the arming hole labelled “GPS 2”, screwing clockwise until the screw no longer turns
- ☐ Arm the Blue Raven altimeter by waving a large magnet over the side of body tube labelled “Blue Raven”
- ☐ Confirm the Blue Raven is armed by listening for its buzzer to indicate arming
- ☐ Verify using the Featherweight ground station that both Featherweight trackers are transmitting the correct location for the rocket
- ☐ Verify using the SRAD launch console receiver that the Josh flight computer has a GPS fix and is transmitting the correct rocket location

Payload

- ☐ Ensure personnel handling cells are wearing appropriate PPE
    - ☐ Dispose of PPE in separate garbage bags, tag and note for disposal.
- ☐ Ensure cells are inserted into bottom 2U of payload
- ☐ Ensure no line cutters or energetics are connected to the payload
- ☐ Insert assembled 3U payload into payload bag
- ☐ Insert disarming pull tab through payload bag and disarm payload
- ☐ Connect piranha line cutter to payload
- ☐ Insert piranha line cutter into line cutter holder on the bag
- ☐ Attach bag and payload to remainder of the recovery system
- ☐ Arm payload camera
- ☐ Ensure personnel responsible for assembling payload into the vehicle has appropriate PPE - including gloves and long sleeves.
- ☐ Execute a nice lil’ smooch of the payload
- ☐ Rest payload gently on mounting feets
- ☐ Disarm payload through body tube payload disarming pull tab locations at both locations immediately

Motor Assembly:

- ☐ Ensure Run Tank is fully assembled including vent bulkhead with torqued vent valve, and fill bulkhead with torqued -16 fitting
- ☐ Ensure Thrust Chamber assembly is fully completed before proceeding, found here: Thrust Chamber Assembly

Aft Section Assembly:

- ☐ Fasten Run Tank vent bulkhead to coupler with 4 x M6 countersunk bolts
- ☐ Fasten stringer onto the Run Tank fill bulkhead with 2 x M6 bolts
- ☐ Fasten stringer onto Thrust Chamber injector bulkhead with 2 x M6 bolts
- ☐ Torque -16 inter-tank fittings to 60 ft-lb
- ☐ Fasten remaining two stringers to the Run Tank fill bulkhead and Combustion Chamber Injector Bulkhead with 4 x M6 bolts
- ☐ Slide the fincan assembly over the Thrust Chamber, aligning with the bolt holes in the stringers, secure with 3 x M4 bolts
- ☐ Secure the metal boat tail to the bottom of the Thrust Chamber with 4 x M2 bolts

Full Assembly:

- ☐ Slide the forward assembly over the break coupler of the aft assembly, ensuring shear pin holes are aligned
- ☐ Insert 6 nylon shear pins into the shear pin holes, apply masking tape over the shear pins to be removed before arming

E Mass Budget

The following table shows the relative confidence of estimated mass coupled with measured masses to create a holistic mass estimate.

SUBSYSTEM	ASSEMBLY (OR GROUP)	COMPONENT	QTY	MASS PER ITEM (kg)	TOTAL MASS (kg)	CONFIDENCE IN MASS	ADDED MASS MARGIN (%)	CORRECTED MASS (kg)	WET MASS FRACTION (%)
AEROSTRUCTURES	**FORWARD SECTION**								
		Nosecone Tip	1	0.0974	0.0974	MEASURED	*0%*	0.097	0.19%
		Nosecone Body	1	1.5520	1.5520	LOW CONF.	*20%*	1.862	3.71%
		Coupler	1	0.2360	0.2360	HIGH CONF.	*5%*	0.248	0.49%
		Avionics/Payload Support Ring	1	0.0710	0.0710	MEASURED	*0%*	0.071	0.14%
		Avionics/Payload Mount	1	0.1420	0.1420	MEASURED	*0%*	0.142	0.28%
		Assembly Total			***2.0984***				
	CENTER SECTION								
		Center Body Tube	1	1.6420	1.6420	HIGH CONF.	*5%*	1.724	3.44%
		Recovery Support Ring	1	0.1110	0.1110	MEASURED	*0%*	0.111	0.22%
		Recovery Mount	1	0.1480	0.1480	MEASURED	*0%*	0.148	0.30%
		Salad Plate (Composite pressure seal)	1	0.0590	0.0590	HIGH CONF.	*5%*	0.062	0.12%
		Hybrid Retention Mount	1	0.3930	0.3930	MED CONF.	*10%*	0.432	0.86%
		Coupler	1	0.3270	0.3270	MED CONF.	*10%*	0.360	0.72%
		Adapter Body Tube	1	0.3940	0.3940	MED CONF.	*10%*	0.433	0.86%
		Eye Bolt	2	0.0680	0.1360	LOW CONF.	*20%*	0.163	0.33%
		Eye Bolt Nut	2	0.0078	0.0155	LOW CONF.	*20%*	0.019	0.04%
		Assembly Total			***3.2255***				
	AFT SECTION								
		Inter-Tank Panels	3	0.0850	0.2550	LOW CONF.	*20%*	0.306	0.61%
		Inter-Tank Stringer	3	0.0750	0.2250	LOW CONF.	*20%*	0.270	0.54%
		Aft Body Tube	1	0.3940	0.3940	MED CONF.	*10%*	0.433	0.86%
		Centering Rings	0	0.0032	0.0000	LOW CONF.	*20%*	0.000	0.00%
		AL Adaptor Coupler	1	0.7177	0.7177	MED CONF.	*0.1*	0.789	1.57%
		Fincan Coupler	1	0.2610	0.2610	HIGH CONF.	*5%*	0.274	0.55%
		Aluminum Boat Tail	1	0.2080	0.2080	MEASURED	*0%*	0.208	0.41%
		Fincan	1	1.7670	1.7670	LOW CONF.	*20%*	2.120	4.23%
		Assembly Total			***3.8277***				
	Airframe Fasteners								
		Fasteners	30	0.0021	0.0644	MED CONF.	*10%*	0.071	0.14%
		Shear Pins	6	0.0011	0.0064	MED CONF.	*10%*	0.007	0.01%
		Airframe Fasteners Total			***0.0644***				
	Other Components								
		Paint	1	0.3000	0.3000	LOW CONF.	*20%*	0.360	0.72%
		Other Total			***0.3000***				
	Aerostructures Subsystem Total				**9.5161**				

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SUBSYSTEM	ASSEMBLY (OR GROUP)	COMPONENT	QTY	MASS PER ITEM (kg)	TOTAL MASS (kg)	CONFIDENCE IN MASS	ADDED MASS MARGIN (%)	CORRECTED MASS (kg)	WET MASS FRACTION (%)
PROPULSION	**Oxidizer Assembly**								
		Casing	1	7.699	7.699	MEASURED	*0%*	7.699	15.35%
		Fill Bulkhead	1	0.621	0.621	MEASURED	*0%*	0.621	1.24%
		Vent Bulkhead	1	0.617	0.617	MEASURED	*0%*	0.617	1.23%
		Oxidizer Mass	1	14.629	14.629	HIGH CONF.	*5%*	15.360	30.63%
		Assembly Total			***23.566***				
	Combustion Chamber Assembly	**There is about a kilo of discrepancy in the openrocket file**							
		Chamber	1	2.250	2.250	MEASURED	*0%*	2.250	4.49%
		Injector Bulkhead	1	1.195	1.195	MEASURED	*0%*	1.195	2.38%
		Injector Plate	1	0.216	0.216	MEASURED	*0%*	0.216	0.43%
		O-Ring	4	0.004	0.016	MEASURED	*0%*	0.016	0.03%
		Fuel grain	1	2.430	2.430	MEASURED	*0%*	2.430	4.85%
		Nozzle	1	0.773	0.773	MEASURED	*0%*	0.773	1.54%
		Nozzle Carrier	1	0.490	0.490	MEASURED	*0%*	0.490	0.98%
		Liner	1	0.620	0.620	MEASURED	*0%*	0.620	1.24%
		Mixing Plate Liner	2	0.2800	0.560	MEASURED	*0%*	0.560	1.12%
		Mixing Plate	1	0.6800	0.680	MEASURED	*0%*	0.680	1.36%
		Assembly Total			***7.9900***				
	Propulsion Fastener Section :)								
		Radial Injector Bulkhead Fasteners	16	0.0015	**0.0240**	MEASURED	*0%*	0.024	0.05%
		Injector Fasteners	6	0.0020	**0.0120**	MEASURED	*0%*	0.012	0.02%
		Nozzle Carrier Fasteners	12	0.0015	**0.0180**	MEASURED	*0%*	0.018	0.04%
	Propulsion Subsystem Total Wet		1	31.5443	**31.5443**				
	Propulsion Subsystem Total Dry		1	14.2653	**14.2653**				
AVIONICS	**Nosecone Avionics**								
		Avionics from El Blasto:	1	1.0481	1.0481	MEASURED	*0%*	1.048	2.09%
		Extras	1	0.2500	0.2500	HIGH CONF.	*5%*	0.263	0.52%
		Assembly Total			***1.2981***				
	Avionics Subsystem Total				**1.2981**				

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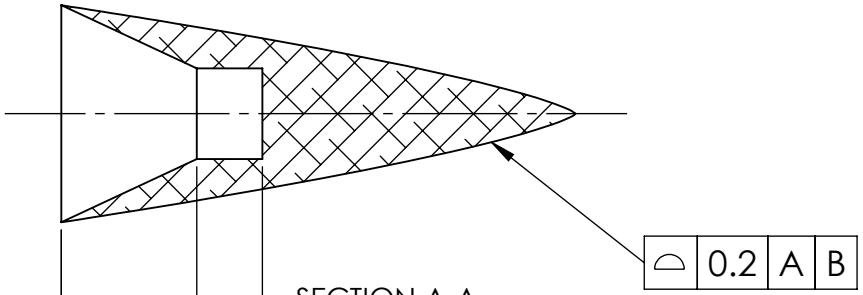
SUBSYSTEM	ASSEMBLY (OR GROUP)	COMPONENT	QTY	MASS PER ITEM (kg)	TOTAL MASS (kg)	CONFIDENCE IN MASS	ADDED MASS MARGIN (%)	CORRECTED MASS (kg)	WET MASS FRACTION (%)
RECOVERY	**Raptor Ejection System**								
		RAPTOR + CO2 Canister	2	0.2500	0.5000	LOW CONF.	*20%*	0.600	1.20%
		E-MATCH	4	0.0020	0.0080	LOW CONF.	*20%*	0.010	0.02%
		Assembly Total			***0.5080***				
	Parachute Assembly								
		Transmitter Box 3D Print	1	0.0640	0.0640	MEASURED	*0%*	0.064	0.13%
		Transmitter Box Nuts & Bolts	1	0.0090	0.0090	MEASURED	*0%*	0.009	0.02%
		Transmitter Box PCB	1	0.0130	0.0130	MEASURED	*0%*	0.013	0.03%
		Transmitter Box Screw Switch	1	0.0030	0.0030	MEASURED	*0%*	0.003	0.01%
		Transmitter Box Wires	1	0.0040	0.0040	MEASURED	*0%*	0.004	0.01%
		Transmitter Box Battery EX ICR186500 2000mAh	1	0.0480	0.0480	MEASURED	*0%*	0.048	0.10%
		Receiver Box 3D print	2	0.0600	0.1200	MEASURED	*0%*	0.120	0.24%
		Receiver Box PCB	1	0.0130	0.0130	MEASURED	*0%*	0.013	0.03%
		Receiver Box Easy Mini	1	0.0110	0.0110	MEASURED	*0%*	0.011	0.02%
		Receiver Box Battery EX ICR186500 2000mAh	2	0.0480	0.0960	MEASURED	*0%*	0.096	0.19%
		Receiver Box Screw Switch	2	0.0030	0.0060	MEASURED	*0%*	0.006	0.01%
		Receiver Box Line Cutter	2	0.0930	0.1860	MEASURED	*0%*	0.186	0.37%
		Reciver Box Nuts & Bolts	2	0.0150	0.0300	MEASURED	*0%*	0.030	0.06%
		Receiver Box E match	2	0.0040	0.0080	MEASURED	*0%*	0.008	0.02%
		Receiver Box Black Powder	2	0.0060	0.0120	MEASURED	*0%*	0.012	0.02%
		Shock Cord (Rocket to Swivel)	1	0.1550	0.1550	MEASURED	*0%*	0.155	0.31%
		Shock Cord (Swivel To Chute)	1	0.0500	0.0500	MEASURED	*0%*	0.050	0.10%
		Swivel	1	0.1360	0.1360	MEASURED	*0%*	0.136	0.27%
		Main Chute (chute+ suspension lines + reefing rings	1	1.6140	1.6140	MEASURED	*0%*	1.614	3.22%
		Parachute Bag	1	0.1450	0.1450	MEASURED	*0%*	0.145	0.29%
		Shock Cord (Swivel To Nosecone)	1	0.0500	0.0500	MEASURED	*0%*	0.050	0.10%
		Pilot Chute	1	0.1910	0.1910	MEASURED	*0%*	0.191	0.38%
		Parachute Bay(body Bag)	1	0.2000	0.2000	MEASURED	*0%*	0.200	0.40%
		Assembly Total			***3.1640***				
	Recovery Subsystem Total				**3.6720**				
PAYLOAD	**Payload Assembly**								
		Payload	1	3.0000	3.0000	MED CONF.	*10%*	3.300	6.58%
		Assembly Total			***3.0000***				
	Payload Subsystem Total				**3.0000**				
	Total Rocket Dry			**No Propellant**	*31.751*		**No Propellant**	***32.364***	***64.53%***
	Total Rocket Wet				*47.891*	Added Margin:	4.73%	***50.155***	
				Dry No Margin (lb)	70.000		Dry (lb)	71.351	
				Wet No Margin (lb)	105.582		Wet (lb)	110.572	

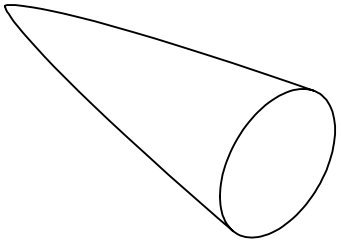
F Drawings

F.1 Aerostructure Drawings

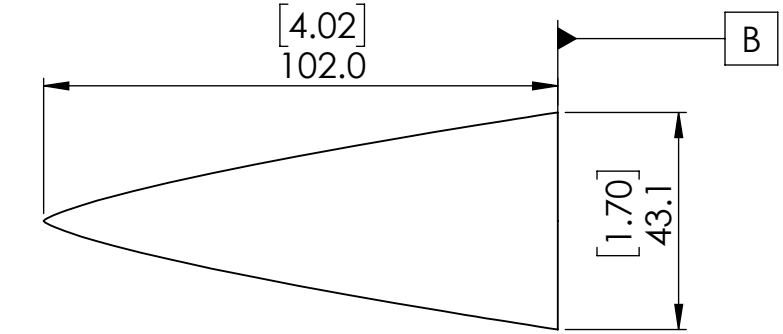
1. MATERIAL: 6061 T6 ALUMINUM
2. SECTION PROFILE TO BE VON KARMAN CURVE (EQUATION DRIVEN SEE CAD FILE)

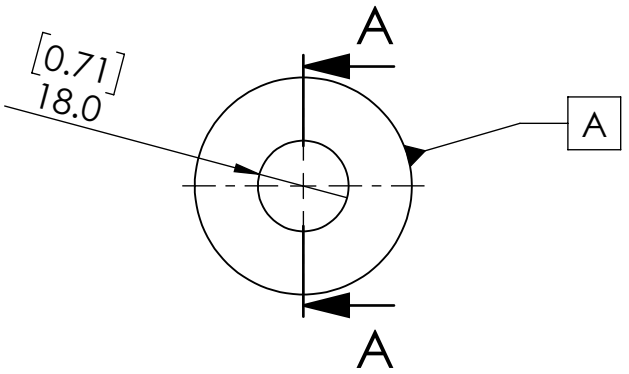


SECTION A-A



NTS





UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm [in] TOLERANCES ARE: ANGLES: ±0.5° X ±1 [0.04] X.X: ±0.1 [0.004] X.XX: ±0.05 [0.002] X.XXX: ±0.005 [0.0002]	DESIGNED RORY WALINGA	2024/12/19
	DRAWN RORY WALINGA	2024/12/19

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE
NOSECONE TIP

ACADEMIC YEAR	QTY	PART NUMBER
2024-2025	1	CR25-01-12-R00
AIRFRAME	SIZE A	SCALE 2:3
CR25		SHEET 1 OF 1

Final Project Report for CU InSpace CR25H
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Revision: R0
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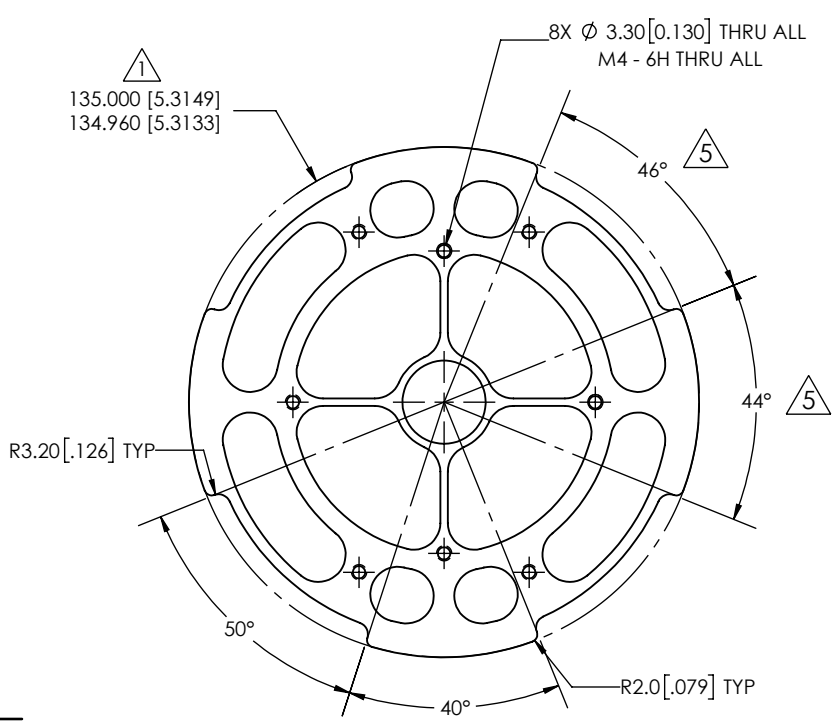
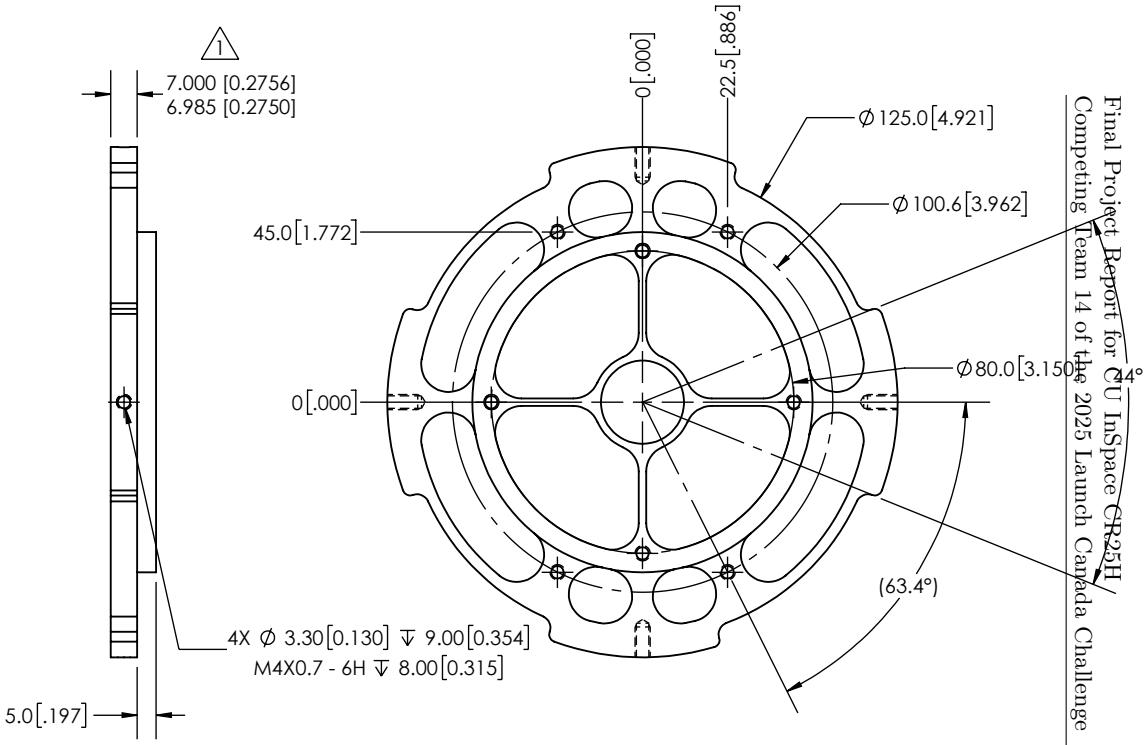
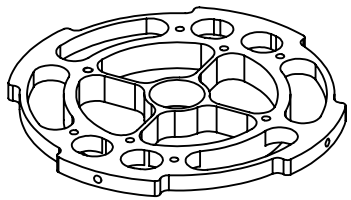
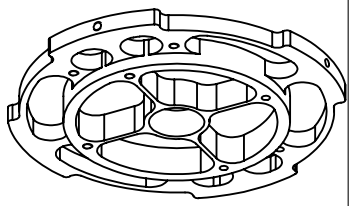


VIEW LOOKING ON TOP SURFACE







NOTES:

1. LIMITS FOR SLIDING CLEARANCE FIT h7/G7 WITH SUPPORT RING.
2. FITS AND TOLERANCES PER ISO 286.
3. MATERIAL: ALUMINUM 6061-T6.
4. SEE ACCOMPANYING .STEP FILE.
5. TYPICAL ANGLES.

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm [in] UNLESS OTHERWISE SPECIFIED TOLERANCES ARE: ANGLES: ±0.5° X ±1 X.X: ±0.1 X.XX: ±0.05 X.XXX: ±0.005	DESIGNED J.GU	2024-2025
	DRAWN J.GU	2024-03-30
	CHECKED	
	APPROVED	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME CR25	

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CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: PAYLOAD / AVIONICS MOUNT

QTY	PART NUMBER	REVISION
1	CR25-01-21-R01	

SIZE A | SCALE 1:2 | SHEET 1 OF 1

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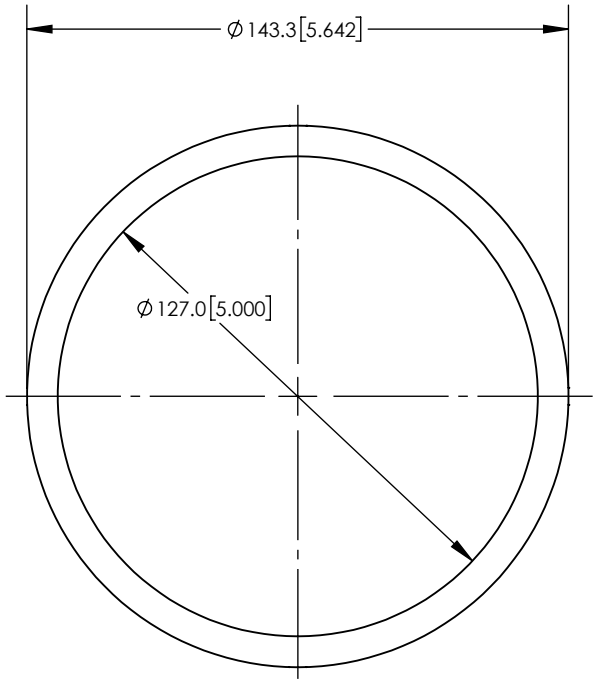
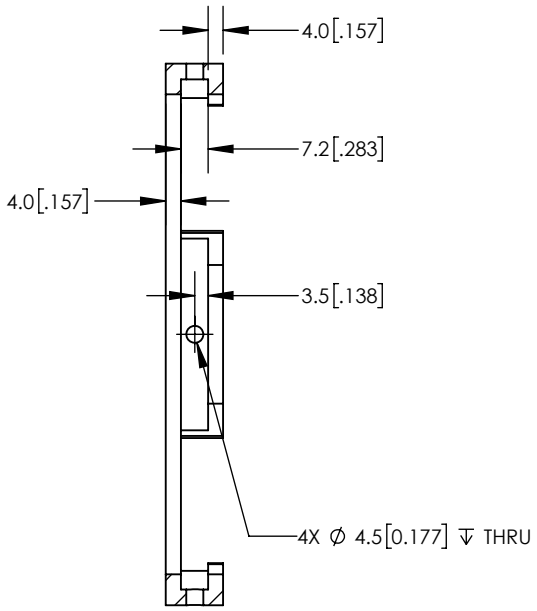
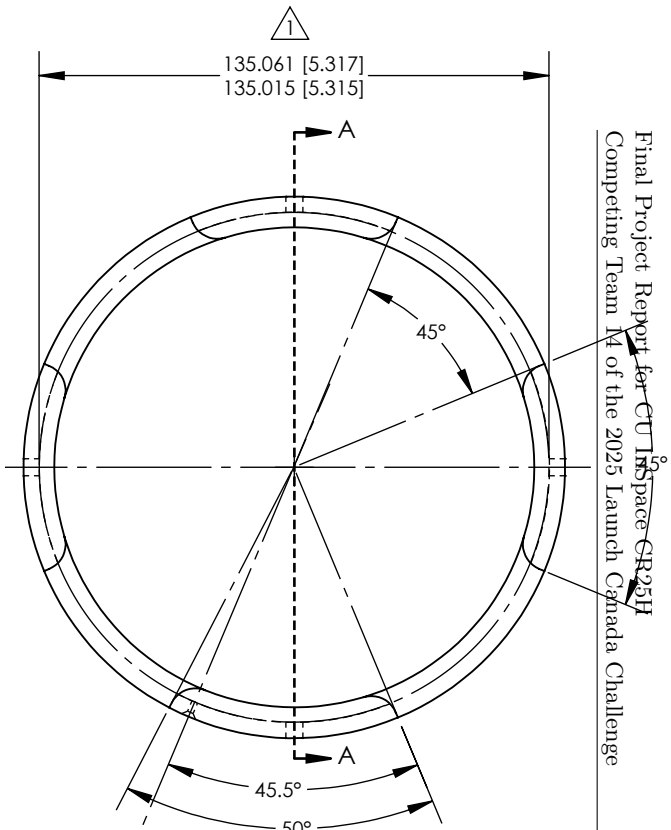
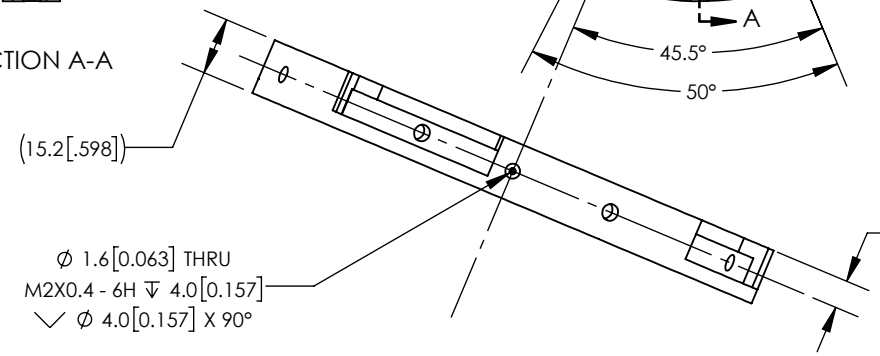


VIEW LOOKING ON TOP SURFACE



SECTION A-A





NOTES:

1. LIMITS FOR SLIDING CLEARANCE FIT h7/G7 WITH MOUNT.
2. FITS AND TOLERANCES PER ISO 286.
3. MATERIAL: ALUMINUM 6061-T6.
4. SEE ACCOMPANYING .STEP FILE.
5. NO COUNTERBORE ON RADIAL HOLES.

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm [in] UNLESS OTHERWISE SPECIFIED TOLERANCES ARE: ANGLES: ±0.5° X ±1 X.X: ±0.1 X.XX: ±0.05 X.XXX: ±0.005	DESIGNED J.GU	2024
	DRAWN J.GU	2025-03-20
	CHECKED	
	APPROVED	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME CR25	

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CARLETON UNIVERISTY
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TITLE: PAYLOAD / AVIONICS SUPPORT RING

QTY	PART NUMBER	REVISION
1	CR25-01-22-R01	5

SIZE A | SCALE 1:2 | SHEET 1 OF 2

Final Project Report for CU INSPACE CR25H
Competing Team M of the 2025 Launch Canada Challenge
Effective Date: 9/8/2025

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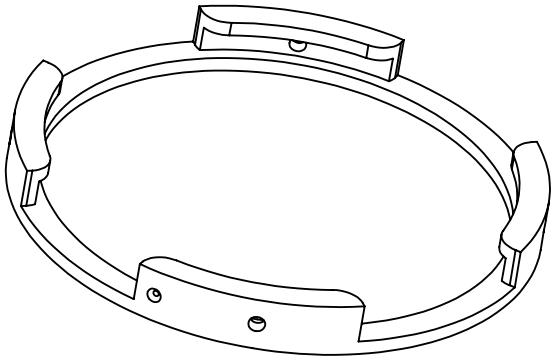

VIEW, UP SIDE DOWN

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm [in] TOLERANCES ARE: ANGLES: ±0.5° X ±1 X.X: ±0.1 X.XX: ±0.05 X.XXX: ±0.005	DESIGNED J.GU	2024-2025
	DRAWN J.GU	2025-03-20
	ACADEMIC YEAR 2024-2025	
	AIRFRAME CR25H	

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OTTAWA, ON, CANADA K1S 5B6

TITLE
PAYLOAD / AVIONICS SUPPORT RING

QTY	PART NUMBER	REVISION
1	CR-01-22-R01	A

SIZE	SCALE	SHEET
A	1:2	2 OF 2

Effective Date: 5/8/2025

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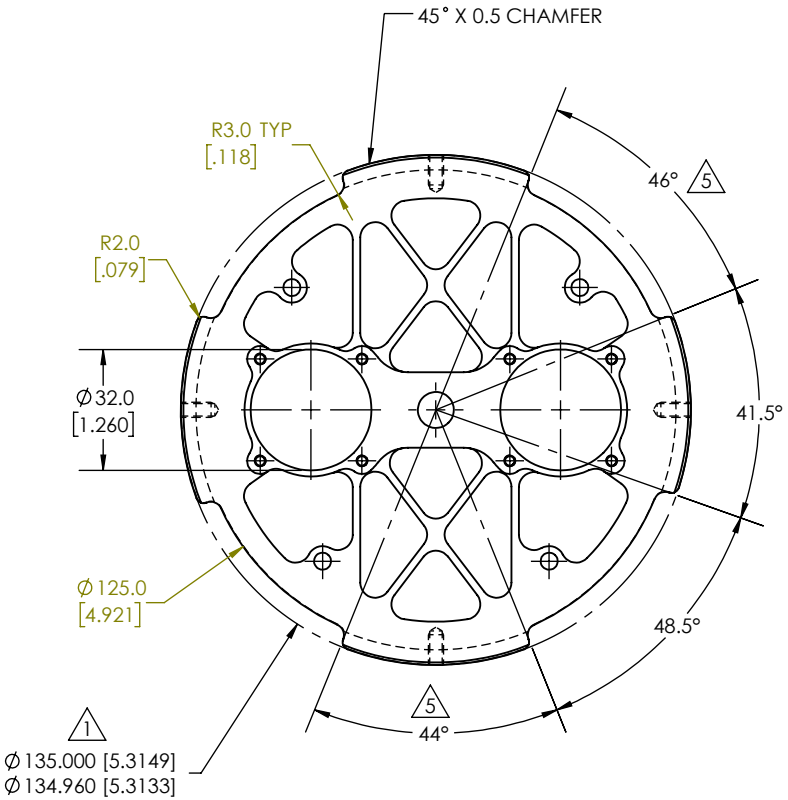
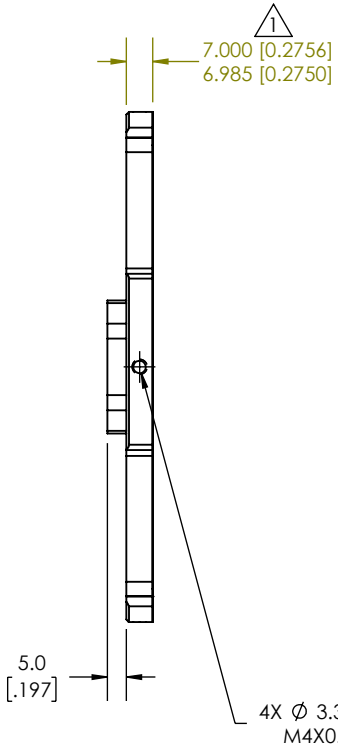
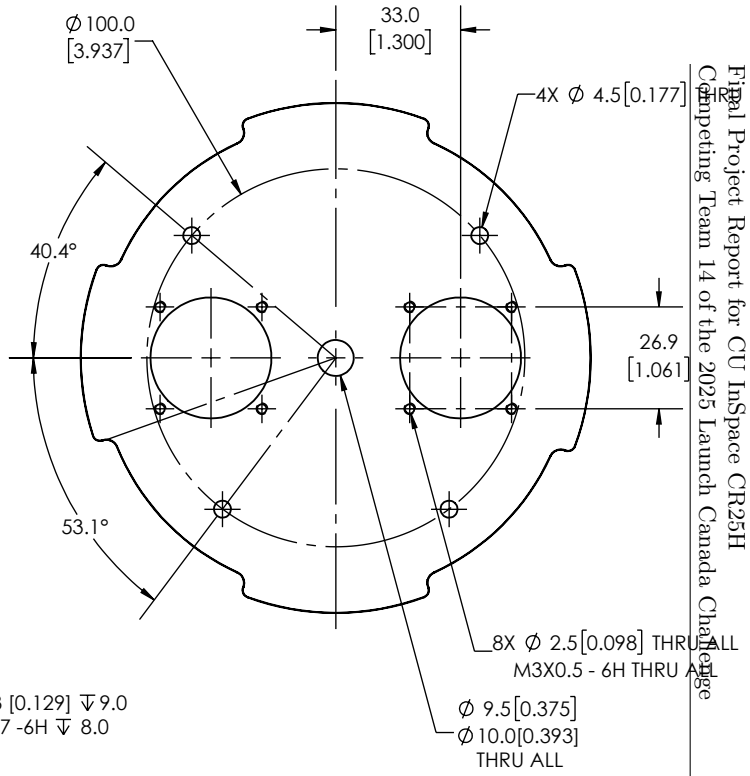
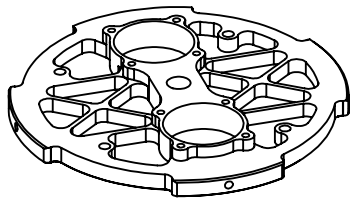
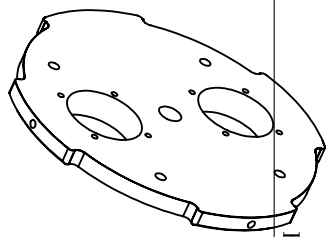


VIEW LOOKING ON TOP SURFACE









NOTES:

1. LIMITS FOR SLIDING CLEARANCE FIT h7/G7 WITH SUPPORT RING.
2. FITS AND TOLERANCES PER ISO 286.
3. MATERIAL: ALUMINUM 6061-T6.
4. SEE ACCOMPANYING .STEP FILE.
5. TYPICAL ANGLES.

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm [in] UNLESS OTHERWISE SPECIFIED TOLERANCES ARE: ANGLES: ±0.5° X ±1 X.X: ±0.1 X.XX: ±0.05 X.XXX: ±0.005	DESIGNED J.GU	2024
	DRAWN J.GU	2025-02-26
	CHECKED	
	APPROVED	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME CR25	

CU INSPACE

CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: RECOVERY MOUNT

QTY	PART NUMBER	SIZE	SCALE	SHEET
1	CR-02-22-R01	A	1:2	1 OF 1

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Revision: R0

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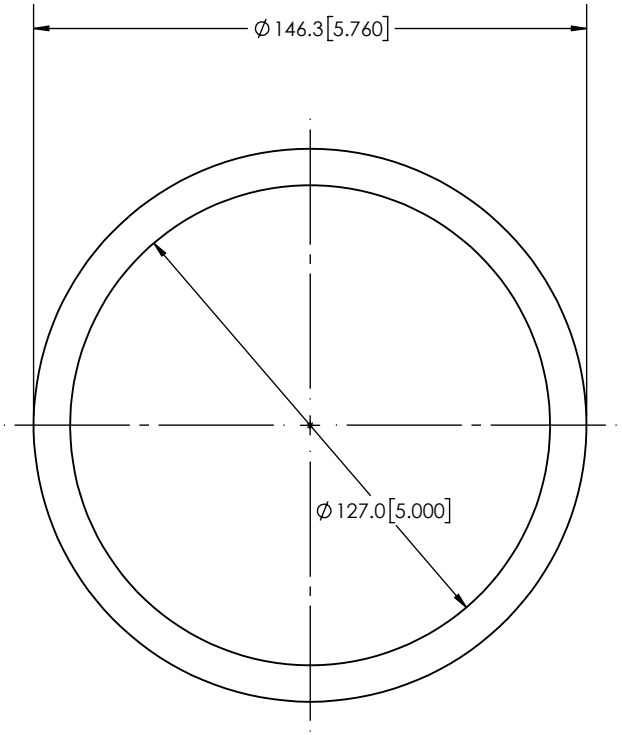
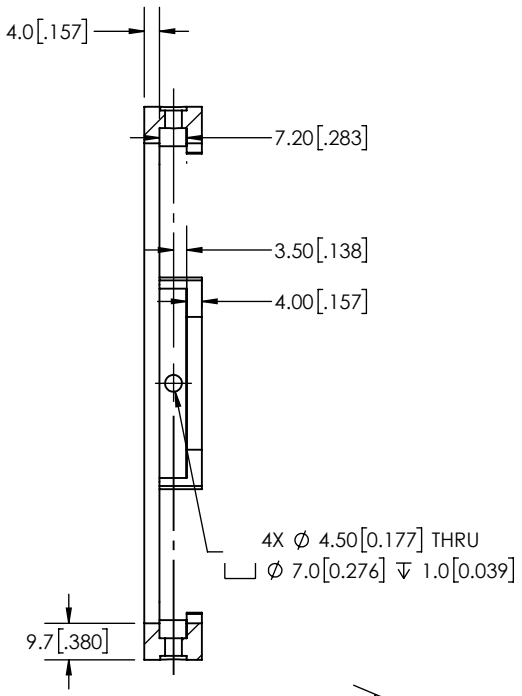
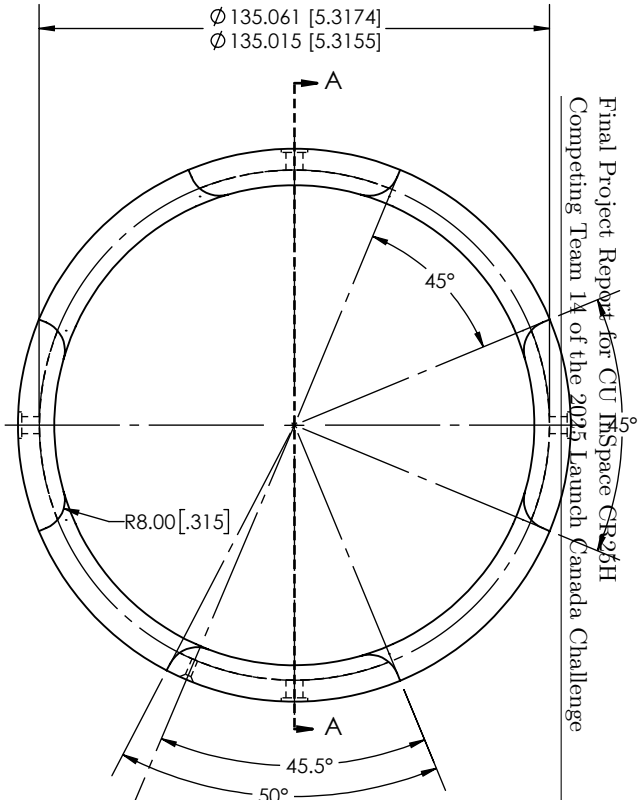
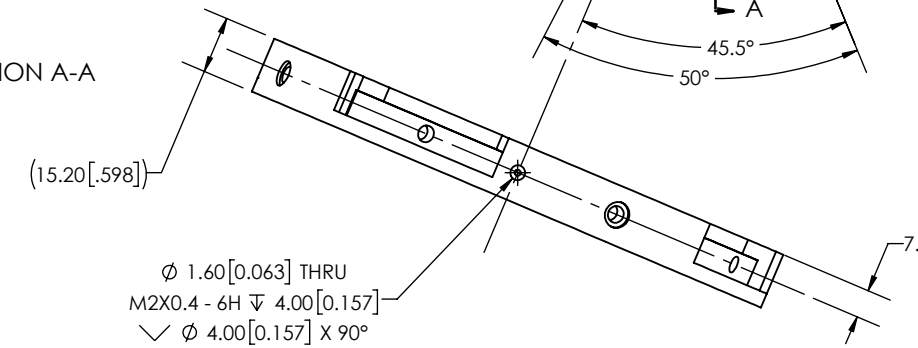


VIEW LOOKING ON TOP SURFACE



SECTION A-A





NOTES:

1. LIMITS FOR SLIDING CLEARANCE FIT h7/G7 WITH MOUNT.
2. FITS AND TOLERANCES PER ISO 286.
3. MATERIAL: ALUMINUM 6062-T6
4. SEE ACCOMPANYING .STEP FILE.

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm [in] UNLESS OTHERWISE SPECIFIED TOLERANCES ARE: ANGLES: ±0.5° X ±1 X.X: ±0.1 X.XX: ±0.05 X.XXX: ±0.005	DESIGNED J.GU	2024
	DRAWN J.GU	2025-03-10
	CHECKED	
	APPROVED	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME CR25	

CU INSPACE

CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: RECOVERY SUPPORT RING

QTY	PART NUMBER	
1	CR25-02-22-R01	
SIZE A	SCALE 1:2	SHEET 1 OF 2

Final Project Report for CU INSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge

Revision: R0
Effective Date: 5/8/2025

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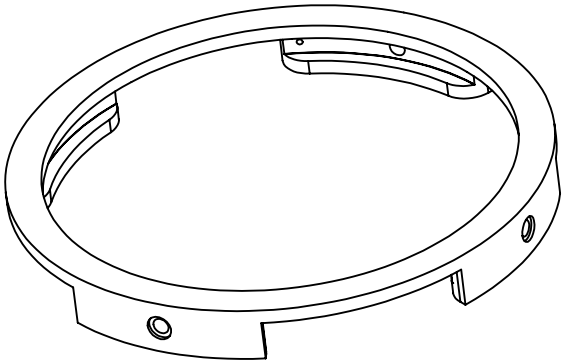
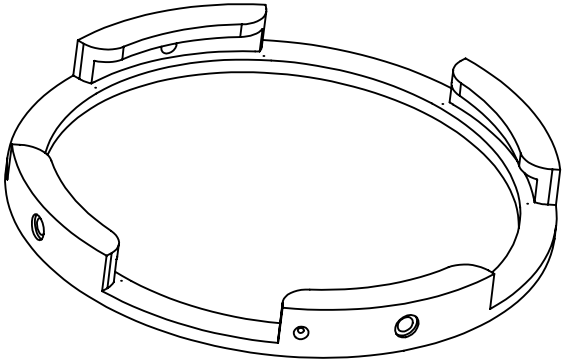


VIEW, RIGHT SIDE UP



VIEW, UP SIDE DOWN

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm [in] TOLERANCES ARE: ANGLES: ±0.5° X ±1 X.X: ±0.1 X.XX: ±0.05 X.XXX: ±0.005	DESIGNED J.GU	2024
	DRAWN J.GU	2025-03-10
	ACADEMIC YEAR 2024-2025	
	AIRFRAME CR25H	

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE
RECOVERY SUPPORT RING

QTY	PART NUMBER
1	CR25-02-22-R01

SIZE	SCALE	SHEET
A	1:2	2 OF 2

Revision: R0
Effective Date: 9/8/2025

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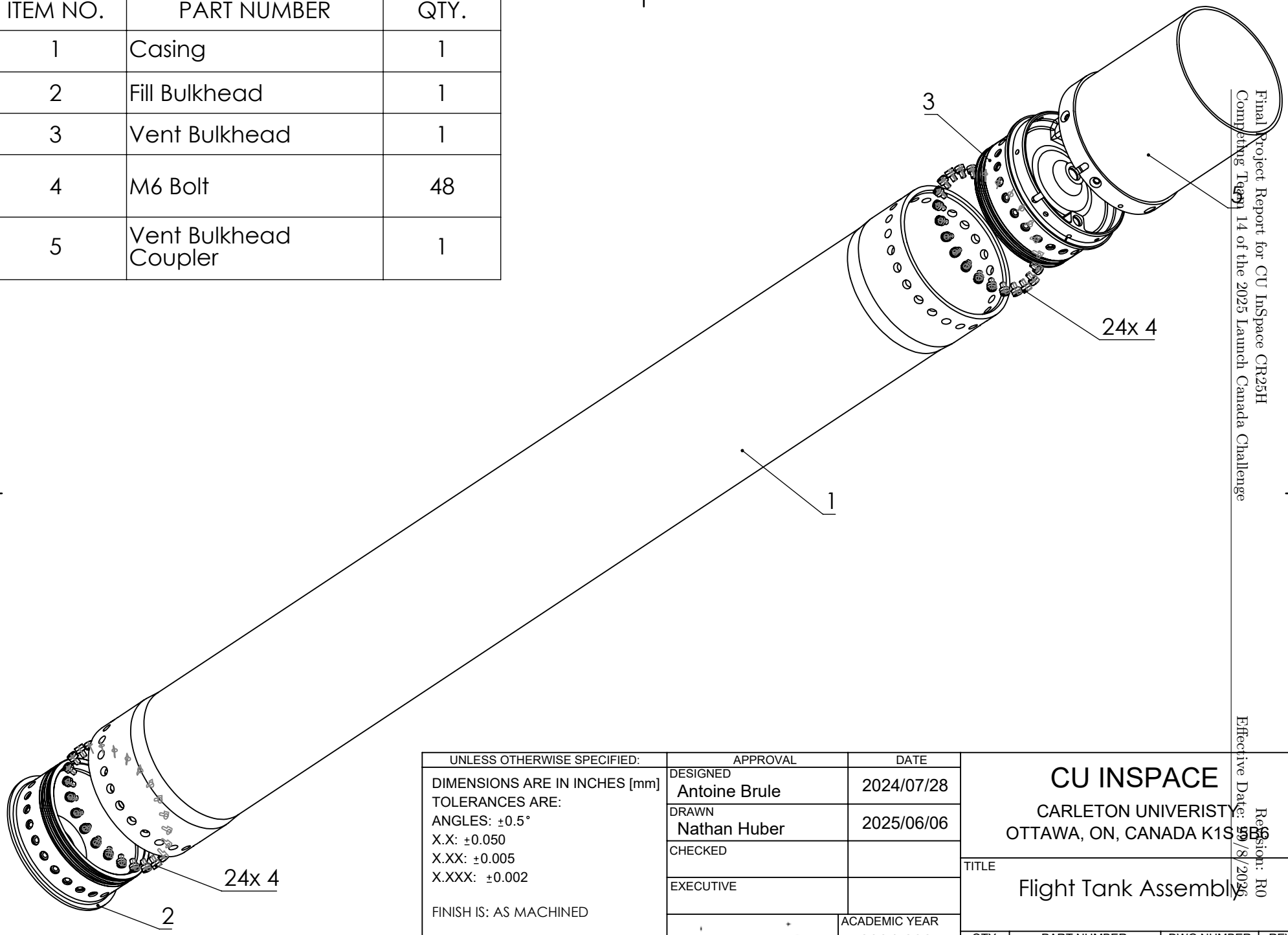
F.2 Propellant Tank Drawings

ITEM NO.	PART NUMBER	QTY.
1	Casing	1
2	Fill Bulkhead	1
3	Vent Bulkhead	1
4	M6 Bolt	48
5	Vent Bulkhead Coupler	1



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Effective Date: 5/8/2026 Revision: R0

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN INCHES [mm] TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002 FINISH IS: AS MACHINED DEBURR & BREAK SHARP EDGES	DESIGNED Antoine Brule	2024/07/28
	DRAWN Nathan Huber	2025/06/06
	CHECKED	
	EXECUTIVE	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME	

CU INSPACE
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OTTAWA, ON, CANADA K1S 5B6

TITLE
Flight Tank Assembly

QTY	PART NUMBER	DWG NUMBER	REV
1			1

SIZE A | SCALE 1:5 | SHEET 1 OF 1

CU INSPACE DRAWING TEMPLATE **V5**
A. THOMAS 2021 auresthomas@cmail.carleton.ca

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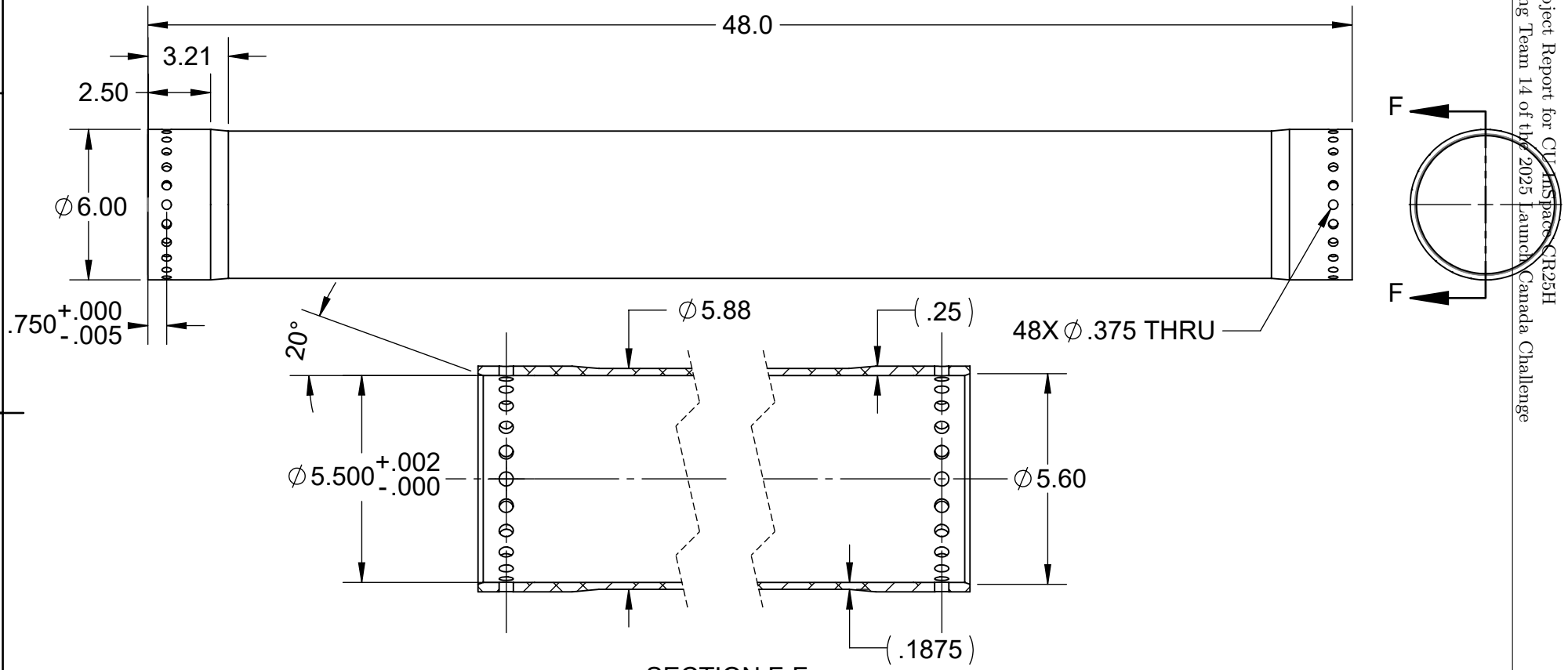
NOTES:

1. MATERIAL: AL 6061-T6
2. BOLT HOLES ARE EQUALLY SPACED
3. SURFACE FINISH OF THE INNER DIAMETER SHOULD BE 32√ WITHIN 2.1" OF ENDS
4. PART IS SYMMETRICAL



SECTION F-F
SCALE 1 : 4

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN INCHES [mm]	DESIGNED ANTOINE BRULE	2025/02/19
TOLERANCES ARE:	DRAWN ANTOINE BRULE	2025/03/21
ANGLES: ±0.5°	CHECKED JACK HARTMAN	2025/03/21
X.X: ±0.050	EXECUTIVE ANTOINE BRULE	025/03/21
X.XX: ±0.005	ACADEMIC YEAR 20YY-20YY	
X.XXX: ±0.002	AIRFRAME	
FINISH IS: AS MACHINED		
DEBURR & BREAK SHARP EDGES		

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE
FLIGHT TANK CASING

QTY	PART NUMBER	DWG NUMBER	REV
1	CRYY-####	YY###	0

SIZE A | SCALE 1:12 | SHEET 1 OF 1

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge
Effective Date: 5/8/2025 Revision: R0

CU INSPACE DRAWING TEMPLATE **V5**
A. THOMAS 2021 auresthomas@cmail.carleton.ca

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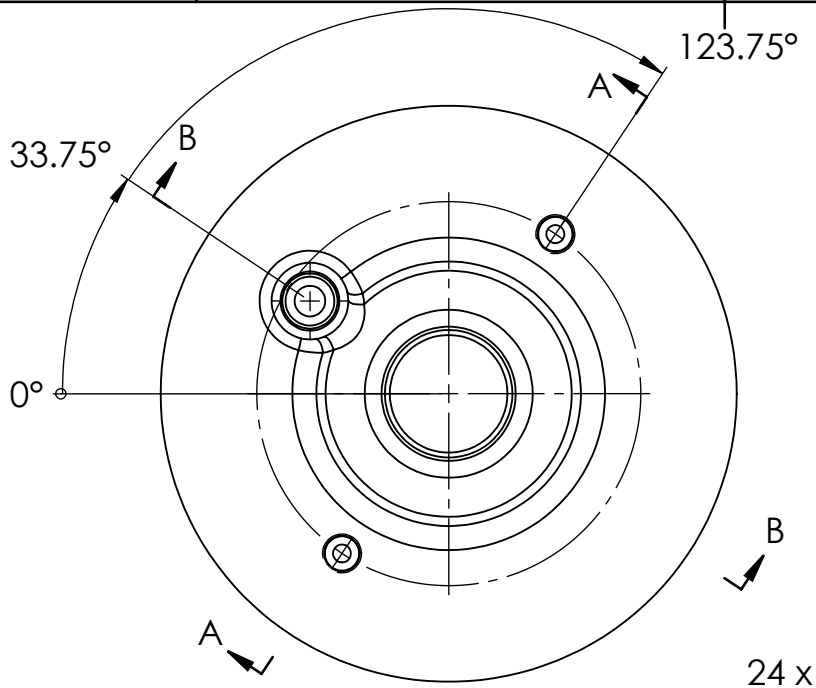

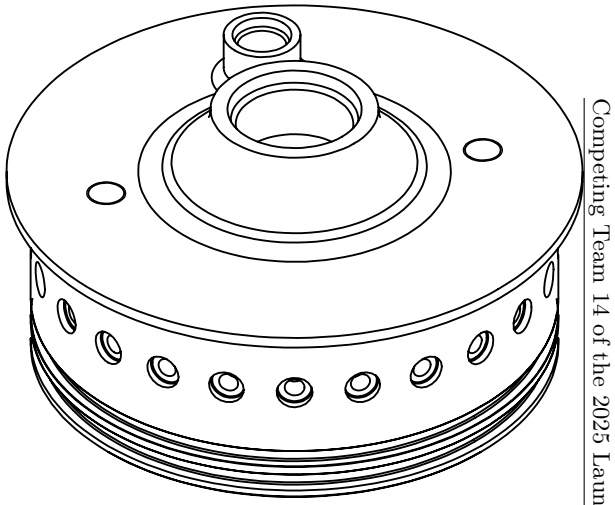

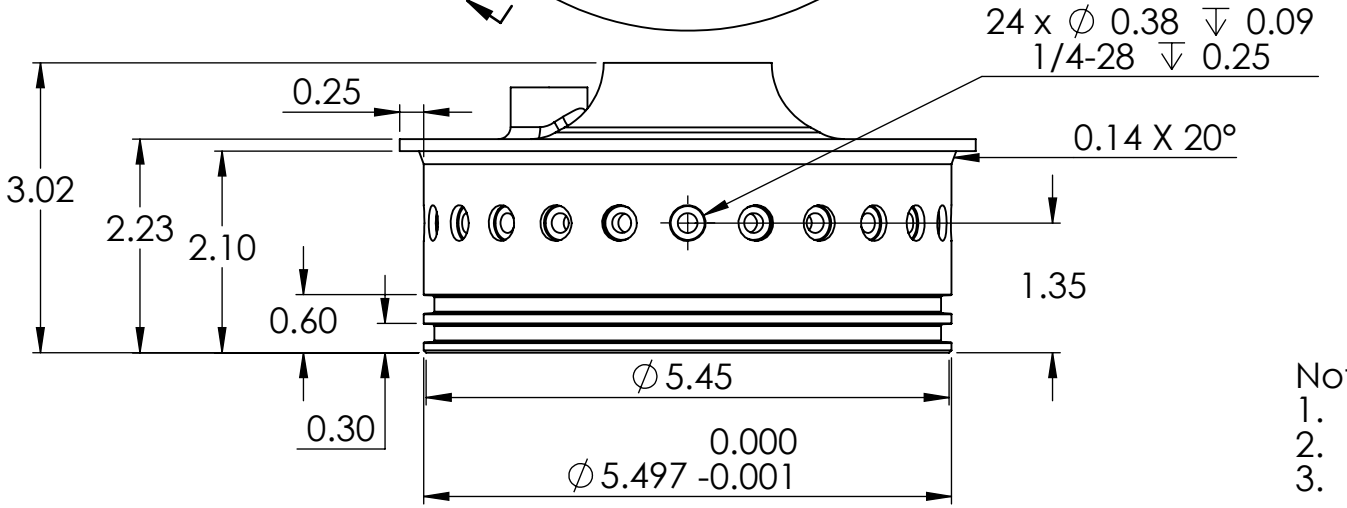

Notes:
1. MATERIAL: ALLUMINIUM T6061-T6
2. BOLT HOLES ARE EQUALLY SPACED
3. BOTH O-RINGS ARE IDENTICAL

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Effective Date: 5/8/2026
Revision: R0

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN INCHES [mm] TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002 FINISH IS: AS MACHINED DEBURR & BREAK SHARP EDGES	DESIGNED: Antoine Brule	2024/07/28
	DRAWN: Nathan Huber	2025/03/19
	CHECKED: Antoine Brule	2025/04/01
	EXECUTIVE: Antoine Brule	2025/04/01
	ACADEMIC YEAR: 2024-2025	
	AIRFRAME	

CU INSPACE
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TITLE: Fill Bulkhead

QTY	PART NUMBER	DWG NUMBER	REV
1			4

SIZE A | SCALE 1:2 | SHEET 1 OF 2

CU INSPACE DRAWING TEMPLATE **V5**
A. THOMAS 2021 auresthomas@cmail.carleton.ca

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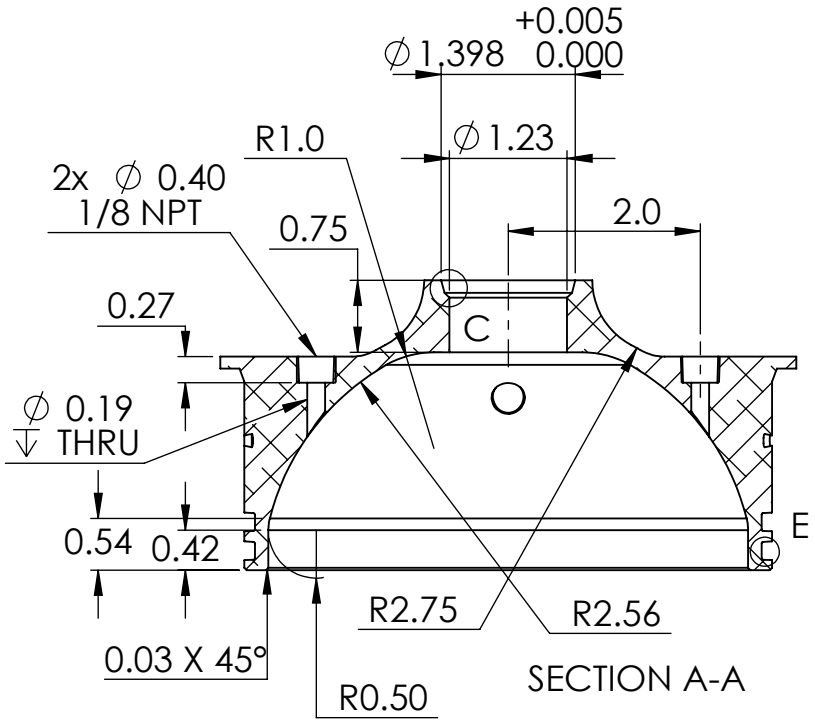
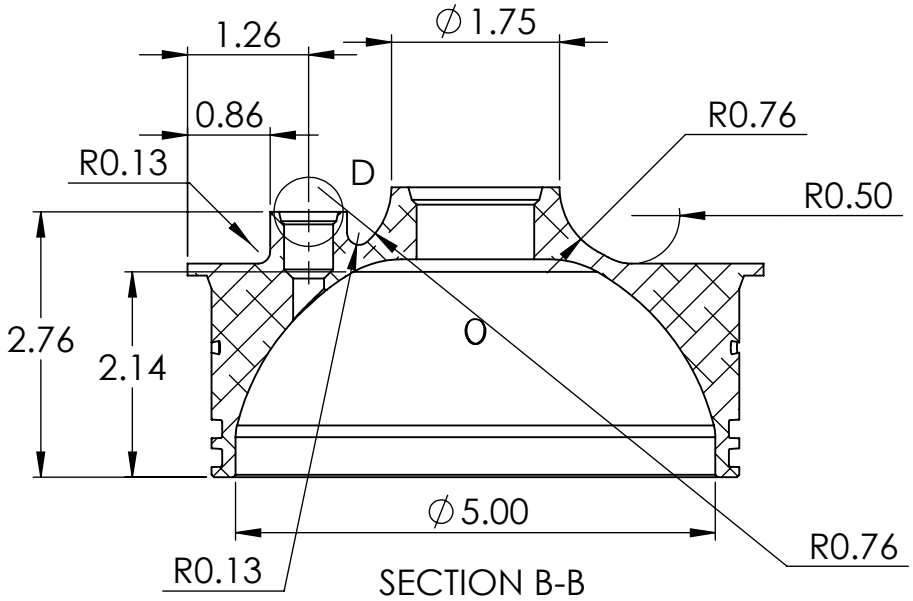


SECTION A-A



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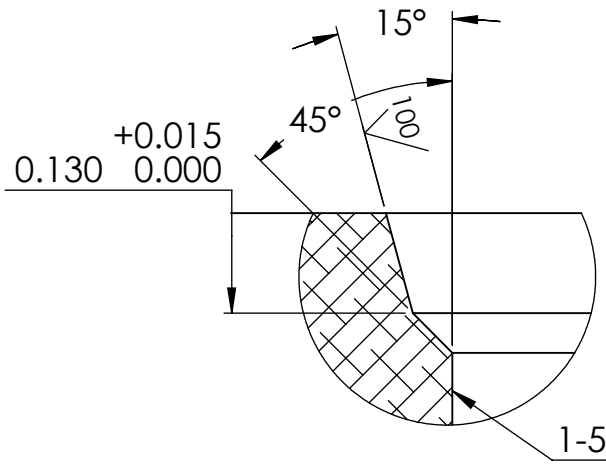
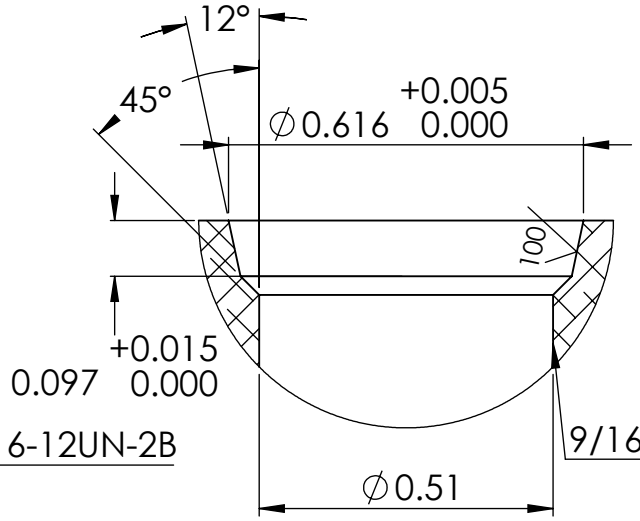
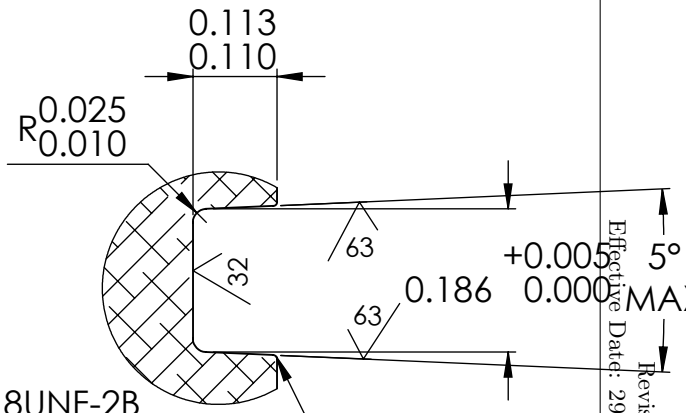


DETAIL C
SCALE 4 : 1



DETAIL D
SCALE 3 : 1



DETAIL E
SCALE 4 : 1

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge
Effective Date: 29/4/2025 Revision: A0

	DWG NUMBER	REV
	YY###	4
SIZE A	SCALE 1:2	SHEET 2 OF 2

CU INSPACE DRAWING TEMPLATE **V5**
A. THOMAS 2021 auresthomas@cmail.carleton.ca
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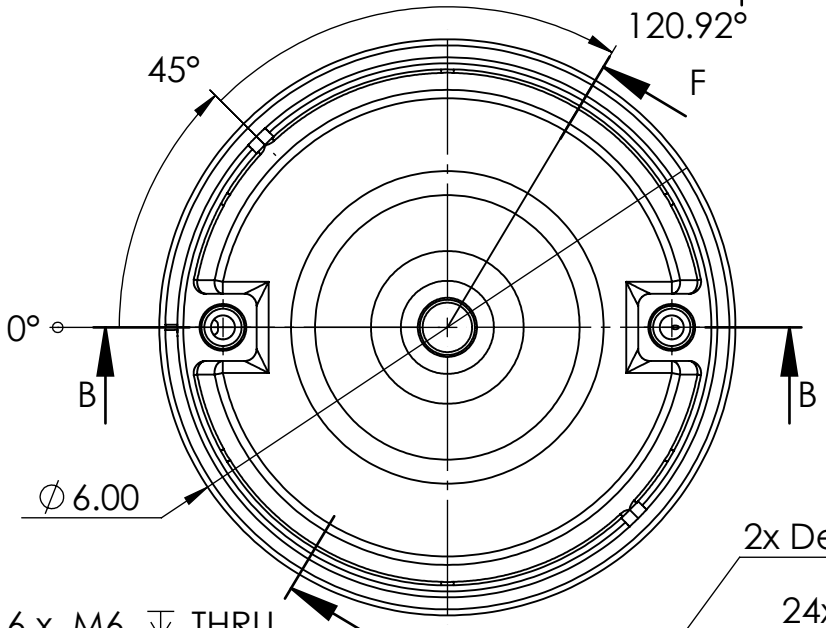

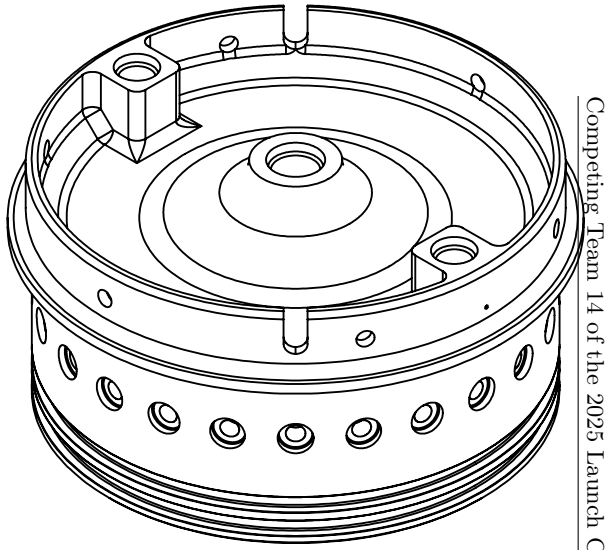

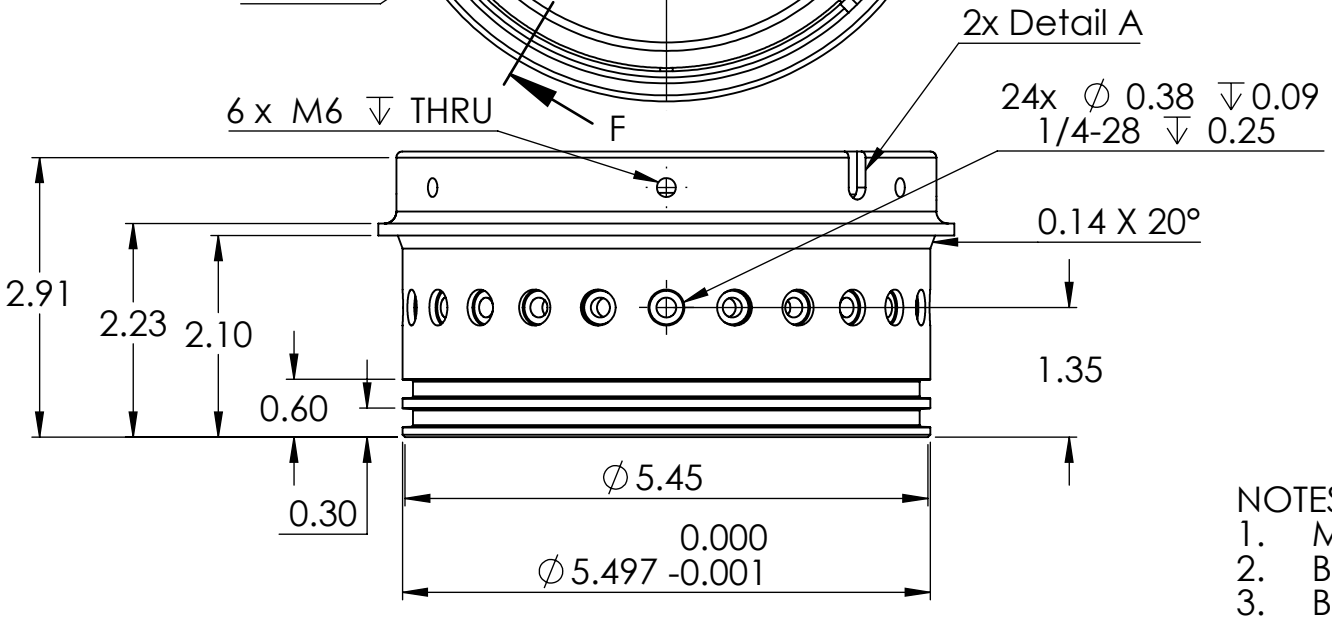

NOTES:
1. MATERIAL: ALLUMINIUM T6061-T6
2. BOLT HOLES ARE ALL EQUALLY SPACED
3. BOTH O-RINGS ARE IDENTICAL

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge
Effective Date: 5/8/2026
Revision: R0

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN INCHES [mm] TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002 FINISH IS: AS MACHINED DEBURR & BREAK SHARP EDGES	DESIGNED: Antoine Brule	2024/07/28
	DRAWN: Nathan Huber	2025/03/19
	CHECKED: Antoine Brule	2025/04/01
	EXECUTIVE: Antoine Brule	2025/04/01
	ACADEMIC YEAR: 2024-2025	
	AIRFRAME	

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: Vent Bulkhead

QTY	PART NUMBER	DWG NUMBER	REV
1			8

SIZE A | SCALE 1:2 | SHEET 1 OF 3

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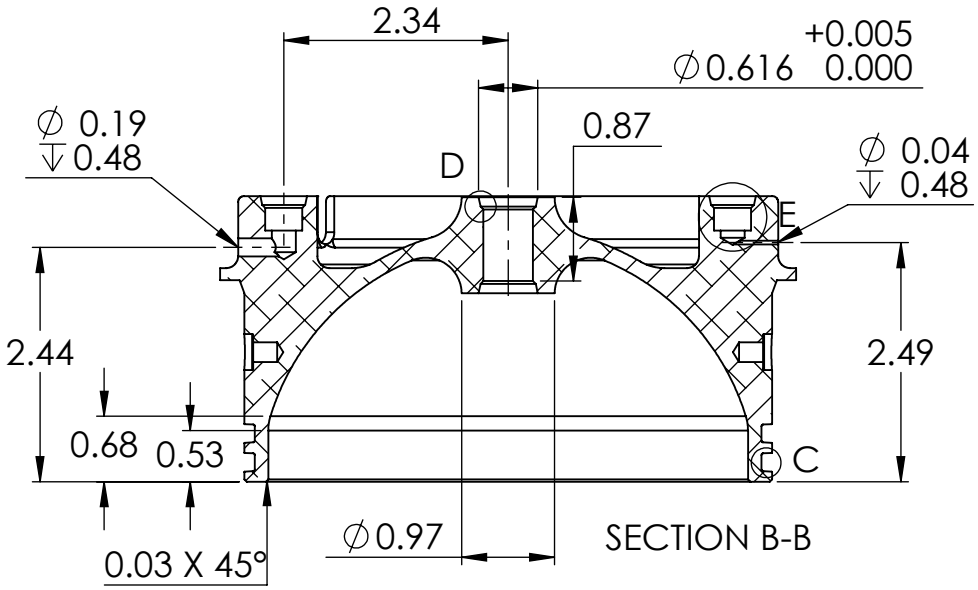
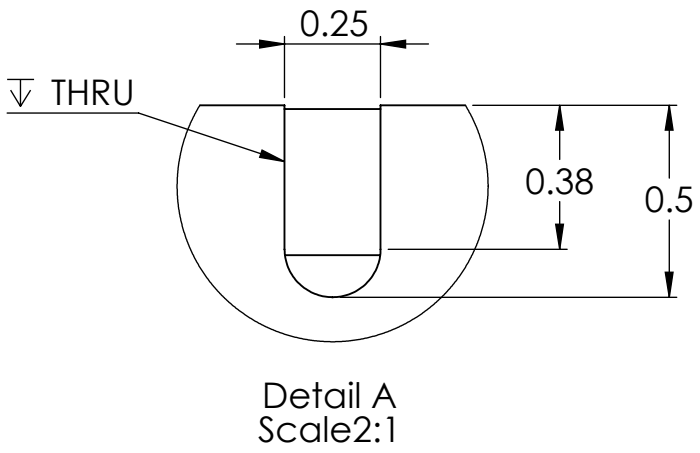
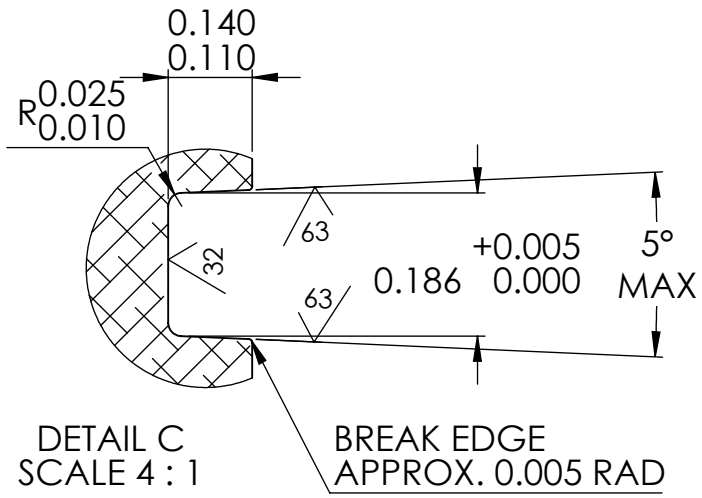
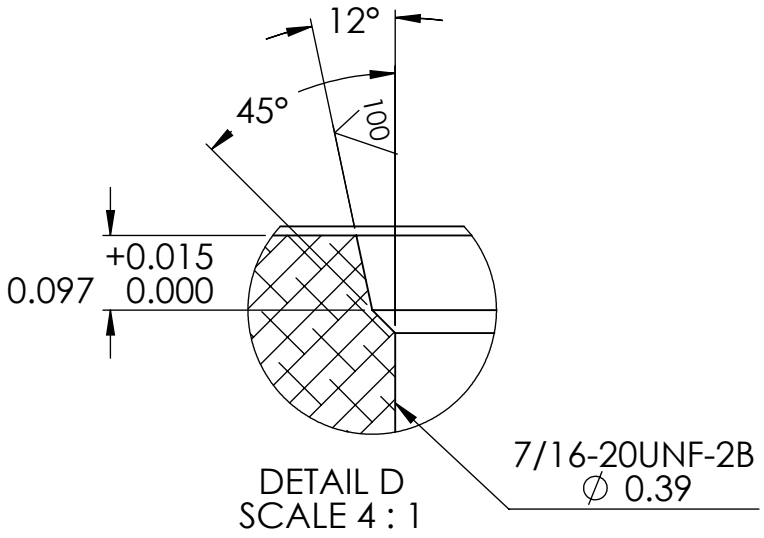




Detail A
Scale2:1



DETAIL C
SCALE 4 : 1



DETAIL D
SCALE 4 : 1

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge

Revision: R0
Effective Date: 29/8/2026

	DWG NUMBER	REV
	YY###	8
SIZE A	SCALE 1:2	SHEET 2 OF 3

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A. THOMAS 2021 auresthomas@cmail.carleton.ca
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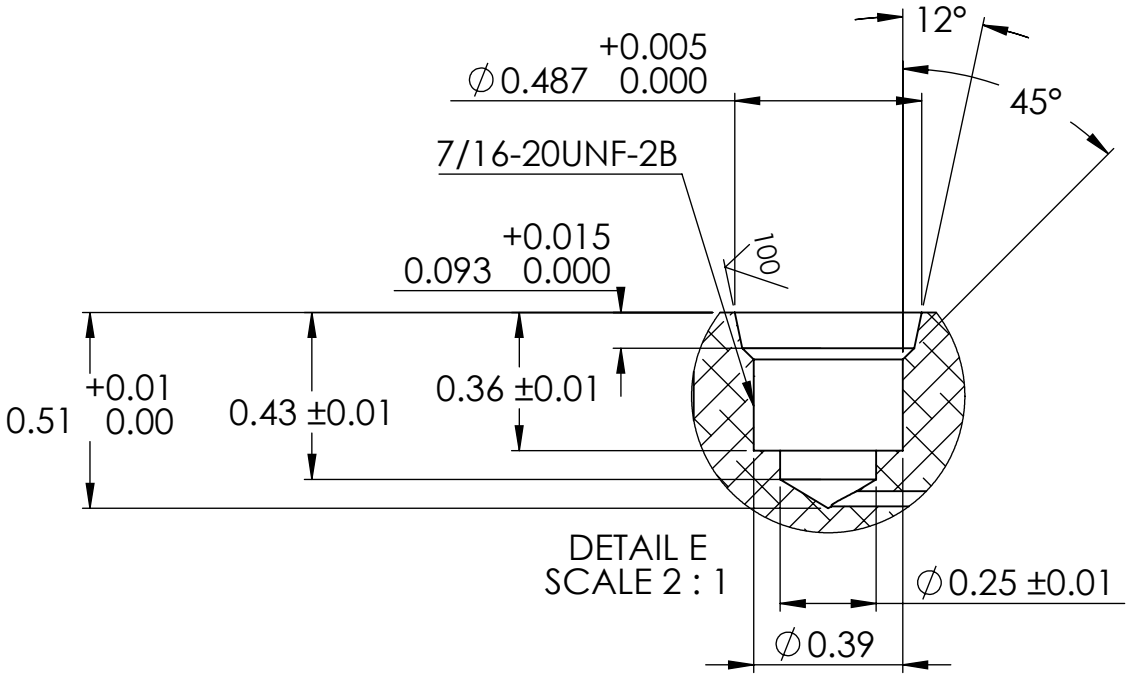
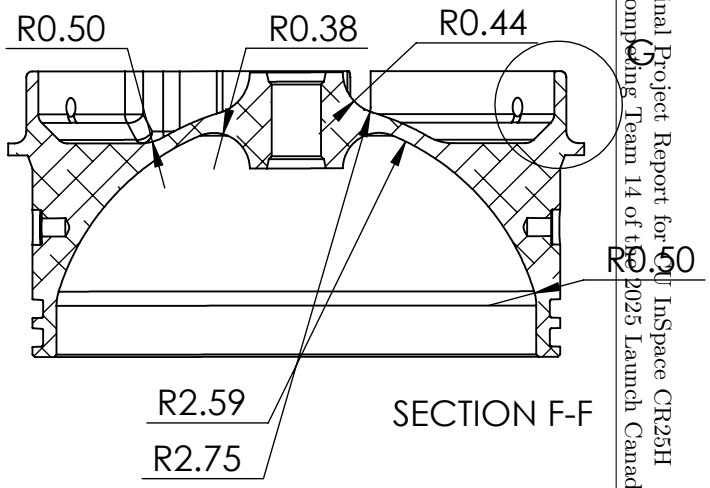
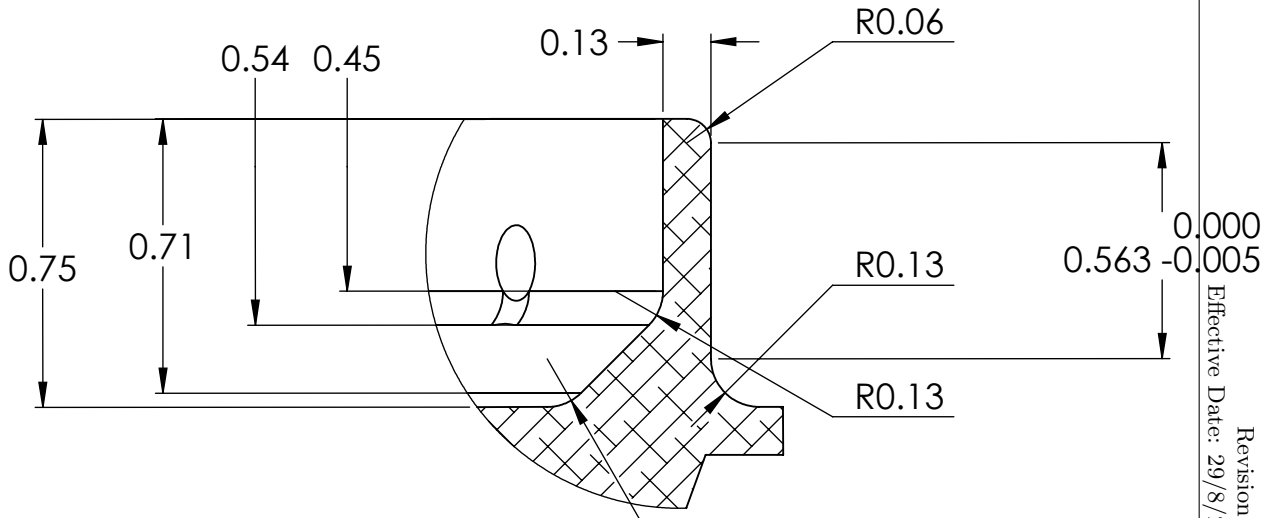


DETAIL E
SCALE 2 : 1



SECTION F-F



DETAIL G
SCALE 2 : 1

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge

Revision: R0
Effective Date: 29/8/2026

	DWG NUMBER YY###	REV 8	
SIZE A	SCALE 1:2	SHEET 3 OF 3	

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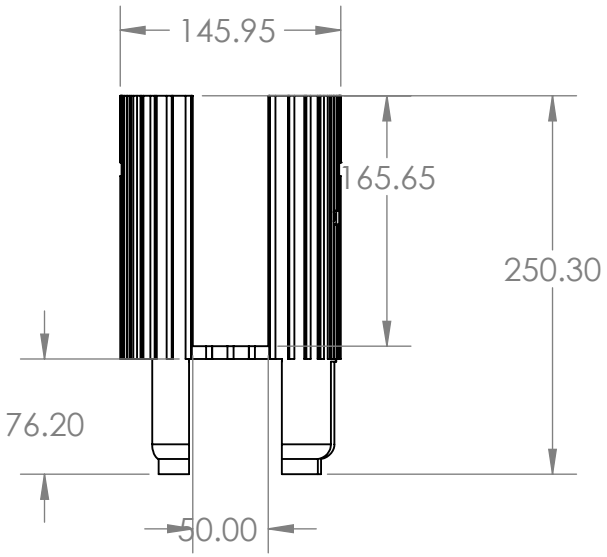
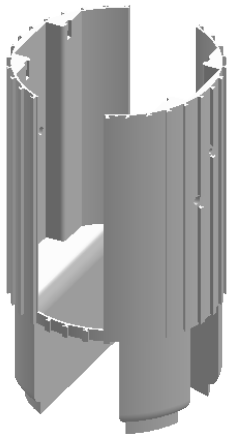
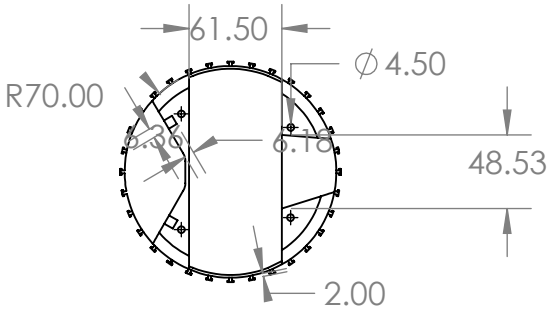
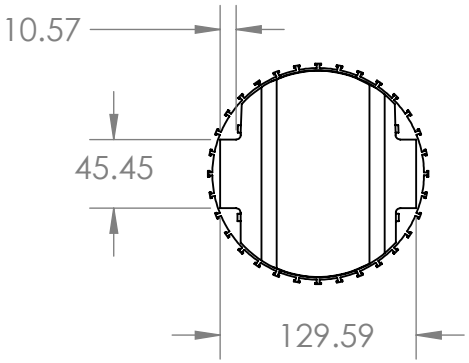
F.3 Recovery Housings Drawings









Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge

Revision: R0
Effective Date: 5/8/2026

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002	DESIGNED: Aaron	2024/09/28
	DRAWN: Kashish Ralli	2025/04/24
	CHECKED: Kashish Ralli	2025/04/24
	EXECUTIVE	
FINISH IS: PETG	ACADEMIC YEAR: 2024-2025	
DEBURR & BREAK SHARP EDGES	AIRFRAME	

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: Parachute Bay

QTY	PART NUMBER	DWG NUMBER	REV
1			1

SIZE A | SCALE 1:5 | SHEET 1 OF 1

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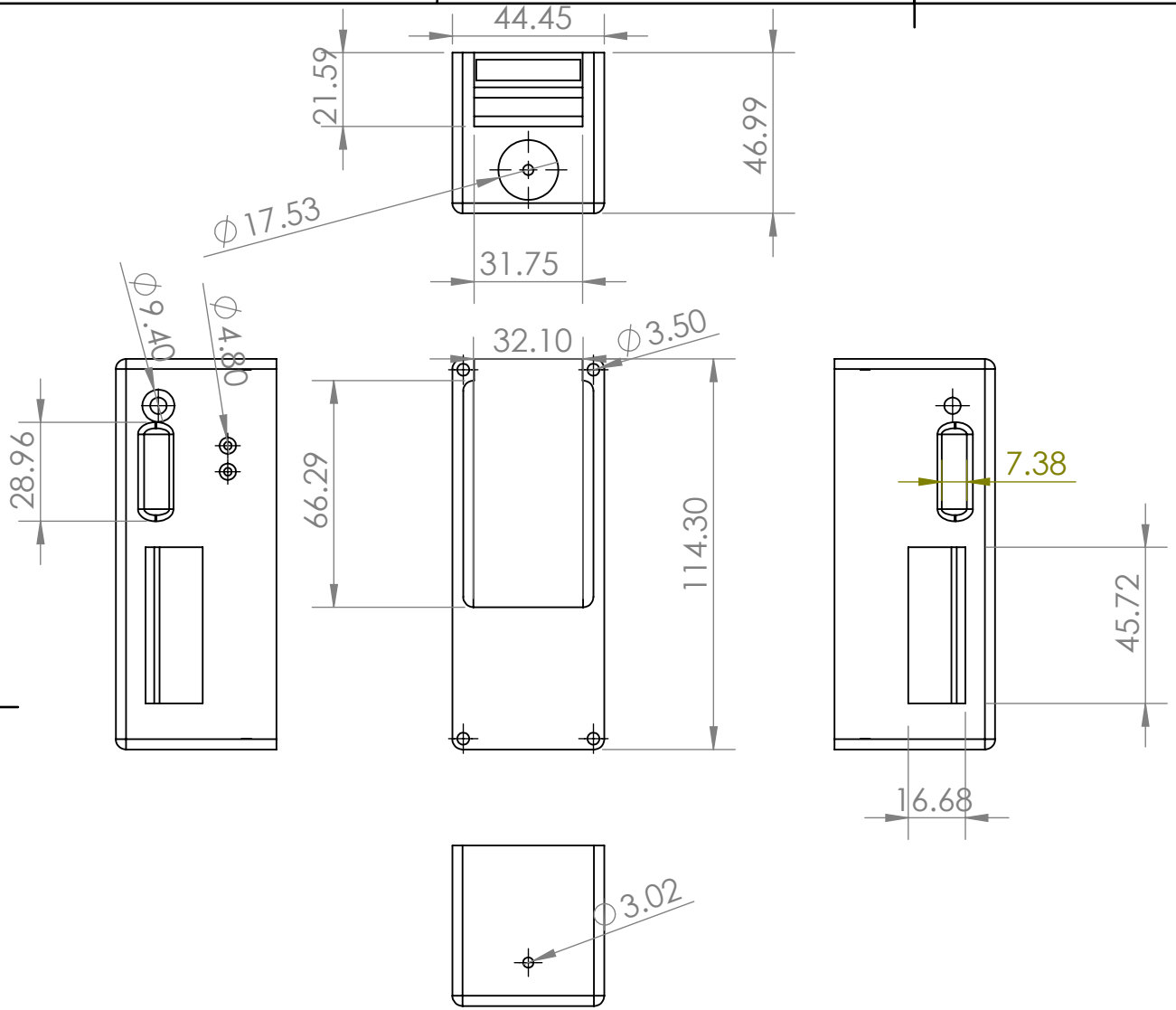

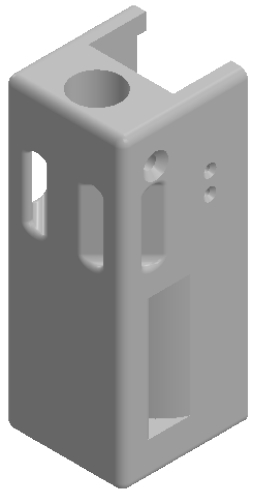

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Competing Team 14 of the 2025 Launch Canada Challenge

Revision: R0
Effective Date: 5/8/2026

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN INCHES [mm] TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002	DESIGNED Jonathan	2025/11/27
	DRAWN Kashish Ralli	2025/06/27
	CHECKED Kashish Ralli	2025/06/27
	EXECUTIVE	
FINISH IS: PLA	ACADEMIC YEAR 2024-2025	
DEBURR & BREAK SHARP EDGES	AIRFRAME	

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: Receiving Box

QTY	PART NUMBER	DWG NUMBER	REV
1			1

SIZE A | SCALE 1:2 | SHEET 1 OF 1

CU INSPACE DRAWING TEMPLATE **V5**
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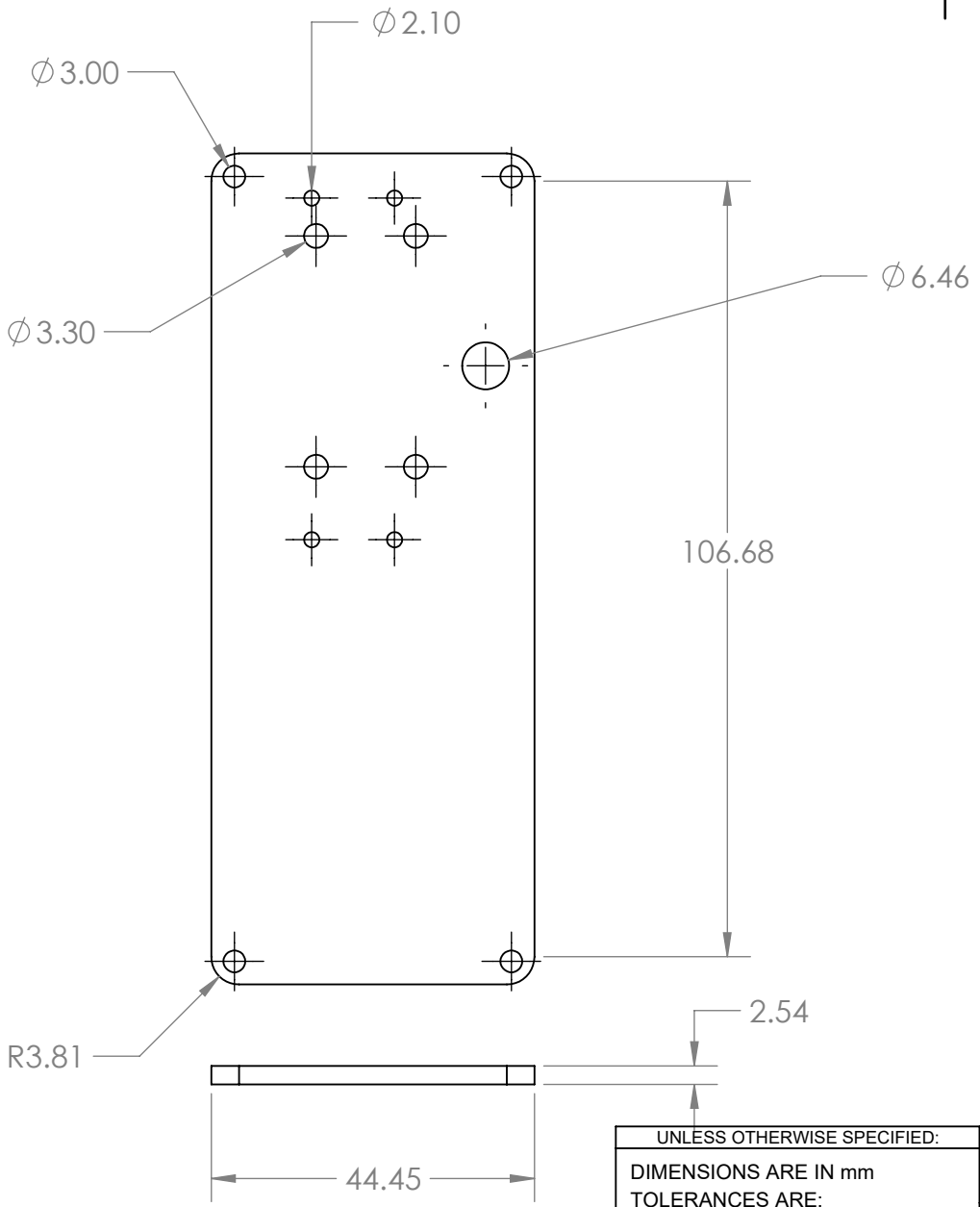

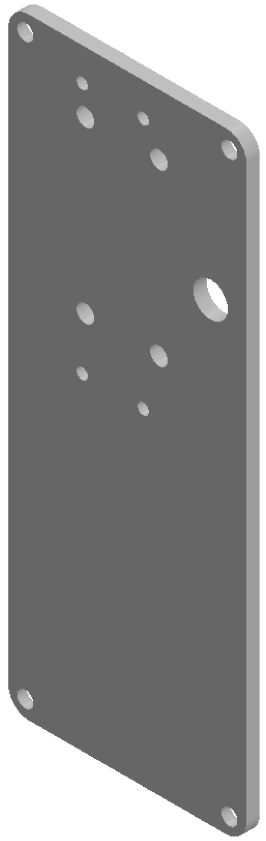

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge

Effective Date: 5/8/2026
Revision: R0

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002 FINISH IS: PLA DEBURR & BREAK SHARP EDGES	DESIGNED Jonathan	2024/11/14
	DRAWN Kashish Ralli	2025/05/17
	CHECKED Kashish Ralli	2025/05/17
	EXECUTIVE	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME	

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE
Receiving Box Lid

QTY	PART NUMBER	DWG NUMBER	REV
1			1

SIZE A | SCALE 1:1 | SHEET 1 OF 1

CU INSPACE DRAWING TEMPLATE **V5**
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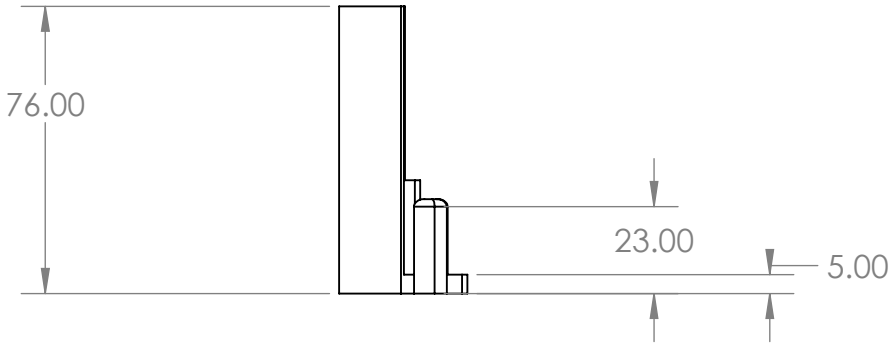

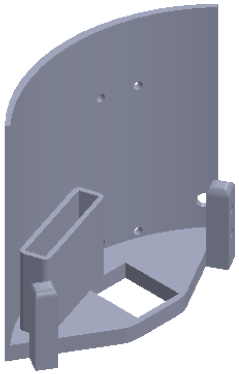

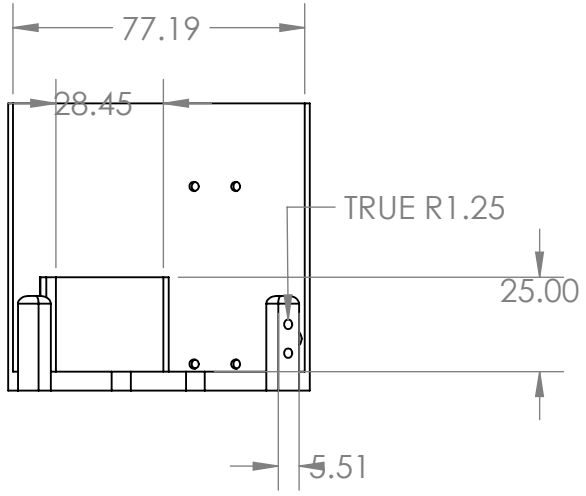

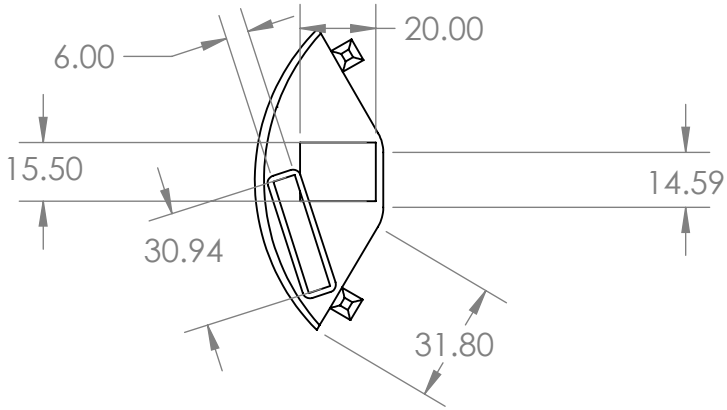

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge

Revision: R0
Effective Date: 5/8/2026

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN INCHES [mm] TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002 FINISH IS: PLA DEBURR & BREAK SHARP EDGES	DESIGNED Anuj	2025/01/13
	DRAWN Kashish Ralli	2025/06/16
	CHECKED Kashish Ralli	2025/06/16
	EXECUTIVE	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME	

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: TGD

QTY	PART NUMBER	DWG NUMBER	REV
1			1

SIZE A | SCALE 1:2 | SHEET 1 OF 1

CU INSPACE DRAWING TEMPLATE **V5**
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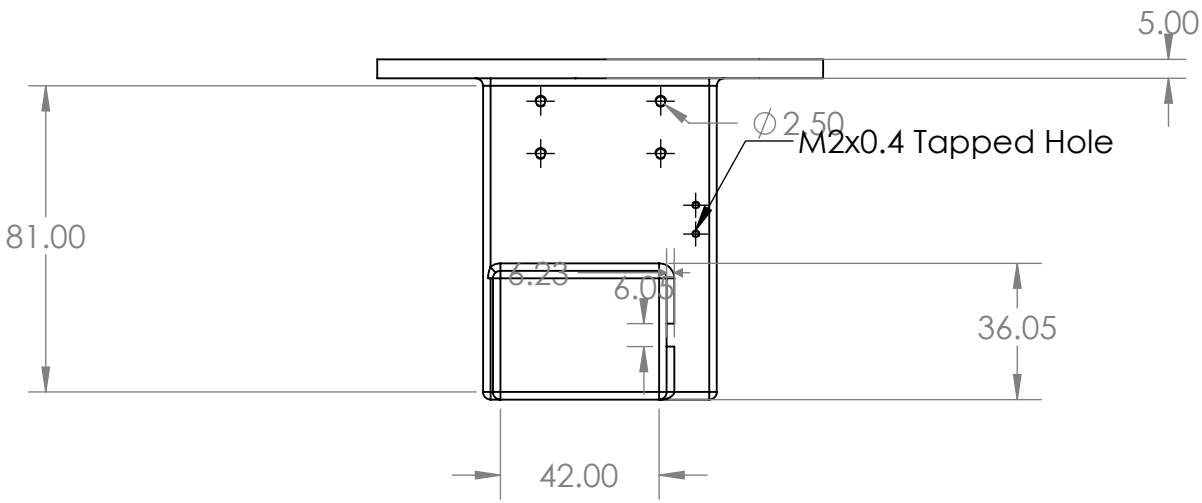

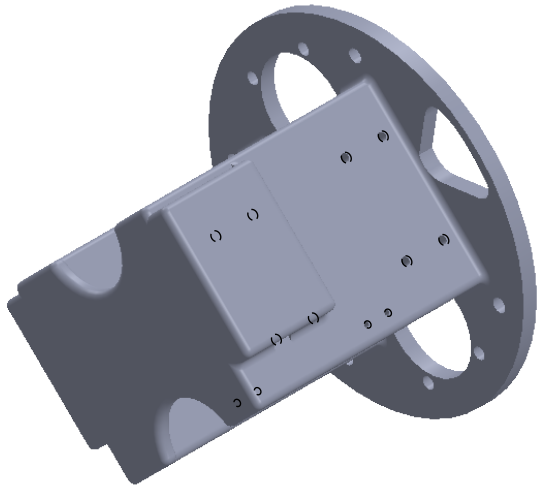

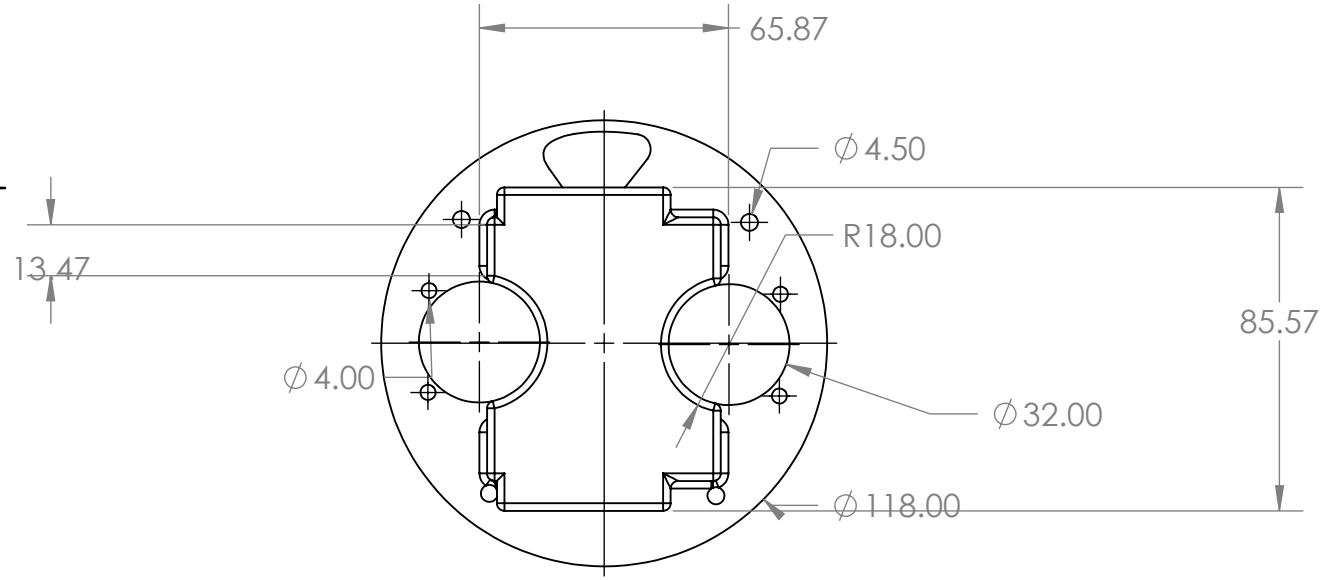

Final Project Report for CU InSpace CR25H
Competing Team 14 of the 2025 Launch Canada Challenge

Effective Date: 5/8/2026
Revision: R0

UNLESS OTHERWISE SPECIFIED:	APPROVAL	DATE
DIMENSIONS ARE IN mm TOLERANCES ARE: ANGLES: ±0.5° X.X: ±0.050 X.XX: ±0.005 X.XXX: ±0.002 FINISH IS: PLA DEBURR & BREAK SHARP EDGES	DESIGNED Brendan	2025/05/12
	DRAWN Kashish Ralli	2025/06/27
	CHECKED Kashish Ralli	2025/06/27
	EXECUTIVE	
	ACADEMIC YEAR 2024-2025	
	AIRFRAME	

CU INSPACE
CARLETON UNIVERISTY
OTTAWA, ON, CANADA K1S 5B6

TITLE: Avionic Stack

QTY	PART NUMBER	DWG NUMBER	REV
1			1

SIZE A | SCALE 1:2 | SHEET 1 OF 1

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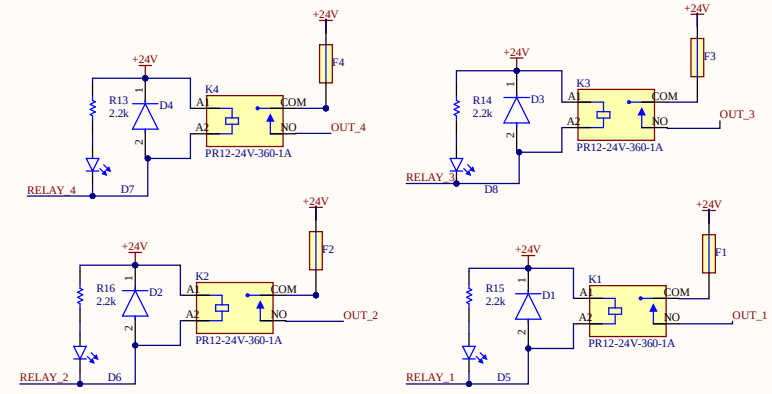
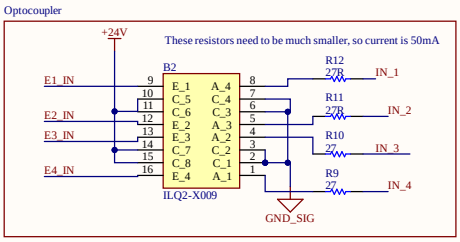
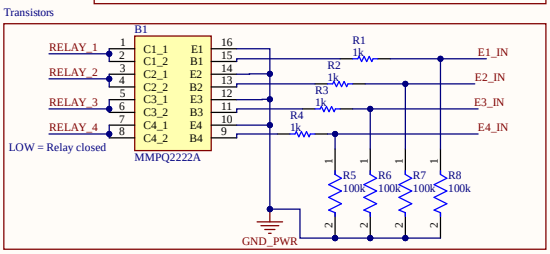
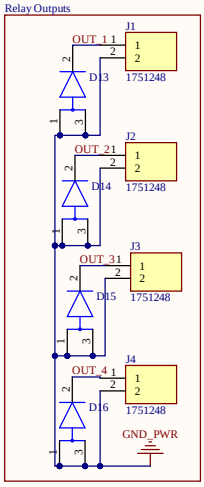
G Electrical Schematics

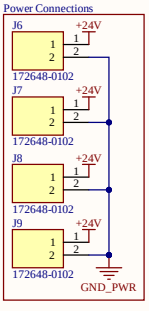
G.1 Hybrid Engine Control

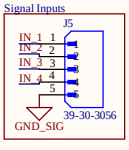












Use JLCPCB JLC-0416H-7628 stackup for impedence control

APPROVAL	DATE
DESIGNED Roland Neill	2024-04-24
CHECKED Matteo Golin	2024-11-15
APPROVED Emma Ricketts	2024-11-23
NOTES	

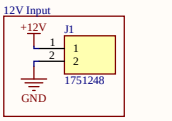
CU INSPACE

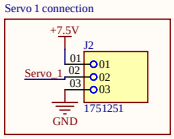
Carleton University
Ottawa, Ontario, Canada K1S 5B6

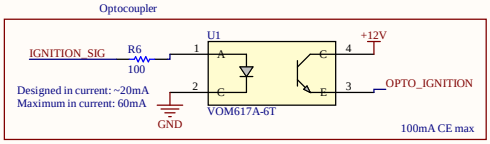
Title:
Test_Stand_Relay.SchDoc

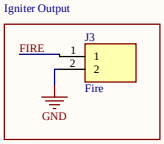
ACADEMIC YEAR	PROJECT	REV
2024-2025	Test_Stand_Relay.PrjPcb	A

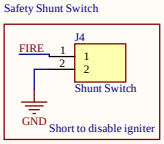
SIZE	MODIFIED	SHEET
A	5/19/2025	1 OF 1

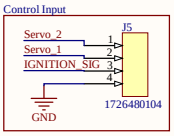


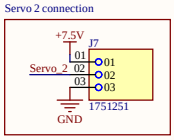


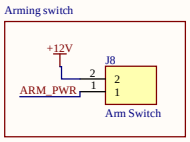


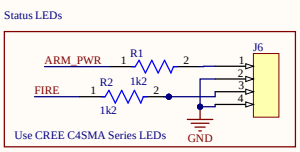


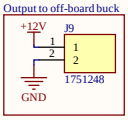


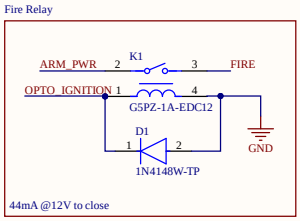


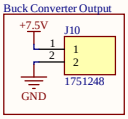


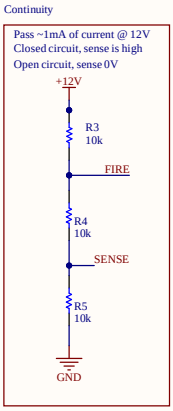


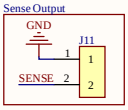












APPROVAL	DATE
DESIGNED Roland Neill	2024-07-20
CHECKED Matteo Golin	2025-03-16
APPROVED Matteo Golin	2025-03-16
NOTES	

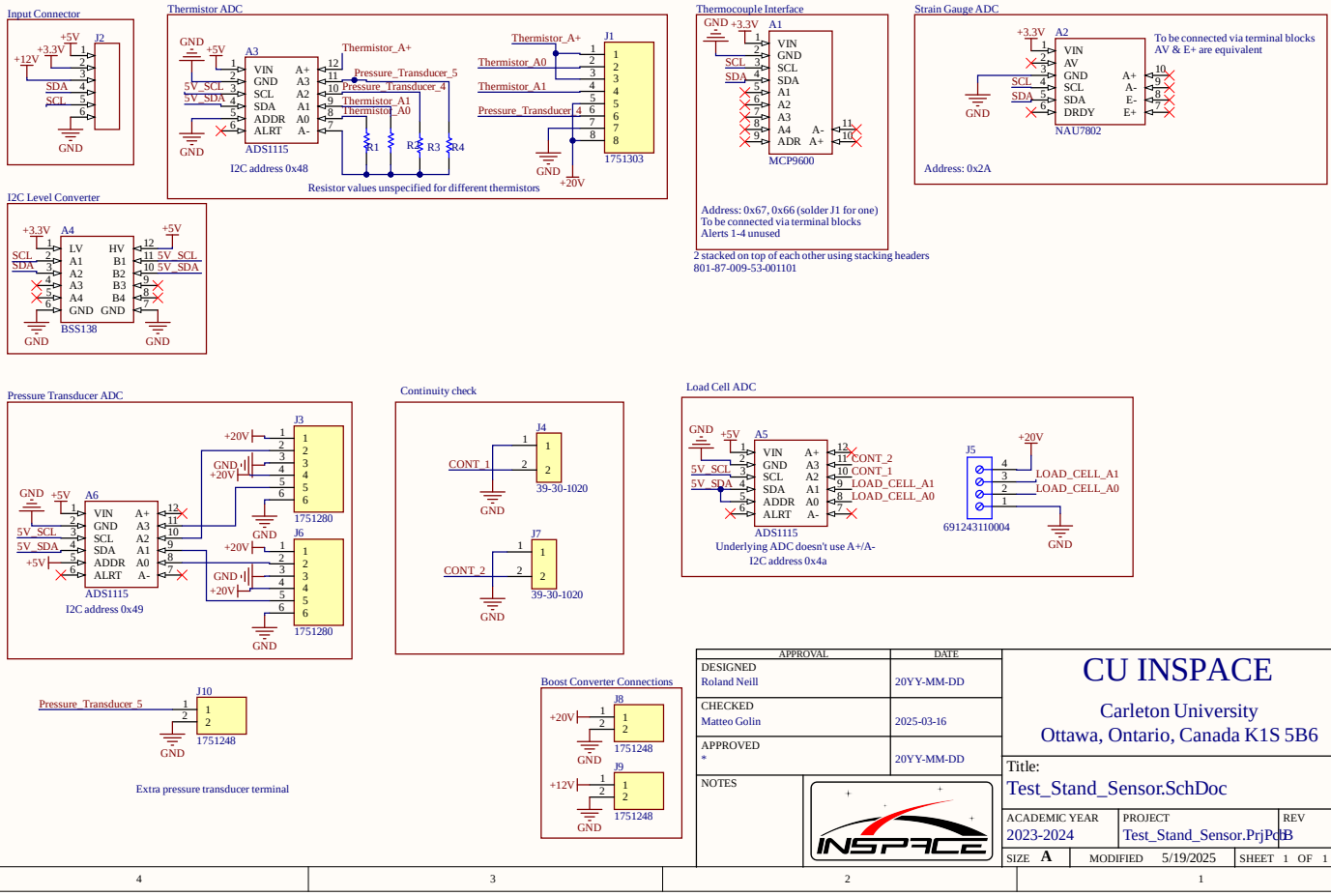
CU INSPACE

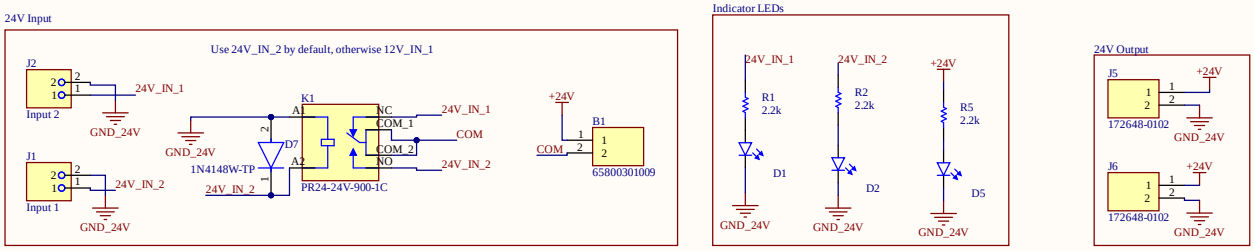
Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
Test_Stand_Fire_Control.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-2025	Test_Stand_Fire_Control.PrjPcb	B

SIZE	MODIFIED	SHEET
A	3/17/2025	1 OF 1


Input Connector
Thermistor ADC
Thermocouple Interface
Strain Gauge ADC
I2C Level Converter
Pressure Transducer ADC
Continuity check
Load Cell ADC
Boost Converter Connections
Extra pressure transducer terminal
I2C address 0x48
Resistor values unspecified for different thermistors
Address: 0x67, 0x66 (solder J1 for one)
To be connected via terminal blocks
Alerts 1-4 unused
2 stacked on top of each other using stacking headers
801-87-009-53-001101
To be connected via terminal blocks
AV & E+ are equivalent
Address: 0x2A
I2C address 0x49
Underlying ADC doesn't use A+/A-
I2C address 0x4a
CU INSPACE
Carleton University
Ottawa, Ontario, Canada K1S 5B6
Title:
Test_Stand_Sensor.SchDoc
DESIGNED
Roland Neill
CHECKED
Matteo Golin
APPROVED
*
DATE
20YY-MM-DD
2025-03-16
20YY-MM-DD
NOTES
ACADEMIC YEAR
2023-2024
PROJECT
Test_Stand_Sensor.PrjPcb
REV
B
SIZE
A
MODIFIED
5/19/2025
SHEET 1 OF 1



Use JLCPCB JLC-0416H-7628 stackup for impedence control

	APPROVAL	DATE
DESIGNED	Roland Neil	2024-07-15
CHECKED	Matteo Golin	2024-12-02
APPROVED	*	20YY-MM-DD
NOTES		

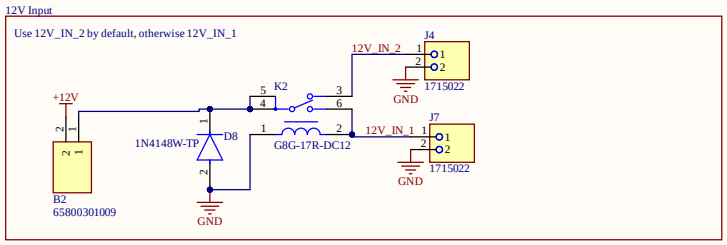
CU INSPACE

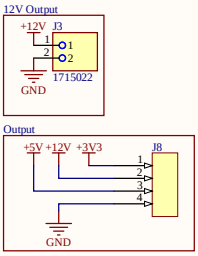
Carleton University
Ottawa, Ontario, Canada K1S 5B6

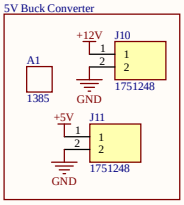
Title:
Test_Stand_Power_24V.SchDoc

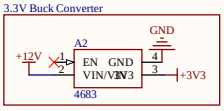
ACADEMIC YEAR	PROJECT	REV
2024-2025	Test_Stand_Power.PrjPcb	B

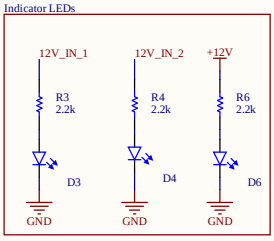
SIZE A | MODIFIED 2/18/2025 | SHEET 1 OF 2











APPROVAL	DATE
DESIGNED Roland Neil	2024-07-15
CHECKED Matteo Golin	2024-12-02
APPROVED NAME	20YY-MM-DD
NOTES	

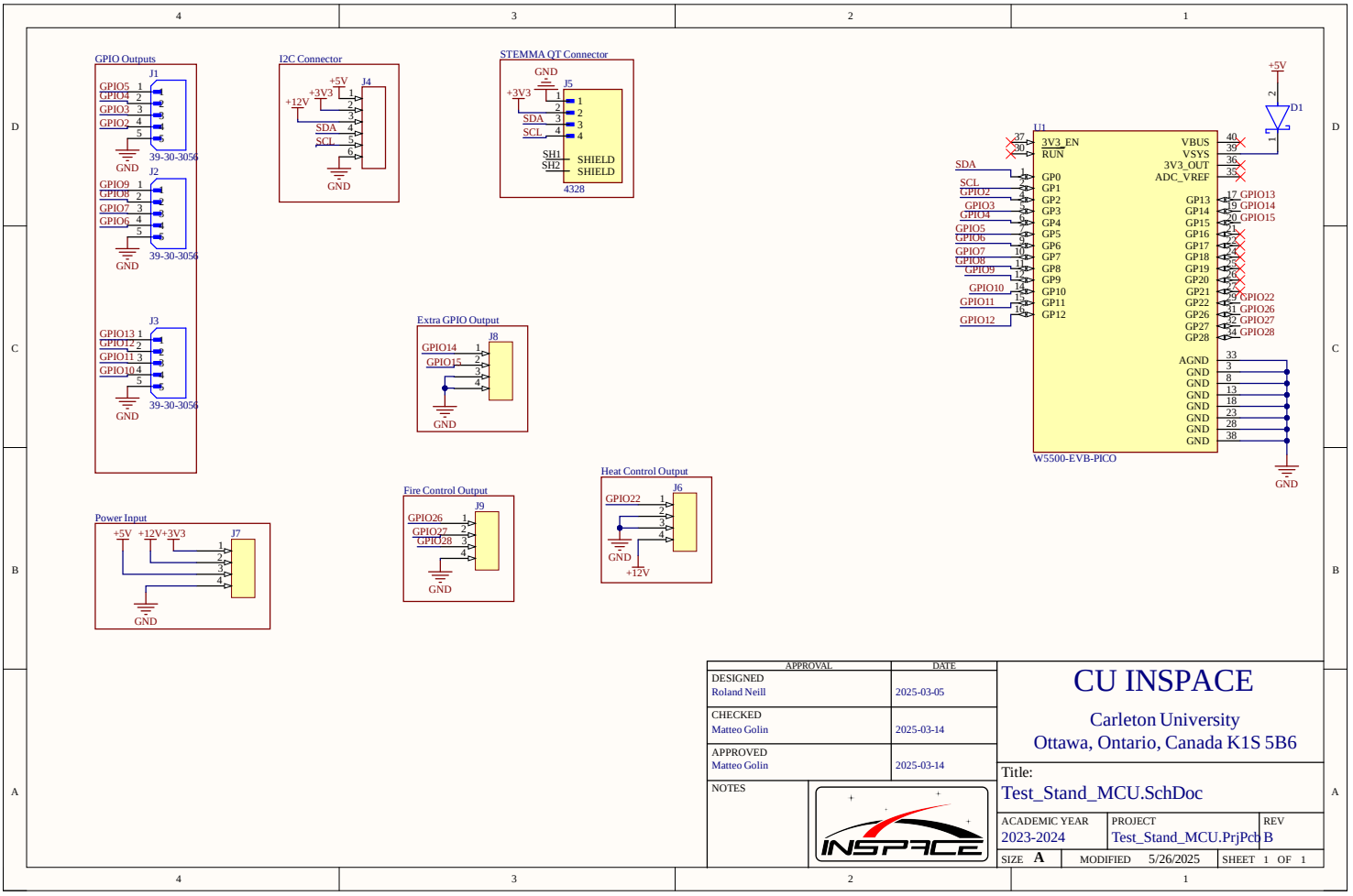
CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
Test_Stand_Power_12V.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-2025	Test_Stand_Power.PrjPcb	A

SIZE **A** | MODIFIED 2/18/2025 | SHEET 2 OF 2



APPROVAL	DATE
DESIGNED Roland Neill	2025-03-05
CHECKED Matteo Golin	2025-03-14
APPROVED Matteo Golin	2025-03-14
NOTES	

CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

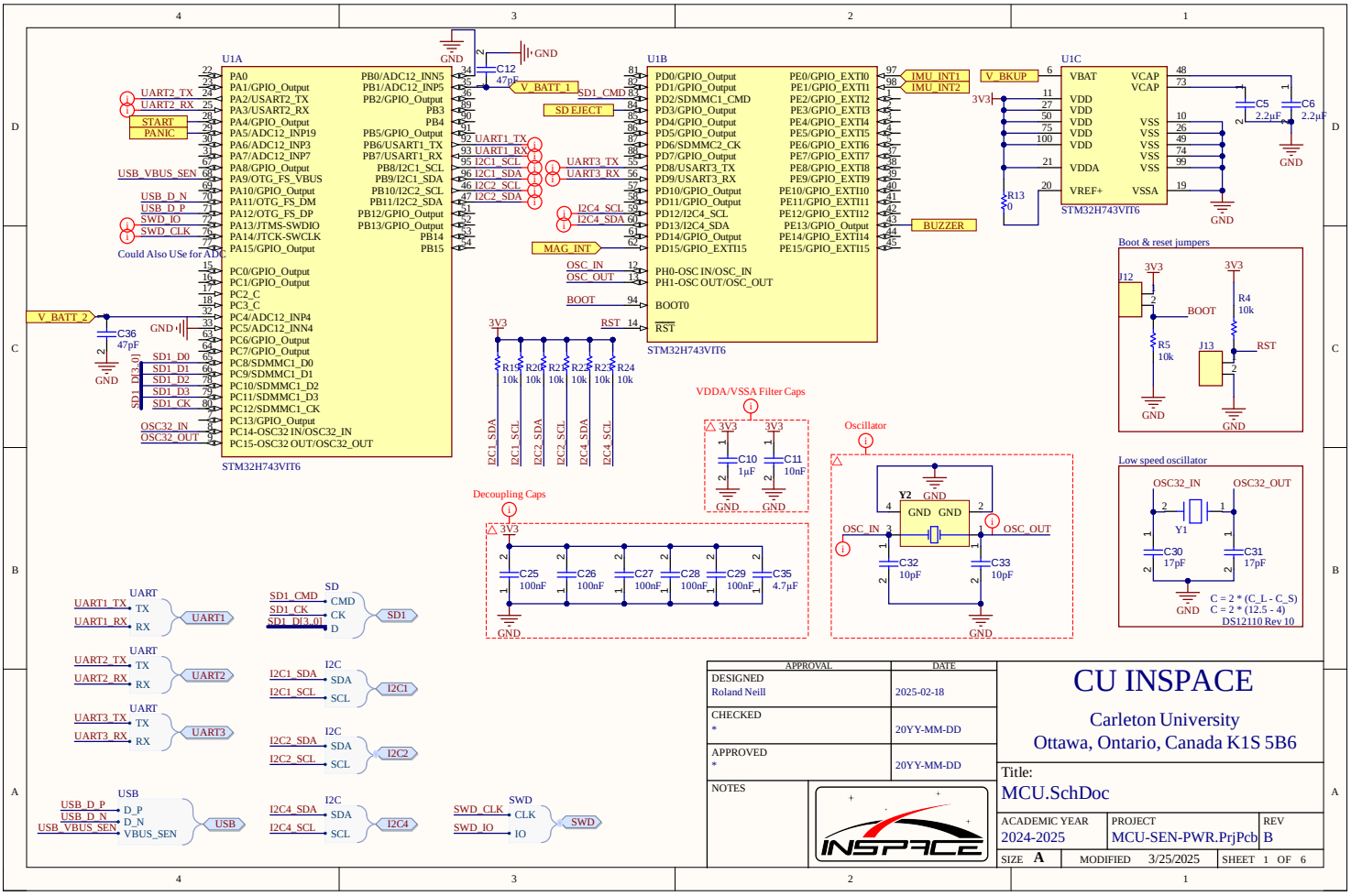
Title:
Test_Stand_MCU.SchDoc

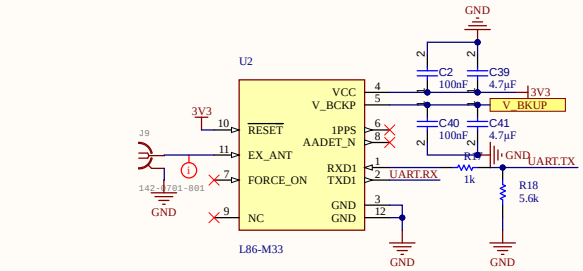
ACADEMIC YEAR	PROJECT	REV
2023-2024	Test_Stand_MCU.PrjPcb	B

SIZE A | MODIFIED 5/26/2025 | SHEET 1 OF 1

The PCB design for the *Valve-O-Matic* can also be found in this GitHub repository. The schematic and board layout can be viewed in the browser using the link provided on the repository.

G.2 Telemetry System


U1A
STM32H743VIT6
U1B
STM32H743VIT6
U1C
STM32H743VIT6
Could Also USe for ADC
Boot & reset jumpers
Low speed oscillator
Decoupling Caps
VDDA/VSSA Filter Caps
Oscillator
C = 2 * (C_L - C_S)
C = 2 * (12.5 - 4)
DS12110 Rev 10
CU INSPACE
Carleton University
Ottawa, Ontario, Canada K1S 5B6
MCU.SchDoc
2024-2025
MCU-SEN-PWR.PrjPcb
B
3/25/2025
1 OF 6



APPROVAL	DATE
DESIGNED Roland Neill	2024-11-15
CHECKED *	20YY-MM-DD
APPROVED *	20YY-MM-DD
NOTES	

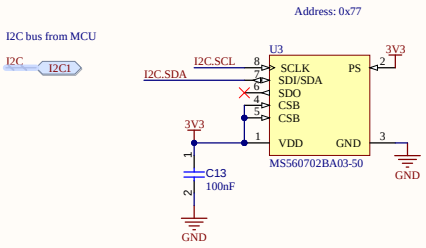
CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
GPS.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-25	MCU-SEN-PWR.PrjPcb	A

SIZE **A** | MODIFIED 3/25/2025 | SHEET 2 OF 6



APPROVAL	DATE
DESIGNED Daniel	2024-11-02
CHECKED Roland Neill	2024-11-04
APPROVED NAME	20YY-MM-DD

NOTES
Datasheet p.14

I2C Address 0x77

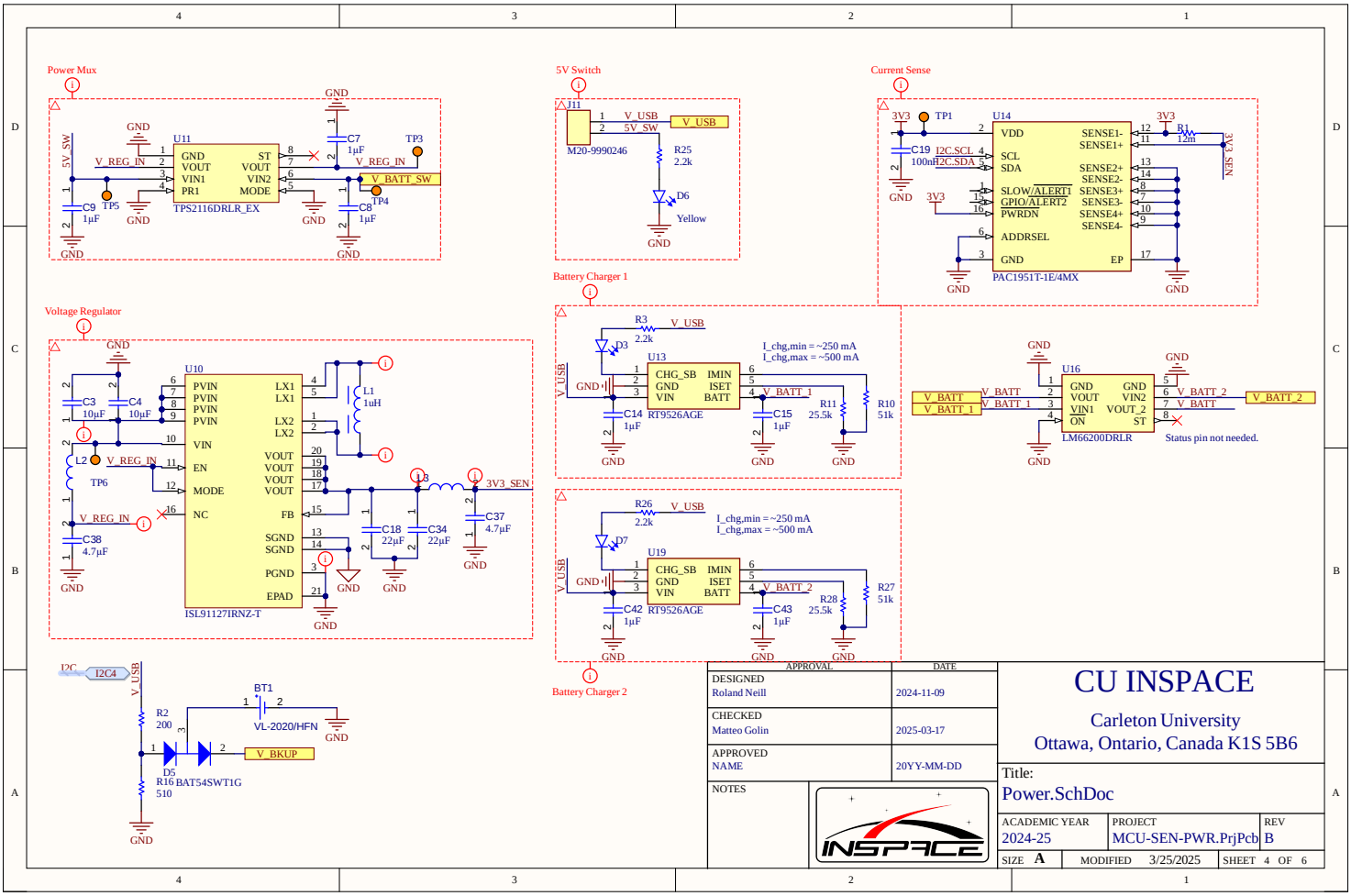
CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
Altimeter.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-25	MCU-SEN-PWR.PrjPcb	A

SIZE **A** | MODIFIED 3/25/2025 | SHEET 3 OF 6



APPROVAL	DATE
DESIGNED Roland Neill	2024-11-09
CHECKED Matteo Golin	2025-03-17
APPROVED NAME	20YY-MM-DD
NOTES	

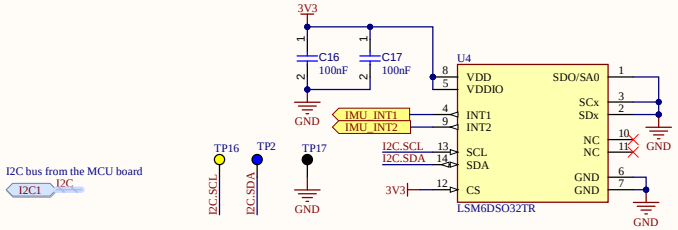
CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
Power.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-25	MCU-SEN-PWR.PrjPcb	B

SIZE A | MODIFIED 3/25/2025 | SHEET 4 OF 6



APPROVAL	DATE
DESIGNED Matteo Golin	2024-11-14
CHECKED Roland Neill	2024-11-15
APPROVED NAME	20YY-MM-DD

NOTES
Datasheet App Mode 1, Sec 7.1

I2C Address: 0x6B

CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

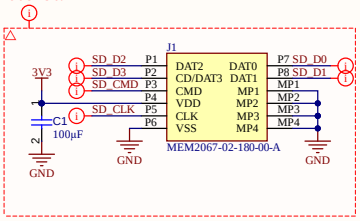
Title:
IMU.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-25	MCU-SEN-PWR.PrjPcb	A

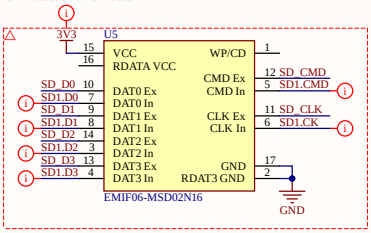
SIZE A | MODIFIED 3/25/2025 | SHEET 5 OF 6

Off-Sheet Connections

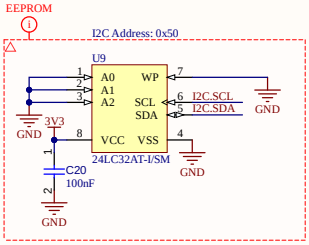
Micro SD Slot



ESD Protection TVS Diodes



EEPROM



APPROVAL	DATE
DESIGNED Roland Neill	2024-10-30
CHECKED NAME	20YY-MM-DD
APPROVED NAME	20YY-MM-DD
NOTES	

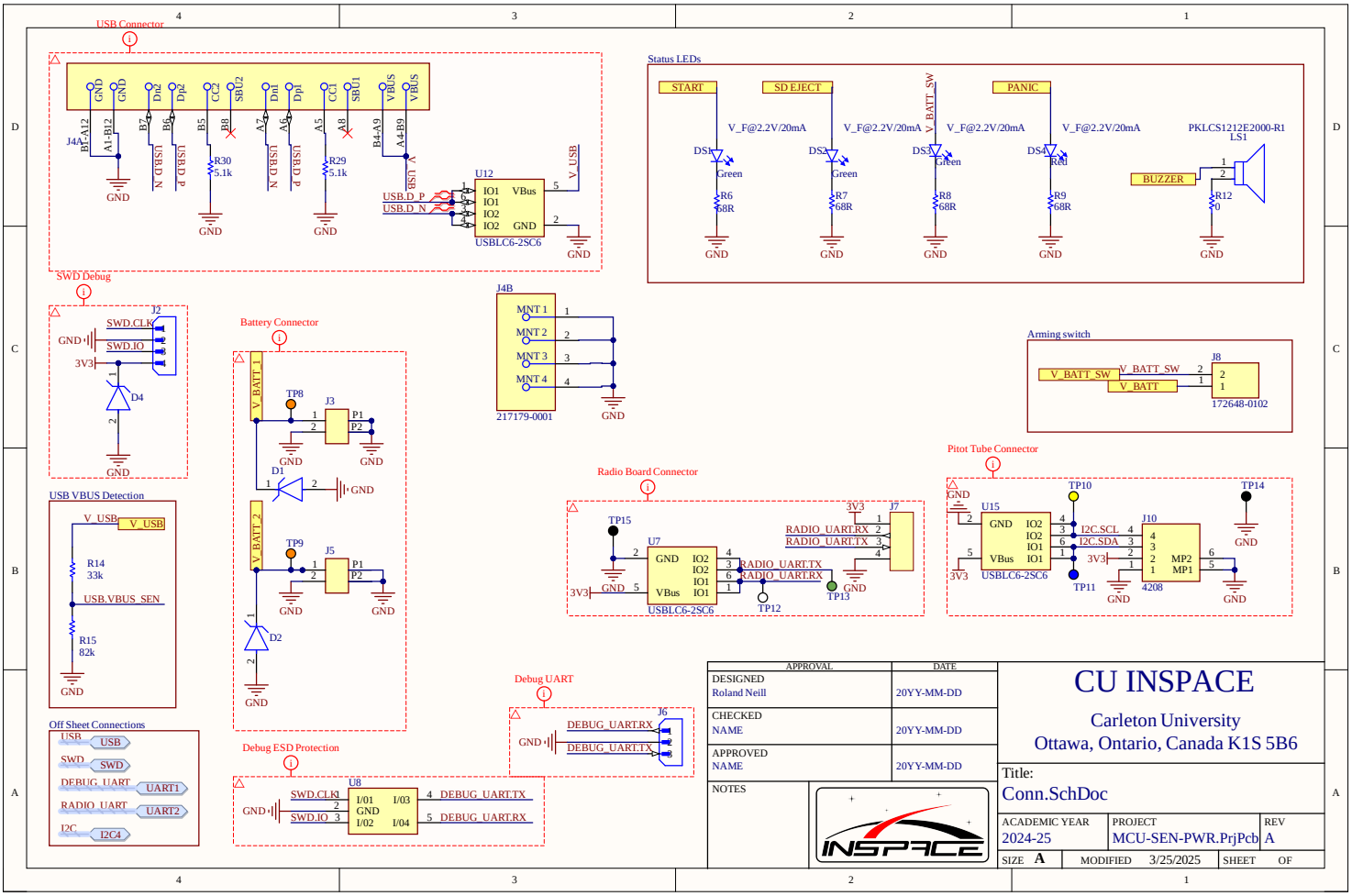
CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
Storage.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-25	MCU-SEN-PWR.PrjPcb	A

SIZE A | MODIFIED 3/25/2025 | SHEET 6 OF 6



APPROVAL	DATE
DESIGNED Roland Neill	20YY-MM-DD
CHECKED NAME	20YY-MM-DD
APPROVED NAME	20YY-MM-DD
NOTES	

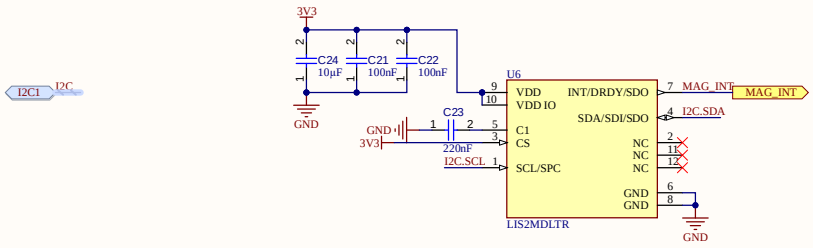
CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
Conn.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-25	MCU-SEN-PWR.PrjPcb	A

SIZE	MODIFIED	SHEET
A	3/25/2025	OF



APPROVAL	DATE
DESIGNED Matteo Golin	2024-11-14
CHECKED Roland Neill	2024-11-15
APPROVED NAME	20YY-MM-DD

NOTES

LIS2MDL Datasheet, Sec 5

I2C Address: 0x1E

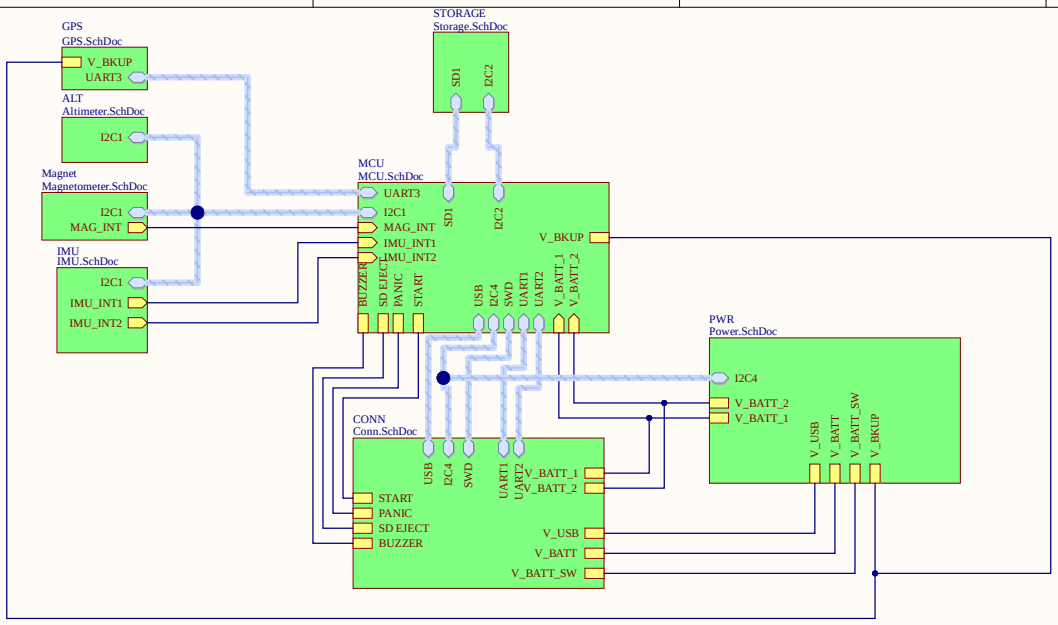
CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
Magnetometer.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-25	MCU-SEN-PWR.PrjPcb	A

SIZE **A** | MODIFIED 3/25/2025 | SHEET | OF



APPROVAL	DATE
DESIGNED NAME	20YY-MM-DD
CHECKED NAME	20YY-MM-DD
APPROVED NAME	20YY-MM-DD
NOTES	

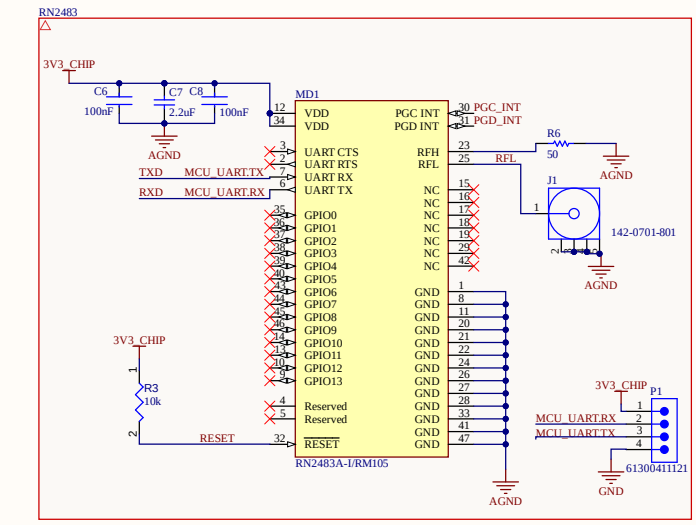
CU INSPACE

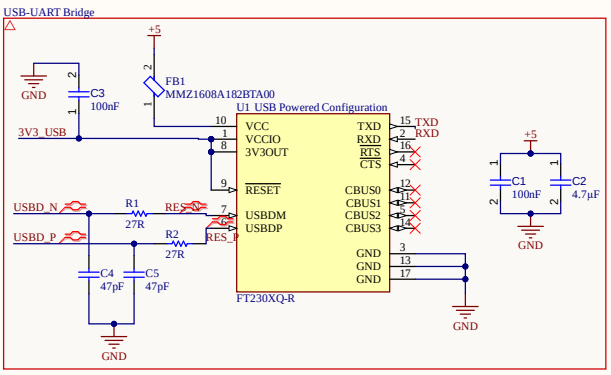
Carleton University
Ottawa, Ontario, Canada K1S 5B6

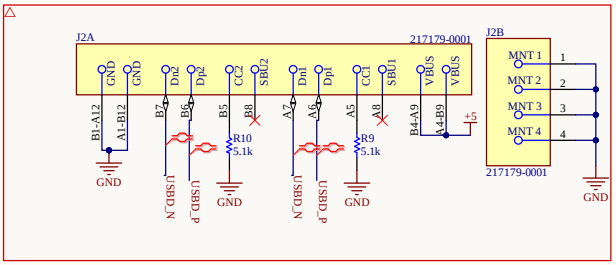
Title:
Overview.SchDoc

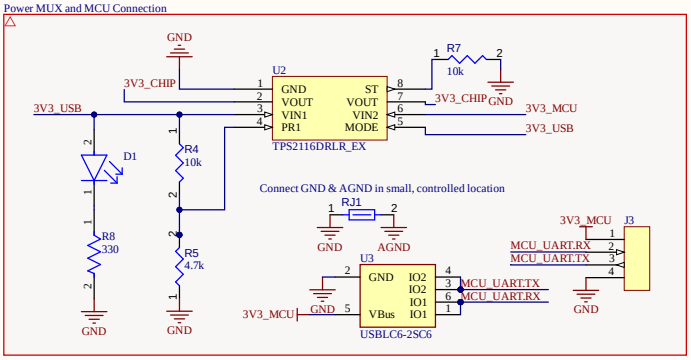
ACADEMIC YEAR	PROJECT	REV
20YY-20YY	MCU-SEN-PWR.PrjPcb	A

SIZE A | MODIFIED 3/25/2025 | SHEET OF









Use JLC04161H-7628 Stackup

APPROVAL	DATE
DESIGNED Amaan Khan	
CHECKED Roland Neil	
APPROVED Matteo Golin	
NOTES	

CU INSPACE

Carleton University
Ottawa, Ontario, Canada K1S 5B6

Title:
RF_Rev0.SchDoc

ACADEMIC YEAR	PROJECT	REV
2024-2025	RF_SRAD.PrjPcb	B
SIZE	MODIFIED 7/2/2025	SHEET 1 OF 1

G.2.1 PCB Design Considerations

The radio board is designed to be a four layer PCBPCB. This allows top and bottom copper signal routing, with separate analog and digital ground in the middle. It is important to separate digital and analog ground not only on two separate layers, but also on different sides of the board, especially for high speed PCB's with RF components. Separate grounds will reduce any potential crosstalk as well as EMI caused by the digital signals interfering with RF signals.

The analog ground plane will be the layer immediately under the top copper, as it provides a reference plane which is beneficial for radio frequency traces. The trace will be a microstrip line, which is a copper trace running above a ground plane, separated by an impedance controlled dielectric.

Our radio frequency traces are impedance controlled to 50 ohms, which is a standard impedance for radio equipment, including the Subminiature A (SMA) connectors and antennas used by CU InSpace.

All radio frequency traces on the board are also kept below the critical length for the 433MHz frequency, which prevents them from acting as an antenna themselves. PCB trace lengths should not exceed 1/12th of the wavelength in the dielectric. **critical-length**

Our manufacturer's dielectric constant for the prepreg layer we use on our impedance controlled boards is 4.4. **jlc-pcb-impedance**. Equation G.2.1 is used to calculate the wavelength of the radio signal in the PCB trace.

$$\lambda = \frac{c}{f\sqrt{\epsilon_R}} \tag{1}$$

Then, given this wavelength, the critical length is calculated using Equation G.2.1.

$$L = \frac{1}{12} \times \lambda \tag{2}$$

The critical length of the trace for our operating frequency of 433.5MHz is approximately 2.75 centimetres. All of our RF traces carrying 433.5MHz signals will therefore be designed to have a total length less than 2.75 centimetres.

$$\lambda = \frac{300 \times 10^6}{(433.5 \times 10^6)\sqrt{4.4}}$$
$$\lambda = 0.32991785095003$$

$$L = \frac{1}{12} \times (0.32991785095003)$$
$$L = 0.027493154245836$$
$$L \approx 0.0275$$

H Software Repositories

The CU InSpace fork of Apache NuttX can be found here. It is where sensor drivers are staged during development and testing before being contributed back to the upstream repository.

The board support firmware for NuttX on the *Josh* flight computer can be found here. This firmware allows NuttX to run on the flight computer and is responsible for bringing up the kernel and device drivers.

The repository for application level logic running on the *Josh* flight computer can be found here. This repository contains both the main telemetry application (under development) as well as other applications such as sensor mocking for system testing.

The packet specification for messages sent over telemetry from the rocket flight computer to the ground station can be found here.

The code for the ground station/launch console telemetry visualization dashboard can be found here.

The code for the ground station/launch console telemetry reception and parsing can be found here. This is the code that directly interfaces with the SRAD radio receiver.

The repository organizing the ground station/launch console telemetry reception, parsing and visualization code in addition to a basic HTTP server that serves the map tile images for the real-time flight map into a single docker application can be found here.

The repository which contains the application logic for the control client, a sample telemetry client and the pad server of the hybrid control system can be found here. This code compiles both for POSIX desktop operating systems and Apache NuttX.

The repository containing the code for the hybrid control system user interface can be found here.

The repository containing the code for the hybrid engine *Valve-O-Matic* can be found here.

The specification for the communication protocol between network nodes on the hybrid control system can be found here. It provides detailed diagrams of the packet format and a specification for each of the different network nodes and their contractual behaviour.

The preliminary design report from the fall semester on the 2024-2025 avionics systems can be found here. It provides some more detail on the avionic system design, but some sections are currently outdated.

I Calculations and Design Trade-offs

I.1 Propellant Tank Calculations

The methodology for the design and analysis of the oxidizer tank was taken from Half-cat's "How to Design Pressure Vessels, Propellant Tanks, and Rocket Motor Casings" [1]. The mechanical properties of aluminum 6061-T6 alloy were obtained from Reference [31]. The 1/4-28 bolts' mechanical properties were obtained from the vendors website [32]. The M6 bolts' mechanical properties were obtained from [33] based on the grade marking on the bolt head. Nominal bending stress of the bulkhead is the max theoretical bending stress for a bulkhead without presence of a port [34]. With

the exception of the bulkhead and casing pressure stresses the same procedure as the Half-cat guide was followed [1]. The tables below give the inputs used and the results with some intermediate steps.

Table 21: Casing Geometry Inputs

Parameter	**Value**	**Unit**
MEOP	1000	psi
Interface Outer Diameter	6	in
Interface Wall Thickness	0.25	in
Interface Inner Diameter	5.5	in
Center-Edge Distance	0.75	in
Nominal Outer Diameter	5.875	in
Nominal Wall Thickness	0.1875	in
Nominal Inner Diameter	5.5	in

Table 22: Bulkhead Geometry Inputs

Parameter	**Value**	**Unit**
Shoulder Depth	0.094	in
Tapped Length	0.25	in
Force on Bulkhead	23758	lb
Spherical Outer Diameter	5.62	in
Spherical Thickness	0.16	in
Spherical Inner Diameter	5.31	in

The shoulder depth is the distance the bolt heads are sunken into the bulkheads. The force of the bulkheads includes 6kN or 1350lb for recovery loads.

Table 23: Bolts Inputs

Parameter	**Value**	**Unit**
Number of Bolts	24	Int
Thread	1/4-28	N/A
Minor Diameter	0.21	in
Nominal Thread Dia.	0.25	in
Pitch Diameter	0.23	in
Thread Pitch	28	TPI
Head Diameter	0.375	in
Head Height	0.25	in
Tensile Strength	180000	psi
Shear Strength	90000	psi
Socket Equivalent Diameter	0.2135	in

The socket equivalent diameter is the diameter of the circle that circumscribes the hex of the bolt.

Table 24: 6061-T6 Mechanical Properties Inputs

Parameter	**Value**	**Units**
Tensile Strength(Yield)	40000	psi
Tensile Strength (Ultimate)	45000	psi
Bearing Strength (Yield)	54000	psi
Bearing Strength (Ultimate)	64000	psi
Shear Strength	30000	psi

Table 25: Bulkhead Failure Modes

Bulkhead Bearing Failiure		
Parameter	**Value**	**Units**
Bearing area	1.1	$\hat{\text{in2}}$
Bearing stress	22821	psi
Safety Factor Yield	2.3	N/A
Safety Factor Ultimate	2.7	N/A

Bulkhead Spherical Pressure Stress		
Parameter	**Value**	**Units**
K	1.06	N/A
Hoop Stress	17504.69	psi
Radial Stress	-1000.00	psi
Von Misses Stress	18504.69	psi
Safety Factor Yield	2.1	N/A
Safety Factor Burst	2.4	N/A

The bulkhead bearing area is calculated by taking the area of the shoulders and a quarter of the area from the threaded parts of the bolts.

Table 26: Casing Failure Modes

Casing Interface Pressure Stress		
Parameter	**Value**	**Units**
K	1.067	N/A
Hoop Stress	11521	psi
Axial Stress	5260	psi
Radial Stress	-1000	psi
Von Misses Stress	10844	psi
Safety Factor Yield	3.4	N/A
Safety Factor Burst	3.9	N/A

Casing Bearing Failure		
Parameter	**Value**	**Units**
Bearing area	1.39	in$\hat{2}$
Bearing stress	17065	psi
Safety Factor Yield	3.1	N/A
Safety Factor Burst	3.7	N/A

Casing Tensile Failure		
Parameter	**Value**	**Units**
Minimum Tensile Area	2.27	in$\hat{2}$
Maximum Tensile Stress	10485	psi
Safety Factor Yield	3.8	N/A
Safety Factor Burst	4.2	N/A

Casing Bolt Tear-out		
Parameter	**Value**	**Units**
Shear Area	0.28	in$\hat{2}$
Tear-out Force	990	lb
Tear-out Stress	3519	psi
Safety Factor	8.5	N/A

Casing Nominal Pressure Stress		
Parameter	**Value**	**Units**
K	1.07	N/A
Hoop Stress	15183.15	psi
Axial Stress	7091.58	psi
Radial Stress	-1000.00	psi
Von Misses Stress	14015.02	psi
Safety Factor Yield	2.63	N/A
Safety Factor Burst	2.96	N/A

Casing Tensile Stress Concentrations		
Parameter	**Value**	**Units**
Center-Center Distance	1.58	in
Concentration Factor	1.58	N/A
Peak Stress	17537	psi
Safety Factor Yield	2.2	N/A
Safety Factor Burst	2.5	N/A

The bearing area for the casing is calculated from the diameter of the bolt heads and the height of the bold heads minus the shoulder depth. The stress concentration factor was obtained from Peterson's Stress Concentration Factors [35].

Table 27: Bolts Failure Modes

Bolt Shear		
Parameter	**Value**	**Units**
Force on Each Bolt	990	lb
Bolt Shear Area	0.03	$\hat{\text{in}2}$
Bolt Shear Stress	29644	psi
Safety Factor	3.0	N/A

Bolt Head Shear		
Parameter	**Value**	**Units**
Head Shear Area	0.07	$\hat{\text{in}2}$
Head Shear Stress	13261	psi
Safety Factor	6.79	N/A

Bolt Head Hoop		
Parameter	**Value**	**Units**
K	1.76	N/A
Hoop Stress	33434	psi
Axial Stress	8185	psi
Radial Stress	17065	psi
Von Misses Stress	22185	psi
Tensile Safety Factor	5.3	N/A

The shear area of the bolts is determined from the minor diameter of the bolts. For the bolt heads the shear area is the area of disk bounced by the diameter of the bolt head and the socket equivalent diameter. The bolt head hoop calculation treats the bolt as a thick walled cylinder under pressure equal to the bearing stress to determine the hoop stress.

I.2 Combustion Chamber Structural Calculations

I.2.1 Safety Factors

The tables below show the factors of safety for the thrust chamber's mechanical design, relative to the MEOP. Explanations for each of the calculations are included in the next section.

Table 28: Chamber Casing Factors of Safety

Failure Mode	**Yield**	**Ultimate**
Hoop Stress Failure	4.92	5.54
Longitudinal Stress Failure	9.85	11.08

Table 29: Injector Bolt Circle Factors of Safety

Failure Mode	**Yield**	**Ultimate**
Bolt Shear	N/A	7.81
Casing Bearing Failure	2.59	3.07
Bulkhead Bearing Failure	3.24	3.85
Bolt Tear-Out	N/A	3.24
Casing Tensile Failure	7.74	8.70

Table 30: Nozzle Bolt Circle Factors of Safety

Failure Mode	**Yield**	**Ultimate**
Bolt Shear	N/A	5.86
Casing Bearing Failure	1.95	2.31
Retainer Bearing Failure	2.44	2.89
Bolt Tear-Out	N/A	2.43
Casing Tensile Failure	5.62	6.32

I.2.2 Calculation Inputs

The material properties for aluminum 6061-T6 were obtained from the Metallic Materials Properties Development and Standardization (MMPDS) as well as Matweb [36], [31]. The mechanical properties of the fasteners used in the chamber are from the vendor's website, as well as online thread charts [37], [38].

Table 31: Aluminum 6061-T6 Material Properties

Parameter	Value	Unit
Tensile Strength (Yield)	40,000	psi
Tensile Strength (Ultimate)	45,000	psi
Bearing Strength (Yield)	54,000	psi
Bearing Strength (Ultimate)	64,000	psi
Shear Strength (Ultimate)	30,000	psi

The bearing strength values for the chamber were taken assuming a pin edge distance to diameter ratio of 1.5, which is more conservative and appropriate for the chamber's bolted joint geometry.

Table 32: M4 Fastener Mechanical Properties

Parameter	Value	Unit
Tensile Strength (Ultimate)	170,000	psi
Minor Diameter	3.002	mm
Major Diameter	3.838	mm
Head Diameter	7	mm
Head Height	4	mm
Hex Drive Size	3	mm

The major and minor diameters of the fasteners are taken as their minimum possible values, to be conservative.

Table 33: Chamber Casing Geometry

Parameter	Value	Unit
Inner Diameter	4.5	in
Wall Thickness	1/8	in

Table 34: Injector Bolt Circle Geometry

Parameter	Value	Unit
Number of Bolts	16	N/A
Bolt Head Protrusion	2.75	mm
Casing Clearance Hole Diameter	7.1	mm
Thread Engagement	8	mm

Note that bolt head protrusion refers to the distance the bolt head extends past the bulkhead and into the casing. It is the depth of the counter bores in the bulkhead subtracted from the height of the bolt head.

I.2.3 Recessed Bolt Head Calculations

The design and analysis of the thrust chamber structure closely followed the methods outlined in Half Cat Rocketry's "How to Design Pressure Vessels, Propellant Tanks, and Rocket Motor Casings" [1]. However, the injector bolt circle with the recessed bolt heads adds some nuance to

Table 35: Nozzle Bolt Circle Geometry

Parameter	**Value**	**Unit**
Number of Bolts	12	N/A
Bolt Head Protrusion	2.75	N/A
Casing Clearance Hole Diameter	7.1	mm
Thread Engagement	8	mm

the calculations. These nuanced calculations are explained below, all other calculations were done in accordance with Half Cat Rocketry's guide.

I.2.4 Bolt Shear

In the typical case of determining the shear stress in the radial bolts, the following formula is used:

$$\sigma_{boltshear} = (\tfrac{\pi}{4} D_{i,casing}^2 \times MEOP)/(N \times \tfrac{\pi}{4} d_{bolt,minor}^2)$$

Where $D_{i,casing}$ is the casing inner diameter, $d_{bolt,minor}$ is the fastener minor diameter, and N is the number of bolts used in the bolt circle. However, in the case of the bolt head being in shear, the cross sectional area of the bolt head must be considered. As a result, the formula becomes

$$\sigma_{boltshear} = (\tfrac{\pi}{4} D_{i,casing}^2 \times MEOP)/(N \times A_{bolt,shear})$$

where

$$A_{bolt,shear} = \tfrac{\pi}{4} d_{bolt,head}^2 - \tfrac{\sqrt{3}}{2} W_{drive}^2$$

Note that $d_{bolt,head}$ is the fastener's head diameter and W_{drive} is the width of the fasteners hex drive size.

I.2.5 Casing Bearing Stress and Bolt Tear-Out

For both the casing bearing stress and bolt tear-out calculations, the modifications to the formulas are very simple. The thickness of the casing is simply replaced with the bolt head protrusion, or the height of the bolt head that is within the casing wall. Note that when the bolt heads are recessed, the tear-out failure mode becomes less likely, as an extra shear plane is added in line with the top of the bolt head. This however is neglected in the calculations to be more conservative.

I.2.6 Bulkhead Bearing Stress

The bearing stress on the injector bulkhead and nozzle retainer was calculated to verify the geometry of the bulkhead's counter bored bolt holes. Calculating bearing stress is not outlined in the Half Cat Rocketry design guide as in most cases, bearing stress generated in the bulkhead/retainer from the bolts will be far less than that developed in the casing. Despite this, the use case of recessed bolts is not clearly addressed in the design guide, so the calculation was done to confirm the bulkhead bearing stress would not be an issue.

The calculation is done using the following formula:

$$\sigma_{bearing,bulkhead} = (\tfrac{\pi}{4} D_{i,casing}^2 \times MEOP)/(N \times A_{bolt,bearing})$$

where

$$A_{bolt,bearing} = L_{counterbore} \times d_{bolt,head} + \frac{1}{2} L_{thread} \times d_{bolt,major}$$

$L_{counterbore}$ is the depth of the counter bore in the bulkhead, L_{thread} is the engaged thread length in the bulkhead, and $d_{bolt,major}$ is the major diameter of the bolt threads. The above calculation assumes the bearing area of the bulkhead is combined between the head and shank of the bolt. However, only half of the bolt's shank is considered to yield a more conservative estimate.

I.3 Bolt Shear Calculations

The shear calculations done for the 4 radial fasteners securing the nose cone support ring and upper mount with the nose cone and coupler are shown here. Even though these four radial bolts will only endure whip loading during recovery events, the inertial load from the payload and avionics stack upon launch will likely exceed the whip load and provide a conservative analysis. Assumptions for this analysis include that no epoxy is assisting in taking the shear load from the inertia of the components housed within the forward section (payload and avionics). 60 per cent of the tensile strength of the fasteners was used as the ultimate tensile strength of the fasteners. The predicted shear loads on the bolts was determined by summing the masses of the components housed within the section and multiplying it by the maximum acceleration of the rocket. Through the calculation a factor of safety for the nose cone bolts in shear was determined to be 32.1.

Bolt Shear (Nosecone):

$$F_{total} = (m_{avionics} + m_{payload} + m_{support\ ring} + m_{hardpoint}) a_{rocket}$$

$$F_{total} = (4.5111)(229.6) = 1035.74856\ N \rightarrow 1.1\ kN$$

$$Shear\ Force = S = 275 \frac{N}{bolt}$$

$$\tau_{max,bolt} = 0.6 \times \sigma_{bolt}$$

$$\sigma_{bolt} = 1.172\ GPa$$

$$\tau_{max,bolt} = 0.6 \times 1.172\ GPa = 0.7032\ GPa = 703.2\ MPa$$

$$A_{bolt} = \frac{\pi}{4} D^2 = \frac{\pi}{4} (4\ mm)^2 = 0.0000126\ m^2$$

$$Shear\ Stress\ per\ Bolt = \frac{S}{A_{bolt}} = \frac{275 \frac{N}{bolt}}{0.0000126\ m^2} = \mathbf{21.88 \frac{MPa}{bolt}}$$

$$FoS = \frac{\tau_{max,bolt}}{Shear\ Stress\ per\ Bolt} = \frac{703.2}{21.88} = \mathbf{32.1}$$

Figure 171: Avionics Bolt Shear Calculations

The shear calculations done for the 4 radial fasteners securing the recovery support ring. These four bolts will endure significant shock loading as they support the aft section of the vehicle during the recovery event. Assumptions for this analysis include that no epoxy is assisting in taking the shear load from the deployment event. 60% of the tensile strength of the fasteners was used as

Bolt Shear (Recovery):

$$F_{total,recovery} = 4\ kN$$

$$Shear\ Force = S = 1000\frac{N}{bolt}$$

$$\tau_{max,bolt} = 0.6 \times \sigma_{bolt}$$

$$\sigma_{bolt} = 1.172\ GPa$$

$$\tau_{max,bolt} = 0.6 \times 1.172\ GPa = 703.2\ MPa$$

$$A_{bolt,head} = \frac{\pi}{4}D^2 = \frac{\pi}{4}(7\ mm)^2 = 0.0000385\ m^2$$

$$Shear\ Stress\ per\ Bolt = \frac{S}{A_{bolt,head}} = \frac{1000\frac{N}{bolt}}{0.0000385\ m^2} = \mathbf{79.58}\frac{\boldsymbol{MPa}}{\boldsymbol{bolt}}$$

$$FoS = \frac{\tau_{max,bolt}}{Shear\ Stress\ per\ Bolt} = \frac{703.2}{79.58} = \mathbf{8.84}$$

Figure 172: Recovery Bolt Shear Calculations

the ultimate tensile strength of the fasteners. The shear loads on the bolts was determined by the predicted recovery shock force on the rocket during parachute deployment. Through the calculation a factor of safety for the bolts in shear was determined to be 8.8.

J Test Stand Calculations

J.1 Ground Anchoring

J.1.1 Ground Anchors

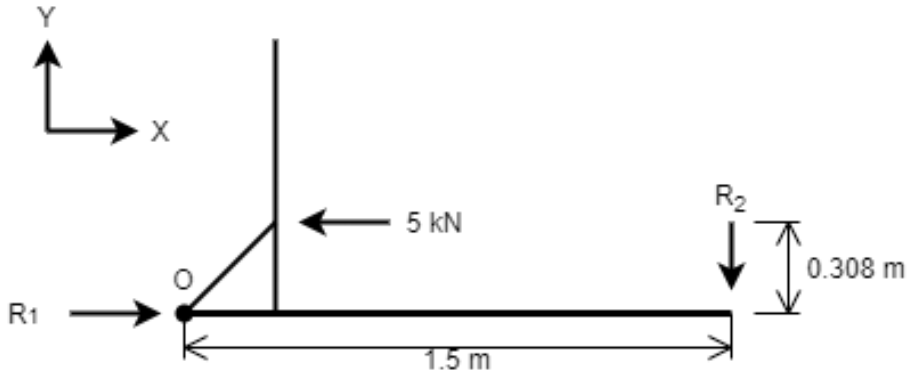


Figure 173: Test-stand Free Body Diagram

By inspection:

$$R_1 = 5 \text{ kN}$$

Assuming that the reaction at R_1 is spread evenly between the 6 anchors at the front of the test-stand. This gives an applied force of 0.833 kN per ground anchor. Since the lateral load capacity of each anchor is 4.89 kN this gives a safety factor of 5.87.

Sum of the moments about point O:

$$0 = 5 \text{ kN} \times 0.308 \text{ m} - R_2 \times 1.5 \text{ m} \rightarrow R_2 = \frac{5 \text{ kN} \times 0.308 \text{ m}}{1.5 \text{ m}} = 1.027 \text{ kN}$$

Assuming that the reaction at R_2 is spread evenly between the two ground anchors at the rear of the test-stand. This gives an applied force of 514 N per ground anchor. Since the vertical load capacity of each anchor is 4.22 kN this gives a safety factor of 8.21.

J.1.2 Forward Anchor Brackets Bearing

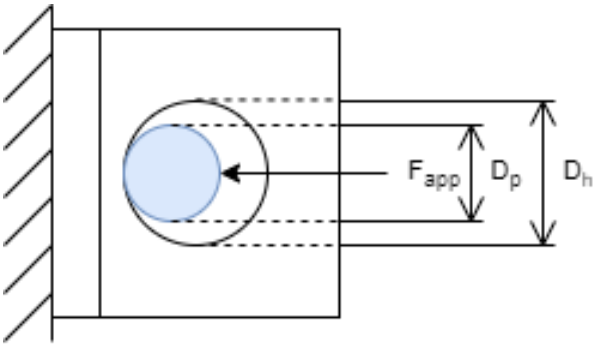


Figure 174: Forward Bracket Diagram

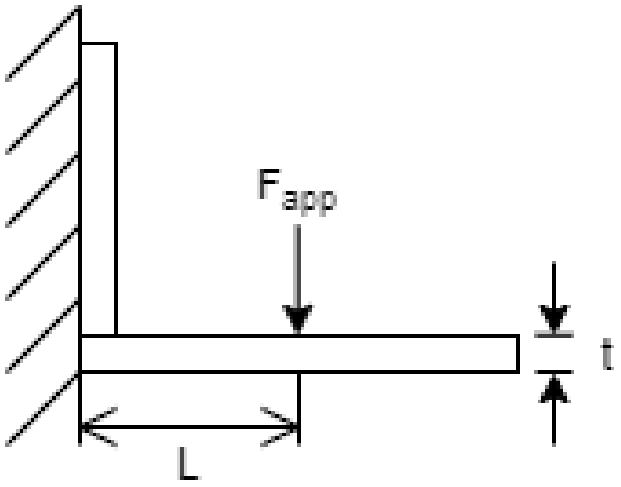


Figure 175: Rear Brackets Bending Diagram

$$F_{app} = 0.833 \text{ kN}, D_p = 25.4 \text{ mm}, D_h = 44.5 \text{ mm}, t = 6.35 \text{ mm}, N_d = 2.0$$

Strength Reduction Factor:

$$C_r = 1 - 0.275\sqrt{1 - \frac{D_p^2}{D_h^2}} = 1 - 0.275\sqrt{1 - \frac{25.4 \text{ mm}^2}{44.5 \text{ mm mm}^2}} = 0.77$$

Bearing Strength:

$$P_{p,u} = 1.25 C_r S_{ty,min} A_p$$

$S_{ty,min}$ is the smaller of the lug and pin tensile yield strength. In this case, it is the pin which is made of A365-T6 aluminium with a yield strength of 165 MPa [39]. A_p is the pin bearing area calculated as follows:

$$A_p = D_p t = 25.4 \text{ mm} \times 6.35 \text{ mm} = 161.29 \text{ mm}^2$$

$$P_{p,u} = 1.25 \times 0.77 \times 165 \text{ MPa} \times 161.29 \text{ mm}^2 = 26.4 \text{ kN}$$

$$SF = \frac{P_{p,u}}{F_{app}} = \frac{26.4 \text{ kN}}{0.833 \text{ kN}} = 31.6$$

J.1.3 Rear Anchor Brackets Bending

$$t = 6.35 \text{ mm}, L = 38.1 \text{ mm}, w = 76.2 \text{ mm}, F_{app} = 514 \text{ N}$$

For a cantilever beam, the largest moment is at the root of the fixed joint and is given by the force times the moment arm.

$$M_{max} = F_{app} L = 514 \text{ N} \times 0.0381 \text{ m} = 19.58 \text{ Nm}$$

The maximum bending stress is given by the equation below:

$$\sigma_{b_{max}} = -\frac{M_{max} y}{I_c}$$

Where $y = t/2$ and I_c is the centroidal moment of inertia given by the following equation for a rectangular cross section:

$$I_c = \frac{h^3 b}{12} = \frac{t^3 w}{12} = \frac{6.35 \text{ mm}^3 \times 76.2 \text{ mm}}{12} = 1625.9 \text{ mm}^4$$

$$\sigma_{b_{max}} = \frac{19.58 \text{ Nm} * 3.175 \text{ mm}}{1626.9 \text{ mm}^4} = \frac{19.58 \text{ Nm} * 0.003175}{1.6269 * 10^{-9} \text{ m}^4} = 382116329 \text{ Pa} = 38.2 \text{ MPa}$$

The brackets are made of 44W steel which have a yield strength of 44 ksi or about 300 MPa which gives a safety factor of [8]:

$$SF = \frac{\sigma_y}{\sigma_{b_{max}}} = \frac{300 \text{ MPa}}{38.2 \text{ MPa}} = 7.8$$

J.2 Thrust Adapter

J.2.1 Bending

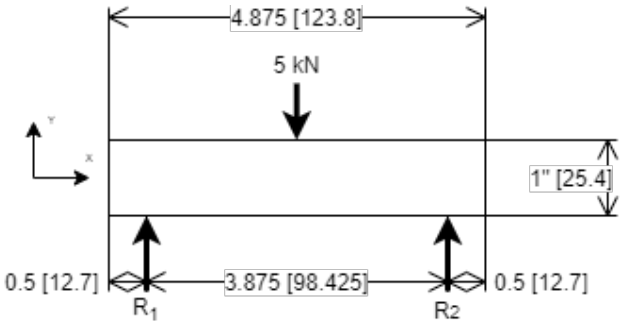


Figure 176: Thrust Adapter Beam Free Body Diagram

By inspection, the reaction forces at R_1 and R_2 are both 2.5 kN which gives the shear diagram below:

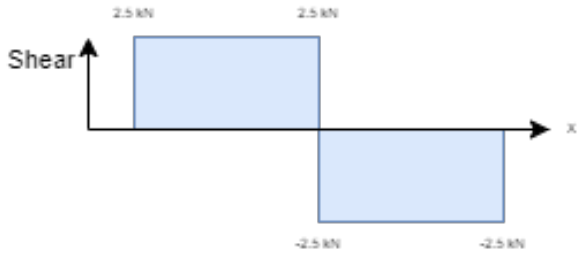


Figure 177: Thrust Adapter Beam Shear Diagram

The maximum bending moment will occur at the center of the beam and is given by the following equation:

$$M_{max} = FL/4 = \frac{5 \text{ kN} \times 98.43 \text{ mm}}{4} = 123 \text{ Nm}$$

This gives the following bending moment diagram:

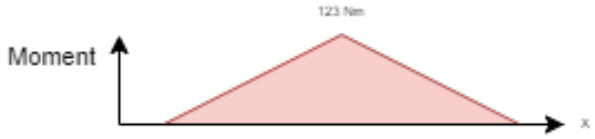


Figure 178: Thrust Adapter Beam Bending Moment Diagram

The maximum bending stress is given by the equation below:

$$\sigma_{b_{max}} = -\frac{M_{max} y}{I_c}$$

Where $y = 12.7$ mm and I_c is the centroidal moment of inertia given by the following equation for a square cross-section:

$$I_c = a^4/12 = (25.4 \text{ mm})^4/12 = 3.468 \times 10^{-8} \text{ m}^4$$

$$\sigma_{b_{max}} = -\frac{123 \text{ Nm} \times 0.0127 \text{ m}}{3.468 \times 10^{-8} \text{ m}^4} = 45 \text{ MPa}$$

The Thrust Adapter Beam is made of 6061-T6 aluminium alloy which has a yield strength of 276 MPa [31].

$$SF = \frac{\sigma_y}{\sigma_{b_{max}}} = \frac{276 \text{ MPa}}{45 \text{ MPa}} = 6.1$$

J.2.2 Column Bending

Using the same worst case off axis angle of 5 degrees as before results in a maximum off axis thrust component of 435.8 N or 217.9 N per column. A free body diagram of the problem is included below.

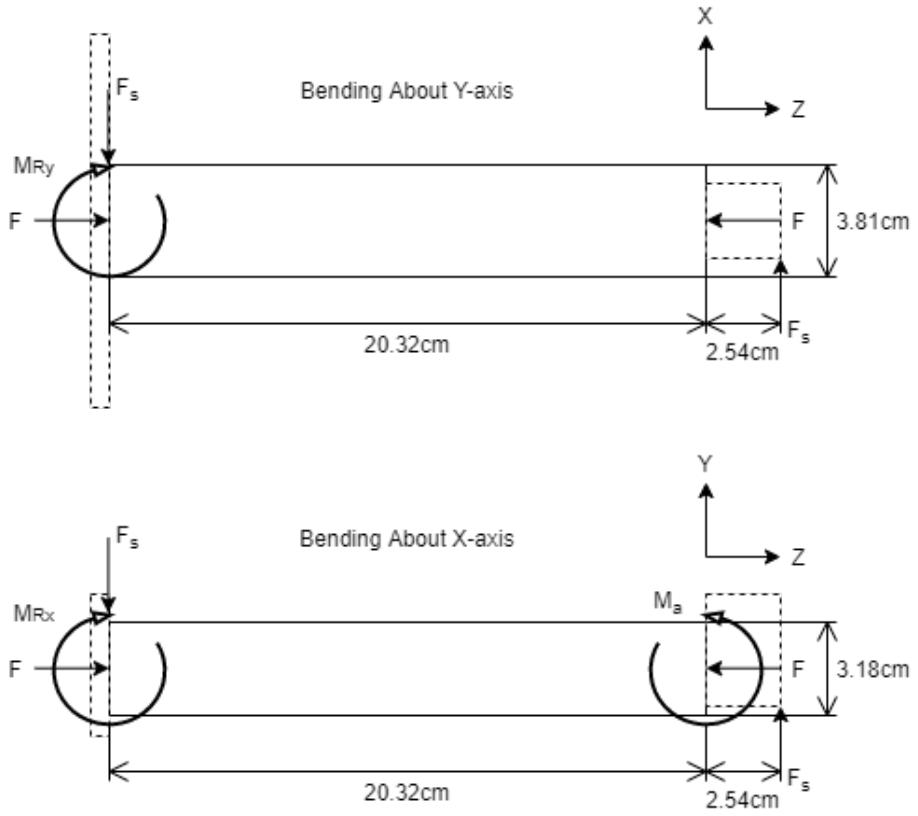


Figure 179: Thrust Adapter Column Free Body Diagram

$$F = 2.5 \text{ kN}, F_s = 217.9 \text{ N}, A = 11.81 \text{ cm}^2, L = 22.86 \text{ cm}$$

$$c_x = 1.588 \text{ cm}, c_y = 1.905 \text{ cm}, I_x = 10.16 \text{ cm}^4, I_y = 14.63 \text{ cm}^4$$

The moment M_a applied on the right end (top) of the column is from the column being eccentrically loaded. The load from the beam onto the column is concentrated towards the inside of the thrust adapter (x-dir). This load is approximated as a triangular distribution which means that the resultant force is located two thirds of the centroidal distance from the X axis. This creates a moment on the column which is calculated as follows:

$$M_a = Fc_x\frac{2}{3} = 2500 \text{ N} \times 0.01588 \text{ m} \times \frac{2}{3} = 26.47 \text{ Nm}$$

The moment applied on the left end (bottom) of the column is the reaction moment of the base of the thrust adapter. The worst case is for the column where M_a and M_{Rx} have the same direction. In the opposite case, the moments M_a and M_{Rx} work against each other which reduces the stress on that column. This is true even when taking into account an extra 1 kN in compressive forces through the column from the moment created by the side force on the Thrust Adapter as a whole. The size of this moment can be calculated by taking the sum of the moment at the center of the left end of the column as follows:

$$M_{Rx} = M_a + LF_s = 26.47 \text{ Nm} + 0.2286 \text{ m} \times 217.9 \text{ N} = 76.28 \text{ Nm}$$

For a cantilever beam bending problem the moment is the largest at the base, and the maximum bending stress can be calculated from the reaction moment as follows:

$$\sigma_b = -\frac{M_{Rx} c_x}{I} = -\frac{76.28 \text{ Nm} \times 0.01588 \text{ m}}{10.16 \text{ cm}^4} = -11.92 \text{ MPa}$$

Including the stress from the thrust load gives the maximum compressive stress in the column. Assuming that the thrust load is evenly distributed through the inner half of the cross-section of the column gives the following maximum compressive stress:

$$\sigma_{c_{max}} = \sigma_b + \sigma_c = 29.67 \text{ MPa} + \frac{2F}{A} = 11.92 \text{ MPa} + \frac{2.5 \text{ kN}}{5.08 \text{ cm}^2} = 16.16 \text{ MPa}$$

The column is made of Aluminium 6061-T6 alloy which has a yield strength of 276 MPa and a shear strength of 207 MPa [31]. Since at the column-base interface the the load will only be on the inner half of column, the maximum compressive stress is doubled.

$$SF_{bending} = \frac{\sigma_y}{2\sigma_{c_max}} = \frac{276 \text{ MPa}}{32.32 \text{ MPa}} = 8.54$$

J.2.3 Bolt Failure

Since the reaction moment at the column-base interface is rather large, this puts a significant load on the bolt connecting the two. This assumes the reaction moment is countered by the moment created by the bolt being loaded in tension and the resultant force of the contact between the two parts is approximated as a linearly distributed load. Taking the sum of the moment about the center-line of the base of the column gives the following:

$$M_{Rx} - F_R \frac{2c_x}{3} = 0 \rightarrow F_R = \frac{M_{Rx}}{\frac{2c_x}{3}}$$

$$F_R = \frac{M_{Rx}}{\frac{2c_x}{3}} = \frac{76.28 \text{ Nm}}{\frac{2 \times 0.01588 \text{ m}}{3}} = 7205 \text{ N}$$

Now taking the sum of the forces along the axis of the column gives:

$$F_R - F_{bolt} - F = 0 \rightarrow F_{bolt} = F_R - F$$

$$F_{bolt} = F_R - F = 7205 \text{ N} - 2500 \text{ N} = 4705 \text{ N}$$

Since the bolt is tapped into an aluminum part, the pullout strength of the internal threads need to be considered. The pullout strength of threads can be calculated as follows [40]:

$$F_{pullout} = \frac{3\pi D_{major,ext} \sigma_{shear} L_e}{4}$$

Where $D_{major,ext}$ is the major diameter of the mating thread and L_e is the thread engagement length. To err on the side of caution, the pitch diameter (d_m) of the mating bolt is used. The bolt is tapped 17 mm into the column and the minimum pitch diameter of an M6 bolt is 5.25 mm.

$$F_{pullout} = \frac{3\pi D_{major,ext}\sigma_{shear}L_e}{4} = \frac{3\pi \times 0.00525 \text{ m} \times 207 \text{ MPa} \times 0.017 \text{ m}}{4} = 43530 \text{ N}$$

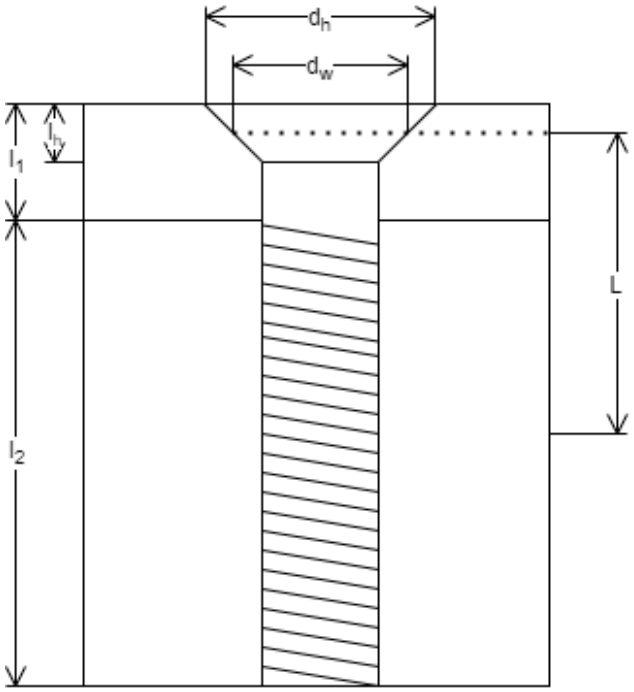


Figure 180: Thrust Adapter Column-Base Joint

$$d_h = 0.012 \text{ m}, l_h = 0.003 \text{ m}, d_{nom} = 0.006 \text{ m}, l_1 = 0.00635 \text{ m}, l_2 = 0.017 \text{ m}$$

Using the methodology outline in the NASA "Preloaded Joint Analysis Methodology for Space Flight Systems" Technical Memorandum the analysis of a tapped joint with a flat-head bolt is as follow [40]:

$$d_w = \frac{d_h + d_{nom}}{2} = \frac{0.0012 \text{ m} + 0.006 \text{ m}}{2} = 0.009 \text{ m}$$

$$L = \left(l_1 - \frac{l_h}{2}\right) + \left(\frac{l_2}{2}\right) = 0.01335 \text{ m}$$

$$K_b = \frac{AE_b}{L}$$

Where K_b is the bolt stiffness, A is the nominal area of the bolt and E_b is the elastic modulus of the bolt. From the nominal diameter of the bolt the nominal area is 2.827×10^{-5} m^2. The elastic modulus of the bolt is assumed to be 200 GPA which is typical for alloy steels [41].

$$K_b = \frac{2.827 \times 10^{-5} \text{ m}^2 \times 200 \text{ GPa}}{0.01335 \text{ m}} = 423.5 \times 10^6 \frac{\text{N}}{\text{m}}$$

$$K_j = \frac{\pi E_j d_{nom}}{\ln\left(5\frac{2L+0.5d_{nom}}{2L+2.5d_{nom}}\right)} =$$

Where K_J is the joint stiffness and E_J is the elastic modulus of the joint. Since both parts of the joints are made of 6061-T6 aluminum alloy they will have the same elastic modulus as each other and in turn the joint[40].

$$K_j = \frac{\pi \times 68.9 \text{ GPa} \times 0.006 \text{ m}}{\ln\left(5 \times \frac{2\times 0.01335 \text{ m}+0.5\times 0.006 \text{ m}}{2\times 0.01335 \text{ m}+2.5\times 0.006 \text{ m}}\right)} = 1022 \times 10^6 \frac{\text{N}}{\text{m}}$$

From this the joint stiffness factor can be obtained as follows[40]:

$$\phi = \frac{K_b}{K_b + K_j} = \frac{423.6\frac{\text{MN}}{\text{m}}}{423.5\frac{\text{MN}}{\text{m}} + 1022\frac{\text{MN}}{\text{m}}} = 0.293$$

$$n = \frac{l_1 - \frac{l_h}{2} + \frac{l_2}{2}}{l_1 + l_2} = \frac{0.00635 \text{ m} - \frac{0.003 \text{ m}}{2} + \frac{0.017 \text{ m}}{2}}{0.00635 \text{ m} + 0.017 \text{ m}} = 0.5$$

Where n is the loading plane factor[40].

From this we can obtain the addition in load on the bolt from the external load on the bolt (F_{bolt} from earlier) as follows[40]:

$$\Delta P_b = n\phi F_{bolt} = 0.5 \times 0.293 \times 4705 \text{ N} = 689.3 \text{ N}$$

The separation load of the joint can also be calculated as follows[40]:

$$P_{sep} = (1 - n\phi)F_{bolt} = (1 - 0.5 \times 0.293) \times 4705 \text{ N} = 4016 \text{ N}$$

This gives us the minimum preload value required to avoid joint separation. Applying a safety factor of 1.5, assuming 10% relaxation, and 25% uncertainty in bolt torque gives a target preload of 8925 N [40]. Taking into account the uncertainty in bolt torque once more gives us a maximum preload of 13387 N. The total axial force sustained by the bolt can be determined as follows:

$$P_b = P_{max,preload} + SF \times \Delta P_b = 13387 \text{ N} + 689.3 \text{ N} = 14076 \text{ N}$$

A grade 10.9 M6 bolt has a tensile strength of 20900 N [42].

$$SF_{bolt} = \frac{20900 \text{ N}}{14076 \text{ N}} = 1.48$$

$$SF_{pullout} = \frac{43530 \text{ N}}{14076 \text{ N}} = 3.09$$

$$SF_{separation} = \frac{TargetPrelaod \times TorqueUncertainty \times Relaxation}{SeparationForce} = \frac{8925 \text{ N} \times 0.75 \times 0.9}{4016 \text{ N}} = 1.5$$

Assuming a nut factor of 0.15 for a lubricated bolt gives the following installation torque [40].

$$T = P_{preload} K d_{nom} = 8925 \text{ N} \times 0.15 \times 0.006 \text{ m} = 8 \text{ Nm}$$

J.2.4 Load cell Contact Stress

The interface between the load-cell and the thrust adapter beam is a circle with a diameter of 8.12 mm. This gives a contact area of 51.78mm^2.

$$\sigma_{Bearing} = \frac{F}{A} = \frac{5000\text{ N}}{51.78\text{ mm}^2} = 96.6\text{ MPa}$$
$$SF = \frac{\sigma_y}{\sigma_{Bearing}} = \frac{276\text{ MPa}}{96.6\text{ MPa}} = 2.85$$

J.2.5 Slipping

Assuming a coefficient of static friction between the aluminum beam and the stainless steel load cell of 0.4, this can be used to obtain a coefficient of static friction as follows [43][44]:

$$F_f = F_N * C_f = 5000\text{ N} \times 0.4 = 2000\text{ N}$$

For a maximum off-axis angle of 5 degrees this gives a safety factor of:

$$SF = \frac{F_f}{F_L} = \frac{2000\text{ N}}{435\text{ N}} = 4.59$$

K P&ID

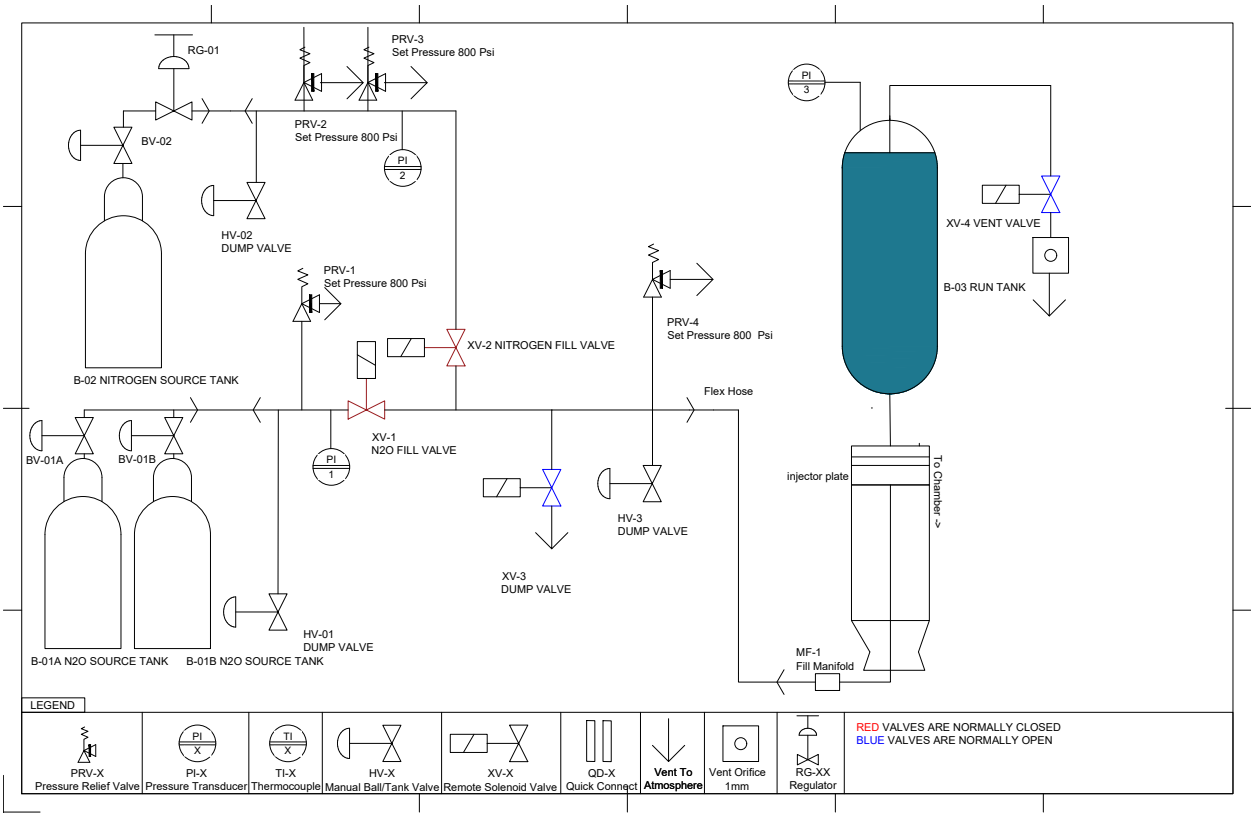


Figure 181: P&ID Diagram for the Test Stand Oxidizer System

	Working Pressure Rating (psi)	Smallest Orifice (in)
1/4” stainless tubing	3600	0.19
3/8” stainless tubing	2900	0.305
3/4” stainless tubing	2900	0.625
1” Aluminum tubing	2100	0.87
MV-100 Valves	3000	0.1
MV-74 Valves	3000	0.1
Process Valves	3000	0.125
Manifold Block	3000	0.3
SS-63 series ball valve	2500	0.5
Oxidizer Tank	2000	N/A
Pressure Relief Valve	6000	0.19
5/16” Nylon Tubing	425	0.21

Relief Valve Sizing

The regulator for the Nitrogen Tank has a Cv of 0.115,obtained from McMaster carr part 6677A13

The nitrogen tank has a pressure of 3000psi when initially filled, so a worst-case scenario will be considered with a full pressure nitrogen tank

The flow rate in nitrogen from a failed-open regulator is calculated using the Swagelok Cv calculator for valve sizing, using the regulator Cv of 0.115, a downstream pressure of the relief valve's fully open position of 1000 psi from their 800 psi setpoint. This results in a value of 163 scfm.

The flow rate from a fully open R3A series pressure relief valve is 100 ft^3/min scfpm. This is converted to a value of nitrogen

$$\frac{100 ft^3}{\min} * \frac{\rho_{air}\ @20c}{\rho_{nitrogen}\ @20c} = 100 \frac{ft^3}{\min} * \frac{0.07516 \frac{lbm}{ft^3}}{0.0724 \frac{lbm}{ft^3}} = 103.81 \frac{ft^3}{\min} N_2$$

This means that two relief valves will be able to vent a failed regulator and keep the system at meop in the case of a regulator failure. To address this, an extra relief valve has been placed on the part of the oxidizer system that is attached to the outlet of the regulator. This insures that if the regulator is to fail, there will always be an adequate amount of relief valves to vent the pressure and present an overpressure.

L Blast Fragmentation

Blast fragmentation calculations were preformed using the method outlined in [45] to calculate the Hazardous Fragmentation Distance (HFD). Calculations were performed assuming 100% decomposition of 25.4 kg of N_2O. This number represents the entire mass of N_2O present at the test site in both the run tank and the source cylinder. The energy of the paraffin fuel is not considered as the nature of hybrid motors prevent propellants mixing until engine firing begins, and because this is the method recommended by [46] for determination of safe distance from a hybrid motor.

The enthalpy of formation of N_2O is known to be 82.05 kJ/mol and the molar mass of N_2O is known as 44.01 g/mol[47]

Calculation for the amount of moles of N_2O:
$25.4 \text{ kg} * \frac{1000 \text{ g}}{\text{kg}} \frac{1}{44.01 \frac{\text{g}}{\text{mol}}} = 577.1 \text{ mol of } N_2O$
Calculation for the amount of energy released by decomposition of 577.1 mol of N_2O:
577.1 mol∗82.05 kJ/mol= 47655kJ
Using a conversion: 4.184 GJ= 1 kg of Trinitrotoluene (TNT) [48]
Converting from KJ of energy to kg of TNT:
$47656 * 10^{-6} \text{ GJ} * \frac{kg \ of \ TNT}{4.184 \text{ GJ}} = 11.48 \text{ kg of TNT}$
Using Equation V3.E3.T2-10 from [49]
$\text{HFD} = -251.87 + [118.56 * ln(11.48 \text{ kg of TNT})] = 166.8 \text{ m}$

The team plans to use a standoff distance of 170 m to avoid the possibility of injury to members in the event of a violent failure of the engine.